

The Seven Wonders of the World

Lecture notes on 21st-century multiphysics

P.G.L. Porta Mana

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pglpm.github.io/7wonders

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$$\frac{dN}{dt} = \mathcal{J} + \mathcal{R} \quad \frac{d\Phi}{dt} = -\mathcal{E} \quad \frac{dQ}{dt} = \mathcal{I}$$

$$\frac{d\mathbf{P}}{dt} = \mathbf{F} + \mathbf{G} \quad \frac{dE}{dt} = W + \mathcal{A} \quad \frac{d\mathbf{L}}{dt} = \mathbf{M} + \mathcal{T}$$

$$\frac{dS}{dt} \geq \Pi$$



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Typeset with L^AT_EX.

No large language models were used in the preparation of this document.

Cover: the *Pale Blue Dot* image of the Earth,
taken by Voyager 1 from right outside Pluto's orbit.

Credit: NASA/JPL-Caltech

<https://science.nasa.gov/resource/voyager-1s-pale-blue-dot/>

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Preface to the student

I don't know what's the matter with people: they don't learn by understanding; they learn by some other way – by rote, or something. Their knowledge is so fragile!

Finally, I said that I couldn't see how anyone could be educated by this self-propagating system in which people pass exams, and teach others to pass exams, but nobody knows anything.

R. P. Feynman 1989

These notes are aimed at undergraduate students in Physics and in Engineering programmes, but graduate students may also find them useful.

You are probably aware of the variety of physics branches, such as mechanics, thermodynamics, chemistry, electromagnetics, fluid mechanics, statics, nuclear physics, and many others. Maybe you are acquainted with some of them. Probably you are also aware of the existence of different physical theories, like Newtonian Mechanics, General Relativity, Quantum Theory, which give different explanations and formulae for the same physical phenomena. Some physical theories are said to be more exact or more approximate than others; Newtonian mechanics, for instance, is an approximation of General Relativity. Does this wild variety of branches and theories also mean a wild variety of principles, methods, mathematical formulae – which you'll have to learn if you want or need to study any particular branch or theory?

Yes, it does.

But there is also a core of very few principles that apply *universally* to every branch and every theory known today, be it mechanics or thermodynamics, Newtonian Mechanics or General Relativity. If you learn these principles, you'll be able to *immediately* work with, and understand, at least the general features of *every* new physical discipline, phenomenon, or technology that you might meet.

The main goal of these notes is to make you acquainted with this core of few physical principles.

How few are these principles? Around *seven* (the exact number depends on how we arbitrarily group or separate them). This is the reason for the title of these notes. These seven principles are quite amazing for various reasons:

- They apply to *every* physical phenomenon, as already mentioned.
- Their meaning is very intuitive: each of them expresses a sort of budget.
- They are responsible for, so to speak, “driving the universe forward in time”. More precisely, they are the basic principles that allow us to simulate and make predictions about physical phenomena.
- Their mathematical formulation is *exactly the same* in all our main approximate and exact theories, like Newtonian mechanics and General Relativity. This means that if you learn how to apply them to a tennis ball, then you are also able to apply them to a black hole.

These universal principles are, obviously, expressed mathematically. Their consequences and applications can be studied, to some degree, by using analytical methods; that is, by methods involving mathematical operations that we can do by hand. But their most fascinating and practical applications need numerical methods, that is, methods involving programming and computer simulation.

In my opinion, if you learn how to apply these universal principles in simple simulations, then you also better understand their meaning and the way they work. For this reason these notes mainly take a computational approach. We shall learn how to implement these universal principles in simple computer code, and see the fun variety of physical phenomena they lead to. Scripts that accompany these notes can be found at

<https://pglpm.github.io/7wonders/>

Thanks

I am deeply indebted to the students attending the courses ING 175, ING 174, ING 162 in Bergen. They allowed me to experiment presenting physics as done in these notes, and gave invaluable feedback about what’s difficult and what’s easy, what’s unclear and what’s clear, with many

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Overview

事事無礙

§ 0.1 The plan of these notes

After some very general remarks about physics, we recall the notions of **physical quantity**, unit, and physical dimension. This is really just a reminder; it's assumed that you are already familiar with these general notions.

We then survey two cardinal physical notions: **time** and **space**. We briefly examine their fascinating nature as we today understand it. The notion of **coordinate system** is also introduced.

Thereafter we take an overview of the main seven physical quantities we shall work with: **matter**, **energy-mass**, **momentum**, **angular momentum**, **entropy**, **electric charge**, **magnetic flux**, discussing some features common to all of them. Other important quantities such as temperature are also introduced.

The most important feature of the main seven quantities is that we can measure their amount within any volume, and the amount that passes through any surface. We therefore study the intuitive ideas of **control volume** and **volume content**, **control surface** and **flux**, and **supply**.

At this point the stage is ready for the introduction of seven physical laws which are just expressions of **balances** of volume contents, fluxes, and supplies of the seven quantities introduced earlier. These seven balance laws are **universal**. They are connected by mathematical expressions called **constitutive relations**, which are not universal but depend on the particular physical phenomenon, and on the particular physical theory we use to describe it.

The balance laws are the ones that allow us to predict how the values of different quantities change with time. We show this by exploiting them to code numerical simulations that can in principle be done for any physical

system. The role that constitutive relations have in connecting the seven balances also becomes quite clear in the simulation code.

The seven universal balances are then examined in turn. For each balance, we study some constitutive relations that are often used together with it, as well as some of its physical applications.

§ 0.2 Prerequisites

Physics

Just some vague reminiscences of secondary/high-school physics should be enough. It can be beneficial if you are familiar with basic physics notions like *velocity*, *mass*, *force*, and similar ones.

Maths

- Working familiarity with algebra, its operations and their properties.
- Working familiarity with solving equations and inequalities, linear and non-linear.
- Working familiarity with the study of functions of one real variable.
- Working familiarity with derivatives.
- Understanding of what an integral is, even if you won't be required to solve integrals.
- Working familiarity with vector calculus.
- Some familiarity with functions of many variables.
- Understanding of what partial derivatives are.

Programming

- Knowing what a computer program is.
- Working familiarity with variables in a computer program.
- Working familiarity with for- and while-loops.
- Working familiarity with outputting and plotting the results of a computer program.

§ 0.3 Features of the text

Graphical devices

The text includes the following graphical devices:

- Important concepts and definitions are also given in **boldface**.
- Important-concept boxes:



Some important notion or definition

This is a definition or explanation of Something.

- Warnings and important points that require careful thinking:



Careful!

Something you must be careful about.

- Exercises:



Exercise 0.1

This isn't really an exercise

- Discussions and connections with more advanced physics:



How things really are in quantum physics

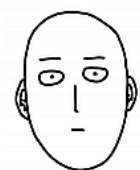
Just for your curiosity.

Side figures and quotes

Figures, graphs, or quotes related to the material are displayed on the right.

Cross-references

The text gives cross-references to the main section where a topic was or will be discussed. Cross-references appear as an eye icon followed by sectioning number and page in round brackets, like this cross-reference to [conservation laws](#) ▶ §5.4 p. 133.



This is an image of Saitama, which actually has nothing to do with the text on the left.

The symbol '§' stands for 'section'.

Hyperlinks and bibliography

Some pieces of text are hyperlinks, like this one about [One Punch Man](#)¹. You recognize them from their different colour and from the little footnote number that follows them. The links' URLs are also listed at the end of each chapter, in case you're reading a printed copy of these notes.

The text gives bibliographic references, like "Einstein [1905a](#)", to scientific literature. The references are listed in the final Bibliography on page [427](#). Bibliographic references are given for two reasons:

- For your own curiosity.
- To back up what's written in the text. In science you should not believe something just because you've read it somewhere. You should, as much as possible, *go and check for yourself the logic and the experimental evidence behind the statement.*

"Believe nothing, O monks, merely because you have been told it, or because it is traditional, or because you yourselves have imagined it. Do not believe what your teacher tells you merely out of respect for the teacher."

(attributed to Gautama Buddha)

§ 0.4 Code for numerical simulations

Why code?

As we learn about physical laws and equations, we shall explore and apply them both *analytically*, that is with pen & paper, and *numerically*, that is by writing and running code in a programming language. Why using code? For three different reasons.

First, a larger and larger part of physics and engineering research and applications requires the numerical solution of physics equations, by means of code in different programming languages. Code is used even for problems that could be solved with pen & paper, because doing so minimizes the possibility of human calculation errors. So it's good if you get familiar with "translating" physics problems into code and solving them by running code.

Second, many – we can say the majority – of physics phenomena cannot be fully studied analytically. But many of them are easy to simulate and study numerically. Therefore using code allows you to explore many additional interesting physical phenomena.

Third, writing code allows us to see more clearly the different roles that different physical laws have in physical phenomena, as well as the interrelationship among them.

The third reason above is the most important for these notes. This is also why our goal is not to write *efficient* code, but to write code that allows us to *understand the physics*.

Octave

The scripts for numerical simulations presented in the text are written in *Octave*². Why *Octave* for these notes? Several reasons:

- It's free and open source.
- It runs on all major operating systems.
- It includes a graphical user interface and development environment, where you can edit your code and see resulting plots.
- You can run it immediately, without needing to compile it first. This means that it's slower than some other programming languages; but as stated above, our goal is not efficient code, but code that makes us understand physics.
- You can use it directly, without loading extra libraries.
- No need to worry about "environments" or version dependencies.
- Its syntax is very similar to MATLAB's, which is a language used in some engineering courses; typically you can run Octave's code in MATLAB and vice versa.
- Its syntax is very close to physics's mathematical notation. For instance, a sum of two vectors expressed mathematically as

$$[1, 3, 2] + [5, 7, 6]$$

is written in Octave like this: `[1, 3, 2] + [5, 7, 6]`.

Compare the same expression in other programming languages:

Python: `np.array([1, 3, 2]) + np.array([5, 7, 6])`
(you must load the numpy library first).

JavaScript: `math.add([1, 3, 2], [5, 7, 6])`
(you must load the math.js library first).

R: `c(1, 3, 2) + c(5, 7, 6)`.

Rust: `Array1::from(vec![1, 3, 2]) + Array1::from(vec![5, 7, 6])`
(you must load ndarray).

C, Visual Basic, Perl, Go: between 5 and 10 lines of code.

Keep in mind that we are not saying that Octave is the best programming language for physics; what's best depends on the specific physics application. We're just saying that Octave is particularly convenient for the goals we have in these notes.

You can find many introductions to Octave online. Check out its own [documentation](#)³ for example.

Code colours

The scripts usually present two colours:

- [Blue lines](#) are strictly related to the physical simulation.
- [Grey lines](#) take care of auxiliary operations, like plotting the results or saving them to file.

Other programming languages

Feel free to translate the Octave scripts used in these notes into your favourite programming language; that's a great exercise! Please do get in touch if you want to make them available in these notes. For example, some [Python](#)⁴ versions are also given for some of the scripts, linked on the margin. Note that they need the [NumPy](#)⁵ and [Matplotlib](#)⁶ packages. The Python scripts should work on [matplotlib.online](#)⁷.

§ 0.5 Additional exercises

Additional exercises, and accompanying solutions for some of them, are available at

<https://pglpm.github.io/7wonders/>

§ 0.6 Large Language Models

Refer to *The Onion*'s excellent [tips for using AI](#)⁸.

§ 0.7 Notation and terminology

Mathematical notation follows as much as possible the standards given by the [International System of Units \(SI\)](#)⁹, listed for example in [iso 2009](#)

and ISO 2019. One slight exception are [physical dimensions](#) ^{§1.7 p.36}, whose notation will be explained in that section.

In physics, engineering, and most other scientific fields *there is no universal standard for symbols*. It's therefore important that you **focus on the concept** that a symbol stands for, rather than the symbol itself. Otherwise you'll face many difficulties reading literature and other scientists' reports.

The present notes try to use as much as possible the symbols recommended by the [International Organization for Standardization \(ISO\)](#)¹⁰, which are often also adopted by other international bodies like the [International Bureau of Weights and Measures \(BIPM\)](#)¹¹, the [Institute of Electrical and Electronics Engineers \(IEEE\)](#)¹², or the [International Union of Pure and Applied Chemistry \(IUPAC\)](#)¹³. Some changes are unavoidable, because different disciplines often use the same symbol for different quantities. For instance, the symbol '*E*' is used in many fields for 'energy', in mechanics for the 'Young modulus' (a mechanical characteristic of elastic materials), in electromagnetics for the 'magnitude of the electric field'.

The most common symbols used in the present notes are listed in [table 0.1](#) on [page 22](#).

Table 0.1 List of common symbols

(Greek symbols)

κ	(kappa)	gravitational constant
μ	(mu)	viscosity coefficient
μ_k		coefficient of kinetic friction
μ_s		coefficient of static friction
Π	(capital pi)	entropy flux
ρ	(rho)	molar mass
Φ	(capital phi)	magnetic flux
ξ	(xi-dot)	rate of conversion
Ω	(capital omega)	angular-velocity matrix
ω	(omega)	angular-velocity vector

(Latin symbols)

A		usually: area; in some contexts: just a coefficient
$\mathcal{A}$	(script capital a)	energy-mass supply
C		usually: molar heat capacity; in some contexts: just a coefficient
c		usually: speed of light; in some contexts: just a coefficient
E		energy-mass content
$\mathcal{E}$	(script capital e)	electric tension (line integral of electric field strength)
$\mathbf{F}$		momentum flux (surface force)
$\mathcal{F}$	(script capital f)	stress vector, traction.
$\mathbf{G}$		momentum supply (volume force)
$\mathbf{g}$		acceleration vector of free fall
h		surface coefficient of heat transfer
$\mathcal{I}$	(script capital i)	electric current
J_c	(capital j sub c)	moment of inertia with respect to mass-centre
$\mathcal{J}$	(script capital j)	matter flux
k		elastic constant
$\mathbf{L}$		angular-momentum content
$\mathbf{M}$		angular-momentum flux (surface torque)
m		mass-energy content, typically at rest
N		matter content (amount of substance)
$\mathbf{P}$		momentum content
p		pressure
$\dot{Q}$	(q-dot)	heat flux
Q	(script capital q)	electric-charge content
R		usually: molar gas constant; in some contexts: radius
$\mathcal{R}$	(script capital r)	matter supply
$\mathbf{r}$		coordinate-position vector
S		entropy content
T		thermodynamic temperature (in kelvins)
$\mathcal{T}$	(script capital t)	angular-momentum supply (volume torque)
t		coordinate time
U		internal energy
V		volume
$\mathbf{v}$		coordinate-velocity vector
W		energy-mass flux
x		position coordinate, typically horizontal
y		position coordinate, typically horizontal
z		position coordinate, typically vertical

URLs for chapter 0

1. <https://onepunchman.fandom.com>
2. <https://octave.org>
3. <https://docs.octave.org/latest/>
4. <https://www.python.org>
5. <https://numpy.org>
6. <https://matplotlib.org>
7. <https://matplotlib.online>
8. <https://theonion.com/tips-for-using-ai/>
9. <https://www.nist.gov/pml/special-publication-811>
10. <https://www.iso.org>
11. <https://www.bipm.org>
12. <https://www.ieee.org>
13. <https://iupac.org>

Physics, quantities, units

Philosophy is written in this grand book, the universe, which stands continually open to our gaze. But the book cannot be understood unless one first learns to comprehend the language and read the letters in which it is composed. It is written in the language of mathematics, and its characters are triangles, circles, and other geometric figures without which it is humanly impossible to understand a single word of it.

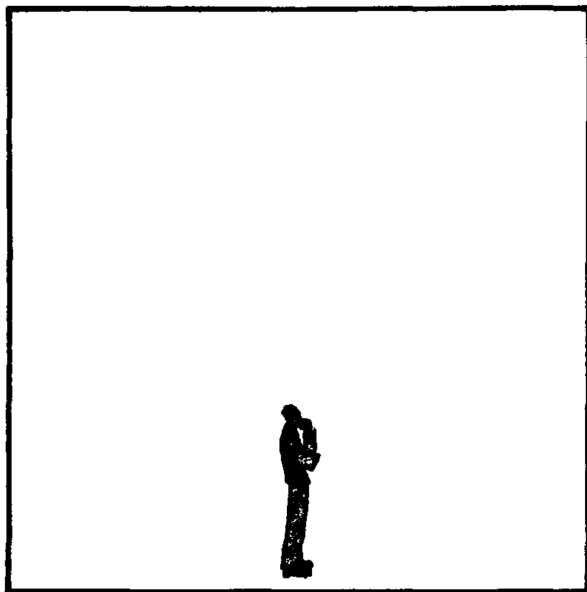
G. Galilei 1623

§ 1.1 Physics?

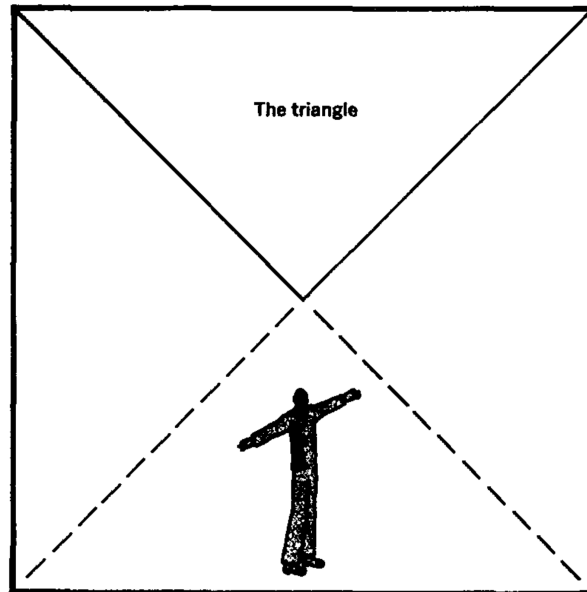
If you think about it, many things we ordinarily do every day are some sort of magic. Think of how you can instantaneously see and speak with a person living on another continent, in real time, using just a small widget in the palm of your hand. Think of how you can instantaneously see where you are on the Earth, using the same widget. Think of how fast you can go to another country, by flying in a huge metal thing. Think of how you can command and interact with a purely fictitious animated world when you play on your computer. The list can go on forever. Other things are luckily less ordinary, but still inspire a lot of awe: think of the devastating power unleashed by something roughly as small as a tennis ball, in an atomic bomb.

We can do these astonishing things thanks to our understanding of how the world works. That's Physics.

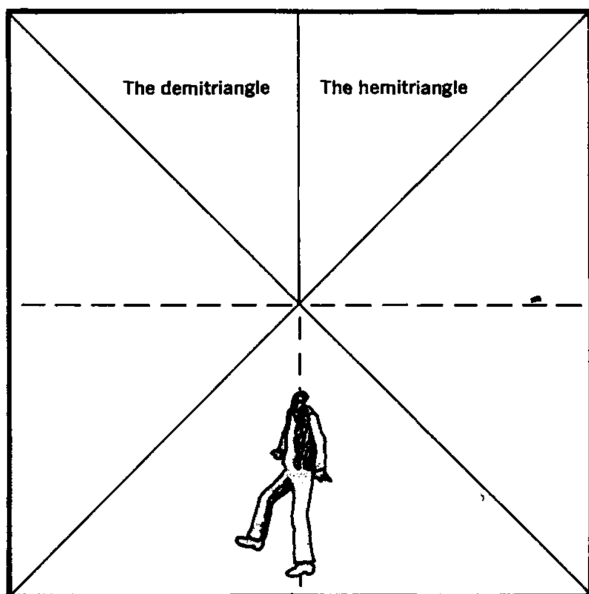
Many things can be said and have been said about science and physics. Rather than repeating what's been already written in many excellent books, I invite you to take a break here and go read their introductions. Choose as you please; compare what they say; don't limit yourself to popular books.



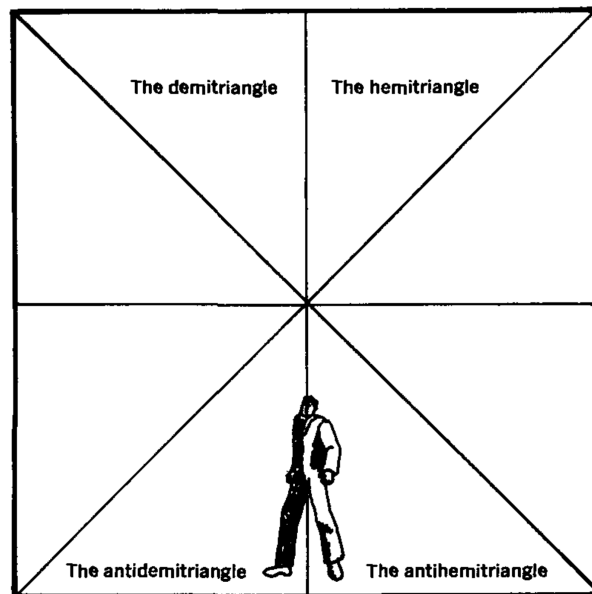
10 000 BC. The inhabitants of the paper square have no conception of the true nature of the universe they inhabit.



1900 AD. Physicists of the square discover a basic subdivision of their universe. They call it the "triangle" and consider it to be the fundamental building block of the universe.

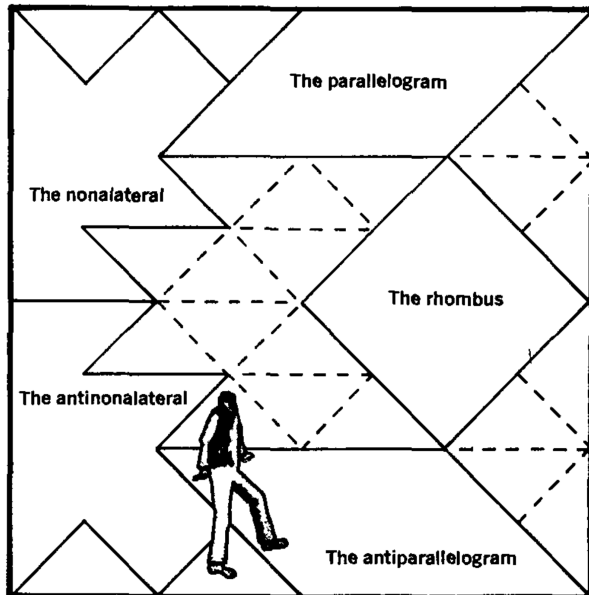


1930 AD. Physicists discover that the triangle can be split. Its parts are termed the "hemitriangle" and the "demitriangle." These are thought to be the fundamental building blocks of the universe.

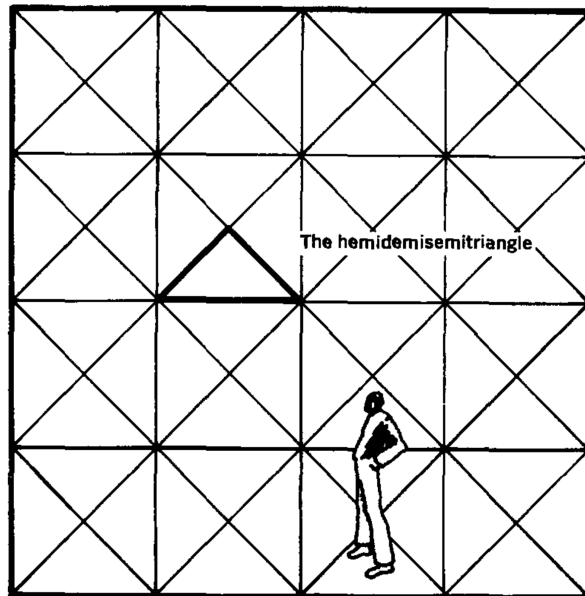


1950 AD. Mirror images of the hemitriangle and the demitriangle are discovered. These are termed "antihemitriangle" and "antidemitriangle."

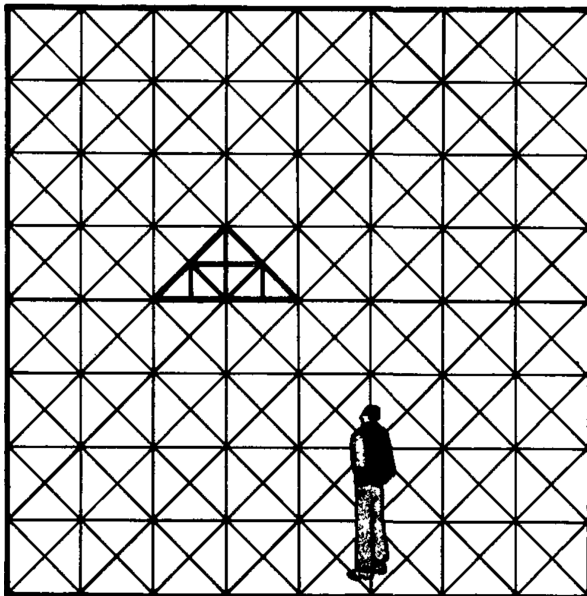
Figure 1.1 (Continues on p. 27) *The progress of "fundamental" physics*, from Chew 1970 as reproduced in Truesdell 1987



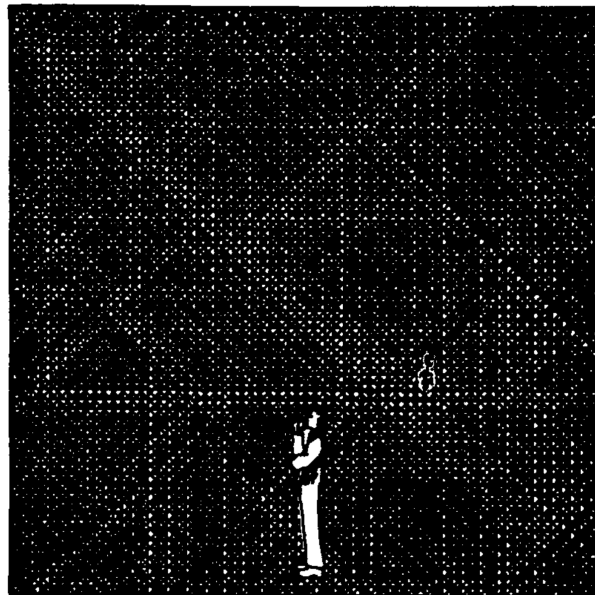
1960 AD. Physicists' conception of their universe is further clouded by new discoveries: the rhombus, the parallelogram, the antiparallelogram, the nonalateral and many others. It is unclear what these discoveries signify.



1970 AD. A new configuration, the "hemidemiseitriangle," is hypothesized, out of which all known configurations of the universe can be constructed. The hemidemiseitriangle is thought to be the fundamental building block of the universe.



1975 AD. The hemidemiseitriangle is discovered. The following year the hemidemiseitriangle is split.



2000 AD. The inhabitants of this paper square have no conception of the true nature of the universe they inhabit.

§ 1.2 What is “fundamental” physics?

But what’s the “ultimate” goal of physics? What’s “fundamental” physics? The answer to this question is again subjective – also in this case physics lets you express your proclivities and personality. In the history of physics one can probably identify two main conceptions of “fundamental” physics.

For some physicists it is about finding the ultimate building blocks, so that one day we can say “. . . and these are the constituents, and they obey these equations”. The history of physics seems to show that this goal is overturned every few generations. And yet every generation says “*Now* we almost have the complete picture – it’s right behind the corner. It’s true that previous generations thought they almost had it, and turned out to be wrong. But *this time* is different, this time we have the real deal!”. The theoretical and particle physicist [Geffrey Chew](#)¹ depicted this situation as in fig. 1.1. For this reason some physicists are a little sceptical about this goal; maybe it’s a never-ending structure, with surprises at every deeper look.

So for other physicists fundamental physics is about finding some point of view or mathematical structure that is rich enough to make useful predictions, and yet flexible enough to accommodate any new patterns or objects that we might discover. In a manner of speaking, it is about finding “patterns of patterns” or “laws about physical laws”.

The two conceptions above are not mutually exclusive, and both are always pursued, even if time-changing fashions may emphasize the one or the other.

In these notes we take a point of view slightly closer to the second conception. This will also be reflected in the main division between two kinds of physical laws that we’ll draw in chapter 5.

§ 1.3 Physical laws

What’s a physical law?

Our lives rely on many kinds of patterns and regular experiences – so common that we take them for granted and don’t even think about them most of the time. If you leave an object in your room in the morning, say a book or a pair of keys, and go out, then when you come back you expect the object to be exactly where you saw it, if you know that nobody entered your room in the meantime. While you go on the street you don’t expect

to suddenly levitate or flay away. While you write on your notepad, you're not afraid that it might suddenly disappear into thin air. The way we use everyday devices of all kinds relies on predictable behaviours and responses on their part.

All these patterns and regular behaviours that we observe and use are the essential core of physical laws.

A **physical law**, in the more technical meaning of this term, goes beyond the simple observation or use of such experiences in at least two ways:

First, a physical law goes from qualitative to **quantitative** observations and expressions. Instead of saying "the book is still on the table, where I left it", it might state "the mass-centre of the book was at position $(x, y, z) = (0.6, 3.1, 1.2)$ m at times $t_0 = 07:50:31$ and $t_1 = 17:14:40$ ". The use of numbers allow us to convey information in a concise and precise way. Imagine you have to tell someone, who doesn't know Bergen, where in Bergen you are right now, to within 10 m. You can do that with a description: "... and there's a building called so-and-so which looks like so-and-so. . .", which would be lengthy and tricky. Or you can just give two numbers: latitude and longitude

60.369 40, 5.3518 .

And in these two numbers all digits are important; for instance, the latitude is not 60.369 47. The use of mathematics leads to amazing precision and predictive powers. It is this precision, which increases every day, that allows for the operation of technologies so important in our everyday lives: from mobile phones to aeroplanes, from laptops to cars.

Second, a physical law goes from very specific experiences to more **general** patterns, which may encompass seemingly very diverse phenomena. Instead of saying "this book is still here" it might say "this kind of matter is conserved in any volume and for any amount of time". Instead of seeing the falling of an apple and the reciprocal movement of planets and sun as two different kinds of phenomena, it summarizes both together as examples of only one physical phenomenon. This generalization leads to the recognition and understanding of many new patterns.

In being more quantitative and general, physical laws often become more **abstract** than everyday observations. Objects like a book and a body of water get included into the more abstract notion of 'matter'. The movement of different kinds of objects and the propagation of light get included into the more abstract notion of 'momentum'.



Some things we don't expect to be possible (image: *Flying Lesson* by S. & R. ParkeHarrison²).

"There is nothing that can be said by mathematical symbols and relations which cannot also be said by words. The converse, however, is false. Much that can be and is said by words cannot successfully be put into equations, because it is nonsense."

Truesdell 1966

To study and use physical laws we must therefore become familiar with and proficient in seeing and describing physical situations in a more abstract way, and to translate them into mathematical terms. This is what we shall learn in the next chapters 2, 3, 4, so that in chapter 5 we shall finally learn and start to use physical laws.

Many different mathematical formalisms

The mathematics used to express physical laws can come in wildly diverse forms, using wildly diverse principles. This leads to what we may call “different physics languages”; a more technical name is “physics formalisms”. One may approach a physics phenomenon or problem in terms of *Lagrangians*, or *Hamiltonians*, or *fibre bundles*, or *categories*, or *action principles*, or many other formalisms. These formalisms or languages are not completely separated; we know how to translate among them. In “doing” physics, we may jump among formalisms, because some physical situations may be easier to express, or some results easier to find, in one formalism than another. No matter which physics formalism you choose, the results and the concrete applications are still the same. The choice is to a great extent subjective, based on your aesthetic tastes. In “doing” physics you can actually express your personality and put your own artistic touch; this is why it’s such a cool subject; and other scientific subjects are like this too.

In these notes I choose one particular formalism: the one that for me is the most easily *visualizable*; because I believe that visualization can be beneficial in learning new things. Or maybe I choose it just because I like it best. I encourage you to explore how the physics you’ve learned is expressed in other physics formalisms; maybe you’ll like another physics formalism better.

The formalism we’ll use might be called “field theory”. Roughly speaking it takes as starting point the ideas of space and time, or better spacetime, in which there are different kinds of “stuff”. It expresses the regularity and patterns that we observe in physical phenomena as “budgets” about the different kinds of stuff, and as relations between them. Please don’t take the description just given too literally; it’s just meant to give you a very vague idea of the field-theoretical viewpoint.

$$\delta \int L dt = 0 \quad L = \frac{1}{2}mv^2$$

$$\mathbf{F} = \frac{d}{dt}m\mathbf{v} \quad \mathbf{F} = 0$$

Example of two different formalisms (red, blue) expressing the same physical phenomenon.

§ 1.4 Physical quantities

As discussed in the previous section, in order to formulate and use physical laws we need to view all physical phenomena in and around us from a more abstract point of view, which more easily lends itself to quantification. This is done by using the concept of **physical quantity**.

The [Joint Committee for Guides in Metrology \(JCGM\)](#)³ defines ‘quantity’ as follows⁴:

property of a phenomenon, body, or substance, where the property has a magnitude that can be expressed by means of a number and a reference.

As you see it’s a somewhat vague definition, but it clearly refers to the possibility of quantification. Some physical quantities are connected with familiar concepts that we use every day, like time, position, distance, velocity, temperature, energy-mass – but we must be careful because their use in physics is more rigorous and often has technical features that the everyday familiar use has not. Other physical quantities, like momentum, magnetic flux, entropy, are more abstract and disconnected from everyday concepts. Their intuition and use therefore require practice and carefulness.

As stated in the definition above, all physical quantities are expressed as a number together with a *reference*, also called a **unit of measurement** or simply *unit*. The unit is a basic standard for comparing the measurement of a quantity. For example, when we say that a table has a *length* of two metres, written ‘2 m’, we mean that it is as long as two of those standard lengths that we call ‘metre’. We shall further discuss units of measurement in §1.7. The number and unit together are handled by usual mathematical rules; so we can for example add or multiply physical quantities. Not all mathematical operations are allowed on all quantities, though.

§ 1.5 Scalar and vector quantities

Some physical quantities can be specified by giving only *one* number together with a unit. Such a quantity is called a **scalar**.

Other physical quantities require the specification of *several* numbers, corresponding to different spatial directions (and sometimes even to a “time” direction), with associated units. Such a quantity is called a **vector**, and can be graphically represented by an arrow. These numbers and their units are called the **scalar components** or the **coordinates** of the vector. They are usually three, one for each spatial dimension. In some applications we can neglect one or two spatial dimensions, and the vector will therefore have only two or one scalar components. In some applications, time is also treated as a fourth dimension in space-time, and we can therefore have vectors, often called *four-vectors*, having four components. The values of the scalar components of a vector depend on the coordinate system under use; we shall therefore say more about them when we discuss coordinate systems in chapter 2.

This terminology is established in ISO 2019.

Examples of **scalar** quantities:

Time When we check the time on our clock, or read a time lapse on a stopwatch, we only read one number; for instance 16:52:01 or 90 s.

Distance The distance between us and some place along a given path is a property of the path, also called the *length* of the path, and is expressed by one number. It could be, say, 100 m.

Energy The amount of chemical energy in packaged food is given by one number, say 800 000 J (roughly equivalent to 191 kcal). Also the total amount of energy radiated by a light bulb in a given time is given by one number, say 10 J/s.

Temperature When we take our own temperature, or check what’s the temperature outdoors or indoors, we just read one number from the thermometer, say 25 °C.

Examples of **vector** quantities:

Position To specify the position or location of an object we usually need two or three numbers. Imagine for instance a pen lying on the floor in a room. We could specify its position by reporting its distance from the east wall and its distance from the north wall – only one of the distances wouldn’t be enough. For an object floating in air, we would also need to specify its distance from the floor. We could report this position as, say, [1 m, 2 m, 0.3 m].

A GPS device gives our position on Earth also in terms of two or three numbers, called *latitude*, *longitude*, *altitude*; for example, the GPS could read $(60.370^\circ, 5.350^\circ, 74.4 \text{ m})$.

Velocity expresses not only how fast something is moving, but also in which direction. This information can equivalently be expressed by three numbers, as we'll discuss in §2.8.

Force and momentum are also related to motion, as we'll discuss at length in later chapters. They therefore require a direction as well as a number, or equivalently three numbers.

There are also quantities that must be specified by more than three numbers and units, collected together into a sort of matrix or higher-dimensional array. Such a quantity is called a **tensor**.

Position vs distance, velocity vs speed

Mark the difference between *position* and *distance*. The first is a vector, because it must give information about 3D space. The second is a scalar, because it's associated to a specific path between two positions (often implicitly understood to be the shortest path). In some cases there may be a simple mathematical relation between distance and one or more positions.

Mark also the difference between *velocity* and *speed*. The latter is defined as the magnitude of the former, and is therefore always a *positive* scalar.

What's scalar or vector or tensor depends on the theory

Scalar, *vector*, *tensor* have specific and slightly different meanings in different theories. It is therefore important that you don't take the classification used in these notes as universal.

For example, in these notes and in Newtonian mechanics we call *energy density* a scalar, but in General Relativity it isn't a scalar: it is one component of a vector, or even of a tensor.

§ 1.6 Derived and primitive quantities

We must briefly discuss a way of categorizing physical quantities which is important for understanding our approach to study physical laws. It's the distinction between *derived* and *primitive* quantities. To some extent this is an arbitrary distinction; that is, we can decide, within some bounds, which quantities we consider as 'derived' and which as 'primitive'.

A **derived quantity** is one that we decide to define in terms of other quantities. For example, we can define *speed* v (more precisely: average speed) as the ratio between a *distance* d and a *time duration* t :

$$v := \frac{d}{t}$$

where the symbol " $:=$ " means "is defined as" or "is defined by". Note how we are already treating and representing 'speed', 'distance', 'time duration' as mathematical objects, and doing mathematical operations on them.

Treating *speed* as a derived quantity means that we could in principle avoid using the word 'speed' and the symbol ' v ' altogether, and instead always speak about 'distance' and 'duration', always using the symbols ' d ' and ' t ' in the combination ' d/v ', instead of using v . Doing so would probably lead to very long sentences and formulae, and therefore be extremely inconvenient; but in principle it could be done. The definition of a derived quantity often tells us how that quantity can be measured.

A derived quantity is defined in terms of other quantities, and these may in turn be derived quantities, defined in terms of still other quantities, and so on. But at some point this chain of definitions must come to an end, otherwise we would go around in circles. A **primitive quantity** or **base quantity** is one that we decide *not* to define in terms of other quantities. That it is not defined in terms of other quantities doesn't mean that we cannot try to explain it. But such explanation must be taken as informal and heuristic. Primitive quantities are often explained through metaphors or by appealing to intuition; but you must always be a little wary of such explanations, because they may fail you spectacularly in some situations. Primitive quantities are the building blocks from which we define all other quantities.

We have therefore a choice about which quantities to take as primitive and which to take as derived. For instance we can define *energy* in terms of quantities like *work* and *heat*, which in turn need to be defined in terms of others, or to be taken as primitive. Or we can take *energy* as primitive, and define *work* and *heat* in terms of it and of other quantities. The latter choice



"you have to be in some framework that you allow something to be true. Otherwise you're perpetually asking 'why'". (see [video⁵](#)).

can be more convenient for stating a physical law and for developing a physical theory. It often happens that a quantity, if taken as primitive, is very convenient for building a theory, but difficult to understand intuitively. Vice versa, a quantity can be very intuitive, and therefore convenient as a primitive; but it leads to a complicated theory. We also have some choice in deciding *how many* quantities to take as primitive. Taking as few as possible primitive quantities can actually make the development of a physical theory more difficult, requiring very complicated definitions of the derived quantities.

In our study of physical laws we shall take these quantities as primitive:

time
space
matter
magnetic flux
electric charge
energy-mass
momentum
angular momentum
entropy

Time and *space* are taken as primitive quantities – for obvious reasons – in almost all current physical theories. We shall discuss them in chapter 2. The other seven physical quantities, which we shall discuss in chapter 3, are chosen because of several advantages:

- They can be understood physically, and treated mathematically, in a similar way. Therefore as we get familiar in thinking about and handling any one of them, we automatically get also familiar with all the others.
- The way in which they can be understood and handled is intuitive and lends itself to mental visualization.
- They lead to physical laws that have an almost identical expression. And, as mentioned before, this expression represent a sort of “budget” and is therefore intuitive.
- These laws are common to all our physical theories, and they can be expressed in a mathematical form that’s the same in any theory.

The price to pay for the advantages above is that some of these quantities may be less familiar than others; but the advantages seem to outweigh this disadvantage.

We shall also use other quantities, some of which are very familiar, like *temperature* or *pressure*. But the quantities listed above are our fundamental building blocks.

§ 1.7 Physical dimensions and units

The topics of measurement, physical dimensions, and units, which are studied in *metrology* and in *dimensional analysis*, could occupy an entire course by themselves! Here we shall overview the basics. You are welcome to take a look at the information available at the [International System of Units \(SI\)](#)⁶ and at the [NIST guide for the use of the SI](#)⁷ for more details.

Measurement is the process by which we determine the value of a physical quantity. Measurements can be extremely complex, and can be extremely different even if they are about the same quantity. Consider the ways we can measure the mass-energy of a football, compared to the ways we can measure the mass-energy of the Sun.

To each quantity we associate a **physical dimension**. The term ‘dimension’ here has nothing to do with physical extension, as in “the dimensions of this box”; be careful not to confuse the two. Usually it’s clear which one is meant from the context.

Physical dimensions reflect the mathematical relationships that exist between physical quantities. For instance, we take *distance* to have dimension length, and *time interval* to have dimension time; if we now define *speed* as *distance* divided by a *time interval*, then the physical dimension of *speed* is length/time. Physical dimensions help us to avoid doing mathematical operations that don’t make sense with some quantities. For example, it doesn’t make sense to sum up the volume of a glass of water with its temperature.

The [SI](#)⁸ represents the physical dimensions of several base quantities by single capital letters; for instance T stands for the physical dimension ‘time’. In the present notes we use the different, non-standard convention of denoting a physical dimension using the full word in sans-serif font: time. This way we avoid ambiguity and confusion with other one-letter symbols.

With each physical dimension we can associate a unit of measurement, which as mentioned in §1.4 expresses a basic standard for comparing the measurement results for quantities having that physical dimension. Units are very important and must always be written, for several reasons. First, a number without units doesn't tell us anything. If I tell you "the place is at a distance 100 from here", you have no idea how far the place is. "100" what? 100 metres? 100 kilometres? These are completely different distances. Second, units give us useful information about physical quantities and their relationships and measurement. If you see the expression "3 m/s", then there's a strong possibility that that's a velocity. If you see the expression "5 J/m²", then you have a hint that it could be measured by measuring an energy-mass and an area, and then dividing them. Third, keeping track of units often allows us to quickly catch errors in solving a physical problem.

If a physical quantity is defined in terms of other quantities, then its unit is usually given in terms of the defining quantities. For example, if we define *speed* as *length* divided by *time*, then its unit is 'metres per second', 'm/s'. Some combinations of units receive special unit names. For instance, *power* is defined as energy-mass divided by time; its unit is therefore 'J/s' ('joules per second'). But this compound unit is usually called 'watt', symbol 'W', as we discuss in the next section.

How to pronounce combinations of units

According to the International Organization for Standardization (ISO 2009 §7.2.4):

Multiplication: The name of the product of two units is the concatenation of the two names, separated by a space. For instance

$$\text{N} \cdot \text{s} \equiv \text{Ns} \longrightarrow \text{newton metre}$$

Power: The name of the power of a unit is the name of that unit followed by 'to the power n '. However, the powers two and three may be expressed by 'squared' and 'cubed'. For instance

$$\begin{aligned} \text{s}^{-1} &\longrightarrow \text{second to the (power of) minus one} \\ \text{m}^2 &\longrightarrow \text{metre squared} \end{aligned}$$

Division: The name of the quotient of two units is formed by inserting the word 'per' between the two names. A compound name shall never contain more than one 'per'. For instance

$$\text{m/s} \longrightarrow \text{metre per second}$$

For other rules about printing and reporting units take a look at the [NIST Check List](#)⁹.

§ 1.8 Relations between units

Some physical quantities have their own unit, often because of historical reasons. For instance, *energy flux* and *energy supply*, which we shall discuss later, have unit ‘watt’, symbol ‘W’. Similarly, *momentum flux* and *momentum supply*, which we also call *forces* and shall discuss later, have unit ‘newton’, symbol ‘N’.

Some units may satisfy particular mathematical relations, either because their quantities are defined in terms of one another, or because their quantities satisfy important physical relationships or laws (in fact, in many situations the distinction between ‘definition’ and ‘physical law’ is not clear-cut).

For example, since *energy flux* is related to change in *energy* divided by *time*, it has physical dimension energy/time. Thus there’s also an equivalence between the units ‘watt’ and ‘joule per second’: one watt and one joule per second are the same:

$$1 \text{ W} \equiv 1 \text{ J/s} .$$

There is a very important relation of this kind among the units for time (s), length (m), momentum flux or force (N), energy (J), mass (kg). The relation can be written in several equivalent ways; here’s one:

Relation between second, metre, newton, joule, kilogram

$$\text{kg} \cdot \frac{\text{m}^2}{\text{s}^2} = \text{J} = \text{N} \cdot \text{m} \quad (1.1)$$

The first equality expresses the fact that *mass and energy are the same thing*, as we’ll discuss in chapters 3 and 11. The second equality expresses a physical fact that comes from General Relativity: *energy and momentum are different aspects of the same physical quantity*, a quantity called ‘energy-momentum tensor’.

$$E = m c^2$$

With some algebra we obtain other equivalent expressions of the equalities above, for example

$$\text{kg} = \text{J} \cdot \frac{\text{s}^2}{\text{m}^2} = \text{N} \cdot \frac{\text{s}^2}{\text{m}}, \quad \text{or} \quad \text{kg} \cdot \frac{\text{m}}{\text{s}} = \text{J} \cdot \frac{\text{s}}{\text{m}} = \text{N} \cdot \text{s},$$

or many others. Keep the main equality (1.1) always in mind when you do conversion between units.

§ 1.9 Base units

The distinction between base quantities and derived quantities brings about an analogous distinction between **base units** and **derived units**. Any unit can always be rewritten as product of powers of base units.

The SI adopts a set of seven base units:

<i>unit</i>	<i>symbol</i>	<i>quantity</i>
metre	m	length
kilogram	kg	mass
second	s	time
ampere	A	electric current
kelvin	K	thermodynamic temperature
mole	mol	amount of substance
candela	cd	luminous intensity



together with some auxiliary units for quantities related to information theory. The majority of these units is **defined with respect to fundamental physical constants**¹⁰.

SI logo depicting the base units and the fundamental constants they are based upon

In these notes we shall emphasize a slightly different set of base units, related to our seven **primitive quantities**^{§1.4 p. 31}:

<i>unit</i>	<i>symbol</i>	<i>quantity</i>
second	s	time
metre	m	length
mole	mol	amount of matter
weber	Wb	magnetic flux
coulomb	C	electric charge
newton	N	momentum flux
joule or kilogram	J or kg	energy-mass
kelvin	K	thermodynamic temperature

Besides second and metre, related to coordinates, the other seven units apply to almost all our seven primitive quantities or their fluxes; so by using these units we can bring to light the presence and role of those main quantities. There is no real conflict with the SI, because all units above are official base or derived SI units.

We shall speak further about units in the next chapters. The main quantities and units used in the present notes are summarized in table 3.1 on page 89 and table 4.1 on page 96.

§ 1.10 Mathematics with quantities and units

Variables and functions

When a physical quantity is denoted by a symbol or variable, keep in mind that a unit is “contained” in the symbol, so to speak. For example if the variable t denotes a time, then it includes some time unit, say seconds. This becomes apparent when we write the value of the symbol, for instance “ $t = 120\text{ s}$ ”. The unit is not predetermined, but it must correspond to the dimension of that quantity. We could for instance write “ $t = 2\text{ min}$ ” instead; the two expressions are completely equivalent.

This fact must be kept in mind when combining symbols. For example, if d is a distance and t is a time, then writing $v = d/t$ tells us that v is a velocity, and it has appropriate units that come from d and t , for instance m/s.

Units otherwise behave just like *literal constants* for all mathematical purposes, just like the letter ‘ a ’ in the expression ‘ $a x$ ’. This is why they can be simplified; for instance:

$$3\text{ mol/s} \cdot 5\text{ s} = 3 \frac{\text{mol}}{\cancel{\text{s}}} \cdot 5 \cancel{\text{ s}} = 15\text{ mol}.$$

Particular care must be taken with trigonometric and exponential functions, like $\sin()$, $\cos()$, $\tan()$, $\exp()$, $\log()$; **these functions only admit a dimensionless argument** (which for the trigonometric ones corresponds to *radians*). So there cannot be units like ‘s’ or ‘m’ within these functions: we must make sure that any units present within cancel out.

This makes sense, because we wouldn’t know how to interpret the argument otherwise. Suppose you read “ $\cos(60\text{ s})$ ” somewhere: how much is that? If we say “just discard the unit”, we would have

$$\cos(60\text{ s}) \stackrel{?}{=} \cos(60) \approx -0.95$$

but wait: $60 \text{ s} \equiv 1 \text{ min}$, so we could equivalently write “ $\cos(1 \text{ min})$ ”. Then, according to the hypothetical rule “just discard the unit”, we would have

$$\cos(60 \text{ s}) \equiv \cos(1 \text{ min}) \stackrel{?}{=} \cos(1) \approx +0.54$$

a completely different result!

For this reason an expression like ‘ $\cos(t)$ ’, with t denoting time, doesn’t make sense: there’s a time unit in the argument of $\cos()$. If we want to express an oscillation with time, we must write instead something like

$$\cos\left(\frac{t}{T}\right)$$

where T is the period of the oscillation, a symbol which also includes a time unit, which simplifies with the one in t . If the period of the oscillation is $T = 1 \text{ s}$ then we can also simply write

$$\cos(t/\text{s})$$

This expression is now unambiguous: suppose that $t = 60 \text{ s} \equiv 1 \text{ min}$, then

$$\begin{aligned}\cos(t/\text{s}) &= \cos(60 \text{ s}/\text{s}) = \cos(60) \approx -0.95 \\ &= \cos(1 \text{ min}/\text{s}) = \cos(1 \cdot 60 \cancel{\text{s}}/\cancel{\text{s}}) = \cos(60) \approx -0.95\end{aligned}$$

Also remember that **the results of trigonometric and exponential functions are dimensionless numbers**. Therefore an expression such as ‘ $3 \cos(\dots)$ ’ denotes a pure number, with no units. If we want to express that the result is, say, a length, then the appropriate units must appear. We can for instance write

$$L \cos(\dots)$$

where L is a length, and therefore includes some kind of length unit such as ‘m’. If this length is, say, $L = 2 \text{ m}$ we can also simply write

$$2 \cos(\dots) \text{ m}$$

Derivatives

When we follow the rules above, all other mathematical operations automatically take care of everything. The derivative, for instance, is calculated

in the usual way, treating any visible units as literal constants. Let's see a concrete example. This expression

$$x(t) = 2 \cos(t/s) \text{ m}$$

says that the position of some object oscillates with time, between the values -2 m and $+2 \text{ m}$. When $t = 0 \text{ s}$, the position is $x = +2 \text{ m}$. The position $x = -2 \text{ m}$ is reached when the argument of $\cos()$ is π , that is

$$t/s = \pi \quad \Rightarrow \quad t \approx 3.14 \text{ s}.$$

The **velocity** ^{§2.8 p. 62} of the object is given by the derivative of this expression with respect to t . Let's calculate it treating all unit symbols as literal constants:

$$\frac{dx(t)}{dt} = \frac{d}{dt} [2 \cos(t/s) \text{ m}] = 2 \left[\underbrace{-\sin(t/s) \cdot \frac{1}{s}}_{\text{chain rule}} \right] \text{ m} = -2 \sin(t/s) \frac{\text{m}}{\text{s}}$$

and we see that the correct units for velocity, ' m/s ', have appeared automatically.

§ 1.11 Expression of measurement uncertainty

✂ To be written

URLs for chapter 1

1. <https://www.physics.lbl.gov/rememberinggeoffreychew/>
2. <https://www.parkeharrison.com/bodies-of-work/architect-s-brother/earth-egg-legies>
3. Joint Committee for Guides in Metrology (JCGM)
4. <https://jcgm.bipm.org/vim/en/1.1.html>
5. <https://www.youtube.com/watch?v=nYg6jzotiAc&t=893s>
6. <https://www.bipm.org/en/measurement-units>
7. <https://www.nist.gov/pml/special-publication-811>
8. <https://www.nist.gov/pml/special-publication-811>
9. <https://www.nist.gov/pml/special-publication-811/nist-guide-si-check-list-reviewing-manuscripts>
10. <https://www.nist.gov/si-redefinition/meet-constants>

Time and space

If we want to describe the *motion* of a material point, we give the values of its coordinates as a function of time. However, we should keep in mind that for such a mathematical description to have physical meaning, we first have to clarify what is to be understood here by “time”. We have to bear in mind that all our propositions involving time are always propositions about *simultaneous events*. If, for example, I say that “the train arrives here at 7 o’clock”, that means, more or less, “the pointing of the small hand of my clock to 7 and the arrival of the train are simultaneous events”.

A. Einstein 1905a

§ 2.1 Time and proper time

Time is a primitive quantity. We understand the notion of time intuitively, even if it is difficult to explain – that’s why it is taken as primitive. In 1905, with the Theory of Relativity, part of our everyday intuition about the notion of time was seriously shaken. For many years afterwards our old intuition could still be used in practice and in applications. But the new, correct intuition is becoming more and more important in everyday life and technologies. GPS navigation, for example – which we use every day in leisure activities like hiking or sightseeing, as well as in more critical ones like aeroplane landing – crucially depends on the correct notion, intuition, and measurement of time. Luckily the new intuition of time is also becoming more and more widespread thanks to films and mass-media; think of movies like *Interstellar*¹.

Let’s see, by means of a thought experiment, how our traditional intuition about time goes astray. Here’s *Alice*, *Bob*, and *Charlie*. They have extremely precise clocks, built in exactly the same way. They synchronize their clocks to, say, 12:00 and stay very close to one another. You can follow their trips in fig. 2.1, which is an example of **spacetime diagram**. The figure represents the temporal dimension and one spatial dimension merged

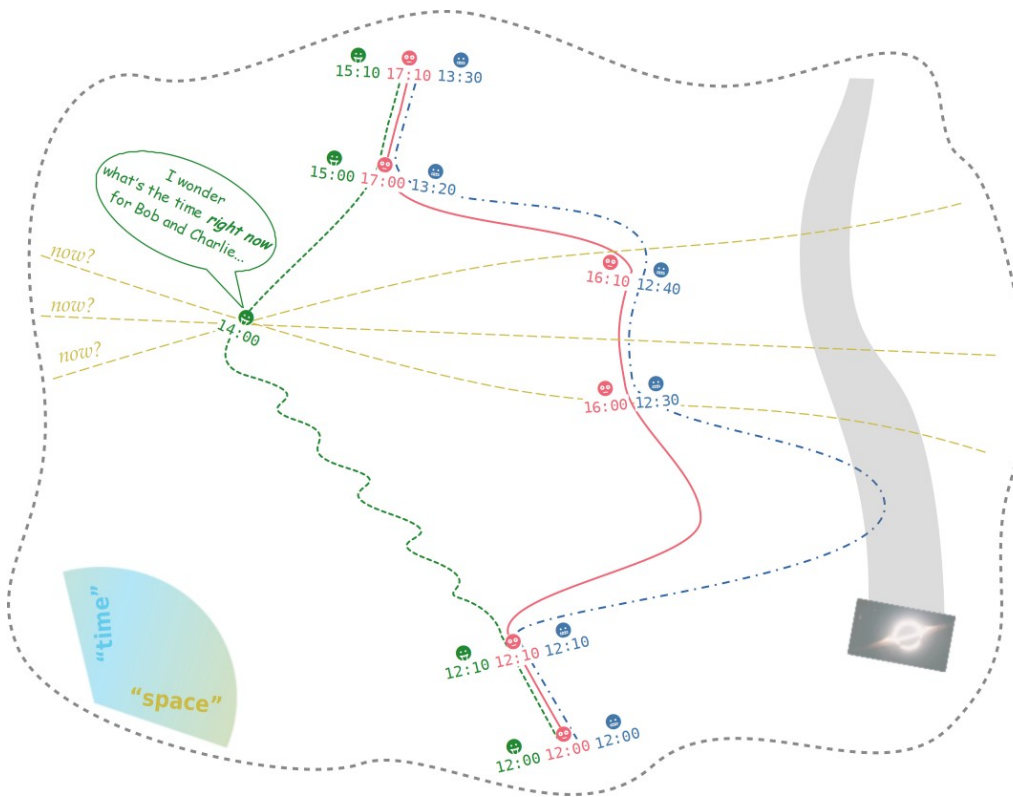


Figure 2.1 A spacetime diagram that illustrates the experiences of Alice (dashed green 🕒), Bob (solid red 🕒), Charlie (dot-dashed blue 🕒) with time. The area within the dotted grey line represents a two-dimensional spacetime. Space is more horizontal than vertical; time is more vertical than horizontal, and flows upward. Note that we can't precisely say, for instance, "the vertical direction is time", because our problem is to see whether we can really separate time and space at all.

Lower part: Alice, Bob, Charlie stay close and observe their clocks are perfectly synchronized from 12:00 to 12:10. Then they separate.

Right: Charlie visits a region near a strong mass-energy source. Upon meeting again with Bob, the two notice their clocks differ: 16:00 for Bob, 12:30 for Charlie. Yet this clock difference stays the same while they travel together for 10 min.

Left: Alice wanders around, travelling at high speed with respect to the fixed stars. When her clock displays 16:00, she wonders what's the time "right now" for Bob and Charlie. But this question doesn't make sense, because (1) when Bob and Charlie are together their clocks differ – impossible to say what's "the" time at their position; (2) it is not clear which instant in Bob & Charlie's trajectory should be considered as "now" for Alice (dashed yellow lines).

Upper part: When Alice, Bob, Charlie meet again, their clocks have completely different readings; and they have aged differently. But their clocks run again at the same rate as long as they stay close again.

together in a two-dimensional image. The motions of Alice, Bob, Charlie 'in space' are therefore represented as lines in this two-dimensional spacetime. They are called *worldlines*. Each point in a worldline has an associated time, the time measured by the person or object following that worldline.

Alice, Bob, Charlie stay very close together at first. They go around, maybe on an aeroplane or on a space ship, and they constantly compare their three clocks. They notice that their clocks stay perfectly synchronized all the time, no matter where they go together and what they do.

At time 12:10 they separate, and each one goes around independently. One of them might stay in place, another might take a helicopter, and another might go for a trip to Mars or to another star and back.

Bob and Charlie at some point meet again. They compare their clocks, and see that they are not synchronized anymore; the difference could be as small as microseconds, or as large as years. In fact, if this time discrepancy is large, Bob and Charlie notice that they have also aged differently: the time difference is not only apparent in their clocks, but also in their bodies. Let's say for concreteness that Bob's clock shows 16:00 and Charlie's shows 12:30 of the same day. So Bob's clock is ahead of Charlie's, or equivalently Charlie's clock is behind Bob's. Note the following aspects:

First, both Bob and Charlie can say "my clock has been working fine, so it should be correct": neither has noticed anything strange about the passage of time.

Second, if they now stay together, no matter where they go, they see that their clocks remain exactly synchronized, besides the time discrepancy they noticed when they met again. This discrepancy doesn't increase or decrease. When Bob's clock shows 16:10, Charlie's shows 12:40.

Let's focus on Alice on her trip. At some point, let's say when her clock shows 14:00, she wonders what the time might be on Bob's and Charlie's clocks. Bob and Charlie are far away. Alice could decide to contact each of them, say via radio broadcast, and ask "what does your clock show, right *now*, at your position?". But consider how weird this question actually is:

First, there would be a delay, even if extremely small, in the radio transmission; so it's unclear to what time Bob's and Charlie's answers would apply. If we say "let's account for the radio-signal speed", we see that there's a logical problem: speed is distance divided by *time*, and here we are trying to determine what's the "correct" time! So we would be reasoning in circles.

Second, suppose that Bob and Charlie receive Alice's radio broadcast when they happen to be together again. Bob's clock shows, say, 16:04,

and Charlie's 12:34. What should they reply to Alice regarding *the time at their position*? They are close together, so both of their clocks refer to that position, but the clocks show different times!

It may happen thereafter that Alice also meets Bob and Charlie again. Her clock will generally show a time different from theirs, and she will show a corresponding different ageing as well. As long as the three stay close together, they again notice that their clocks stay perfectly synchronized, besides the discrepancy at the moment of meeting. So if Alice's clock shows 15:00, Bob's 17:00, and Charlie's 13:20 when they meet (Charlie and Bob had gone apart again), then when Alice's shows 15:10 Bob's will show 17:10 and Charlie's 13:30.

The experiences just described will occur again any time Alice, Bob, Charlie, or any two of them, meet or go apart. There could be a hundred observers like Alice, Bob, Charlie, initially at the same place and with synchronized clocks. Whenever two or more of them meet after having been separated, they will notice discrepancies in their clocks (and in the ageing of their bodies). But their clocks will have exactly the same time lapses as long as they stay together.

Consider for a moment an imaginary world in which these experiments had given a different kind of result. According to Newtonian mechanics, whenever two or more initially synchronized observers like Alice, Bob, Charlie were to meet again, their clocks would always show identical times. If one year, 23 days, 8 hours, 9 minutes, and 3.045 399 283 240 992 663 02 seconds had passed for Bob since he last met Alice, he would see that exactly the same amount of time had passed for Alice since their previous meeting. If you think about it, in this case it would have been somewhat natural for them to think "right now, the clock of far-away Charlie must show the same time as ours" (even though they have no real experimental way of confirming that).

But that's an imaginary world. In our world what occurs is the more complicated situation with time discrepancies described initially. Only one conclusion can be drawn from these experimental results: **Time is not some sort of universal quantity. It is, so to speak, "local" to a person or clock, or to a group of persons or clocks that stick together.** This also means that it does not make sense to ask questions like "what is the time for far-away Charlie, *right now*?", or "what is happening at some other place *right now*?". There is no universal 'now'; the notion of *now* is local.

The time measured by a specific observer is called the **proper time** of

that observer. Luckily we know how to calculate how much the proper times of separated observers will differ when the observers meet again. According to our current understanding, it turns out that the time differences depend, roughly speaking, on how fast the observers are moving with respect to one another and with respect to the distribution of energy-mass in the universe, and on how much energy-mass is contained in the regions they travel across. The general theory of relativity gives us the equations that determine the proper-time differences.

Time discrepancies can be measured, for example, by comparing initially synchronized clocks that have been put in aeroplanes flying in different directions. The first measurement of this kind was made by Hafele and Keating in 1971. They synchronized four caesium atomic clocks with a reference clock, and then flew the four atomic clocks around the world on commercial jet flights, first eastward, then westward. At the end of the eastward trip, the clocks showed a time *behind* the reference one by around 6×10^{-8} s. At the end of the westward trip, their time was *ahead* the reference one by around 3×10^{-7} s. These measurements were in agreement with General Relativity's prediction, within experimental error.

Most importantly, these time discrepancies affect everyday technologies such as the Global Positioning System. Formulae from General Relativity appear in your phone's GPS software; see for instance §20.3.3.3 of the Interface Control Document IS-GPS-200 at <https://www.gps.gov/technical/icwg/>. Time discrepancies must also be taken into account in the establishment and synchronization of time in our everyday equipment:

In 1976, the International Astronomical Union introduced relativistic concepts of time and the transformations between various time scales and reference systems. [...] Now [...] it is necessary to base all astrometry, reference systems, ephemerides, and observational reduction procedures on consistent relativistic grounds. This means that relativity must be accepted in its entirety, and that concepts, as well as practical problems, must be approached from a relativistic point of view.

(Kovalevsky & Seidelmann 2004)

Note therefore the following curious situation. In setting up a time to meet a friend, you don't need to worry about the discrepancies between your and your friend's proper times: if your friend walks 1000 m away from you and then immediately back to you, at 1 m/s, then the time elapsed for you will be 1000 s (around 17 min), but the time elapsed for your friend will be 999.999 999 999 999 994 437 s. That's a difference of less than 10^{-14} s,



Hafele, Keating, and their clocks aboard aeroplane (from *Time*, October 1971²)

clearly negligible for the two of you, so you don't need to worry with General Relativity formulae in setting up your meeting time. Yet, if you set up a meeting place via GPS, then the true nature of time and General Relativity formulae become important: if they were not accounted for, you and your friend might end up off your meeting place by 100 m.

Time lapse or duration has SI dimension time, and we shall usually measure it using the unit *second*, symbol 's'.



Exercise 2.1

Consider two clocks: one at rest on the Earth's surface, at a distance r_e from the Earth's centre; the other on a GPS satellite or, say, on the [International Space Station](#)³, right above the first clock, at a distance r_s from the Earth's centre. An observer by the clock on Earth measuring a time lapse Δt_e will see that the clock on the satellite has run for a time lapse Δt_s . The relation between two time lapses is approximately given by

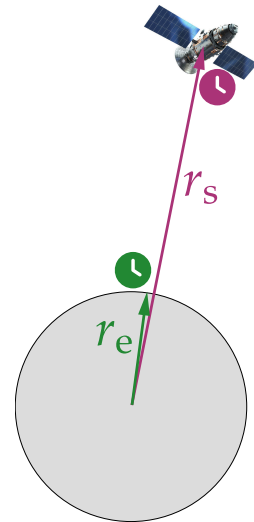
$$\frac{\Delta t_s}{\Delta t_e} = \frac{\sqrt{1 - 2\frac{\kappa}{c^2} \frac{M}{r_s}}}{\sqrt{1 - 2\frac{\kappa}{c^2} \frac{M}{r_e}}}$$

where $\kappa \approx 6.7 \times 10^{-11} \text{ m}^3/(\text{kg s}^2)$, $c = 3.0 \times 10^8 \text{ m/s}$, and the Earth's mass-energy $M = 6.0 \times 10^{24} \text{ kg}$. In this approximation we're neglecting the velocities of the two clocks in order to simplify the maths. The velocities can be very relevant, so don't take the results you'll find too literally.

1. Take the case of a GPS satellite, with $r_e = 6.4 \times 10^6 \text{ m}$ and $r_s = 2.6 \times 10^7 \text{ m}$ ([NASA data](#)⁴). If you, on the ground, measure a time lapse of $\Delta t_e = 10 \text{ years}$, what's the difference, in seconds, with the time lapse Δt_s you see on the satellite?
2. If the time lapses are large compared with the time needed to go from ground to orbit or vice versa, then $\Delta t_s/\Delta t_e$ is also the ratio between the real *ageing* of a person who's been in orbit and one who's been on the ground, when they meet again.

Now consider the case with a black hole instead of Earth. The formula above can still be applied as an approximation.

In the film *Interstellar*⁵, two astronauts travel to Miller's planet, at a distance r_e from the black hole Gargantua, while leaving a third



astronaut in orbit at a distance $r_s \approx \infty$ (the distance is large enough that it can be approximated as infinity). The two astronauts stay on Miller's planet for 3 hours. When they meet the latter astronaut in orbit again, the latter has aged 23 *years*.

Given that Gargantua's mass-energy is $M = 2.0 \times 10^{38}$ kg, calculate the distance r_e of Miller's planet from the black hole.

§ 2.2 Coordinate time

The fact that the time elapsed for you can be different from that elapsed for a satellite can therefore make it difficult to coordinate activities and to operate some technologies.

But luckily there's a way to bypass proper-time discrepancies. Instead of referring to my proper time or to your proper time, we can agree on assigning a somewhat arbitrary numerical time label to every event: this is called a **coordinate time**.

Coordinate time is generally different from the proper times measured by different observers. It can nevertheless be used for “doing physics”, and it is the time we shall most often use in our equations. The coordinate time commonly used for Earth-physics purposes is **Universal Coordinated Time UTC**⁶, or the **International Atomic Time TAI**⁷ for astronomy purposes:

International Atomic Time (TAI) is based on more than 250 atomic clocks distributed worldwide that provide its stability, whereas a small number of primary frequency standards provide its accuracy. Universal Coordinated Time, which is the basis of all legal time scales, is derived from TAI. To allow the construction of TAI and the general dissemination of time, clocks separated by thousands of kilometres must be compared and synchronized. [...] The achieved performances of atomic clocks and time transfer techniques imply that the definition of time scales and the clock comparison procedures must be considered within the framework of General Relativity.

(Petit & Wolf 2005)

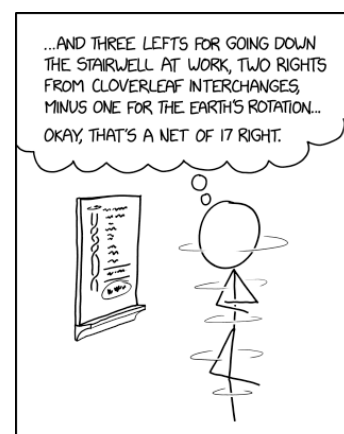
UTC and TAI have the same time lapse, but UTC differ by irregular readjustments of its “zero”, the **leap seconds**⁸.

The clock on your phone, and on devices synchronized via internet, shows UTC, not your proper time. An observer on Earth at 0 m over sea level, and not moving, measures a proper-time lapse equal to UTC or TAI (besides small variations coming from the irregularity and internal motions

of the Earth). But observers at other altitudes and observers moving with respect to Earth's surface can measure that their proper-times lapses are slightly different from UTC and TAI.

In applications related to interplanetary spacecraft navigation and cosmological observations, the Barycentric Coordinate Time TCB is used. Its time lapse is slightly faster than the one of UTC: for each second that passes in UTC, 1.000 000 014 8 s pass in TCB on average. Two coordinate times to be used around an on the Moon, **Lunar Coordinate Time TCL**⁹ and Lunar Time LT, are on the making in the year 2025. Lunar Time will lapse faster by 5.6×10^{-5} s per day compared to our UTC. The establishment of all these kinds of coordinate times shows how important the difference between coordinate and proper time is for our current technology.

When we use coordinate time, some important physics formulae turn out to be the same no matter whether we use General Relativity or an approximate theory such as Newtonian Mechanics. Thanks to this fact, for the most part of these notes we will not need to deal with proper-time details. But it is important for you to keep in mind how time really works, and the small time discrepancies that exist and occur all the time along your *worldline*.



SPACETIME HEALTH TIP: REMEMBER TO CANCEL OUT YOUR ACCUMULATED TURNS AT THE END OF EACH DAY TO AVOID WORLDLINE TORSION.

<https://xkcd.com/2882>

§ 2.3 Space, length, distance

Together with the notion of time, also the notions of space, length, distance lose some of their traditional intuition. Traditionally when we speak of the distance or the length of a moving object at a given time, we mean the distance 'at the same instant of time'. But we have seen that it does not make sense to ask "what is the time for the object, right now?". And when we speak of lengths or distances, we mean measurements 'on a straight line'. But spacetime is curved.

To see the complications in defining 'distance' and 'length', imagine two objects or people that are moving with respect to each other, getting closer and farther away. Call them A and B; let's say that A represents you. You have a clock, and B is equipped with another clock identical to yours. These clocks were synchronized (to time 06:00) in the past when you and B were in contact with each another.

We can represent the motions of A and B as *worldlines* in spacetime, as in the side figure (similarly to what we did in fig. 2.1). The figure shows the points in spacetime where your clock displays the times 06:00, 08:00,

and 08:10, and where B's clock displays 06:00 and 08:00. Note in particular that you and B touch each other and are synchronized at 06:00; then you touch each other again later, when your clock displays 08:10 and B's clock displays 08:00.

Suppose that when your clock displays time 08:00 you ask "how distant is B from me *right now*?". How can we measure or define such distance?

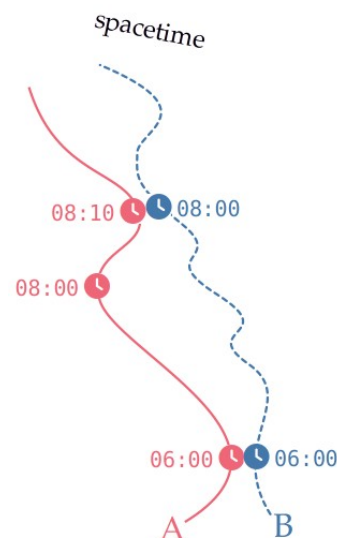
One possibility is to say that we measure the length of a straight line joining you at *your* time 08:00, with B at *its* time 08:00. Could we define this as the distance between you and B at 08:00? Well, there's something bizarre: when B's clock shows 08:00, B is in contact with you! So we would essentially be measuring the distance between you and yourself at two different times of your clock. This doesn't seem to be very sensible. Also, should this distance be zero, then? But when your clock displayed 08:00, B was *not* in contact with you, so it wouldn't make sense to say that its distance from you was zero.

So let's discard the measurement procedure above, which seems to inconsistent results. Another possibility is to measure the length of a straight line joining you at your time 08:00, with B at some other time on its clock. But which time should we choose? Any choice seems arbitrary.

And there's one more problem. Let's say that we decide to measure the distance between you when your clock shows 08:00, and B when its clock shows 07:50 or some other arbitrary time when it is not in contact with you. We should measure this distance on a straight line. But what's a 'straight line'? Spacetime is curved. A straight line in the figure above is not necessarily a straight line in spacetime. The notion closest to a 'straight line' in a curved space is that of a *geodesic*¹⁰. It turns out that there may be several geodesics connecting you (at 08:00) and B (at 07:50).

Similar problems appear if we ask questions about *lengths*. Suppose there is a rubber band stretched between you (remember you're A) and B. When your clock shows 08:00 you ask "how long, *right now*, is the rubber band joining me and B?" We encounter the same difficulties as above in trying to answer this question.

A spectacular case of the effect of spacetime curvature is the phenomenon of *gravitational lensing*¹². Owing to the curvature of spacetime, light and other electromagnetic radiation emanating from the object reach us along different paths in spacetime. All these paths are "straight lines" (geodesics). Coming from different directions, to us these look like distorted, duplicated images of the object, as schematized in the side figure. The



From *The ABC's of Distances*¹¹.

curvature is generated by some large distribution of energy-mass, such as a galaxy, between us and the object. A beautiful example of this phenomenon is given by the [quasar RX J1131-1231](#)¹³, side image. Spacetime curvature separates the light arriving to us from this quasar into four [yellow & red](#) spots, three at the top and one at the bottom in the image. The curvature is generated by a galaxy visible as the [blue spot](#) at the centre.



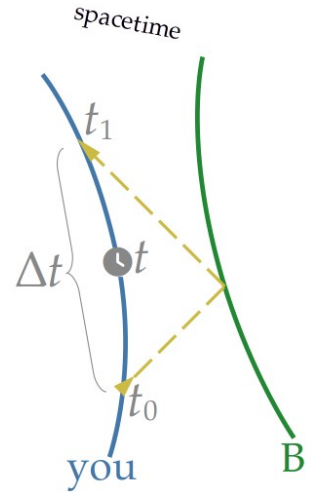
From [ESA/Webb](#)¹⁴.

A consequence of these peculiar situations and of the curvature of spacetime is that many notions of distances and length can be defined and measured, which are generally *not* equivalent to one another. Cosmology, for example, uses a [plethora of different distances](#)¹⁵. One must therefore be careful about which definition of distance is being used. In the next sections we shall focus on two of them: *radar distance* and *coordinate distance*.

§ 2.4 Radar distance

The definition of distance that's regarded as the most "physical" is *radar distance*, defined as follows.

Consider again a situation in which an object B is moving with respect to you. Yours and B's motions in spacetime, around the time when your clock displays time t , are illustrated in the next side figure. Your *worldline* in spacetime is the [blue](#) one on the left; the worldline of object B is the [green](#) one on the right. At your proper time t_0 you send a light pulse towards object B. The pulse travels in empty space, and upon hitting object B it immediately bounces back to you. It reaches you at your proper time t_1 . The worldline of the light pulse is shown in [dashed yellow](#) in the figure.



A proper time $\Delta t = t_1 - t_0$ has elapsed for you between emission and reception of the light pulse, and the time exactly in between emission and reception is $t = (t_1 + t_0)/2$. The **radar distance** d of object B from you at time t is defined as

$$d := \frac{1}{2}c \Delta t \quad (2.1)$$

where c is the speed of light in vacuum, a universal physical constant:

$$c = 299\,792\,458 \text{ m/s} \quad (\text{exactly}). \quad (2.2)$$

The SI unit of length, the [metre](#)¹⁶, is based on the measuring procedure above. Common laser distance meters also work by the same procedure, and therefore yield radar distance. As the name indicates, this is also the distance measured by radars.



A laser distance meter (the light beam is not visible in reality).

In using radar distance, however, we must be wary of the following peculiarities:

- Radar distance makes sense only if the time lapse Δt is small enough compared to variations in the relative motion of the observer and the object, so that this motion is approximately uniform. For this reason this distance cannot be used if the object is too far away: the farther away it is, the longer it takes for a light beam to travel to and fro. Radar distance can be used between the Earth and other Solar System planets; but it cannot be used for galaxies or other distant cosmological objects.

- **Radar distance is not symmetric:** the radar distance of B from A at A's time t is generally different from the radar distance of A from B at B's time t .

- **The value of radar distance depends on the relative motion** between A and B. Imagine that a friend of yours is located very close to you at time t , but is moving with respect to you. Upon measuring object B's radar distance, your friend will generally find a value different from yours. The discrepancy between you and your friend's measured values will be the larger, the higher is the relative velocity between you two. Several observers in motion with respect to one another will generally disagree on the dimensions of an approximately rigid objects in their vicinity.

The dependence on relative motion also affects, at high speeds, how we *see* objects, which appears more and more deformed. You can find beautiful visualizations, both static and animated, at [Relativity visualized](#)¹⁸.

Luckily, for relative speeds that are not too high compared to the speed of light, the radar distances measured by different observers differ by amounts that are negligible in everyday circumstances. As an example, consider a car moving on a road at 100 km/h, that is around 28 m/s. The car's driver measures the length of the car to be 4 m by radar distance. A pedestrian that sees the car passing by instead measures its length to be 3.999 999 999 999 98 m by radar distance. This is obviously a negligible difference.

Length and distance have SI dimension length, and we shall usually measure them using the unit *metre*, symbol 'm'.



How a street in Tübingen would look like (except for colour and some other features) if we travelled through it at around 240 000 000 m/s (from [Relativity visualized](#)¹⁷)



Exercise 2.2

Imagine that you and a friend of yours are measuring your distance from a wall, using a laser distance meter each. You and the wall are static with respect to Earth's surface. Your friend is moving with a speed v towards the wall, and is right beside you at the exact moment of the measurement.

In this specific situation, if d_{you} is the distance measured by you, and d_{friend} the distance measured by your friend, the two are related by the formula

$$d_{\text{friend}} = d_{\text{you}} \cdot \sqrt{1 - v^2/c^2},$$

where c is the speed of light, given in eq. (2.2). Note that this formula is also valid if your friend is moving away from the wall, rather than towards it.

1. Suppose you find that the distance of the wall from you is 200 m. Your friend's speed is 300 m/s. How much is the distance from the wall to your friend, who's right beside you, as measured by your friend? (You'll need a high-precision calculator and 18 significant digits.)
2. Now instead suppose you find that the distance of the wall from you is 500 m. Your friend measured (when right beside you) a distance of 499 m. How fast was your friend moving?

§ 2.5 Coordinate systems

From our discussion about time and space we conclude that physical events happen in spacetime, and there is no unique way to attribute a universal time or a universal position in space to a physical event.

In the previous sections we used the word 'event', informally taking its meaning for granted. Let's be more precise now. We call **event** or **spacetime point** a very small region of space that lasts for a very short lapse of time, so that it can be considered as a point in a four-dimensional space. When we say 'small region' or 'short time lapse', it doesn't matter which definition of distance or time lapse we're using.

The word 'event' is used because typically we identify such a spacetime point by means of a physical phenomenon of limited spatial extension and duration. How "limited" should these extension and duration be? It

"Henceforth, space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality."

Minkowski 1908

depends on the kind of physical phenomenon we're interested in. The sudden burst of a soap bubble can be considered as an event in comparison to geological distances and times; but it cannot be considered as an event if we're studying subatomic particles.

The peculiarities of spacetime can make it difficult to communicate the positions of objects and events by relying on distances. In giving indications about the location of a shop we can say "it's 200 metres down the road" without ambiguity. But in situations where much higher precision is needed and extreme motions or gravitational fields are involved, we would need to know the velocity of the person we're talking to, because the distances measured by us and by that person could be very different.

In the case of time, we bypassed the problem that time lapse depends on the observer's motion by introducing a [coordinate time](#) ^{§2.2 p.51}. This way each event gets an arbitrary but agreed-upon time label. We can bypass the analogous problem that length and distance depend on the observer's motion, by introducing a set of arbitrary *spatial coordinates* for each coordinate time.

All these coordinates together form a *coordinate system*, also called *reference system*:



Coordinate system or reference system

A **coordinate system**, also called **reference system**, is the assignment, by agreement, of four numerical labels to every point in spacetime: one *coordinate time* and three *spatial coordinates*. We use symbols such as (t, x, y, z) , or (t, r, θ, ϕ) , or others, for these coordinates. These four coordinates are obviously the same for all observers, because they are decided by agreement.

A coordinate system is usually defined on a limited region of spacetime. The location where all spatial coordinates have value zero is called the **origin** of the spatial coordinates.

Often the coordinates have physical meaning – like the proper time elapsed for a specific clock, or the distance from some event as measured by a specific observer – but they don't need to. Typically we use three spatial coordinates. In special situations, such as locating points on the Earth's surface neglecting altitude, only two or even one spatial coordinate can be enough.

A coordinate system can be visualized as a grid made by a set of lines or planes, one set for each coordinate, which allow us to read the coordinates of any point. The side figure shows an example with spatial coordinates (x, y) in two dimensions, for a specific coordinate time. It is of course assumed that the grid can be refined as much as needed. We are all familiar with the coordinate system (λ, ϕ) of *latitude* and *longitude* to identify locations on Earth's surface, and used internally by the positioning systems of mobile phones.

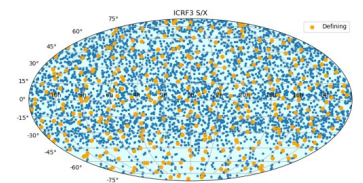
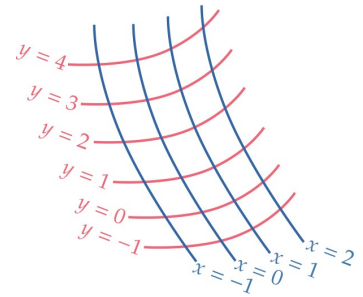
Since a coordinate system is arbitrary, we often choose one adapted to the physical phenomenon under study. It's very common to choose a coordinate system (t, x, y, z) that has time coordinate $t = 0$ s at the beginning of our observation of the phenomenon, and spatial coordinates $(x, y, z) = (0, 0, 0)$ m at a location close to where the phenomenon happens.

The spatial coordinates may be chosen so as to have particular physical properties, which in turn may lead to simpler expressions for some physical laws [as we shall see later](#) ^{§5.9 p. 154}. For instance, the coordinate lines might be straight lines (geodesics), in which case we speak of *rectilinear coordinates*. If they are not, then we call them *curvilinear coordinates*; the coordinates in the previous side figure are curvilinear.

If the spatial coordinate lines are *orthogonal* to one another at every point, that is, they intersect at $\pi/2$ rad, then we call them **orthogonal coordinates**. This is a very useful property, which simplifies many physics formulae. In these notes we shall always use orthogonal coordinates. Note that a coordinate system can be curvilinear *and* orthogonal.

In astronomy, in space and satellite communication, and in [geodesy](#)¹⁹ (the science of accurately measuring and understanding the Earth's geometric shape, orientation in space, and gravity field), [several important coordinate systems](#)²⁰ are used. For example:

- The *International Celestial Reference System (ICRS)*²² is a coordinate system with orthogonal spatial coordinates, which are almost rectilinear. Their origin is close to the centre of the Sun. Several distant cosmological objects, like [quasars](#)²³, have fixed spatial coordinates in this reference system. Its coordinate time is called Barycentric Coordinate Time (TCB).
- The *Geocentric Celestial Reference System (CGRS)* is a coordinate system with orthogonal spatial coordinates, which are almost rectilinear. Their origin is close to the centre of the Earth. Distant cosmological

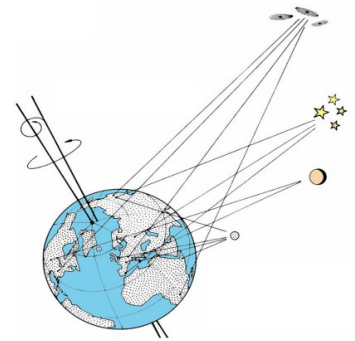


Map of some distant astronomical objects used to define the International Celestial Reference Frame (from [The ICRF](#)²¹).

objects have almost constant spatial coordinates in this reference system; this means that the Earth rotates with respect to it. Its coordinate time is called Geocentric Coordinate Time (TGB), which is very similar to the TAI.

- The *International Terrestrial Reference System (ITRS)*²⁴ is a coordinate system similar to the CGRS, but with the important difference that the Earth is approximately static in this reference system; this means that distant cosmological objects rotate with respect to it. It has the same coordinate time as the ICRS.
- The *World Geodetic System 1984 (WGS 84)*²⁵ is very similar to the ITRS, besides a discrepancy of some centimetres. It is used by the system of GPS satellites.

But how do we determine the position and time of an event in these coordinate systems? The procedure can be extremely complicated in fact. The starting point is the assignment of some predefined coordinates to objects that seem to have fixed spatial coordinates in the coordinate system; for instance distant cosmological objects like *quasars*²⁶ in the case of the ICRS and CGRS, or *particular reference stations*²⁷ on Earth's surface in the case of the ITRS and WGS 84. The coordinate of other events are then calculated from measurements of proper times, radar distances, and angles from the reference objects, using formulae from general relativity. As we mentioned in §2.1, even your mobile phone participates in these complex calculations.



The assignment of coordinates on and around Earth depends on nominal values assigned to distant astronomical objects (image from Capitaine 2010)



Exercise 2.3

On Earth's surface we often use the system of two coordinates called *latitude* λ and *longitude* ϕ , measured in degrees. Coordinate lines of constant latitude are called *parallels*; those of constant longitude are called *meridians*.

1. Find what's at these three coordinate pairs:

$$(\lambda, \phi) = (60.369\,002^\circ, 5.350\,336^\circ)$$

$$(\lambda, \phi) = (35.658\,587^\circ, 139.745\,424^\circ)$$

$$(\lambda, \phi) = (-13.163\,069^\circ, -72.545\,265^\circ)$$

2. Are latitude and longitude *orthogonal* coordinates? Explain why or why not.

§ 2.6 Spatial coordinate distance and length

If we have chosen a coordinate system, we can define a notion of distance called *coordinate distance* between two points at any coordinate time t . The idea is simple: we measure the length of the shortest path in spacetime joining the two points, and all intermediate points on the path must have the same coordinate time t . Of course it can happen that there is more than one path having shortest length.

Coordinate distance is different from radar distance. It has two advantages: it doesn't depend on relative motion, and can be defined also between objects that are very far apart. In regions of spacetime with low curvature and slow relative motions, however, radar distance and coordinate distance based on standard coordinate times are approximately the same, and approximately independent of any motions.

Note also that the speed of light defined with respect to coordinate distance need not have the value c , or even be constant! You might have heard or read about distant cosmological objects, like quasars, that are said to be receding from us at speeds faster than light's. How is that possible? The reason is that the 'speeds' they're talking about are defined with respect to coordinate distance, not radar distance.



Cartesian coordinates

We call **Cartesian coordinates** a set of spatial coordinates (x, y, z) with a very special property: the coordinate distance between two locations A and B having position vectors (x_A, y_A, z_A) and (x_B, y_B, z_B) is simply given by

$$d_{AB} = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}. \quad (2.3)$$

Perfect Cartesian coordinates do not exist, because spacetime is curved. But it is possible to choose coordinates that are approximately Cartesian in limited regions of spacetime. The coordinate systems ICRS, GCRS, ITRS discussed in §2.5 are *not* Cartesian: they include the effects of curvature generated by the Earth and other bodies in the Solar System.

In most of these notes we will not have to worry about the discrepancies between radar distance and coordinate distance, and about the motion of the observer or instrument that is measuring a distance. So we shall use the

term ‘distance’ without specifications. And we shall often use Cartesian coordinates.



Exercise 2.4

In the Geocentric Celestial Reference System, defined by the International Astronomical Union, the distance between two very close locations A and B having positions (x_A, y_A, z_A) and (x_B, y_B, z_B) close to Earth’s surface is approximately given by

$$d_{\text{GCRS},AB} = (1 + gR/c^2) \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2} \quad (2.4)$$

where $g \approx 9.8 \text{ m/s}^2$ is the gravitational acceleration, $R \approx 6371 \times 10^3 \text{ m}$ is Earth’s radius, and c is the speed of light. This is also the radar distance of B from A, if A is not moving with respect to the coordinate system.

1. Verify that the formula above is dimensionally correct: both left and right side should have dimension length.
2. Assume that your coordinates are (x_A, y_A, z_A) , and consider an object B at a x -coordinate difference of 100 m from you, that is, with coordinates

$$x_B = x_A + 100 \text{ m}, \quad y_B = y_A, \quad z_B = z_A.$$

How much is the difference between the distances d_{AB} and $d_{\text{GCRS},AB}$, calculated between you and the object?

(The inaccuracy in the specification of g and R is actually much larger than the difference you just found.)

§ 2.7 Coordinate position

Let us agree on some notation and terminology that will be used in these notes.

We shall often denote the four coordinates of a coordinate system by the letters

$$(t, x, y, z).$$

Unless stated otherwise, the coordinate time t will be taken to be UTC ^{§2.2 p.51}, and the spatial coordinates (x, y, z) will be taken to be

Cartesian ^{§2.6 p.60}. As mentioned in the previous section, the definition of the spatial coordinates is usually different from problem to problem; so it is always important to specify how the coordinate system you're using is defined.



Position vector

The triplet of spatial coordinates is called the **position** vector and is often denoted by $\mathbf{r}$:

$$\mathbf{r} := (x, y, z) \quad \text{or} \quad \mathbf{r} := [x, y, z] \quad \text{or} \quad \mathbf{r} := \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

use round brackets '()' or square brackets '[]', and horizontal or vertical notation as you prefer.

Whenever we speak of a “region of space” or of a “surface in space”, we mean a 3D or 2D region at some specific coordinate time t .

Some physical phenomena happen approximately along a line, in one dimension; think for instance of a small falling object. Other phenomena happen approximately on a surface, in two dimensions; think for instance of a swinging pendulum. In these cases we can omit two or one of the spatial coordinates, and simply assume that the omitted ones have some constant, unimportant values. In such cases we can simply write, for instance, (t, x) or (t, x, y) as our coordinates.

§ 2.8 Coordinate velocity and acceleration

In some situations the spatial coordinates $\mathbf{r} = (x, y, z)$ may be functions of the time coordinate t . A typical example is when we describe how the spatial position of a small volume or body changes with coordinate time. We can write this functional dependence in different ways, for instance

$$\mathbf{r}(t) \quad \text{or} \quad [x(t), y(t), z(t)].$$

So $\mathbf{r}$ is a vector function of time, which simply means that we have a collection of three functions of time.

If we take the derivative of each coordinate with respect to the time t , we obtain the *coordinate velocity*:



Coordinate velocity

The **coordinate velocity** is a vector defined as the derivative of the position vector $\mathbf{r}(t)$ with respect to coordinate time t :

$$\mathbf{v}(t) := \frac{d}{dt}\mathbf{r}(t) \quad \text{or} \quad \begin{bmatrix} v_x(t) \\ v_y(t) \\ v_z(t) \end{bmatrix} := \begin{bmatrix} \frac{d}{dt}x(t) \\ \frac{d}{dt}y(t) \\ \frac{d}{dt}z(t) \end{bmatrix}.$$

The word **speed** means the *magnitude* of the velocity:

$$|\mathbf{v}| \equiv \sqrt{v_x^2 + v_y^2 + v_z^2}.$$

The derivative of some quantity with respect to coordinate time is often denoted by a **dot** over the quantity. So we can also write

$$\mathbf{v}(t) = \dot{\mathbf{r}}(t) = [\dot{x}(t), \dot{y}(t), \dot{z}(t)]$$

The coordinate velocity is usually different from the *physical velocity*, which an observer would measure using proper time and space, for instance using bouncing light rays. In many everyday situations the difference between coordinate and physical velocity is so small that it can be neglected, so we shall simply use the word ‘velocity’. But in situations involving subatomic particles at high speed, for example, one must take into account that the two velocities are different.

Taking the derivative of the velocity with respect to coordinate time, that is, the second derivative of position, we obtain the *coordinate acceleration*:



Coordinate acceleration

The **coordinate acceleration** is a vector defined as the derivative of the velocity vector $\mathbf{v}(t)$ with respect to coordinate time t :

$$\mathbf{a}(t) := \frac{d}{dt}\mathbf{v}(t) = \frac{d^2}{dt^2}\mathbf{r}(t) = \left[\frac{d^2}{dt^2}x(t), \frac{d^2}{dt^2}y(t), \frac{d^2}{dt^2}z(t) \right].$$



Acceleration in relativity theory

In relativity theory, acceleration has a special physical significance because it includes the effect of gravity. We distinguish between *physical acceleration* and *coordinate acceleration*. The calculation of physical acceleration requires more than a time derivative.

For instance, let's say that you are standing still on the ground, and let's use a coordinate system where x points in front of you, y to your left, and z upward. Then your coordinate velocity is $\mathbf{v} = (0, 0, 0)$ m/s, also according to relativity theory. But your spatial *physical* acceleration is approximately $(0, 0, 9.8)$ m/s², not zero!

The definitions and values of physical acceleration according to relativity theory and to Newtonian mechanics are therefore quite different even in everyday situations. In these notes we'll mean coordinate acceleration, unless stated otherwise. This is also the meaning in Newtonian mechanics.

Integration of velocity and acceleration

Since velocity is the time derivative of position, we can find the position from the velocity by the inverse operation of integration. So we have

$$\mathbf{r}(t) = \mathbf{r}(t_0) + \int_{t_0}^t \mathbf{v}(t) dt \quad \text{or} \quad \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} x(t_0) \\ y(t_0) \\ z(t_0) \end{bmatrix} + \begin{bmatrix} \int_{t_0}^t v_x(t) dt \\ \int_{t_0}^t v_y(t) dt \\ \int_{t_0}^t v_z(t) dt \end{bmatrix}.$$

Note from the formulae above that if we know the expression of the velocity $\mathbf{v}(t)$, we still cannot know the position $\mathbf{r}(t)$ unless we *also* know the value $\mathbf{r}(t_0)$ of the position at some particular time t_0 .

In an analogous way we can find the velocity $\mathbf{v}(t)$ from the acceleration $\mathbf{a}(t)$ by integration, provided we also know the value $\mathbf{v}(t_0)$ of the velocity at some particular time t_0 :

$$\mathbf{v}(t) = \mathbf{v}(t_0) + \int_{t_0}^t \mathbf{a}(t) dt \quad \text{or} \quad \begin{bmatrix} v_x(t) \\ v_y(t) \\ v_z(t) \end{bmatrix} = \begin{bmatrix} v_x(t_0) \\ v_y(t_0) \\ v_z(t_0) \end{bmatrix} + \begin{bmatrix} \int_{t_0}^t a_x(t) dt \\ \int_{t_0}^t a_y(t) dt \\ \int_{t_0}^t a_z(t) dt \end{bmatrix}.$$



Exercise 2.5

- Here are the three components of a time-dependent velocity vector
The variable t is the time, and therefore has dimension time. Introduce

units 's' and 'm' in such a way that the expression is dimensionally correct:

$$\mathbf{v}(t) = [4, \cos(3t), -\exp(8/t)]$$

2. The position vector of a satellite is given below. Calculate the satellite's velocity and acceleration vectors:

$$\mathbf{r}(t) = \begin{bmatrix} 2.0 \times 10^7 \cos\left(\frac{t}{13751\text{s}}\right) \\ 2.2 \times 10^7 \sin\left(\frac{t}{13751\text{s}}\right) \\ 0 \end{bmatrix} \text{ m}$$

3. What is the satellite's velocity at $t = 6875\text{ s}$? What is the magnitude of the velocity (that is, the speed)?
4. Suppose you know the expression of the *speed* $v(t) \equiv |\mathbf{v}(t)|$ of an object, and also the object's position $\mathbf{r}(t_0)$ at time t_0 . From this information, can you find the position $\mathbf{r}(t)$ at any other time t ?

§ 2.9 Angles

 To be written.

URLs for chapter 2

1. <https://www.imdb.com/title/tt0816692/>
2. <https://time.com/vault/issue/1971-10-18/page/93>
3. <https://www.nasa.gov/international-space-station/>
4. <https://www.nasa.gov/directorates/somd/space-communications-navigation-program/gps/>
5. <https://www.imdb.com/title/tt0816692/>
6. <https://www.nist.gov/pml/time-and-frequency-division/time-realization/utc-nist-time-scale-0/>
7. https://gssc.esa.int/navipedia/index.php?title=Atomic_Time
8. <https://webtai.bipm.org/ftp/pub/tai/Circular-T/cirhtml/cirt.442.html>
9. <https://doi.org/10.1103/Physics.17.140>
10. <https://mathworld.wolfram.com/Geodesic.html>
11. <https://www.astro.ucla.edu/~wright/distance.htm>
12. <https://science.nasa.gov/mission/hubble/science/science-behind-the-discoveries/hubble-gravitational-lenses>
13. <https://www.nasa.gov/image-article/distant-quasar-rx-j1131>
14. <https://esawebb.org/images/potm2406a>
15. <https://doi.org/10.48550/arXiv.astro-ph/9905116>
16. <https://www.nist.gov/si-redefinition/meter>
17. <https://www.spacetime.travel.org/>
18. <https://www.spacetime.travel.org/>
19. <https://oceanservice.noaa.gov/geodesy>
20. https://gssc.esa.int/navipedia/index.php?title=Reference_Systems_and_Frames
21. <https://hpiers.obspm.fr/icrs-pc/newwww/icrf/index.php>
22. https://aa.usno.navy.mil/faq/ICRS_doc
23. <https://esahubble.org/wordbank/quasar>
24. <https://itrf.ign.fr/en/background>
25. <https://earth-info.nga.mil/?dir=wgs84&action=wgs84>
26. <https://esahubble.org/wordbank/quasar>
27. <https://itrf.ign.fr/en/solutions/ITRF2020-u2023#frame-definition>

Main physical quantities

For Euler, clarity was the hallmark of truth. [...] To him we owe also the brilliant imagination of the internal pressure in generality [...] I remark upon it in emphasis of the role of imagination and the importance of quantities which can only be thought of and cannot in themselves be measured.

C. A. Truesdell 1956

§ 3.1 Seven primitive quantities

The discovery, formulation, and use of [physical laws](#) ^{§1.3 p.28} requires us to look at the world from a more abstract and quantifiable point of view. As [previously discussed](#) ^{§1.4 p.31}, we shall achieve this by interpreting all physical phenomena around and within us in terms of time & space, which we have studied in chapter 2, and of seven physical quantities:



Seven primitive quantities

matter
magnetic flux
electric charge

energy-mass
momentum
angular momentum
entropy

We take these seven quantities as primitive, and shall build our physical laws upon them. Each of these seven quantities satisfies a universal physical law; other physical laws express relationships among these quantities.

Recall that [primitive quantities cannot be defined](#) ^{§1.4 p.31}: we can only try to understand them intuitively. This is the goal of present chapter: to make a brief acquaintance with the seven primitive quantities and their main properties.

§ 3.2 Two basic properties

Our seven primitive quantities have two basic properties in common:

First property: three measurements

For each quantity – with a slight change for the magnetic flux – we can ask about or measure three kinds of amount:

Three basic measurements of the seven quantities

- M1. How much of this quantity is **contained** in a particular three-dimensional region of space at a particular time instant?
- M2. How much of this quantity **flows** through a particular two-dimensional surface, in a given direction, during a particular time lapse?
- M3. How much of this quantity is **produced** in a particular three-dimensional region of space during a particular time lapse?

We can ask these questions for any region of space, any time instant, any time lapse. The regions of space can be moving and deforming. The results of the three measurements above are scalars for scalar quantities, and vectors for vector quantities.

Questions M2. and M3. can also be asked in a different way. Consider a very short lapse of time, and divide the net flow and the net production by that lapse of time. This way we have an alternative form of the second and third measurements, as flow or production divided by time:

Second and third measurement: new version

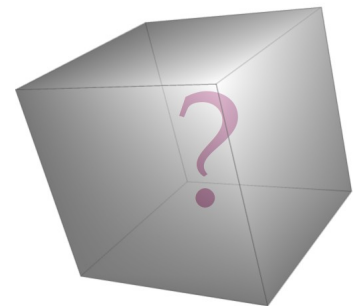
- M2b. At which rate is this quantity flowing through a particular two-dimensional surface in a given direction, at a particular time instant?

We call this the **flux** or **current** or **flow rate** of the quantity through that surface.

- M3b. At which rate is this quantity being produced in a particular three-dimensional region of space, at a particular time instant?

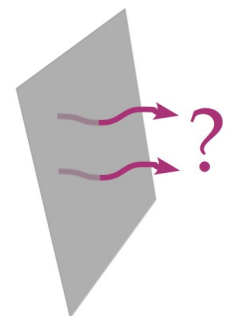
We call this the **supply** or **source** of the quantity in that region.

$t = 19:32:15$



For six main quantities we can ask: how much of it is in a given volume, at a given time?

between $t = 19:32:15$
and $t = 19:32:45$



For six main quantities we can ask: how much of it is flowing through a given surface in a given direction, during a given time lapse?

In the case of magnetic flux we can ask the three questions above in 2 and 1 dimensions, rather than in 3 and 2 dimensions, as we shall see in chapter 8.

The second property common to all seven quantities tells how the measurements above combine for several regions of space and several surfaces:



Extensivity

- If we consider two or more non-overlapping volumes, the amount of quantity contained in the total volume is equal to the sum of the amounts contained in the individual volumes.
- If we consider two or more non-overlapping surfaces, the amount of quantity flowing through the total surface is equal to the sum of the amounts flowing through the individual surfaces.
- If we consider two or more non-overlapping volumes, the amount of quantity produced in the total volume is equal to the sum of the amounts produced in the individual volumes.

We say that each of the seven quantities is **extensive**; some literature uses the term *additive*.

The basic measurements above cannot in general be made, and do not even make sense, for some other quantities. For instance, we cannot ask “what’s the total amount of temperature in this region?”, or “how much velocity is flowing through this surface?”.

Thanks to the two properties above, each of the seven quantities can be intuitively visualized as some kind of “stuff” that fills regions of space or flows through surfaces. This visualization is useful, but also comes with some warnings which we shall discuss later.

What’s remarkable about matter, magnetic flux, electric charge, energy-mass, momentum, angular momentum, and entropy, is that *they are common to all our main physical theories*, approximate or not: from Newtonian mechanics to General Relativity and Quantum Theory; from subatomic scales to cosmological scales. And in all these theories they possess the two basic properties discussed above. The mathematical characterization of these quantities can be slightly different depending on the physical theory and spatial or temporal resolution. For example, in quantum theory a quantity is mathematically represented by a so-called ‘operator’. And

on molecular scales, entropy has a meaning connected with probability theory. Yet, these seven quantities are universal in our present way of doing physics and of describing and understanding physical phenomena all around and within us.

Let us make a first acquaintance with these seven quantities. The discussion that follows is meant as an introduction. We shall repeat and say more about each quantity in later chapters. But **remember**^{§1.4 p.31} that it is very difficult, if not impossible, to answer questions like “what is *really* the quantity. . . ?”.

§ 3.3 Matter

Matter: units and notation

Matter, also called *amount of substance*² in chemistry, is a *scalar* quantity. The amount of matter in a volume will be denoted N . Its *unit*³ is the *mole*, mol. In chemistry we usually specify what kind of matter we are speaking about, writing for instance $N_{\text{Ca}} = 5.3 \text{ mol}$, to indicate an amount of 5.3 mol of *calcium*⁴ atoms.

Flux of matter will be denoted $\mathcal{J}$. Supply of matter will be denoted $\mathcal{R}$. Their unit is the *mole per second*, mol/s.

In statistical mechanics and particle physics, matter is often simply counted and thus measured in dimensionless units, rather than in moles.

Matter is probably the easiest quantity to grasp intuitively: it is what we ordinarily call “stuff”. It is usually classified into several kinds. The classification depends on the physical phenomena and theory one works with. A building engineer, for instance, could classify “matter” into different kinds of *materials* – such as wood, concrete, steel, sand, plastic, and so on – keeping track of the amount of each material in different regions of space, its movement, its rate of production and transformation. Each material has different physical properties.

A chemist could classify matter into different *substances* – such as water, hydrogen, oxygen, carbon dioxide, and so on – again keeping track of their amounts, movements, production. According to this classification, the “materials” of the building engineer would be mixtures of the different substances. But note that there is no clear boundary between one classification and the other.



One mole of different substances (image: NIST¹).

A chemist could also classify matter into different kinds of *atoms* – such as [hydrogen](#)⁵, [helium](#)⁶, [lithium](#)⁷, and the other kinds that appear in the [periodic table](#)⁸ – and seeing substances and materials as combinations of these different atomic kinds of matter. This classification is special because these different kinds have, at least approximately, the property of being [conserved](#) ^{§5.3 p. 130}: their amounts in a container or in a region of space can only change if these kinds are entering through an opening in the container or through the boundary of the region of space. In other words, they cannot be created or destroyed. This conservation property is only approximate, however. [Radioactive atoms](#)⁹ can transmute from one kind to another. This possibility is crucial and must be taken into account in phenomena involving [radioactivity](#)¹⁰ and [nuclear energy-mass](#)¹¹.

A chemist or a particle physicist may classify matter into fewer different kinds: protons, neutrons, electrons, anti-protons, anti-neutrons, anti-electrons (also called *positrons*), seeing the different atomic kinds as being made of these six basic ones. These kinds may be conserved even when kinds of atoms are not.

But a nuclear or particle physicist knows that the conservation properties of the six kinds above is also only approximate, and there are also other kinds, produced only in special circumstances.

We therefore go down into more and more subtle classifications. This kind of research is still open, but it seems that the total amount of [baryonic](#)¹², (including protons and neutrons) and [leptonic](#)¹³ (including electrons) matter is always conserved.

According to the definition of matter that we're adopting, the total amount of some kind of matter in a region can in principle be *negative*. A negative amount simply denotes the presence of [anti-matter](#)¹⁴. Anti-matter appears in small amounts in everyday life, for example in connection with common radioactivity processes. It is also created and used in medicine, in [positron-emission tomography \(PET\)](#)¹⁵ scans. In ordinary chemical applications, however, all amounts of matter within a region are usually positive or zero.

Why do we need to worry about how matter gets classified depending on the application? Because for describing a physical system and predicting its behaviour we usually have to use at least one physical law *for each kind of matter*. So the more kinds of matter we have to keep track of, the more equations we will have.



In positron-emission tomography there is creation of amounts of matter (leptonic matter) that can be considered negative: their *lepton number* is negative ([image: Helse Bergen](#)¹⁶).



Ambiguity of the term ‘matter’

In these notes we use the term ‘matter’ in the generic sense discussed above. But be aware that in some disciplines this term may have a much more specialized and slightly different meaning. It may *not* even be used at all. In chemical applications, for instance, one typically speaks specifically of ‘compounds’, ‘mixtures’, ‘substances’, ‘elements’, rather than ‘matter’. A particle physicist speaks of *matter* and *anti-matter*, but in the present notes the term ‘matter’ refers to both.



Matter is different from mass-energy

It is important to clearly distinguish *matter* from *mass-energy*. Mass-energy can be considered a property of matter, but the two are different. In nuclear reactions, for instance, the mass-energy of some amount of matter may change, while the amount of matter stays the same.

As far as we know, the total amount of mass-energy associated with an amount of matter is always positive, whether the amount of matter is positive or negative (antimatter). This is the reason why antimatter “falls downward” just like positive matter, a fact that has been experimentally confirmed (Anderson et al. 2023).



Exercise 3.1

According to statements on [symmetrismagazine.org](https://www.symmetrismagazine.org)¹⁷ and [quantumdiaries.org](https://www.quantumdiaries.org)¹⁸,

The average banana (rich in potassium) produces a positron roughly once every 75 minutes.

Unfortunately the original site where this statement was discussed, and the corresponding calculation made, seems not to exist anymore.

1. Do a little research and find out whether this statement is true.
2. From your research, approximately quantify the flux of positrons around an ordinary banana, expressing it in particles/s.



How many positrons do bananas produce?

§ 3.4 Magnetic flux

Magnetic flux: units and notation

Magnetic flux is a *scalar* quantity.

The **unit for magnetic flux**¹⁹ is the *weber*, Wb. The “flux of magnetic flux” is called *electric potential difference* or *electric tension*; its **unit**²⁰ is the *volt*, $V = \text{Wb/s}$.

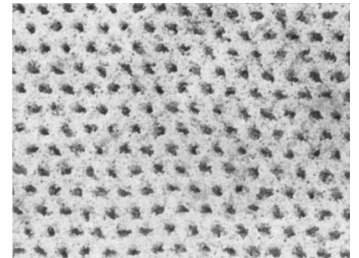
The magnetic flux is usually calculated by means of a *vector* quantity called *magnetic flux density*; its **unit**²¹ is the *tesla*, $T = \text{Wb/m}^2$.

The electric potential difference is usually calculated by means of a *vector* quantity called *electric field strength*; its unit is the *volt per metre*, V/m .

As we shall see in more detail in chapter 8, magnetic flux differs from the other six main quantities in that it answers the two “how much?” questions *in one lower dimension*: “How much magnetic flux is in this surface?” and “How much magnetic flux crosses this curve in the unit of time?”. It also requires a slightly different notion of orientation of a surface. The “flux of magnetic flux”, or *electric potential difference*, is therefore a flow connected to a curve.

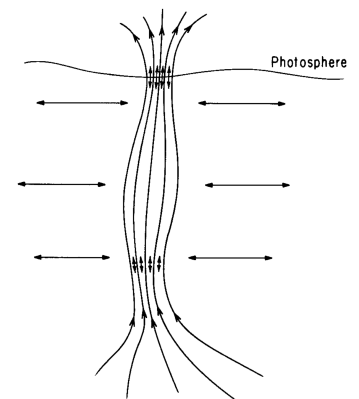
The magnetic flux density and the electric field strength together are usually called “electromagnetic field”, which is therefore commonly represented by two vectors associated to each point in space. But it can also be interpreted and visualized as a collection of spinning magnetic *tubes*, either closed or extending indefinitely, which move around. This visualization is somewhat analogous to how we visualize matter and charge, as moving blobs, but with one more dimension. This interpretation goes back to Faraday (1846), Maxwell (1855), and later Dirac (1955) among others, and today is conveniently used in some fields such as **solar physics**²², for example to study **sunspots**²³ (see Ryutova 2018). In particular situations, for example in some superconductors subjected to external magnetic fields, the magnetic-flux tubes can literally be seen and even tracked as they move around (see previous side figure).

✂ To be completed in a later version



“magnetic-flux lines emerging from the surface of a Type II superconductor”

Essmann & Träuble 1971



“sketch of the magnetic lines of force in a magnetic filament extending up through the photosphere.”

Parker 1974a

§ 3.5 Electric charge

Electric charge: units and notation

Electric charge is a *scalar* quantity.

The **unit for the amount of electric charge**²⁴ is the *coulomb*, C. The flux of electric charge is called *electric current*, its **unit**²⁵ is the *ampere*, $A = C/s$.

✂ To be completed in a later version

§ 3.6 Energy-mass

Energy-mass: units and notation

Energy is a *scalar* quantity.

The amount of energy in a volume is usually denoted E . Its unit for the is the *joule*, J.

Flux of energy will be denoted W . Supply of energy will be denoted $\mathcal{A}$. Their unit is the *joule per second*, J/s, also called *watt*, $W = J/s$.

Equivalently we can speak of mass. The amount of mass in a volume is usually denoted m . Its **unit**²⁶ is the *kilogram*, kg.

We shall not use special symbols for the flux or supply of mass. Their unit is the *kilogram per second*, kg/s.



The *chemical* energy-mass content in an ordinary AA battery is around 10 000 J. The *total* energy-mass content, including rest energy-mass, is around 10^{15} J.

The notion of energy is extremely important today, and central in many world-wide discussions and worries – think of today’s “energy crisis”, the need for “renewable energy”, and so on. It is somewhat funny that despite the importance of this notion it is actually difficult to answer ‘what is energy, really?’. Often we speak about energy as something that “flows”, is “transported”, “converted”, “stored”, and similar visualizations. This intuition will be enough in these notes. The notion of *mass* is also very intuitive in our everyday life; we associate it with the “resistance” we feel when setting objects into motion or when stopping them from moving, or with the weight of objects.

From Relativity Theory, and experimentally, we know that *energy and mass are the same quantity*, and in these notes we shall emphasize this experimental fact.

Energy is mass, mass is energy

Let's see some examples of why it is impossible to make a distinction between energy and mass. The following examples have been simplified in some of their aspects, but their main point is valid.

Heated gas. Imagine we have a box with a given amount of gas, say 1 mol of oxygen molecules. Using an extremely precise weighing scale, we observe that the mass of the gas is, say, exactly

0.031 999 540 000 000 000 kg .

Now we heat the gas, providing 60 J of energy, while making sure that not a single molecule of oxygen gets into or out of the box. The temperature of the gas increases by around 3 K. We observe that the weight measured by the scale increases while we heat the gas, reaching the new value

0.031 999 540 000 000 668 kg .

Clearly the mass has increased, but no molecules were added! The additional mass of around 7×10^{-16} kg is exactly the energy (60 J) that we provided to the gas by heating. Energy has weight, energy is mass.

Stretched or moving rubber band. Take a common rubber band, and imagine again that we have an extremely precise weighing scale. The rubber band, unstretched, has a mass of exactly

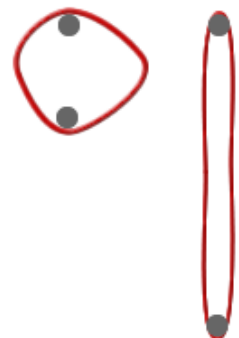
0.000 500 000 000 000 000 000 kg .

Now we stretch the band a little. By doing so we give energy to the band, which is said to acquire 'elastic energy'. Let's say we have given 0.3 J to the band this way. Now we weigh the rubber band again, while stretched. We measure a mass of

0.000 500 000 000 000 003 338 kg .

The extremely small difference of around 3×10^{-18} kg from the initial mass is exactly the elastic energy (0.3 J) that we provided by stretching. Energy has weight; energy is mass.

Now set the unstretched band in motion. Owing to the motion, the band is said to have acquired ‘kinetic energy’; let’s say an amount 0.3 J.



When we stretch a rubber band, its mass increases slightly – even if the amount of rubber remains exactly the same.

If we could weigh the band while in motion (but without moving the weighing scale), we would observe again a mass of approximately

$$0.000\,500\,000\,000\,000\,003\,338\text{ kg}.$$

The small difference from the initial mass is the additional kinetic energy of the band. Energy has weight; energy is mass.

Silicon crystal and gas At a temperature of 0 K, take a number of **silicon-28 atoms**²⁸ forming a crystal of mass exactly 1 kg. Then *that same* number of atoms, in dilute gas form, always at 0 K, has mass

$$1.000\,000\,000\,18\text{ kg}.$$

The equivalent mass of the cohesive energy of a crystal of silicon or graphite reduces the mass of the crystal by parts in 10^{10} as compared to the mass of the same number of unbound atoms, assuming in both cases that the atoms are at rest and in their ground state. (Davis & Milton 2014)



Prototype of a silicon-28 single crystal (Leibniz-Institut für Kristallzüchtung²⁷)

Fission and atomic bombs. The **atomic bomb**³⁰ is a dark example of the fact that mass is energy. In phenomena of nuclear fission, we notice a decrease in the weight, measured at rest, of nuclear material before and after the phenomenon of fission. But we also observe that a great amount of (kinetic) energy is released. This amount is exactly equal to the apparently missing weight.

Electric heater. As a final example consider a 1000 W electric heater, which is radiating 1000 J in one second. The heater is also losing around 0.000 000 000 000 01 kg of mass every second owing to this heat radiation – although it's also acquiring the same amount of mass as electromagnetic energy.



Hydrogen Bomb Test, 1954 (National Museum of Nuclear Science & History²⁹)

The practical use of the words 'mass' and 'energy'

From the examples above it becomes clear that energy and mass are two names for the same thing. The equivalence between energy and mass is given by the famous formula $E = mc^2$, where c is the speed of light, eq. (2.2). In their respective units this gives

$$1\text{ kg} = 8.987\,551\,787\,368\,176\,4 \times 10^{16}\text{ J} \quad (\text{exactly})$$

$$1\text{ J} \approx 1.112\,65 \times 10^{-17}\text{ kg}$$

To grasp these numbers, consider that the mass of the rubber band in the

"we are led to the more general conclusion: The mass of a body is a measure of its energy content; if the energy changes by L , the mass changes in the same sense by $L/9 \cdot 10^{20}$, if the energy is measured in ergs and the mass in grams." Einstein 1905b

example above, 0.5 g, is comparable to the energy released by the **atomic bomb over Hiroshima**³¹.

But in our daily experience we deal with energy-mass in two different ways:

On the one hand, we deal with huge (atom-bomb-like) amounts of energy-mass packed in very small volumes: the huge amounts of energy-mass that go together with objects and stuff like pens, keys, bicycles, cars, houses, water, and so on. We move, push, pull these huge energy-mass amounts from one place to another, and even put them in our pockets. We ourselves are huge bundles of energy-mass moving around. These amounts of energy-mass change a little, all the time, as in the example above with the rubber band. But these changes are so small as to be undetectable with ordinary weight scales, and negligible for practical purposes. We use the word ‘mass’ for any such huge amount of energy-mass, and measure it with a unit – kg – that doesn’t lead to ridiculously large numbers. And we also agree to neglect the imprecision and fluctuation in its measurement, say any imprecision **under 0.000 01 %**³². So we say “the rubber band has a mass of 0.0005 kg”, rather than “the rubber band has an energy-mass of 45 000 000 000 000 J”.

On the other hand, we also deal with the small energy changes and exchanges in all these objects. These energy exchanges are very important for our daily life: they keep us warm, keep our cells active, make our laptops work. In dealing with these energy exchanges, we don’t care about the huge energy reservoirs they come from. So we agree to measure them with a unit – J – that doesn’t lead to ridiculously small numbers. And we also agree not to be precise about the total amount in the reservoir from which these bits of energy come.

As an analogy, think of when we speak about the amount of people in different countries. We can say that in Norway there are 5 millions, and in India 1500 millions, so in India there are 300 times more people. By this we don’t mean that in Norway there are *exactly* 5 000 000 people and that India has *exactly* 300 times more people. These numbers are approximate, and change slightly all the time. We don’t care if these approximate numbers are off by 10 or even 100 000 people. At the same time, if we have three dear friends or relatives visiting us from abroad, then the amount of 3 people is now for us very important, and we know it exactly – even if it is a very small amount compared to the total population of the country we live in.

“although an exact relationship between mass and amount of substance is not fundamentally challenged by the technologies used to weigh chemicals (which will subsequently be improved), a point will be reached when the equivalent masses of molecular bond energies can no longer be considered negligible.”

Davis & Milton 2014

The distinction above is of course not clear-cut. In dealing with some physical phenomena, for example with few molecules or with subatomic particles, the but pragmatic but fictitious distinction between mass and energy becomes too blurry and not useful anymore. In discussing these phenomena, indeed, one often uses the terms ‘mass’ and ‘energy’ interchangeably, as well as a common unit for both, such as the *electronvolt*³³.

In these notes we shall often use the expressions ‘energy-mass’ and ‘mass-energy’ to remind ourselves that these two words denote the same physical entity.

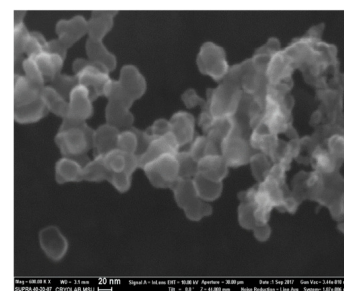
Different ‘forms’ of energy-mass

We often speak of different *forms* of energy-mass. The most important forms for us will be **internal**, **kinetic**, **gravitational**, **electromagnetic**.

The differences among these forms of energy-mass arise from the way they are calculated from other quantities, as we shall see later. For example, if in a volume there’s an amount of a particular kind of matter, then in that volume there must also be an amount of energy-mass, given by a formula that involves the amount of matter. And if that matter is moving, then we have to add an extra amount of energy-mass given by another formula which involves the velocity. And if in that volume there’s a gravitational field (that is, a particular kind of spacetime curvature), then another extra amount of energy-mass must be added, given by yet another formula involving the gravitational field. Similarly if we know that an electromagnetic field is in that volume.

We also speak of different forms of flux of energy-mass. The most important for us will be **heat** and **mechanical power**. The difference is again in how these fluxes are calculated depending on whether there are also fluxes of matter and of other quantities.

The distinctions between different forms of energy-mass also depend on the observation scale and the theory used. Take for instance some water in a glass resting on a table. We can observe and describe that water on a macroscopic scale of centimetres, seeing it as a still, uniform fluid. On this macroscopic scale we say that the water has some internal energy-mass, or that there is internal energy-mass in the glass. But we can also observe and model that same water as a collection of molecules, on a microscopic scale of nanometres (10^{-9} m). On this microscopic scale we say that water has internal energy-mass *and* kinetic energy-mass, because the molecules are in constant motion. The *total* amount of energy-mass is the same on



Hydrocarbon fuel particles³⁴. The small blobs have size of around 2×10^{-8} m.

the centimetre-scale and on the nanometre-scale, but its partition into different “forms” depends on the scale: only internal on the centimetre scale, and internal plus kinetic on the nanometre scale.

The same is true for flux of energy-mass. What we call ‘heat’ on one observation scale appears as a flux of energy-mass not associated with the motion of matter. But on a finer scale it is instead called ‘work’, and it appears as an energy-mass flux associated with the microscopic motion of matter.



Exercise 3.2

In an hour, 14 people exit through a door. Taking the average human weight to be 62 kg (Walpole et al. 2012), what’s the average flux of *energy-mass*, in J/s, through that door?

§ 3.7 Momentum



Momentum: units and notation

Momentum, also called *linear momentum* or *translational momentum* to distinguish it from angular momentum, is a *vector* quantity.

The amount of momentum in a volume is usually denoted **P**. It can be expressed in several equivalent units; we shall keep in mind especially these three:

$$\begin{array}{ccccc} \text{newton second} & \equiv & \text{kilogram metre per second} & \equiv & \text{joule second per metre} \\ \text{N} \cdot \text{s} & & \text{kg} \cdot \text{m/s} & & \text{J} \cdot \text{s/m} \end{array}$$

Flux of momentum is also called *contact force* or *surface force*, usually denoted **F**. Supply of momentum is also called *body force* or *volume force*, and will be denoted **G**. Flux and supply can be expressed in several equivalent units:

$$\begin{array}{ccccc} \text{newton} & \equiv & \text{kilogram metre per squared second} & \equiv & \text{joule per metre} \\ \text{N} & & \text{kg} \cdot \text{m/s}^2 & & \text{J/m} \end{array}$$

Since momentum and momentum flux are vector quantities, they are usually expressed with three numbers, typically their *x*-, *y*-, and *z*-components.



A walking person has, with respect to the ground, an amount of horizontal momentum of around 70 N s (image: Antonio Romei³⁵).



A 10 cm beam of light from a 60 W torch has an amount of momentum around 10^{-16} N s.

Momentum is a subtle quantity, maybe subtler than energy-mass. Textbooks that focus on Newtonian mechanics *define* it as the product of the mass and the velocity of a body, usually written “ $\mathbf{p} = m\mathbf{v}$ ”. This relation, however, is only valid in special circumstances, and cannot be used in many technological applications, especially when electromagnetism or high speeds are involved. For example, in a small region where there are matter and electromagnetic fields, momentum is approximately given by

$$\mathbf{P} = m\mathbf{v} + \epsilon_0 \mathbf{E} \times \mathbf{B} ,$$

where the quantities $\mathbf{E}$, called electric field strength, and $\mathbf{B}$, called magnetic flux density, are related to the electromagnetic properties of matter. If we were describing a volume V of matter composing a star, we would use a formula for momentum like the following:

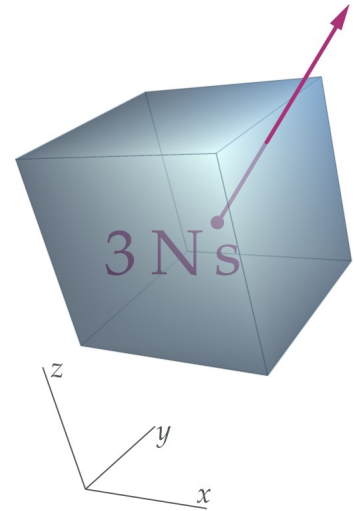
$$\mathbf{P} = m_r \mathbf{v} + \frac{1}{c^2} m_r \mathbf{v} \left(\frac{1}{2} v^2 + U/V - 2G \right) + \frac{1}{c^2} (\mathbf{\Phi} + p \mathbf{v})$$

where m_r is the energy-mass of that matter if it were at rest, U its internal energy, $\mathbf{\Phi}$ the heat flux through it, p its pressure, and G a term describing gravitational effects. Still other, different formulae for momentum are used in other contexts.

It is therefore beneficial to separate our idea of momentum from the “mass-times-velocity” formula, keeping in mind that the latter is just a particular case of momentum. Instead, think of momentum as *something associated with translational motion* of matter and of electromagnetic fields. Translational motion is the kind of motion that leads to a new position in space. For instance, when you walk from one place to a different one, you have performed translational motion (note that translational motion doesn’t need to be in a straight line). So if something – be it matter or a magnetic-flux tube – is changing its position in space, then that something has momentum. Indeed in many languages – for example Chinese, French, Italian, Japanese, Norwegian, Spanish, Swedish – the term for momentum is literally ‘quantity of motion’.

Given a particular volume at a particular instant in time, and given a coordinate system, we can speak of the total amount of momentum within that volume. This amount is represented by a *vector*. You can imagine a continuous collection of vectors filling the volume, possibly with different directions and small magnitudes; the total momentum is the sum of all these vectors. This visualization obviously comes with many warnings, but it can be very useful if we are careful.

Fluxes and supplies of momentum are what we call **forces**. Whenever we exert a force on an object, for instance when we pull a door or simply hold a bag, we are transferring momentum between us and that object. The force of gravity, or weight, that you feel every day in your body, is a continuous production of momentum that happens because of the gravitational-inertial field of the Earth in the specific coordinate system that's fixed with you. The force that presses you against your seat when you sit in an accelerating car or in an aeroplane taking off, is also an instance of production of momentum by the same gravitational-inertial field. The term 'force' is therefore synonymous with 'flux of momentum' or 'supply of momentum'.



The amount of momentum within a volume at a given instant is represented by a **vector**



Momentum and energy-mass flux are proportional

According to Relativity Theory, momentum is always proportional to energy-mass flux, and energy-mass flux is always proportional to momentum. If we represent the magnitude of momentum in a volume V with P , and the flux of total energy-mass through an area A with W , their proportionality can be approximately expressed as

$$\frac{W}{A} \approx \frac{P}{V} c^2. \quad (3.1)$$

Compare this formula with $E = mc^2$. From this point of view, you can think of momentum as "energy-mass in motion".



Exercise 3.3

Assume the relation $P = mv$ between the magnitude of momentum P , mass-energy m , and its speed v . When you're walking with a speed of 1 m/s, how much momentum does your body contain?

§ 3.8 Angular momentum

Angular momentum: units and notation

Angular momentum, also called *moment of momentum* or *rotational momentum*, is a *vector* quantity.

The amount of angular momentum in a volume is usually denoted $\mathbf{L}$. It can be expressed in several equivalent units; we shall keep in mind especially these three:

$$\begin{array}{ccccc} \text{newton metre second} & \equiv & \text{kilogram squared metre per second} & \equiv & \text{joule second} \\ \text{N} \cdot \text{m} \cdot \text{s} & & \text{kg} \cdot \text{m}^2/\text{s} & & \text{J} \cdot \text{s} \end{array}$$

Flux of angular momentum is also called *contact torque* or *surface torque*, and usually denoted $\mathbf{M}$. Supply of angular momentum is also called *body torque* or *volume torque* and will be denoted $\mathcal{T}$. Flux and supply can be expressed in several equivalent units:

$$\begin{array}{ccccc} \text{newton metre} & \equiv & \text{kilogram squared metre per squared second} & \equiv & \text{joule} \\ \text{N} \cdot \text{m} & & \text{kg} \cdot \text{m}^2/\text{s}^2 & & \text{J} \end{array}$$

Since angular momentum and its flux are vector quantities, they are usually expressed with three numbers, typically their x -, y -, and z -components.

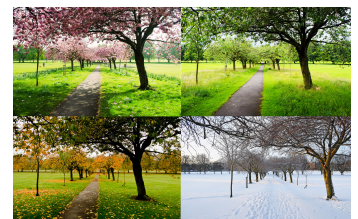
Given a particular volume at a particular instant in time, and given a coordinate system, we can speak of the total amount of angular momentum within that volume. This amount is represented by a vector.

Just as momentum is associated with translational motion, angular momentum is *something associated with rotational motion* of matter and of electromagnetic fields. Rotational motion is the kind of motion that leads to a *new orientation* in space, rather than to a new position. For instance, if you turn to your left or to your right while standing in place, you have performed a rotational motion.

You can get a feeling of angular momentum and torque by playing with a hand-held gyroscope (side figure), typically used for wrist exercise. But the physical role of angular momentum is actually in front of us every day: its balance is the chief reason of such important phenomena as the alternation of day and night and the alternation of the seasons.



A spinning dancer has, with respect to the ground, an amount of angular momentum of around 10 N m s (image: RG-Dance³⁶).

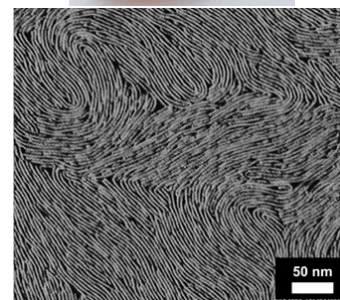


Yet, angular momentum is an even subtler quantity than momentum, and usually it's less in the spotlight than momentum. This happens for several reasons.

One reason is that the distinction between linear momentum and angular momentum is not clear-cut and depends on our choice of coordinates. For instance, what can be considered as a component of linear momentum in a polar coordinate system, can also be considered as a component of angular momentum in a rectangular coordinate system. This reflects the fact that there isn't a clear-cut distinction between translational and rotational motion; usually they involve each other to some degree. A translational motion can be interpreted as a rotation around a point that is very far away; and the rotation of an extended object can be interpreted as a collection of small translational motions of its parts.

Another reason is that the balance law for angular momentum can be expressed in an alternative way: as a set of symmetries between momentum fluxes, and between energy-mass flux and momentum. With this alternative approach we do not even need to introduce the quantity 'angular momentum' at all! But we would still need to include an additional universal law, independent of those of momentum and energy.

This approach is indeed followed for many physical phenomena and applications, for example those that involve fluid motion. But for other phenomena it's more convenient to speak of angular momentum and to use its balance law. For example, in phenomena involving liquid **polymers**³⁷, elementary particles, or electromagnetic radiation it's convenient to use angular momentum and to let it include an additional part, called *spin* or *intrinsic angular momentum*, which is associated with rotational motion that is invisible to the naked eye or to the particular instruments used to measure motion. We shall further discuss these topics in chapter 12.



Some liquid polymers (**top**: Liquid Diethoxymethane Polysulfide) need to be described with a special kind of angular momentum, owing to their molecular structure (**bottom**).



Exercise 3.4

Assume the relation $L = mvR$ between the angular momentum L , the mass-energy m , the speed v , and the orbital radius R of a planet around a star. For the Earth³⁸ we have approximately

$$m = 6.0 \times 10^{24} \text{ kg}, \quad v = 3.0 \times 10^4 \text{ m/s}, \quad R = 1.5 \times 10^{11} \text{ m}.$$

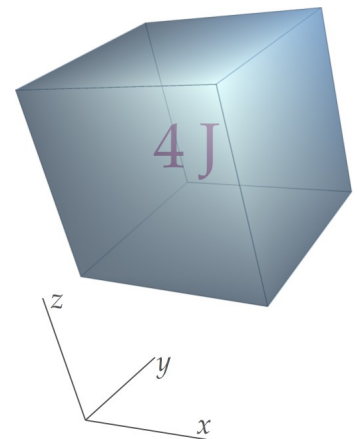
How much is the Earth's angular momentum around the Sun? (Note that we are neglecting the angular momentum coming from the Earth's own rotation, which is more than one million times smaller.)

§ 3.9 Energy-mass, momentum, angular momentum are coordinate-dependent

An aspect of energy-mass, momentum, angular momentum that must always be kept in mind is that **their amounts depends on the coordinate system we're using**. If someone points at a specific region of space at a particular instant, and asks "how much energy-mass is there?", we *cannot* give an answer until a coordinate system is specified. Once the coordinate system has been chosen, then a precise and unambiguous answer can be given. The same is true for the flux of energy-mass through a surface, and for the amounts, fluxes, supplies of momentum and of angular momentum. This coordinate-dependence also means that observers using different coordinates will usually assign different amounts of energy-mass, momentum, angular momentum to the same regions of spacetime.

This is an important difference between energy-mass, momentum, angular momentum on one side, and matter, magnetic flux, electric charge on the other side. *For matter, magnetic flux, electric charge, the questions about their contents and fluxes can be answered unambiguously independently of any spatial coordinate system.*

This coordinate-dependence is not a problem: we must in practice always specify our coordinate system anyway, in order to agree on the time and position of physical events. But it can cause problems if we calculate or measure some amount of energy-mass at a given time and place using a coordinate system, and we calculate or measure some amount at another time or place using a *different* coordinate system. Combining these two amounts, or using them in the same physical law, has no meaning whatsoever.



"How much energy-mass is there in this volume at this instant?" This question cannot be answered until we have specified which coordinate system we're using.

⚠ Amounts of energy-mass, momentum, angular momentum are coordinate-dependent

Never change coordinate system in the middle of calculations about energy-mass, momentum, or angular momentum!



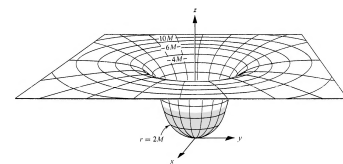
What are energy-mass, momentum, angular momentum?

From the discussions and formulae above, it seems that energy-mass, momentum, angular momentum are quite closely related to one another. For all three, the amount in a volume or through a surface is undefined unless we specify a coordinate system. And

we shall see later that all three satisfy balance laws but not necessarily conservation laws.

Relativity Theory indeed shows that energy-mass, momentum, angular momentum are different aspects of one single geometric object, called *energy-momentum tensor*. They are like its “shadows”, that we can observe by looking at it from different points of view in time and space. This is also why their values get intermixed if we change our system of coordinates. This topic will be discussed in chapter 14.

General Relativity gives a new meaning to these quantities: they are *particular curvatures of spacetime*. They express how spacetime is curved in different directions. So whenever we measure, say, the energy-mass or the momentum of some object or of some electromagnetic radiation, we are actually measuring how much that object or radiation is curving spacetime in a particular way.



Energy-mas, momentum, angular momentum are measures of particular curvatures of spacetime.

§ 3.10 Entropy



Entropy: units and notation

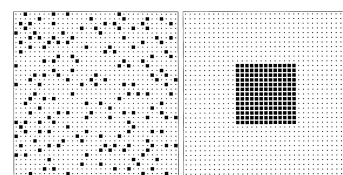
Entropy is a *scalar* quantity. The unit for entropy content is the *joule per kelvin* J/K; the unit for entropy flux is *joule per kelvin per second* J/(K s).

The amount of entropy in a volume is usually denoted S . The flux of entropy through a surface is denoted Π .

From its SI physical dimension and unit, entropy would seem to be derived from temperature. However, although temperature is taken as primitive quantity by the SI, the *definition of temperature*³⁹ actually depends on a fixed value of *Boltzmann's constant*⁴⁰, which has the dimension of entropy.

Entropy is probably the most difficult quantity to grasp intuitively. Many seemingly intuitive descriptions given in some textbooks are, unfortunately, unhelpful and even misleading. One particularly *misleading* description is to say that entropy is a “measure of disorder”. Besides the fact that “disorder” is a very vague and subjective notion, it turns out that some physical phenomena, for example involving *liquid crystals*⁴¹, can be considered more “disordered”, and yet have *lower* entropy, than others. See also the example in the side figure. We shall discuss more about such phenomena later on.

In these notes we shall rely on the idea that *entropy expresses a limit on the flux of a particular kind of energy-mass*. Said in simpler but more imprecise



Two microscopic configurations of a lattice gas. **Left:** configuration coming from a *low*-entropy state. **Right:** configuration coming from a *high*-entropy state (Styer 2000).

words, entropy is a bound on how fast we can heat something up. We shall develop this idea further later.

One reason why entropy is difficult to grasp intuitively is that it has very different physical and mathematical aspects depending on the spatial scales and physical theory that we use to describe physical phenomena.

In many “continuum” phenomena, that is, phenomena where the molecular constitution of matter is not visible or not taken into account, entropy is treated as a “stuff-like” quantity similar to energy-mass or electric charge. But there are difficulties also in this case. For some phenomena, for example involving non-elastic materials such as a simple paper clip, it is possible to introduce several entropies having different values – and not just because of a change in measuring scale – all of which can serve their purpose perfectly fine.

In molecular phenomena involving statistical mechanics, on the other hand, entropy is no longer a physical notion, but a *probabilistic* and *statistical* one, related to guesses and inferences that we make about the physical phenomenon. Yet from many points of view it has roles similar to those of the entropy used in continuum phenomena.

We shall see later that the physical laws for entropy have also a different status with respect to the laws for the other six main quantities: they are, so to speak, “laws about laws”.

§ 3.11 Auxiliary quantities

Besides the seven principal quantities, other auxiliary quantities appear in some physical theories. Important examples are *temperature*, *metric*, *strain*, *magnetization*. Some auxiliary quantities are not extensive; for instance we cannot ask “what’s the total amount of temperature in this region?”.

The dimensions, units, and scalar or vector character of all quantities mentioned so far are summarized in table 3.1 on page 89.

§ 3.12 Thermodynamic temperature

We have an intuitive understanding of the notion of *hotness* and *coldness*. Temperature quantifies these notions. The physical bases and measurement procedures for this quantification are far from trivial, but we shall take them for granted in the present notes.

For some physical phenomena, especially those involving gases, we know that temperature is related to the invisible motion of microscopic

parts of matter, such as molecules. But there are also physical phenomena for which our microscopic understanding of temperature is more complex, and in some cases still unclear.

Temperature is a *scalar* quantity. There are several definitions and scales of measurement for temperature. Of special importance is **thermodynamic temperature**⁴², also called *absolute temperature*, which is measured in *kelvins* (K). Thermodynamic temperature has the special property of being always positive in most physical phenomena (there are exceptions, especially in some phenomena where **statistical mechanics**⁴³ becomes relevant).

In these notes we shall use thermodynamic temperature, denoting it T . Its relation with Celsius temperature T_C , which is measured in *degrees Celsius* °C, is given by

$$T_C = T - 273.15 \text{ K} \quad (3.2)$$

that is, a redefinition of the “zero” value; for instance $25.00^\circ\text{C} = 298.15 \text{ K}$. Note that temperature *differences* are the same for the two temperature scales: $\Delta T_C = \Delta T$, because the constant zero-value cancels out.

Temperature is useful because it enters in many physical laws that involve energy-mass, but it’s easier to measure than energy-mass. Temperature generally depends on the time and place, so it can be a function of the coordinates: $T(t, x, y, z)$.



What is temperature?

Inventing Temperature by H. Chang (2004) gives a brilliant account of the history of invention of temperature, as well as an interesting portrait of how scientific concepts are born and develop.

Is There a Temperature? by T. S. Biró (2011) discusses fascinating physical phenomena for which our microscopic understanding of temperature is still incomplete.

§ 3.13 Metric

A very important quantity is a fundamental building block of all our physical theories: the **metric**. It is quite different from the seven fundamental quantities, from a physical and also from a geometrical point of view.

The metric characterizes our measurements of space and time. It’s the object that allows us to calculate how much *physical* time has elapsed, the *physical* distance between two objects, the volume (say, in cubic metres) of

a three-dimensional region of space, and the area (say, in square metres) of a surface. In General Relativity the metric allows us to calculate the curvature of spacetime.

The metric itself, in the technical sense of the so-called “metric tensor”, is *not* an extensive quantity. We can’t ask “what’s the total amount of metric in this region?”; that’s a meaningless question. There are, however, other quantities which can be derived from the metric and which are extensive. Important examples are the so-called “volume density” and “area density”, which are the ones that allow us to calculate extended volumes and areas.

In the Newtonian approximation, that is, for speeds smaller than the speed of light and low energy-mass densities (hence weak gravitational fields and small spacetime curvature), the metric is just a static, uniform object, the same everywhere in spacetime; and spacetime is *flat*, that is, it has no curvature. This is why we can speak of an ‘absolute time’ and ‘absolute distances’ in this approximation. In these notes we shall for the most part use this Newtonian approximation.

In General Relativity the metric is a dynamic object instead: it can change with coordinate time, and can vary from one point in space to another. These changes are determined by the seven main quantities, and the metric, in turn, determines changes in the seven quantities.

Quantity	SI dimension	Unit
Time	time	<i>second</i> s
Length, distance	length	<i>metre</i> m
Position	length	m
Matter	amount of substance	<i>mole</i> mol
Magnetic flux	$\text{mass} \cdot \text{length}^2 / (\text{current} \cdot \text{time}^2)$	<i>weber</i> Wb
Electric charge	current · time	<i>coulomb</i> C
Energy-mass	$\text{mass} \cdot \text{length}^2 / \text{time}^2$, mass	<i>joule</i> J, <i>kilogram</i> kg
Momentum	mass · length/time	N · s, kg · m/s, J · s/m
Angular momentum	mass · length ² /time	N · m · s, kg · m ² /s, J · s
Entropy	$\text{mass} \cdot \text{length}^2 / (\text{time}^2 \cdot \text{temperature})$	J/K
Temperature	temperature	<i>kelvin</i> K

Table 3.1 SI dimensions and units of the main physical quantities used in these notes. Their fluxes have the dimensions divided by time, and therefore units divided by seconds. Quantities in **boldface** are **vectors**, the others are scalars.

URLs for chapter 3

1. <https://www.nist.gov/image/moleedit2jpg>
2. <https://doi.org/10.1351/goldbook.A00297>
3. <https://www.nist.gov/si-redefinition/redefining-mole>
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6. <https://pubchem.ncbi.nlm.nih.gov/element/Helium>
7. <https://pubchem.ncbi.nlm.nih.gov/element/Lithium>
8. <https://iupac.org/what-we-do/periodic-table-of-elements/>
9. <https://www.ciaaw.org/radioactive-elements.htm>
10. <https://www.iaea.org/newscenter/news/what-are-radioactive-sources>
11. <https://www.iaea.org/newscenter/news/what-is-nuclear-energy-the-science-of-nuclear-power>
12. <http://hyperphysics.phy-astr.gsu.edu/hbase/Particles/hadron.html#c6>
13. <http://hyperphysics.phy-astr.gsu.edu/hbase/Particles/lepton.html>
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16. <https://www.helse-bergen.no/avdelinger/radiologisk-avdeling/senter-for-nukleermedisin-og-pet/tilvising-til-pet>
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31. <https://visual-archives-hiroshima.jp/en/damage/>
32. <https://www.nist.gov/si-redefinition/kilogram-disseminating-new-kilogram>
33. <https://home.cern/tags/13-tev>
34. <https://doi.org/10.4209/aaqr.2019.04.0177>
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37. <https://www.science.org.au/curious/everything-else/polymers>
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42. <https://www.iso.org/obp/ui/#iso:std:iso:80000:-5:ed-2:v1:en:tab:1>
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Content, flux, supply

When we regard energy as residing intrinsically in a body, we may measure its intensity by the amount contained in unit of volume. [. . .] The only way we have of defining the motion of the fluid is by considering it as a flux [. . .]. This distinction is still more necessary when we come to heat and electricity. The flux of heat or of electricity cannot be even thought of in any way except as the quantity which flows through a given area in a given time.

J. Clerk Maxwell 1869

§ 4.1 Content, flux, supply

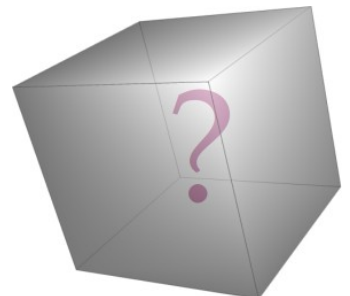
The concept of physical quantity allows us to look at the word in ways that can be quantified and expressed with numbers and mathematics. This is how we can formulate [physical laws](#) ^{§1.3 p.28}. In the previous chapter we made our acquaintance with seven physical quantities. They were chosen as our building blocks because their quantification is easy to grasp intuitively and to visualize. In the present chapter we study this quantification more rigorously, and start developing the necessary mathematics.

For each of the seven primitive quantities, except magnetic flux, we can measure three kinds of amount: **volume content**, **flux**, and **supply**:



Volume content

Volume content or **volume integral** or **storage** is the amount of quantity contained within a three-dimensional region, at a specific time instant. The volume content of a quantity does *not* depend on how the volume is moving. The volume content has the same physical dimension as the quantity.

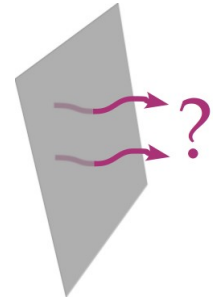




Flux

Flux or **current** or **flow rate** is the amount of quantity flowing through a two-dimensional surface in a given direction, *per time*, at a particular time instant.

The flux of a quantity through a surface depends on how that surface is moving and deforming. The flux has the physical dimension of that quantity divided by time.



Supply

Supply or **source** is the amount of quantity being produced or destroyed within a three-dimensional region *per time*, at a particular time instant.

The supply of a quantity in a region depends on how that region is moving and deforming. The supply has the physical dimension of that quantity divided by time.

For the magnetic flux we shall introduce analogous kinds of amount, but in one less dimension.

The volume content, flux, and supply of the four scalar quantities – matter, electric charge, energy-mass, entropy – are also *scalars*; that is, each is expressed by one number and a unit. The volume content, flux, and supply of the two vector quantities – momentum and angular momentum – are also *vectors*; that is, each is usually expressed by three numbers and units.



Different terminology across disciplines and literature

Keep in mind that many disciplines and their literature use terms different from ‘content’, ‘flux’, ‘supply’, to indicate the *same notions*. It’s therefore important that you understand the *concepts* these terms stand for, without getting too attached to the terms themselves.

The term *content* is very rarely used. In most disciplines, one simply speaks of the quantity *in* the control volume or *stored* in the control volume, without mentioning any ‘content’. For instance, instead of saying “the energy content in the control volume is 2 J”, one simply says “the energy in the control volume is 2 J”. In these notes we use the word

‘content’ to clearly distinguish its notion from those of *flux* and *supply*.

The term *flux* is often used in disciplines like fluid mechanics. In solid state physics and some other disciplines the term *current* is used instead; the most evident example is *electric current*, which is exactly the same as *flux of electric charge*. In physical chemistry and other disciplines, the term *flow rate* is used instead. You may even meet other technical terms for the same concept. In these notes ‘flux’ is chosen because it’s short and unlikely to be misinterpreted in a qualitative sense.

One more confusing aspect about the terms *flux* and *current* is that some disciplines use them to denote the notions we mean in these notes, but *divided by area*. This means that their units will also be different in those disciplines. The [International Organization for Standardization \(ISO\) recommends](#)¹ that the terms ‘areic flux’ or ‘flux density’, and similarly with ‘current’, be used instead to denote the division of a flux by the area of the surface through which it occurs. Our terminology agrees with the ISO.

The term *supply* is often used in disciplines like fluid mechanics. In solid state physics and some other disciplines the term *source* is used instead, and *sink* to denote a supply having a negative value. Some disciplines use *supply* and *source* to denote these notions but *divided by volume*. The ISO recommends instead to use the terms ‘volumic supply’ or ‘supply density’ in this case.

Symbols and notation

The three kinds of amount, for each quantity, are denoted by special symbols. Table 4.1 on page 96 summarizes the symbols, units, and scalar or vector character for the volume content, flux, supply of the seven quantities, as used in these notes. Let’s see some examples.

Consider a bottle on a table. You measure the amount of water molecules in it at a given time, and find that it’s 51.3 mol. We can express this as follows:

$$N_{\text{H}_2\text{O}} = 51.3 \text{ mol} .$$

The symbol ‘*N*’ is used for the volume content of *matter*. The kind of matter in this case is water, so we append its chemical formula H_2O as a subscript; but we could also have used the word ‘water’ or the letter ‘w’. We could



<i>Quantity</i>	<i>Vol. content [unit]</i>	<i>Flux [unit]</i>	<i>Supply</i>
matter	N [mol]	$\mathcal{J}$ [mol/s]	$\mathcal{R}$
electric charge	Q [C]	$\mathcal{I}$ [C/s or A]	
energy-mass	E [J] m [kg]	W [J/s or W] [kg/s]	$\mathcal{A}$
momentum	P [Ns]	F [N]	G
angular momentum	L [Nm s]	M [Nm]	$\mathcal{T}$
entropy	S [J/K]	Π [J/(K s)]	

Table 4.1 Symbols and units for volume content, flux, supply of six main quantities. Vector quantities are in **boldface**. Supplies have the same units as fluxes.

<i>Quantity</i>	<i>Flux [unit]</i>	<i>Circuitation [unit]</i>
magnetic flux	Φ [Wb]	$-\mathcal{E}$ [Wb/s or V]

Table 4.2 Symbols and units for flux and circutation of magnetic flux.

also append the word ‘bottle’ to indicate that we’re speaking about the volume content of the bottle.

As a second example, consider two ordinary batteries inside some device; one is on the left, the other on the right in the battery compartment. You measure the amount of chemical energy-mass in each battery at a given time, and find 9873 J for the battery on the left, and 4221 J for the one on the right. We can express this as follows:

$$E_L = 9873 \text{ J} , \quad E_R = 4221 \text{ J} .$$

The symbol ‘ E ’ is used for the volume content of *energy-mass*. In this case we’re speaking about chemical energy-mass, and we assume this is clear from the context, so we don’t indicate this in the symbols. But we must distinguish the volume contents of the two batteries, so we use the subscripts ‘L’ and ‘R’; but we could have used ‘l’ and ‘r’, or ‘left’ and ‘right’.

As a third example, consider a comb that has acquired electricity, say from rubbing with hair. You measure the net volume content of electric charge in the comb when your stopwatch shows 5 s, and find $-0.000\,000\,4 \text{ C}$ (note that the charge is more concentrated at the boundary of the volume). Then you measure it again 10 seconds later and find that there is no net charge. We can express this as follows:

$$\begin{aligned} t_0 &= 5 \text{ s} & Q(t_0) &= -0.000\,000\,4 \text{ C} , \\ t_1 &= 15 \text{ s} & Q(t_1) &= 0.000\,000\,0 \text{ C} . \end{aligned}$$

The symbol ‘ Q ’ is used for the volume content of *electric charge*. We assume that it’s clear from the context that we’re speaking about the charge of the comb, so we don’t indicate this in the symbols. The volume content Q of the charge is changing with time, that is, it is a *function* of time. Thus we indicate its value at different times by explicitly writing its argument in round brackets.

The examples above show that it depends on the context what kinds of subscript or additional signs that we use together with the symbols for volume content, flux, supply. We have some freedom in which additional signs to use; what’s important is to make the message unambiguous.

Volume content, flux, and supply always refer to some particular time, so strictly speaking they are functions of time and would need a time argument like ‘ (t) ’. But in situations where we are considering one particular time only, or where the quantities are constant in time, we can for brevity omit the time argument.



§ 4.2 Extensivity

The first kind of mathematical operation that we can do with our seven quantities is related to the other important property common to all of them, *extensivity*^{§3.2 p.69} (also called *additivity* in some literature):

- The content in a volume consisting of non-overlapping volumes is equal to the *sum* of the contents in the individual volumes.
- The flux through a surface consisting of non-overlapping surfaces is equal to the *sum* of the fluxes through the individual surfaces.
- The supply in a volume consisting of non-overlapping volumes is equal to the *sum* of the supplies in the individual volumes.

When we speak of sum, we mean an ordinary sum for scalar quantities, and a *vector* sum for vector quantities.

As an example, let's consider again the two batteries. If we take them together, we're considering a new volume, consisting of the volumes of the two batteries. We can denote the content of energy-mass in this volume with E_{tot} to distinguish it from the contents of the left battery E_L and of the right battery E_R . The content in the combined volume is given by

$$\begin{aligned} E_{\text{tot}} &= E_L + E_R \\ &= 9873 \text{ J} + 4221 \text{ J} = 14\,094 \text{ J} \end{aligned}$$

and this equation expresses the principle of extensivity.

§ 4.3 Control volumes and control surfaces

For each of six main quantities we can therefore say how much of that quantity is in a given volume, how much is flowing through a given surface per time, and how much is being created in a given volume per time. But how should we choose such volumes and surfaces? Are we constrained in their choice? Do they need to have particular shapes or sizes or positions? Do they need to follow the contours of particular physical objects?

These volumes and surfaces will play an important role in formulating our main physical laws; so one might think that they must be chosen in very special ways. But that isn't the case!

What's surprising and extremely useful is that **the choice of volume** for a volume content or supply, **and the choice of surface** for a flux, **are completely arbitrary**. Moreover, **these volumes and surfaces can be completely imaginary**.

Since they are under our control, and since they allow us to keep under control how the amounts of our main quantities change, we call them *control volumes* and *control surfaces*:

Control volume

A **control volume** or (3D) **domain** is an arbitrary three-dimensional region of space. This region can have any position, shape, and size, and these can change smoothly in time.

A control volume can also consist of several disconnected three-dimensional regions, and it can be completely imaginary.

Control surface

A **control surface** or **2D domain** is an arbitrary two-dimensional surface through which we can choose a crossing direction. This surface can otherwise have any position, shape, and size, and these can change smoothly in time.

A control surface can also consist of several disconnected surfaces, and it can be completely imaginary.

As explicitly stated in the definitions above, control volumes and control surfaces don't need to be static: they can move and deform.

The fact that we can choose control volumes and control surfaces arbitrarily gives us a lot of power in solving physics problems and in simulating and predicting the behaviour of physical phenomena. Control volumes and surfaces are typically chosen so as to simplify the equations that describe the physical situation and to focus on details of interest.

Up to now our discussion has been somewhat abstract. This abstractness reflects the amazing flexibility of the notions of control volumes and control surfaces. We now explore some more concrete examples.

Examples

Let's start with an informal example of control volume and control surface.

Consider a classroom with some people within. In your imagination you can divide the classroom into two halves, say the front half and the rear half. You effect this division by imagining a two-dimensional, immaterial surface, going from the floor to the ceiling, and from one side wall to the

opposite one. The exact position and shape of this imaginary surface are up to you, as long as the surface has a clear crossing direction.

You can then simply measure how many people are in the rear half at a particular time instant, say time 14:43:27 of a specific date. This measurement could be done for instance by taking a photograph at exactly that instant, and then counting the people in the rear half on the photograph.

What you just did was to choose a *control volume* at a given time, and to measure its “volume content” of people. Note how the control volume in this case was defined partly by real objects (half of the ceiling and floor, half of the side walls, the rear wall) and partly by an imaginary object (your imagined surface dividing the classroom).

You can also focus on your imaginary surface only, rather than the room, and with the help of a video camera you can keep track of how many people cross the imaginary surface between times 14:43:27 and 14:43:30. You could for instance observe that in those three seconds 10 people cross the imaginary surface from the rear half to the front half of the room, and 4 people cross it from the front rear. So a net number of 6 people cross from the rear to the front half in three seconds, or 2 people/second on average.

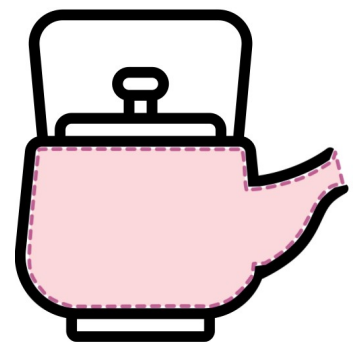
What you just did was to consider a *control surface*, and to measure the “flux” of people through it. In this case the control surface was defined by an imaginary object (your imagined surface). But you could also consider a different, larger control surface, consisting for instance of the imaginary surface and one side wall. This control surface would be defined partly by a real object (the side wall) and partly by an imaginary object (your imagined surface).

In the previous example the control volume and control surfaces had very simple shapes. We can choose more complex shapes if needed. Take for instance a kettle, and suppose we want to keep track of how much water, steam, air, and maybe energy-mass go into or out of it. The interior of the kettle is our control volume. It is delimited by the interior surface of the metal that makes up the kettle, and also by an imaginary surface at the end of the spout. Note that as we move the kettle around, so do these control volume and control surface. If we make a dent on the kettle with a hammer, then the control volume and control surface will follow that dent.

In a similar way we can consider as control volumes more complicated objects and spaces – like a car, a rocket, a bacterium, a planet, the chamber

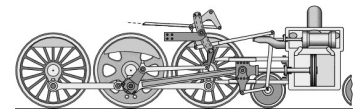


A classroom with an imaginary, slightly skew surface (in red) dividing the front and rear half of the room. (Room design: Raihanali².)



Simplified section of a kettle, with a control volume (in red) delimited by a closed control surface (dashed red line).

in a piston – with appropriate control surfaces that “wrap” them. Keep in mind that these control volumes and surfaces bend and move as needed. The locomotive system illustrated on the side can be considered as a control volume, with a complex movable control surface that perfectly wraps every element of it.



From 009 Lynton and Barnstaple Railway County Gate³

§ 4.4 Choices of control volumes and surfaces

The full freedom we have in choosing control volumes and surfaces is extremely powerful for physics and engineering applications. Let’s discuss two particular kinds of choice that are often made in two broad types of physical situations, and that we shall also often use in these notes.

First choice. When we study solid objects, or objects made of solid parts – like a football, a book, a car, an aeroplane, a rocket, and in some situations a human being or a planet – then we use the object itself as a control volume. The control surface tightly “wraps” the object.

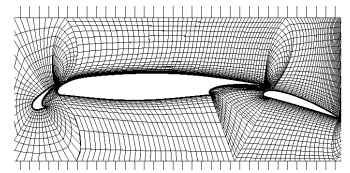
Because of this particular choice, we often say, for instance, ‘the momentum of the tennis ball’, but what it’s meant is ‘the momentum in the volume that encloses the tennis ball’.

There are two main reasons for this choice. First, this choice makes one particular physical law, the [balance of matter](#) ^{chapter 7 p.187}, automatically satisfied. So we don’t need to consider it when describing the physical phenomenon of interest. Second, the movement and deformation of control volumes and surfaces so chosen are not too complicated to represent mathematically; because the object is solid, so its kinds of movement are limited.

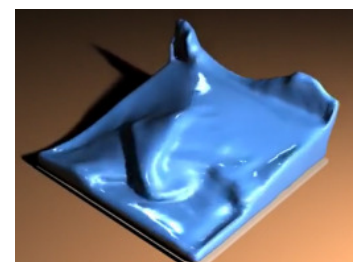
Second choice. When we study something flowing or moving in a more fluid way – like a body of water, the atmosphere, fuel or gas in a cylinder, plasma, or an electromagnetic field – then we divide the space of interest into small, *static* control volumes and surfaces. We essentially construct a *mesh*.

With this choice we do have to explicitly consider the physical law for the balance of matter, making the equations a little more complicated. But at the same time the mathematical representation of these control volumes and surfaces is extremely simple, because they’re static.

There are also hybrid choices between the two special choices above, with some control volumes and surfaces that are static, and others that move and deform in a more fluid way. These hybrid choices are used for



Section view of grid of control volumes and surfaces for studying the flow of air around an aeroplane wing (from Schieffer 2013). The extended white areas are sections of the wing and of its [slat and flap](#)⁴.



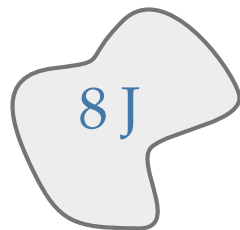
Clever hybrid choices of control volumes and surfaces allow us to simulate complex motions of fluid surfaces. [See also animation](#)⁵ (Wojtan et al. 2009).

instance in simulations of fluids where the behaviour of their surface – waves, detaching drops, and similar – are important.

§ 4.5 Volume content

Scalar quantities

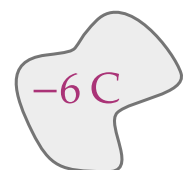
A volume content (or volume integral) for a scalar quantity, for example energy-mass, can be represented like this:



we have eliminated one spatial dimension for simplicity, considering the analogous two-dimensional idea. The volume is in light grey, delimited by a closed darker grey boundary, and we're indicating that the volume content, that is, the amount of energy-mass within, is 8 J.

As a visualization device, this representation can be useful. But let's straighten out some of its aspects:

- Recall that this is a snapshot at a given time instant. So there are 8 J of energy-mass in the volume at that instant, but we don't know the situation earlier or later: there could be a different amount of energy-mass, the region might be at a different position and have a different shape, or it might not even exist.
- Recall that some scalar quantities, like electric charge and in some situations matter (antimatter), can have negative amounts.
- We must not surmise that the amount of quantity is uniformly distributed within the volume. In fact there could be negative amounts of it in some subvolumes and positive in others. In particular, even if there is a zero amount of quantity in a volume, some subvolumes could have non-zero amounts: some positive and some negative, so that the total is zero.



A region with a negative amount of charge

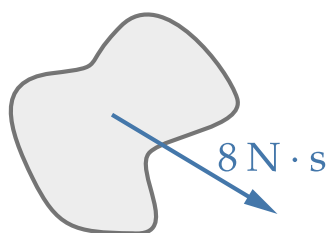


Exercise 4.1

The volume content of matter in a particular volume is equal to 36 mol. Can we conclude that the volume doesn't contain antimatter?

Vector quantities

A volume content for a vector quantity, for example momentum, can be represented as follows (we still simplify our visualization to two dimensions):



Momentum is a vector quantity, so the total amount in the volume above is a vector: the figure shows the direction and orientation of this vector, and the magnitude of $8 \text{ N} \cdot \text{s}$ is explicitly reported.

! Vector magnitudes and opposite vectors

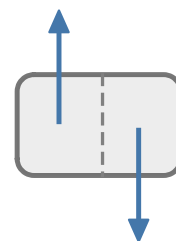
Remember that the *magnitude* of a vector is always positive, and that

$$\nwarrow = -1 \cdot \searrow$$

The visual representation above is useful, if we keep in mind remarks analogous to the scalar case:

- This is a time snapshot.
- The application point of the vector representing the volume content is unimportant: for instance, it doesn't need to be placed at the centre of the volume. The vector refers to the volume as a whole, not to some specific point within.
- Different subvolumes could have amounts represented by different vectors; only the total vector is represented above.

This last remark is especially important when we discuss momentum and angular momentum. As an example, look at the side figure: the volume content for the whole region is *zero*, but its left and right subregions have *non-zero and opposite* volume contents.



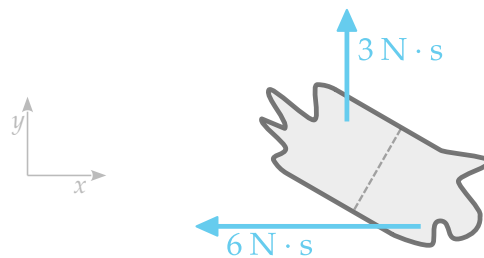
The whole region has zero volume content. The left and right subregions have non-zero and opposite volume contents.



Exercise 4.2

Recall *extensivity*, the second property of our seven primitive quantities: the amount in a volume consisting of separate volumes is equal to the total of the separate amounts.

We have a region consisting of two subregions; the amounts of momentum in each subregion are shown below.



1. Write the total momentum in each subregion in component form, (P_x, P_y) , according to the coordinate system shown.
2. Calculate the momentum in the whole region; represent it graphically as vector and write it in component form.



Adding vectors in General Relativity

We are used to the idea of adding vectors placed at different points in space: we only have to move each vector, keeping it parallel to itself, to a common point; and then add them all at that point with the usual rule.

This operation *cannot* be done so simply in General Relativity: the notion of parallelism doesn't apply anymore in a simple way, owing to the curvature of spacetime. The addition would lead to different results depending on how we transported the vectors. But it is still possible to add the momentum of two different spatial regions, simply because *momentum is defined with respect to a coordinate system*. This coordinate system selects, so to speak, a unique way to transport the momentum vectors to a common point. We are again reminded of the fact that **momentum is a coordinate-dependent quantity** ▶ §3.9 p.84.

In General Relativity momentum is not really a vector, but just a special triplet of quantities.

§ 4.6 Flux of scalar quantities

To get an intuitive grip of the notion of flux, consider a flow of people through an open door. The door is our control surface. We can ask how many people crossed the door during a minute. But one more detail about

this flow is important: *in which direction did the people cross the door?* This detail is important because, for example, the door leads to a classroom and we need to keep track of how many seats are free. We therefore need to know whether each person who crossed the door was actually *entering* or *leaving* the classroom.

In order to do this we can proceed as follows:

1. Assign a crossing direction to the door, for instance the direction from outside to inside the classroom.
2. Count as 'positive' each person who crosses the door in the chosen crossing direction, and as 'negative' each person who crosses the door in the opposite direction.

The total of this counting tells us the *net* number of people who crossed the door in the chosen direction. If we chose a crossing direction *from outside to inside* the classroom, then this total is the net number of people who *entered* the classroom. If we chose a crossing direction *from inside to outside* the classroom, then this total is the net number of people who *left* the classroom.

Therefore a **flux represents only a net amount crossing a control surface**. Note also that this net amount can be *negative*. For instance if we chose a crossing direction from outside to inside the classroom, and the net amount is -3 , then it means that *more people got out than in*: 9 persons may have entered the room during that minute, and 12 persons left. Or maybe no person entered the room, and 3 persons left. In either case, the final situation is that those who got out during that minute were three more than those who got in.

One important aspect of this example and terminology is the following *symmetry*:

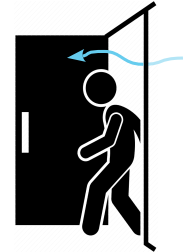
- It is completely arbitrary which crossing direction we choose.
- If we choose the other crossing direction, then the net amount will switch sign.

The physical situation is of course the same. The sentences

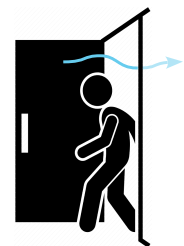
“+5 persons *entered* the room”

and

“−5 persons *left* the room”



According to the crossing direction indicated by the blue wiggly arrow, the person crossing the door counts as '+1'.



According to the crossing direction indicated by the blue wiggly arrow, the person crossing the door counts as '−1'.

are saying exactly the same thing.

Now consider a similar example, but instead of people, think of a quantity that can ordinarily also be *negative*, such as electric charge. Let's choose the door-crossing direction from outside to inside the room. If we're told that a net amount -5 C of charge crossed the door in the chosen direction in one minute, then this could have happened in several ways:

- a charge of -5 C was brought into the room
- a charge of $+5\text{ C}$ was brought out of the room
- a charge of -2 C was brought into the room during the first 30 s, and a charge of $+3\text{ C}$ was brought out in the remaining 30 s
- a charge of -2 C was brought into the room during the whole minute, and a charge of $+3\text{ C}$ was brought out at the same time
- ... and many other possible combinations.

So the statement that "the flux of electric charge into the room was -5 C in one minute" does not tell us which of the situations above occurred.

In fact, ordinary electricity in wires was thought for some time to be associated with movements of negative *and* positive charges in opposite directions. Today we know that it consists in the movement of negative charges only.

The purpose of the previous examples is to make you aware of some fundamental aspects of what we call "flux". These aspects are trivial but important when considering fluxes of physical quantities:

"Fechner [in 1845] supposed every current to consist in a streaming of electric charges, the vitreous charges travelling in one direction, and the resinous charges, equal to them in magnitude and number, travelling in the opposite direction with equal velocity." Whittaker 1951



Fundamental aspects and symmetry of flux

- A flux in a particular surface-crossing direction only tells us the *net* amount of substance that crosses the surface in that direction per time.
- A flux can be *negative*.
- **Symmetry of flux:** A flux in a particular surface-crossing direction is equivalent to a flux of *opposite* sign in the *opposite* crossing direction.

This is called *Cauchy's fundamental lemma* in the technical literature.



What a flux value does not tell

- A flux value does not tell us the amount that crossed during shorter times or through different parts of the surface.

- A flux value does not tell us whether the transfer of the quantity through the surface occurs because the quantity is flowing, or because the surface is moving, or both.

§ 4.7 Representation of scalar fluxes

How can we graphically represent the flux of a quantity, in a way that takes care of all its fundamental aspects?

First let's consider a surface through which we're measuring a flux, at a particular instant of time. Here it is represented as line, removing one spatial dimension to simplify the drawing:



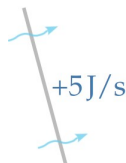
Keep in mind that the surface could have any other shape – as long as it can be given a clear crossing direction – and could also be in motion.

Let's indicate a crossing direction through the surface by one or more wiggly arrows:



Keep in mind that these arrows *are not vectors*! They don't have a 'magnitude' or 'components'. They simply indicate a sense in which we imagine the surface to be crossed. We could also have used only one wiggly arrow or three instead of two.

Let's take a scalar quantity such as energy. A flux of energy $+5\text{J/s}$ through the surface, in the first crossing direction, can then be depicted as follows:



This picture says that a net amount of 5 J is crossing the surface, per second, from the left side to the right side. This also means that per second a net amount of 5 J is “disappearing” from the left side of the surface and “appearing” on the right side.

Now consider the opposite crossing direction, depicted like this:

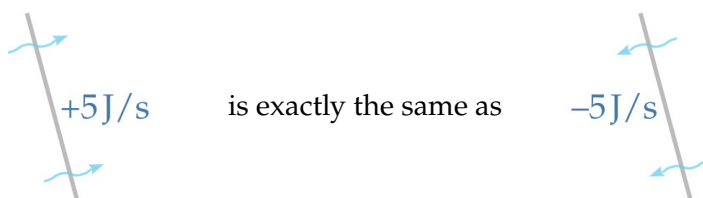


Because of the [symmetry of flux](#) ^{§4.6 p. 106}, we can say that the flux of energy equals -5 J/s in this opposite direction. We depict this as follows:



This picture says that a net amount of -5 has crossed the surface from the right to the left side, in a unit of time. This also means that a net amount of -5 has “disappeared” from the left side of the surface and “appeared” on the right side, in a unit of time.

But this is indeed exactly the same situation as before. *Both pictures therefore represent the same flux:*



It is extremely important that you remember that the two kinds of picture above are completely equivalent. You can always mentally switch from one to the other. A flux in one crossing direction is exactly the same as a flux with opposite sign in the opposite direction.

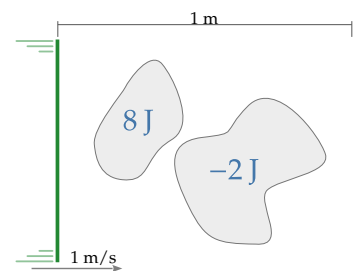
The two equivalent pictures do *not* say that a given amount is *only* moving from left to right, or vice versa. We have seen that in general we don’t know this. Both pictures say that on the left side the amount of quantity is changing at a rate of -5 J per second, and on the right side by $+5 \text{ J}$ per second. In these notes we shall usually display only one of these two equivalent representations.



Exercise 4.3

For each question, answer in an *unambiguous* way and sketch a picture representing the flux.

1. The two sides of a particular surface are called 'up' and 'down'. During 0.2 s, an energy-mass of +3 J flows from the up-side to the down-side, and an energy-mass of -4 J flows from the down-side to the up-side. How much is the flux of energy-mass through the surface?
2. Through the same surface, at a later time, 2 mol of neutrons flow from the up- to the down- side in 0.01 s, and 2 mol of neutrons flow from the down- to the up-side during the same time. How much is the flux of matter through the surface?
3. The two sides of a surface are called 'in' and 'out'. During 0.01 s there is a flow of 1000 electrons from the in-side to the out-side, and also a flow of 1000 *positrons* (anti-electrons) in the same direction. How much is the flux of matter through the surface?
4. The side figure shows a **control surface** moving from left to right at a (constant) velocity of 1 m/s. The space to its right has two static regions with some amount of **energy-mass** as shown (there's no energy-mass behind to the left of the surface). How much is the flux of energy-mass through the surface in 1 s?



§ 4.8 Flux of vector quantities and its representation

The flux of a vector quantity is also a vector, because it is given by an amount of that quantity, which is a vector, divided by time, which is a scalar. We can think of it as three fluxes of three scalar quantities.

The intuition and mental representation of the flux of a vector quantity through a control surface may be less straightforward than for a scalar quantity. Think again of the previous examples with people or electric charges crossing a door or control surface. In the case of flux of a vector quantity, we may imagine that what's crossing the control surface are vectors. We are going to discuss some possible graphical representations.

The remarks about the choice of crossing direction and about the symmetry of flux, which we made for scalar quantities, also apply in

analogous ways to the flux of vector quantities. For instance, if the flux of momentum through a surface in a particular crossing direction is

$$[-3, 4, 0] \text{ N},$$

then if we choose the opposite crossing direction the flux is

$$-[-3, 4, 0] \text{ N} = [+3, -4, 0] \text{ N}.$$

If we think of vectors as arrows, we must only remember that a minus sign changes their orientation:

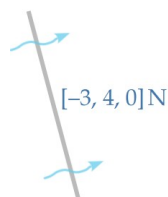
$$\nearrow = -1 \cdot \searrow$$

We can devise a graphical representation of the flux of a vector quantity similar to that [for the flux of a scalar quantity](#) ^{§4.7 p. 107}.

First, it's important to indicate the crossing direction, and we can do that again with one or more wiggly arrows; for instance:



Now we have to indicate how much is the flux. In the case of a scalar quantity we simply reported the value, including the unit. For the flux of a vector quantity we have three values, so one possibility is to simply report them. Suppose we are speaking about momentum and the flux in the given crossing direction is $[3, -4, 0] \text{ N}$; we can then write this explicitly:

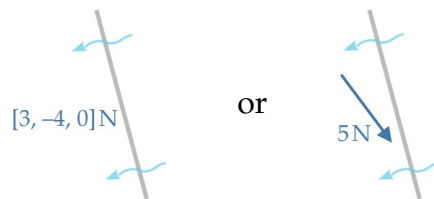


Another alternative is to draw a vector representing these components:



This picture says that a net vector amount of momentum  is crossing, per second, the surface from the left side to the right side. To help our intuition we can imagine the **vector** “moving” across the **control surface** in the direction indicated by the **wiggly arrow**; an animated representation of this can be found [at this link](#)⁶.

In the opposite crossing direction the flux gets a minus sign, because of the symmetry of fluxes. The corresponding graphical representation is



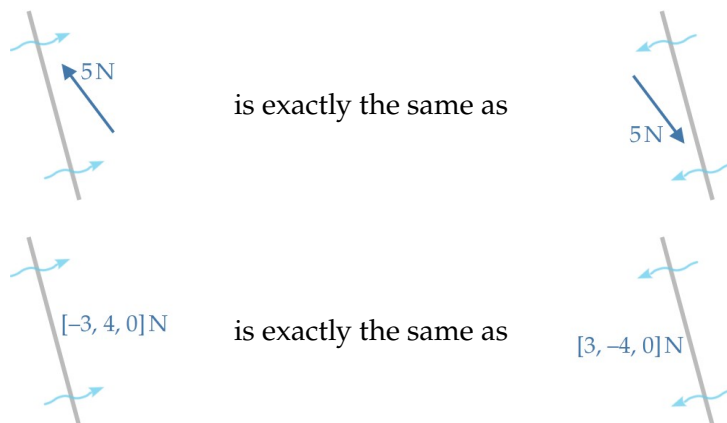
note how the components have flipped sign, and how the vector has flipped direction keeping the same magnitude. The pictures above say that a net vector amount of momentum $[3, -4, 0] \text{ N s}$, or graphically , is crossing, per second, the surface from the right side to the left side.

But this is exactly the same flux as before, because

$$[3, -4, 0] \text{ N s} = -[-3, 4, 0] \text{ N s}$$

$$\begin{matrix} \text{5Ns} \\ \swarrow \end{matrix} = - \begin{matrix} \text{5Ns} \\ \searrow \end{matrix}$$

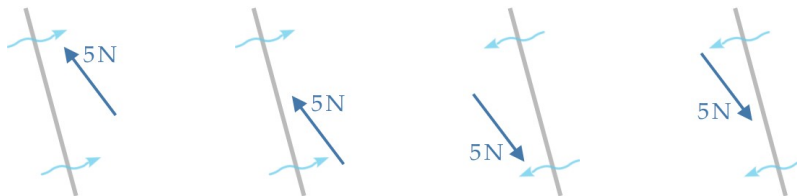
In other words, *the following two pictures represent the same vector flux*:



An important aspect of vector fluxes which we must try not to get confused about is the application point of the vector, that is, the base point of the arrow. Just as for **vector volume contents** ^{§4.5 p. 103}, **the application**

point of the vector representing the flux is unimportant; the vector refers to one side of the surface as a whole. Graphically:

these four pictures represent exactly the same flux



Exercise 4.4

A horizontal surface is given, and there is a flux of a vector quantity through it; for the moment we neglect units:

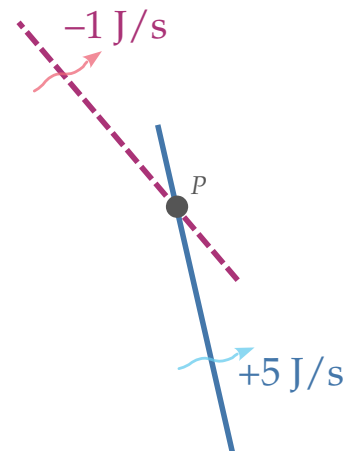
1. If we take the *downward* crossing direction as 'positive', the flux xyz -components are $[5, 5, 0]$. Represent this flux graphically, in the way discussed in the present section. Use the coordinate system  where y points upward.
2. Taking the same crossing direction, represent graphically the flux $[0, -2, 0]$ instead.
3. Taking the same downward crossing direction, we are now told that there is a flux with components $[1, -2, 3]$. What are the components of this flux if we take the *upward* crossing direction as positive?

§ 4.9 Fluxes through different surfaces

What happens to the value of a flux through a surface, if we consider a *different* surface, maybe intersecting the original one? It's important to keep in mind that

- A flux refers only to a specific surface.
- The fluxes through two distinct surfaces can be very different, even if the two surfaces are quite close.
- The flux depends on the motion of the surface. So if we consider the same surface but with a different instantaneous motion, then the flux may be very different.

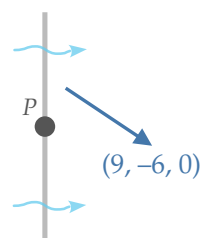
Consider for instance the picture on the side. It depicts two intersecting surfaces (as usual simplified by removing one dimension) and two chosen crossing directions on them. The crossing directions are both roughly rightward. Yet the energy-mass flux through the **solid blue surface** is $+5 \text{ J/s}$, whereas the energy-mass flux through the **dashed red surface** is -1 J/s .



There is, however, a relation between the fluxes through surfaces that share a common point. If we know the flux through *three different, small, static* surfaces having a common point, then we can find the flux through any other *small, static* surface passing through that same point. This possibility leads to the representation of flux through a small surface as a *vector*, called 'flux-density vector'. In these notes we shall not consider flux densities and their vector representation.

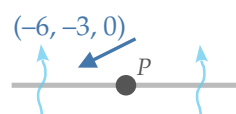
We saw that the flux of a scalar quantity can be very different if we take a slightly different surface, or the same surface with a different motion. The same is true of the flux of a vector quantity: in particular, **the vectors representing the fluxes through two slightly different surfaces can point in completely different directions.**

Here is an example. Take a fixed point P . Now take a small vertical surface passing through P , and choose a crossing direction from left to right. The flux of a vector quantity (momentum for example) through this quantity could be as in this picture:



this flux has components $(9, -6, 0)$, with magnitude around 10.8.

Now forget about that surface, and take instead a small horizontal surface passing through the same point P , and upward crossing direction. The flux of the same quantity through this new surface could be as in this picture:



it has components $(-6, -3, 0)$, with magnitude around 6.7.

Clearly the two vector fluxes through these two surfaces are completely different: they point in different directions and have different magnitudes.



Each surface has a unique flux

A flux, scalar or vector, through a particular surface, in general doesn't tell us anything about the flux through another surface, nor about the flux through the same surface but with a different instantaneous motion.

§ 4.10 Flux and supply of momentum: force

We already mentioned that [fluxes and supplies of momentum are what we call 'forces'](#) ^{§3.7 p.79}. Since the notion of *force* is important in the many branches of physics which rely on Newtonian mechanics, let us discuss this connection in depth. This connection, as well as the connection to Newton's laws, will become even clearer when we discuss the balance of momentum in chapter 10.

Recall that [momentum can be measured in newton-seconds, 'N · s'](#) ^{§3.7 p.79}. Its flux, being a momentum divided by time, is therefore measured in newtons, 'N'.

Surface forces and volume forces

The notion of force is very intuitive. We associate it to the sensations that we feel in our skin, flesh, bones when, for instance, we push against a wall, twist a door knob, push backwards on the ground with our feet to run or jump. This is what we call a **surface force** or **contact force**.

We also associate force to the sensation we feel in our whole body when we sit in an accelerating car, or go forth and back in a swing, or sit on an aeroplane that's taking off; in these cases the sensation is not limited to a surface. This is what we call a **volume force** or **body force**.

Surface forces and volume forces have some properties in common, because both can lead to changes in momentum content. Both are represented by a vector having the direction and orientation of the "push" or "pull", and a magnitude expressing its intensity, which is the rate at which momentum is produced.

But they also differ in important respects: a surface force is a *flux* of momentum through a surface, whereas a volume force is a *supply* of momentum in a volume. Surface forces therefore satisfy the [principle of](#)

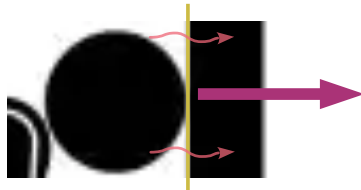
[symmetry of flux](#) ^{§4.6 p.106}, whereas volume forces do not always do, or do so only approximately.

§ 4.11 Flux of momentum is surface force

What we call *surface force* is therefore just a *flux of momentum*. This fact can be illuminating in some physical problems.

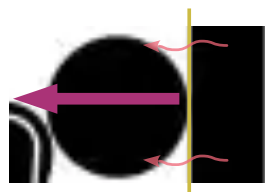
As a concrete example, imagine a person pushing against a wall, as in the side figure. In terms of force, we say that *the person's head is exerting a surface force on the wall*. The vector that represents this pushing force, the **purple arrow** in the figure, has a person→wall orientation.

To understand force in terms of momentum flux, let's consider an imaginary control surface separating the person and the wall; the **yellow line** in the figures below. If we choose a person↔wall crossing direction, indicated by the **wiggly red arrows** in the following figure, then the momentum flux indicated by the **purple arrow** has a person→wall orientation:



This figure says: on the side of the wall, an amount momentum is being added at a given rate; this momentum has a person→wall orientation.

Because of the [symmetry of flux](#) ^{§4.6 p.106}, if we instead choose a wall↔person crossing direction for the control surface, then the momentum flux has opposite orientation:



This figure says: on the side of the person's head, momentum is being added at a given rate; this momentum has a wall→person orientation.

Since contact force and momentum flux are the same thing, the last figure tells us that *the wall is exerting a surface force on the person's head*. The vector that represents this opposite pushing force has a wall→person orientation. – But this is *Newton's Third Law*! This example therefore shows



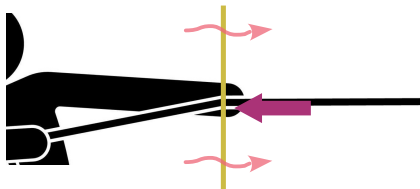
A distressed person pushing against a wall

that Newton's third law is the expression of the symmetry of flux. We shall discuss this fact again soon.

Now let's consider an example involving pulling instead of pushing. Imagine a person pulling a rope fastened somewhere, as in the side figure. In terms of force we say that *the person is exerting a surface force on the rope*; the vector that represents this force has a rope→person orientation.

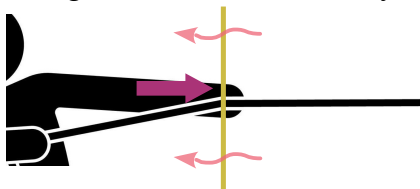


In terms of momentum flux, let's consider an imaginary vertical control surface between the person's hand and the rope; the **yellow line** in the figures below. If we choose an arm↔rope crossing direction, indicated by the **wiggly red arrow**, then the momentum flux has a rope→arm orientation:



This figure says: on the side of the rope, an amount momentum is being added at a given rate; this momentum has a rope→arm orientation. Note the difference from the previous example involving *pushing*; compare the two momentum fluxes.

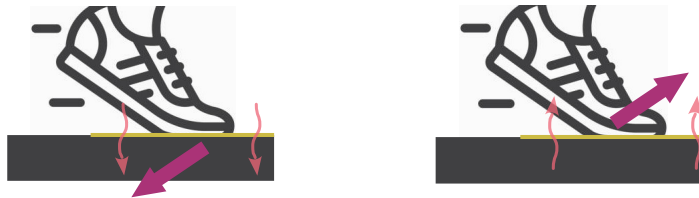
But again this momentum flux and the previous figure are completely equivalent to the following one, because of the symmetry of flux:



This figure says: on the side of the arm, an amount momentum is being added at a given rate; this momentum has an arm→rope orientation. Again we find Newton's third law: the rope is exerting a surface force on the person's arm; the vector that represents this opposite pushing force has a rope→arm orientation.

A final example illustrates a situation in between the previous two. Consider the foot of a running person as it pushes on the ground. In terms of momentum flux, take an imaginary horizontal control surface between the runner's foot and the ground; the **yellow line** in the figures below. If we choose a foot→ground crossing direction, the momentum flux is oriented diagonally, downward and backwards, towards the rear of the foot. Because

of the symmetry of flux, if we choose a ground→foot crossing direction, then the momentum flux has opposite diagonal direction, upward and forward, towards the front of the foot. This momentum flux can be depicted in these two equivalent ways:



The figures say: on the side of the ground, an amount momentum is being added at a given rate; this momentum has an diagonal, downward and backward orientation. On the side of the foot, an amount momentum is being added at a given rate; this momentum has an diagonal, upward and forward orientation. Both are aspects of the same momentum flux.

Newton's Third Law!

From the examples above, we see that thinking of surface force as momentum flux automatically leads to *Newton's third law*: if one side is gaining momentum with a given orientation, the other side by symmetry is gaining momentum with the opposite orientation. In other words, if one side is experiencing a surface force with a given orientation, the other side is experiencing a surface force with the opposite orientation.

We thus see that *Newton's third law is the expression of the symmetry of flux* ^{§4.6 p.106} for the specific case of the flux of momentum. We also realize that this "law" is more general: it applies not only to surface force, but also to the flux of all other six quantities, even scalar ones.

"LAW III. To every action there is always opposed an equal reaction: or, the mutual actions of two bodies upon each other are always equal, and directed to contrary parts."

Newton 1726a



Exercise 4.5

Using your intuition, try to guess the various momentum fluxes (except their magnitudes) between this awesome person and the walls:



(Buster Keaton in *'The Electric House'*⁷)

§ 4.12 Pressure, tension, shear force

The examples of the previous section demonstrated a variety of possible orientations of the momentum-flux vector with respect to the surface through which it occurs. All orientations of momentum flux are possible. But we give special names to momentum flux with three special orientations: *pressure*, *tension*, and *shear force*.

Take a small surface; it doesn't matter whether vertical, horizontal, or with some other inclination. Call the surface's sides A and B . Consider the crossing directions $A \rightsquigarrow B$ and $B \rightsquigarrow A$.

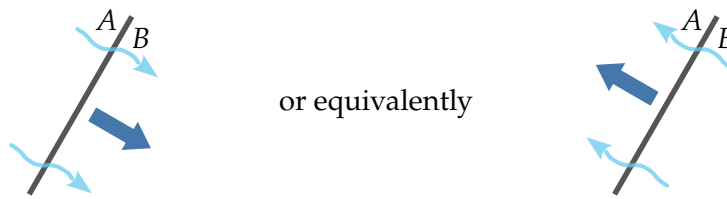


Pressure

Pressure, or *compressive momentum flux* or *compressive force*, is a flux of momentum through a surface, which points *away from* the surface.

So in the crossing direction $A \rightsquigarrow B$, pressure is represented by a vector oriented from A to B . Equivalently, in the crossing direction $B \rightsquigarrow A$

pressure is represented by a vector oriented from B to A :



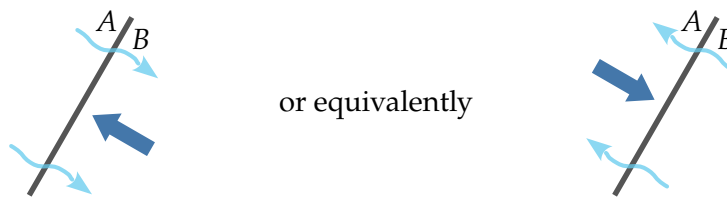
(animated version⁸)

Pressure is the kind of momentum flux that we exert when we *push* on an object, and that air exerts on all objects it surrounds.

Tension

Tension, or *tensile momentum flux* or *tensile force*, is a flux of momentum through a surface, which points *towards* the surface.

So in the crossing direction $A \rightsquigarrow B$, tension is represented by a vector oriented from B to A . Equivalently, in the crossing direction $B \rightsquigarrow A$ tension is represented by a vector oriented from A to B :



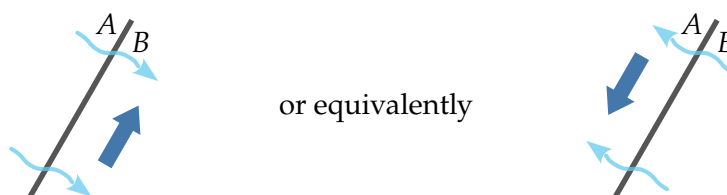
(animated version⁹)

Tension is the momentum flux that we experience in our bones when we *pull* an object, and that occurs in any section of a stretched rubber band.

Shear force

Shear force, or *shearing momentum flux*, is a flux of momentum through a surface, *parallel* to the surface.

In the crossing direction $A \rightsquigarrow B$ and in the crossing direction $B \rightsquigarrow A$, shear force is represented by a vector parallel to the surface. Obviously the orientation is opposite on the two sides, owing to the symmetry of flux:



(animated version¹⁰)

Shear force is the kind of momentum flux that we experience under our feet when we walk or run, and that occurs between a car's wheels and the ground.

In general, a momentum flux won't have any of the three special directions above, but rather a combination of them.



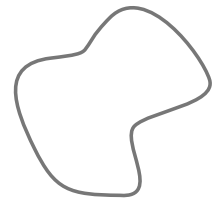
Exercise 4.6

Using your intuition, try to identify the various momentum fluxes that occur in the different beams of a tower crane. Which momentum fluxes are approximately compressive, tensile, and shearing?



§ 4.13 Closed control surfaces, influxes, effluxes

We shall often consider **closed** control surfaces, that is, control surfaces that don't have a rim or border or holes, like the surface of a sphere or of a cube. A closed surface delimits a specific three-dimensional volume, and we can therefore speak of its **interior** and its **exterior**. An example (simplified by removing one dimension as usual) is the surface in the side picture.

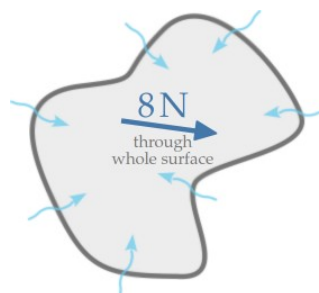


The two crossing directions of a closed surface therefore take on special names: *inward*, from exterior to interior; and *outward*, from interior to exterior. A flux through the surface is usually called **influx** if we are considering the inward crossing direction, and **efflux** or *outflux* if we are considering the outward crossing direction. Obviously, by the symmetry of flux,

$$\text{influx} \equiv -\text{efflux}$$

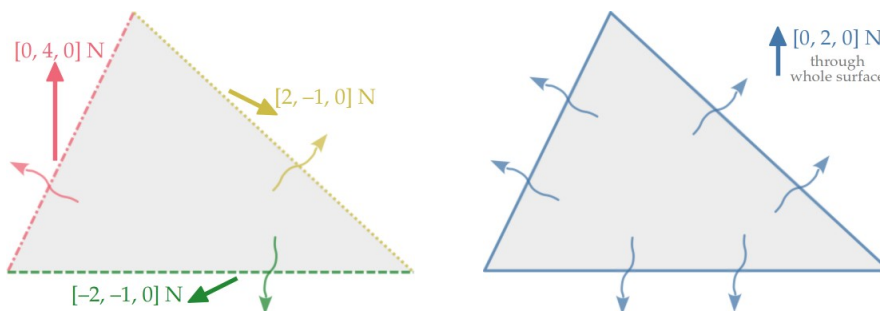
$$\text{efflux} \equiv -\text{influx}$$

The influx and efflux are fluxes *through the whole surface*. Consider for instance these influxes of energy-mass and of momentum:



In the left picture we have a net influx of 5 J/s on the whole inner side of the control surface. Remember that we don't know whether these 5 J/s are being evenly distributed over the surface, or just at particular spots of it, or whether they are the net result of positive and negative amounts on different parts of the surface. In the right picture we have a vector influx of 8 N, the vector pointing approximately rightward. Again we don't know what are the flux vectors for parts of the surface: the vector in the picture is just the grand total.

Let's see another example of this fact. The picture on the left below shows the **outward fluxes** through three parts (dotted yellow, dashed green, dot-dashed red) of a closed control surface. The picture on the right shows the **total efflux** through the same surface:



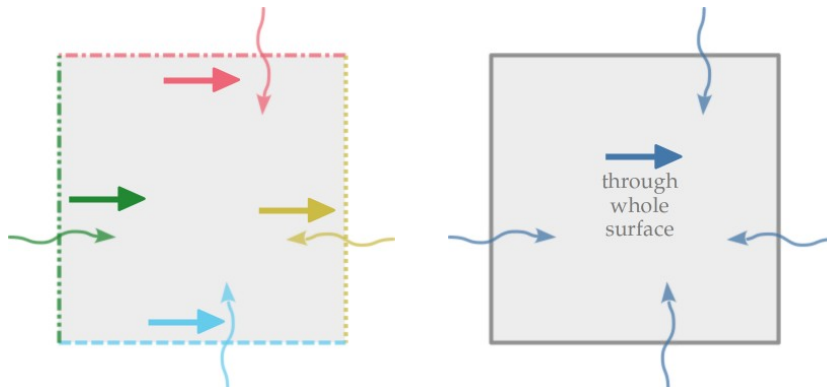
The individual fluxes and the total efflux are consistent because

$$[2, -1, 0] \text{ N} + [-2, -1, 0] \text{ N} + [0, 4, 0] \text{ N} = [0, 2, 0] \text{ N}$$



Exercise 4.7

1. Are the four partial influxes shown on the left (with different colours and line styles) consistent with the **total influx** shown on the right? Why or why not?



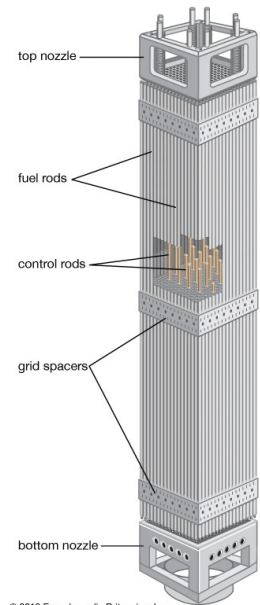
2. Take an imaginary cylindrical surface enclosing one **control rod**¹¹ in a **nuclear-fission reactor**¹² (see side figure). Let's say that in a reactor there are 20 such rods. Approximately 5×10^{19} neutrons are liberated in a second in the whole reactor by the fission fuel, but $2/3$ of these are *absorbed* by the control rods.

How much, on average, is the **efflux** of neutrons (matter) through the surface of one control rod?

Express the result first in neutrons/s, and then in mol/s, using the Avogadro constant

$$N_A = 6.022\,140\,76 \times 10^{23} \text{ particles/mol}.$$

Be careful about the signs!



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§ 4.14 Time-integrated fluxes and supplies

A flux is defined as the amount of a quantity crossing a control surface in a short time lapse Δt , divided by that time lapse. Denoting the flux by, say, $\mathcal{J}$, this definition also means that the amount of quantity crossing the surface in a short time Δt is approximately equal to $\mathcal{J} \Delta t$.

Now consider a control surface between two time instants t_0 and t_1 ; during this time lapse it could be moving and changing shape. Choose a crossing direction through the surface. At each intermediate time instant t we can measure the flux of a quantity crossing the surface in that direction, at that instant; denote this flux by $\mathcal{J}(t)$.

The total amount of quantity that crosses the surface between times t_0 and t_1 can be found by integrating $\mathcal{J}(t)$. That is, we divide the time interval into very short time lapses of length Δt ; for each short time lapse we know that the amount that crosses the surface is $\mathcal{J}(t) \Delta t$; the total is then obtained by adding these small amounts. As we consider shorter and shorter Δt , this sum is by definition an integral:



Time-integrated flux

The total amount of quantity flowing through a control surface in a specified crossing direction, between times t_0 and t_1 , is called the

time-integrated flux and is given by

$$\int_{t_0}^{t_1} \mathcal{J}(t) dt, \quad (4.1)$$

where $\mathcal{J}(t)$ is the flux of the quantity at time t .

The meaning of the integral above should be clear for any scalar quantity, for which the flux is also a scalar. In the case of a vector quantity, for instance momentum, the flux is also a vector, represented by three components. The integral of a vector is obtained by calculating the integral for each component, obtaining three results, which are the components of a new vector. Geometrically this corresponds to summing a large number of very short vectors.

Take the case of momentum, whose flux (force) we denote $\mathbf{F} = [F_x, F_y, F_z]$. The time integral of this flux is then

$$\int_{t_0}^{t_1} \mathbf{F}(t) dt := \left[\int_{t_0}^{t_1} F_x(t) dt, \int_{t_0}^{t_1} F_y(t) dt, \int_{t_0}^{t_1} F_z(t) dt \right]. \quad (4.2)$$

An analogous discussion can be made about the supply of a quantity in a control volume:



Time-integrated supply

The net amount of quantity created in a control volume, between times t_0 and t_1 , is called the **time-integrated supply** and is given by

$$\int_{t_0}^{t_1} \mathcal{R}(t) dt, \quad (4.3)$$

where $\mathcal{R}(t)$ is the supply of the quantity at time t . If the integral is negative, it means that a net amount of quantity has been destroyed.



Time-integrated quantities can be zero even with non-zero flux or supply

The result of the integral defining a time-integrated flux or supply can be zero. This means that no *net* amount of quantity flowed through the surface, or was created in the volume, between t_0 and t_1 . Yet the

flux or supply can be non-zero, even at all times; of course it needs to be positive at some times, and negative at others, in order for the time integral to be zero.



Exercise 4.8

1. What are the physical dimensions of the time-integrated flux and the time-integrated supply of a quantity?
2. Suppose that we calculate the time integral for a particular control surface in the case of matter, finding a total of $\int_{t_0}^{t_1} \mathcal{J}(t) dt = 7 \text{ mol}$. Now we change our mind and choose the opposite crossing direction for that surface. How does the result above change?

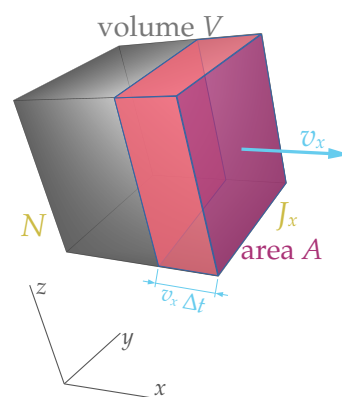
§ 4.15 The relation between fluxes and velocities

The idea of flux evokes the idea of movement, and therefore of *velocity*. Is there a relationship between flux and velocity?

The answer is yes: the velocity of a quantity is essentially defined its flux and its volume content. Consider for example how we measure the velocity of an object: we are actually keeping track of a flow of matter – the matter that makes up the object – from a region of space to another.

The rigorous definition of velocity from flux is somewhat involved, so here we'll just see a simplified and approximate example of how this definition works.

Take a scalar extensive quantity like matter, electric charge, energy-mass, or entropy. For concreteness let's take matter. Choose a coordinate system (t, x, y, z) and consider a point in space at a specific coordinate time. Around this point, choose a very small *static* cuboid region, as in the side figure, delimited by six small rectangular *static* surfaces: two parallel to the yz -coordinate plane, two to the zx one, and two to the xy one. The cuboid has volume V , and the volume content of matter in it is N . The two yz surfaces (one of them is in **dark red** in the picture) have area A , approximately the same for both; and there is a flux of matter $\mathcal{J}_x$ crossing either of these two surfaces in the positive- x direction. These two parallel surfaces have approximately the same area and the same flux because the cuboid region is very small.



This control volume is small, static, and with sides parallel to the coordinate axes.

The x -component of the coordinate velocity of matter in this region is then *defined* as

$$v_x := \frac{J_x/A}{N/V} \quad (4.4)$$

with analogous definitions for the y - and z -components.

The velocity $\mathbf{v} = [v_x, v_y, v_z]$ so defined has the following intuitive property. If you choose *any* very small surface centred at this point, and move it with velocity $\mathbf{v}$, in the direction specified by the velocity, then the matter flux through it is zero. This reflects the intuitive understanding that if a surface is moving together with the matter, at the same speed, then we shouldn't observe any flux through it.



Exercise 4.9

1. Try to prove the formula (4.4) relating flux and velocity in an intuitive way, referring to the picture above. As a starting point, consider this question: if the amount of matter N in the volume V is moving with velocity v_x in the positive- x direction, how much of it will cross the area A during time Δt ?
2. A small cuboid region has a volume of $1 \times 10^{-9} \text{ m}^3$, and its sides parallel to the xy axes have each area $1 \times 10^{-3} \text{ m}^2$. Through each of these sides there is a flux of energy-mass $W_z = 3 \text{ J/s}$ in the positive z -direction. The cuboid region contains $E = 0.5 \text{ J}$ of energy-mass. How much is the velocity of energy-mass in the z -direction?



Velocities of quantities in General Relativity

One consequence of the relationship between velocities and fluxes is that we can define such a velocity for any extensive quantity. So we have a velocity matter from the flux of matter, but also a *velocity of energy-mass* from the flux of energy-mass.

In Newtonian approximation these two velocities are approximately equal, so we do not need to distinguish them. But in situations where the Newtonian approximation is not valid, we have to take into account the velocity of matter and the velocity of energy-mass separately. This difference is important for instance in the study of plasma in stars and in numerical General Relativity.

There is an ongoing discussion as to which of the two velocities is more convenient to use; see for instance Kandus & Tsagas 2008, especially the section *Eckart frame versus Landau frame*, which refers to the choice between these two velocities.

URLs for chapter 4

1. <https://doi.org/10.6028/NIST.SP.811e2008>
2. <https://www.turbosquid.com/3d-models/classroom-1726208>
3. <http://www.009.cd2.com>
4. <https://www.grc.nasa.gov/WWW/k-12/VirtualAero/BottleRocket/airplane/flap.html>
5. <http://www.youtube.com/watch?v=M5xnAdVPbgQ>
6. <https://pglpm.github.io/7wonders/media/vectorfluxanim2.webp>
7. <https://www.imdb.com/title/tt0013099/>
8. <https://pglpm.github.io/7wonders/media/pressure.webp>
9. <https://pglpm.github.io/7wonders/media/tension.webp>
10. <https://pglpm.github.io/7wonders/media/shearforce.webp>
11. https://energyeducation.ca/encyclopedia/Control_rod
12. <https://www.iaea.org/newscenter/news/what-is-nuclear-energy-the-science-of-nuclear-power>

Physical laws

Every branch of physical science is based on two sets of fundamental equations. The first set is that of basic laws of physics, which are postulated to hold valid for all bodies under all conceivable circumstances [...]. The second set of fundamental equations are the constitutive equations: these are relationships which are not supposed to hold for all bodies, but only to describe the behavior of some restricted class of bodies, or possibly of a larger class of bodies for a more restricted class of phenomena.

G. Astarita 1990

§ 5.1 Some classifications of physical laws

Physical laws, very generally speaking, are mathematical relations between physical quantities. Given information about some quantities, physical laws allow us to deduce information about other quantities, or about the same quantities at other times or spatial regions. As [previously discussed](#) ^{§1.3 p.28}, we use many physical laws every day, in a qualitative and approximate way, without even thinking about them.

Physical laws can be classified or categorized in many different ways; for instance by the quantities they involve, or by the kind of mathematics they use. So we speak of ‘laws of mechanics’ and ‘electromagnetic laws’; or of ‘differential laws’ and ‘integral laws’; and so on.

One classification distinguishes between *fundamental* vs *derived* physical laws. This distinction is similar to the one between [primitive and derived quantities](#) ^{§1.4 p.31}. A physical law is ‘fundamental’ if it is taken as empirically valid, and as the starting point to make predictions or calculate other kinds of consequences. A physical law is ‘derived’ if it can be deduced from other fundamental laws: in a manner of speaking, it doesn’t really say anything new that wasn’t already a consequence of the fundamental laws. But it may still be a very useful shortcut.

Here is an extremely simple imaginary example. Suppose we have three different physical quantities denoted by a , b , c . One fundamental physical law states that a and b are equal; another fundamental law states that b and c are equal too:

$$a = b, \quad b = c \quad (\text{fundamental laws})$$

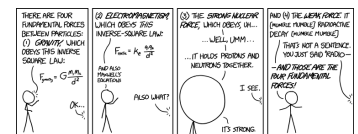
Combining these two fundamental laws we obtain a further law: a is equal to c :

$$a = b, \quad b = c \implies a = c \quad (\text{derived law})$$

This latter physical law doesn't tell us anything new that wasn't already implicit in the first two together. But in some situations it is useful to simply remember directly that $a = c$.

The distinction between fundamental and derived physical laws is not objective, but mostly a matter of convenience and even of personal taste. We can often promote a derived law to fundamental law, demoting some other fundamental law to a derived-law status. In the example above, we could take $a = b$ and $a = c$ as fundamental laws; then $b = c$ becomes a derived law, because it can be obtained from the other two. It is important to be aware of this flexibility in what's fundamental and what's derived. You'll find physics and engineering texts that present a physical theory as consisting in a particular collection of fundamental laws, and other texts that present the same physical theory as consisting in a collection of slightly different fundamental laws. There is no contradiction there: it means that some laws taken as derived in some texts, but as fundamental in others, and vice versa.

Yet, the choice fundamental laws is not without consequence. A particular choice of fundamental laws, rather than others, may cover more physical situations. It may also suggest new physical ideas and generalizations, leading to the discovery of new physical phenomena.



<https://xkcd.com/1489/>

§ 5.2 Universal laws vs constitutive relations

Another important distinction can be made between **universal** physical laws and **constitutive** physical relations. This distinction is determined not by convenience and personal taste, but by experiment:

Universal laws vs constitutive relations

- **Universal laws** represent universal physical patterns that we observe in all possible physical phenomena we manage to investigate
- **Constitutive relations** represent physical patterns and physical properties that are peculiar to – and therefore are only valid for – specific phenomena, or specific scales of time and space, or specific kinds of control volumes and surfaces, or specific physical theories. Constitutive relations are also called *constitutive equations* or *closure equations*.

One of the meanings of the word *constitutive* is ‘that makes a thing what it is’ (*Oxford English Dictionary* 2009).

The distinction above is different from the one between ‘fundamental’ and ‘derived’. Let’s try to understand this difference by means of our previous simple example.

Suppose we observe that the law $a = b$ is always valid in all physical phenomena we explore, under all possible extreme conditions, circumstances, regions of space, and time. Also suppose we observe that $b = c$ is instead only valid in specific physical phenomena and conditions, but not in others. For instance, we may observe that it is only valid when we make experiments with gases and low speeds, but not with solids or high speeds. The conclusion (until we find a disproof) is that the physical law $a = b$ is *universal*, whereas the physical law $b = c$ is *constitutive*: constitutive of gases and low speeds. Yet both laws are fundamental, because we cannot deduce $b = c$ from $a = b$: the law about b and c says something new, although something that is true only in some circumstances. In concrete applications we may therefore need to use both $a = b$ and $b = c$.

But are there really universal physical laws, which can be applied to *every* physical phenomenon, without exclusions or exceptions?

The answer is *yes*. We shall meet them in §5.8, and we shall see that they are tightly connected with the seven primitive quantities discussed in chapter 3. Our study of physics will indeed completely hinge on these universal laws. They are applied with extreme confidence to every new phenomenon we observe, and they often allow us to make predictions of at least a qualitative character without the need of constitutive relations. We would modify these universal laws only as a last resort; so far this has rarely or never been necessary. On the other hand, we have a large freedom in modifying constitutive relations, and in proposing new ones to account for newly observed physical phenomena.

As real examples, the *balance of momentum* ▶ §10.1 p. 227

$$\frac{d\mathbf{P}(t)}{dt} = \mathbf{F}(t) + \mathbf{G}(t)$$

is a universal physical law: it applies – exactly as written – to any control volume, small as an atom or large as a group of galaxies; to all objects and to electromagnetic fields; to General Relativity and to Newtonian mechanics, and with some reinterpretation of the symbols also to Quantum Theory. Instead the *Newtonian relation between momentum content and matter flux* ▶ §10.2 p. 228

$$\mathbf{P} = m\mathbf{v}$$

is a constitutive relation: it applies to enough small control volumes, but not to large ones; it applies to matter, but not to electromagnetic fields; it applies at low speeds and weak gravitational fields, but not at high speeds or strong gravitational fields; it is used in Newtonian mechanics, but not in General Relativity.

In §§5.9–5.11 we shall discuss further important differences between constitutive relations and a particular kind of universal laws called *balance laws*, which we discuss next.

! 'Fundamental' vs 'universal' in other texts

Be aware that the terms 'fundamental' and 'universal' are typically not used in a technically precise sense. Some texts may even use 'fundamental' in the sense of 'universal' as done here. What is important for us to remember is that there are physical laws that are used in *all* situations, and physical laws that can be used only in a restricted range of situations.



Exercise 5.1

In our imaginary example from the preceding sections, the law $a = b$ is fundamental and universal, and the law $b = c$ is fundamental and constitutive. Is the derived law $a = c$ universal? or is it constitutive?

§ 5.3 Balance laws and conservation laws

There are physical laws which have a special meaning and mathematical form: they express a sort of trade-off or “budget” among different amounts.

Their basic idea is therefore quite intuitive. They are called **balance laws**, and a special subgroup of them are called **conservation laws**.

Balance laws concern *extensive* quantities, like our [seven primitive quantities](#) ^{§3.1 p.67}. Recall that for extensive quantities we may ask: ‘how much is there in a particular control volume?’, ‘how much is flowing through a particular control surface?’, ‘how much is being produced in a control volume?’



Balance laws and conservation laws

A **balance law**, or simply *balance*, expresses a relation between the volume content, flux, and supply of an extensive physical quantity.

A **conservation law** is a special and important kind of balance law, in which the supply is always zero.



‘Conservation’ vs ‘balance’ in other texts

Be aware that some physics and engineering texts do *not* distinguish between ‘conservation law’ and ‘balance law’, and often use “conservation” to mean both. This use is unfortunate, because balance is the most general of the two. Always try to infer from the context how these two terms are used.

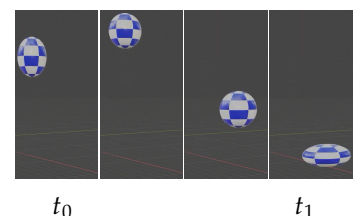
It turns out that *we can formulate almost all known universal laws as balance laws*. For this reason we now study balance laws in detail.

Setup for balance and conservation laws

The general setup to formulate a balance or a conservation law is as follows.

We choose a [closed control surface](#) ^{§4.13 p.120} that exists between coordinate times t_0 and t_1 . Being closed, this control surface delimits a control volume. This surface and volume can move and deform arbitrarily in between the two times; what’s important is that the control surface remains closed: no holes or cuts can form. At the final time t_1 the closed control surface and volume can be very different from how they were at the initial time t_0 , and even at different positions. We are completely free in our choice of location, size, shape, motion of this control surface and volume.

A very simple example could be a spherical surface delimiting a ball-shaped region. But we could as well consider a surface that initially is



Four snapshots of a moving and deforming closed control surface ([animated version¹](#))

spherical and delimits a region the size of a tennis ball, and at the end is cubical and delimits a region the size of our galaxy. The closed control surface doesn't even need to be connected: it could consist of several separate closed control surfaces, each delimiting a separate volume.

Now consider an extensive quantity; specifically take any one of our [primitive quantities](#) ^{§3.1 p.67}, except magnetic flux. For concreteness let's take some kind of *matter*, but keep in mind that it could be energy-mass, or momentum, or any of the other six. Recall that we denote the volume content of matter by N , the flux by $\mathcal{J}$, the supply by $\mathcal{R}$.

Make the following measurements:

- $N(t_0)$: total *volume content* of the quantity in the control volume at time t_0 .
- $\int_{t_0}^{t_1} \mathcal{J}(t) dt$: total *time-integrated influx* of the quantity through the closed control surface between times t_0 and t_1 .
- $\int_{t_0}^{t_1} \mathcal{R}(t) dt$: total *time-integrated supply* of the quantity through the closed control surface between times t_0 and t_1 .
- $N(t_1)$: total *volume content* of this quantity in the control volume at time t_1 .

Note that in the second measurement we measure the **influx**, not the efflux, through the closed control surface. We must pay close attention to the crossing direction and sign of this flux.

Each of the three amounts above is, in this example, a scalar that can be positive, negative, or zero. For instance we could have:

$$t_0 = 0 \text{ s} \quad N(t_0) = 10.5 \text{ mol} ,$$

$$\int_{t_0}^{t_1} \mathcal{J}(t) dt = -9.7 \text{ mol} ,$$

$$\int_{t_0}^{t_1} \mathcal{R}(t) dt = 0 \text{ mol} ,$$

$$t_1 = 18 \text{ s} \quad N(t_1) = 0.8 \text{ mol} .$$

But if the quantity in question were a vector, like momentum, then each of the three amounts above would be a vector, with any combination of positive, negative, or zero components.

With this setup we first study a special kind of balance law.

§ 5.4 Conservation laws

Conservation laws tightly connect *volume contents* and *fluxes* ▶ chapter 4 p. 93:



Conservation law

A quantity is said to satisfy a **conservation law**, or to be **conserved**, if the following equality between its volume content and time-integrated influx holds for *any closed control surface and volume*, and *any coordinate times* t_0 and t_1 :

$$\text{volume content}(t_1) = \text{volume content}(t_0) + \int_{t_0}^{t_1} \text{influx}(t) \, dt \quad (5.1)$$

For example, in the case of matter with volume content N and influx $\mathcal{J}$, the conservation law would be

$$N(t_1) = N(t_0) + \int_{t_0}^{t_1} \mathcal{J}(t) \, dt .$$

The meaning of a conservation law is intuitive. Let's take again the example of some kind of matter. The amount of quantity in the final control volume, $N(t_1)$, must be equal to the amount in the initial control volume, $N(t_0)$, plus the total amount that flowed in through the surface between those times, the integrated flux $\int_{t_0}^{t_1} \mathcal{J}(t) \, dt$. Said otherwise: any amount of quantity that appears in (or disappears from) the control volume, must come in (or go out) through the surface; it can't appear or disappear out of nowhere. Note that these amounts can be positive, negative, or zero.

A conservation law allows us to make several kinds of deductions and predictions. For example:

- If we know the amount of quantity in the control volume at t_0 , and the net amount that entered through the control surface between t_0 and t_1 , then we can predict the amount in the control volume at t_1 : it is given explicitly by the equation above.
- If we know the amount of quantity in the control volume at t_0 , and the one at t_1 , then we can deduce the net amount that entered through the control surface between the times t_0 and t_1 :

$$\int_{t_0}^{t_1} \mathcal{J}(t) \, dt = N(t_1) - N(t_0) .$$

- If we know the amount of quantity in the control volume at t_1 , and the net amount that entered through the control surface between t_0 and t_1 , we can deduce the initial amount in the control volume at time t_0 :

$$N(t_0) = N(t_1) - \int_{t_0}^{t_1} \mathcal{J}(t) dt .$$

These were just examples. Since a conservation law involves different times and a time integral, it also allows us to make deductions about how volume contents or fluxes depend on time, and to predict the time when a volume content or flux has a particular value.

These kinds of predictions and deductions are very powerful because we are fully free to decide the shape and motion of the control surface and volume, as well as the times t_0 and t_1 .

We unconsciously use several conservation laws all the time in our everyday life. If a bike tyre is suddenly deflated, we conclude that that air must have got out of it through its surface, which must therefore have a hole or a defective valve; we don't conclude that air "just disappeared".

Note that the discussion so far is equally valid for a scalar or a vector quantity. We shall consider vector quantities more in detail in later sections.

Example with a moving control surface

The tube within the tyre of a bicycle has, at a given time, 12.6 mol of air. The bicycle and its tyres moves around, and thirty minutes later the tyre is flat, the inner tube having only 0.5 mol of air.

We consider an imaginary, closed control surface wrapping the tube. The control surface deforms just like the tube deforms. We assume that air satisfies a conservation law. From the description above we have

$$t_0 = 0 \text{ s} , \quad t_1 = 1800 \text{ s} , \quad N(t_0) = 12.6 \text{ mol} , \quad N(t_1) = 0.5 \text{ mol} .$$

The conservation law (5.1) allows us to calculate the integrated **influx**:

$$\begin{aligned} \int_{t_0}^{t_1} \mathcal{J}(t) dt &= N(t_1) - N(t_0) \\ &= 0.5 \text{ mol} - 12.6 \text{ mol} \\ &= -12.1 \text{ mol} \end{aligned}$$

The result, with a negative sign, says that 12.1 mol of air have crossed the moving control surface in an *outward* direction, during the thirty minutes.



Obviously this happened because the tyre tube has a hole somewhere. Note that this is a physical hole in the physical tube; our control surface is imaginary and doesn't have any holes. It's just a sort of "border checkpoint" where anything can in principle move through; we only keep track of what's crossing it, how much, and how fast.

The fact that air satisfies, in this example, a conservation law, allows us to determine the integrated flux through the control surface. But it doesn't allow us to determine the flux $\mathcal{J}(t)$ at every time between t_0 and t_1 . For example, we don't know if there was a larger flux at the beginning than at the end; the tyre may have gone completely flat just after 60 s; in that case the flux $\mathcal{J}(t)$ was zero for $t > 60$ s.

Example with a static control surface

In the previous example we chose a closed control surface that moved and deformed with some object of interest (the tyre tube). But we can also choose a closed control surface that is static and doesn't follow or wrap any object.

Consider a block of 53.4 mol of ice having cylindrical shape with 1 m diameter and 1.226 m height. It is falling downward, in a vacuum, owing to gravity. At a particular time t_0 it occupies a particular position. At a time t_1 , 0.5 s later, it is situated for an instant immediately underneath the initial position, as illustrated in the side figure.

Let's arbitrarily choose a static, closed control surface of cylindrical shape, located in such a way that it wraps the block of ice at the final time. At the initial time this control surface is therefore empty. See the side figure below, where this imaginary control surface is represented by the **dashed yellow line**.

Suppose we want to know the net amount of ice that entered this control surface between times t_0 and t_1 . We can calculate this quantity by assuming that ice – as matter – satisfies a conservation law. According to the description above we have

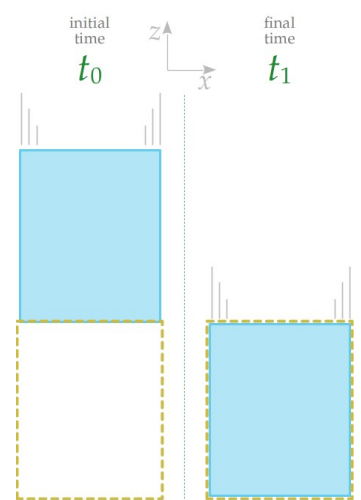
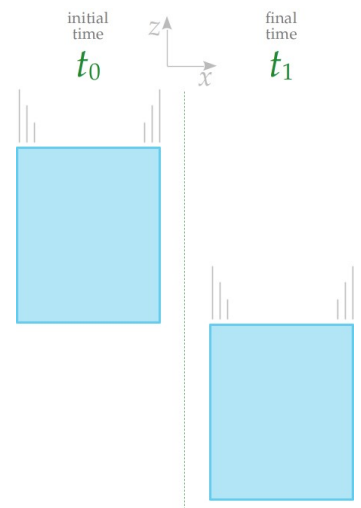
$$t_0 = 0 \text{ s} , \quad t_1 = 0.5 \text{ s} , \quad N(t_0) = 0 \text{ mol} , \quad N(t_1) = 53.4 \text{ mol} .$$

From the conservation law (5.1) we calculate the time-integrated **influx**:

$$\begin{aligned} \int_{t_0}^{t_1} \mathcal{J}(t) dt &= N(t_1) - N(t_0) \\ &= 53.4 \text{ mol} - 0 \text{ mol} \\ &= 53.4 \text{ mol} \end{aligned}$$



There's a physical hole in the physical tube, but no holes in the imaginary control surface that wraps the tube



and this is the net amount of ice that entered the control surface during the 0.5 s.

Note that what we found is the time-integrated influx through the *whole* closed control surface. The conservation law doesn't tell us anything about the integrated influx through *parts* of the surface. In the present case we can divide the surface into three parts: a circular surface at the bottom, a side surface, and a circular surface at the top. If we provide the extra knowledge that the fluxes through the side and bottom surfaces are zero, then by the **property of extensivity** §3.2 p. 69 we have

$$\int_{t_0}^{t_1} \mathcal{J}(t) dt = \int_{t_0}^{t_1} [\mathcal{J}_{\text{top}}(t) + \mathcal{J}_{\text{side}}(t) + \mathcal{J}_{\text{bot}}(t)] dt$$

$$53.4 \text{ mol} = \int_{t_0}^{t_1} [\mathcal{J}_{\text{top}}(t) + 0 \text{ mol/s} + 0 \text{ mol/s}] dt$$

from which we find, as was intuitively clear, that the net amount of ice that crossed the top surface in a downward direction is 53.4 mol.



Exercise 5.2

*Solve the following exercise not just by using intuition, but **by explaining step-by-step how you use a conservation law** to obtain the result:*

- What is the relevant time interval?
- How do you define the closed control surface and its movement, as well as any subdivisions of the surface?
- What are the values of volume content and of time-integrated flux known to you? Which ones do you want to find?

For the present exercise we assume that energy-mass satisfies a conservation law.

An apartment's room has two identical electric heaters along a wall. An electric heater can be considered as a piece of surface across which energy-mass flows into the room: the energy-mass is entering around the electric wires in the form of electromagnetic energy-mass, and is converted into internal energy-mass (mainly of the room's air) by means of the heater. Suppose that each heater corresponds to an influx of 200 J/s.

The room also has a window, which is the only other part of the room's boundary where energy-mass can flow in or out.



In one hour we measure that the total amount of energy-mass in the room has not changed. How much is the integrated energy-mass *influx* through the window during that time?

§ 5.5 Balance laws

Our intuitive understanding of a conserved quantity is that it cannot be “created” or “destroyed”; it can only “move around”. But there are quantities that are not conserved. Such quantities satisfy a *balance* law. They could be produced or disappear in a control volume; recall that we do have a term and measurement for this kind of production or disappearance: the *supply* ▶ §3.2 p. 68.

To formulate a general balance law we use the same *setup* ▶ §5.3 p. 131 as for a conservation law. We take a closed control surface, delimiting a control volume, between coordinate times t_0 and t_1 . Having chosen one of the seven quantities except the magnetic flux, we measure its *volume content* at time t_0 , its *time-integrated influx* between times t_0 and t_1 , its *time-integrated supply* between times t_0 and t_1 , and its *volume content* at time t_1 .



Balance law

A quantity is said to satisfy a **balance law**, or to be **balanced**, if the following equality holds for *any closed control surface and volume*, and *any coordinate times* t_0 and t_1 :

$$\text{volume content}(t_1) = \text{volume content}(t_0) + \int_{t_0}^{t_1} \text{influx}(t) \, dt + \int_{t_0}^{t_1} \text{supply}(t) \, dt \quad (5.2)$$

For example, in the case of energy-mass with volume content E , influx W , supply $\mathcal{A}$, the balance law would be

$$E(t_1) = E(t_0) + \int_{t_0}^{t_1} W(t) \, dt + \int_{t_0}^{t_1} \mathcal{A}(t) \, dt .$$

The meaning of a balance law is intuitive: the amount of quantity in the final control volume: $E(t_1)$, must be equal to the amount in the initial control volume: $E(t_0)$, plus the total amount that flowed in through the surface between those times: $\int_{t_0}^{t_1} W(t) \, dt$, plus the total amount that was created in the volume: $\int_{t_0}^{t_1} \mathcal{A}(t) \, dt$.

We see that a *conservation law* is a special, powerful case of a balance law. Let's make this connection explicit:

Connection between balance and conservation laws

A quantity is said to satisfy a *conservation law* if it satisfies a *balance law* and **its supply is always zero**, for any closed control surface and volume and any coordinate times t_0 and t_1 .

A conservation law is powerful because it means that the supply is always known in advance and has an extremely simple value: zero. The important consequence of this fact is that it allows us to predict the amount of quantity in a final volume by knowing what happens *only on the boundary of the volume* during the time lapse. A general balance law, instead, requires us to know also what happens at every point *within the volume* during the time lapse: we need to know whether some amount of quantity was created or destroyed within the volume.

For these reasons a balance law is in some respects more trivial than a conservation law. If we measure that $E(t_0) - E(t_1) - \int_{t_0}^{t_1} W(t) dt$ is not zero – so there's no conservation law – we can always say that some amount of quantity must have been created or destroyed within the control volume between t_0 and t_1 . An extensive quantity can therefore always be said to satisfy a balance law. A balance is not trivial, however, if we have some other physical law that tells us in advance how the amount created or destroyed at each instant, $\mathcal{A}(t)$, can be calculated.



A conservation law involves only knowledge about the initial and final control volumes (represented by the **red hatched** disks), and about the closed control surface during the time lapse (curved **red** contours); but not about the control volume during the time lapse. A balance law instead involves also this additional information.

Supplies are very different from fluxes

One could object: “why don't you just put the flux $W(t)$ and the supply $\mathcal{A}(t)$ together, adding the two integrals in equation (5.2)? Wouldn't you get

$$E(t_1) = E(t_0) + \int_{t_0}^{t_1} [W(t) + \mathcal{A}(t)] dt = 0$$

which looks like a conservation law?”

Unfortunately the mathematical expression above wouldn't be a conservation law, despite its appearance. The point is this: the flux W involves only what's happening *on the control surface*; the supply $\mathcal{A}$, instead, involves what's happening *within the control volume*. A conservation

law does *not* require us to know what's happening within the control volumes, except at the initial and final time.

Remember, moreover, that **fluxes always satisfy a symmetry principle** §4.6 p. 106, but *supplies in general do not satisfy any analogous symmetry principle*.

Balance law for vector quantities

Let us now consider an extensive vector quantity like *momentum*, which will be very important in our future investigations. Its initial volume content is denoted $\mathbf{P}$; its influx, also called surface force, $\mathbf{F}(t)$; and its supply, also called body force, $\mathbf{G}$. They are time-dependent vectors, defined in a coordinate system (t, x, y, z) :

$$\mathbf{P}(t) = \begin{bmatrix} P_x(t) \\ P_y(t) \\ P_z(t) \end{bmatrix} \quad \mathbf{F}(t) = \begin{bmatrix} F_x(t) \\ F_y(t) \\ F_z(t) \end{bmatrix} \quad \mathbf{G}(t) = \begin{bmatrix} G_x(t) \\ G_y(t) \\ G_z(t) \end{bmatrix}.$$

The balance law for a vector quantity like momentum has exactly the same expression we already know:

$$\mathbf{P}(t_1) = \mathbf{P}(t_0) + \int_{t_0}^{t_1} \mathbf{F}(t) dt + \int_{t_0}^{t_1} \mathbf{G}(t) dt \quad (5.3)$$

with the only difference that the quantities involved are vectors. So it corresponds to a system of three balance equations, one per component:

$$\begin{cases} P_x(t_1) = P_x(t_0) + \int_{t_0}^{t_1} F_x(t) dt + \int_{t_0}^{t_1} G_x(t) dt \\ P_y(t_1) = P_y(t_0) + \int_{t_0}^{t_1} F_y(t) dt + \int_{t_0}^{t_1} G_y(t) dt \\ P_z(t_1) = P_z(t_0) + \int_{t_0}^{t_1} F_z(t) dt + \int_{t_0}^{t_1} G_z(t) dt \end{cases} \quad (5.4)$$

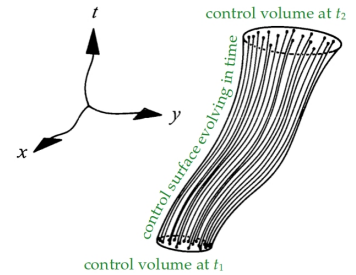


Balance laws in General Relativity

Conservation and balance laws appear simpler from the point of view of Relativity Theory. From a four-dimensional spacetime perspective, a 3D volume is a region,

called *hypersurface*, having one less dimension than spacetime. But a moving 2D surface followed through time is also a spacetime region that has one less dimension than spacetime: two spatial dimensions and one temporal one. Thus the distinctions among the 3D region at t_0 , the moving 2D surface between t_0 and t_1 , and the 3D region at t_1 , disappear: they are seen to be just different parts of the same three-dimensional *hypersurface*. We perceive some parts of this hypersurface as belonging “to the same time”, showing their three dimensions all at once; and other parts of it as extending through time, showing only two dimensions at any time. In fact, different observers make this division in different ways.

And from a spacetime perspective, the amount of a quantity $E(t_0)$ or $E(t_1)$ within a 3D region is seen as a flux through time; so its apparent difference from the flux W also disappear.



Spacetime representation of the evolution of a closed control surface, containing some pointlike objects (adapted from Misner et al. 2017)



What about the magnetic flux?

In the case of magnetic flux, the idea of a conservation law is analogous, but is *formulated with one less spatial dimension*, and with a different notion of orientation: we consider a control surface that exists between times t_0 and t_1 ; this surface has a closed control curve as boundary. The magnetic flux turns out to be a quantity for which it's possible to ask how much of it is intersecting a surface, and how much of it is crossing a closed curve. One way to understand this is to imagine magnetic flux as a bundle of tubes or lines that are either closed or extend to infinity. It will be discussed in depth in chapter 8.

§ 5.6 Examples

Let us now study a couple of example applications of balance laws.

In order to apply a balance law we must choose one or more closed control surfaces and corresponding control volumes. Fix a coordinate system (t, x, y, z) . We previously discussed that [two typical choices of control volumes and surfaces](#) §4.4 p. 101 with respect to a coordinate system are: (a) a moving one, usually “wrapping” some solid object; (b) a static one. Let's see how a balance law is applied in these two cases. For the quantity to study we choose momentum, so that we can get immediately acquainted with handling vector equations.

Moving control surface

Let's say that coordinates x, y have horizontal directions, and z an upward direction. Consider a flying tennis ball. Choose an imaginary, closed control surface that perfectly wraps the tennis ball and moves with it.

At a particular time instant t_0 the tennis ball has momentum $[0, 1.70, 0.98]$ N s; by this we mean that the control volume corresponding to the ball contains that amount of momentum.

While the tennis ball is flying, we assume that the net influx of momentum through the control surface is zero, for instance because the ball is in a vacuum. But there's a supply of momentum within the volume, constant in time, equal to $[0, 0, -0.579]$ N. This supply, as we'll see later, exist because the tennis ball is in the gravitational field of the Earth.

The tennis ball flies for two seconds. How much is the momentum of the tennis ball at the end of this time lapse? Let's call this time t_1 .

From the description above we have these data:

$$t_0 = 0 \text{ s}, \quad t_1 = 2 \text{ s},$$

$$\mathbf{P}(t_0) = \begin{bmatrix} 0 \\ 1.70 \\ 0.98 \end{bmatrix} \text{ N s},$$

$$\mathbf{F}(t) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ N (const. in time)}, \quad \mathbf{G}(t) = \begin{bmatrix} 0 \\ 0 \\ -0.579 \end{bmatrix} \text{ N (const. in time)}.$$

The balance law (5.3) allows us to find the amount of momentum in the tennis ball at time $t_1 = 2$ s:

$$\begin{aligned} \mathbf{P}(t_1) &= \mathbf{P}(t_0) + \int_{t_0}^{t_1} \mathbf{F}(t) dt + \int_{t_0}^{t_1} \mathbf{G}(t) dt \\ &= \mathbf{P}(t_0) + \mathbf{F} \cdot (t_1 - t_0) + \mathbf{G} \cdot (t_1 - t_0) && \text{(because } \mathbf{F}, \mathbf{G} \text{ are constant in time)} \\ &= \begin{bmatrix} 0 \\ 1.70 \\ 0.98 \end{bmatrix} \text{ N s} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ N} \cdot 2 \text{ s} + \begin{bmatrix} 0 \\ 0 \\ -0.579 \end{bmatrix} \text{ N} \cdot 2 \text{ s} \\ &= \begin{bmatrix} 0 \\ 1.70 \\ -0.18 \end{bmatrix} \text{ N s}. \end{aligned}$$



Therefore two seconds later the ball's momentum has the same x - and y -component as it had initially; but now its z -component points downward – which means that the ball is also moving downward, not only horizontally.

Static control surface

Take again the previous [example of a cylindrical block of ice](#) ^{§5.4 p. 135} moving downward, in a vacuum, during a lapse of time of 0.5 s, depicted again in the side figure. In the previous example we discussed the net amount of matter that crosses a static, closed control surface at the final location of the ice block. Now we are instead interested in the net amount of *momentum* that crosses the same surface. We shall therefore use the balance of momentum (5.3).

Suppose we have this information:

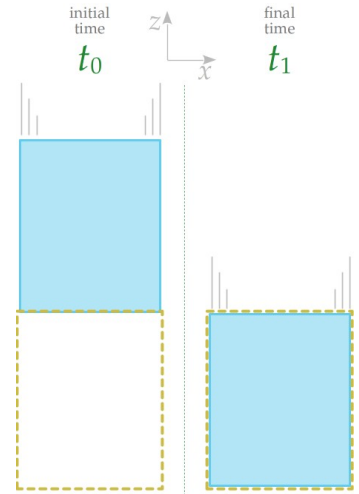
$$t_0 = 0 \text{ s}, \quad t_1 = 0.5 \text{ s}, \quad \mathbf{P}(t_0) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ N s}, \quad \mathbf{P}(t_1) = \begin{bmatrix} 0 \\ 0 \\ -4.72 \end{bmatrix} \text{ N s}.$$

The expression for $\mathbf{P}(t_0)$ says the initial momentum content in the control volume is zero. This makes sense, because the control volume is initially empty of matter. The expression for $\mathbf{P}(t_1)$ says that the final momentum content in the control volume is non-zero: it directed fully downward, with magnitude 4.72 N s. This also makes sense, because at this time in the control volume there's matter moving downward.

Can we find the time-integrated influx or supply of momentum during the time lapse of 0.5 s? Strictly speaking, from the data mathematically expressed above, the answer is no: to find the time-integrated influx we would need the time-integrated supply, and vice versa. This situation illustrates what we said previously: balance laws generally require more information than conservation laws.

Suppose we are told that the time-integrated supply of momentum *within the control volume*, during the time lapse, is directed downward and has magnitude 1.57 N s. That is,

$$\int_{t_0}^{t_1} \mathbf{G}(t) dt = \begin{bmatrix} 0 \\ 0 \\ -1.57 \end{bmatrix} \text{ N s}.$$



❗ Remember again that *this is not a flux*: it isn't momentum that "enters" from the top of the control surface because the ice block is entering there. This is momentum *created* (by gravity) in the parts of the volume where there's matter.

The influx through the top surface can now be found using the balance law (5.3):

$$\begin{aligned}
 \int_{t_0}^{t_1} \mathbf{F}(t) dt &= \mathbf{P}(t_1) - \mathbf{P}(t_0) - \int_{t_0}^{t_1} \mathbf{G}(t) dt \\
 &= \begin{bmatrix} 0 \\ 0 \\ -4.72 \end{bmatrix} \text{Ns} - \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{Ns} - \begin{bmatrix} 0 \\ 0 \\ -1.57 \end{bmatrix} \text{Ns} \\
 &= \begin{bmatrix} 0 \\ 0 \\ -3.15 \end{bmatrix} \text{Ns}
 \end{aligned}$$

This is the momentum that “enters” through the closed control surface. We can intuitively deduce that it is specifically entering through the top part of the control surface.

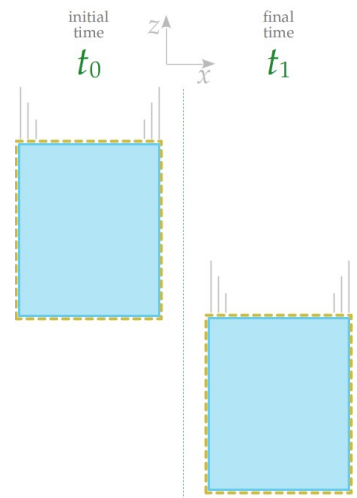
Another example with a moving control surface

Let us consider the same physical situation as in the previous example, but now let’s choose a *moving* control surface instead, as in the first example with the tennis ball. We choose a closed control surface that tightly wraps the block of ice at all times between t_0 and t_1 .

An important remark: since we are now choosing a different control surface and volume from the previous example, the values of momentum contents and fluxes may be different from the previous ones as well – they refer to different regions of space. To avoid getting confused, it’s good to denote the amounts in the present example with different symbols. Let’s underline them for instance; we could also use some other graphical symbol, or simply change the letters themselves.

Suppose that we want to know how much is the time-integrated supply of momentum in the new, moving control volume, between times t_0 and t_1 . In order to find it from the balance of momentum we need to know: (a) the initial momentum content, (b) the final momentum content, (c) the time-integrated influx.

We are told that the ice block initially has zero momentum (because it’s released, and therefore has zero velocity, exactly at that instant), and at the final time it has a downward momentum of magnitude 4.72 N s. We are



also told that there is no net flux of momentum, at any time, through the control surface that moves along with the ice block. Our data are therefore

$$t_0 = 0 \text{ s} , \quad t_1 = 0.5 \text{ s} ,$$

$$\underline{\mathbf{P}}(t_0) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ N s} , \quad \underline{\mathbf{P}}(t_1) = \begin{bmatrix} 0 \\ 0 \\ -4.72 \end{bmatrix} \text{ N s} ,$$

$$\underline{\mathbf{F}}(t) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ N (const. in time)} .$$

We have enough data to find the time-integrated supply of momentum generated within the moving control volume, using the balance of momentum:

$$\begin{aligned} \int_{t_0}^{t_1} \underline{\mathbf{G}}(t) dt &= \underline{\mathbf{P}}(t_1) - \underline{\mathbf{P}}(t_0) - \int_{t_0}^{t_1} \underline{\mathbf{F}}(t) dt \\ &= \begin{bmatrix} 0 \\ 0 \\ -4.72 \end{bmatrix} \text{ N s} - \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ N s} - \int_{t_0}^{t_1} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ N } dt \\ &= \begin{bmatrix} 0 \\ 0 \\ -4.72 \end{bmatrix} \text{ N s} \end{aligned}$$

As we remarked above, even if the physical event is exactly the same, the momentum flux and supply in the present analysis are different from the previous analysis, because *they refer to different imaginary control surfaces and volumes*. Compare the time-integrated influxes of the two analyses:

$$\int_{t_0}^{t_1} \underline{\mathbf{F}}(t) dt = \begin{bmatrix} 0 \\ 0 \\ -3.15 \end{bmatrix} \text{ N s} \neq \int_{t_0}^{t_1} \underline{\mathbf{F}}(t) dt = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ N s} .$$

and the time-integrated supplies:

$$\int_{t_0}^{t_1} \underline{\mathbf{G}}(t) dt = \begin{bmatrix} 0 \\ 0 \\ -1.57 \end{bmatrix} \text{ N s} \neq \int_{t_0}^{t_1} \underline{\mathbf{G}}(t) dt = \begin{bmatrix} 0 \\ 0 \\ -4.72 \end{bmatrix} \text{ N s}$$



Fluxes and supplies depend on the control surface and volume

The time-integrated flux and time-integrated supply that appear in balance laws are strictly dependent on the closed control surface and volume that we choose.

Therefore, if we analyse the *same* physical phenomenon with a *different* set of control surfaces and volumes, we cannot expect the results of calculations with the old set to be valid for the new one.



Exercise 5.3

Consider once more the block of ice analysed in the previous two examples.

This time choose a *static* closed control surface that coincides with the *initial* position of the block. This surface therefore includes all ice at time t_0 , but is empty at time t_1 . The initial and final momentum contents are zero.

Calculate the integrated supply for this new control surface between t_0 and t_1 .

(Hint: consider the results from the example of §5.6, and use the **symmetry of flux** to find the flux through the bottom part of the control surface of the present exercise. Recall also that the fluxes through the side and top surfaces are zero.)

§ 5.7 Balance laws: differential expression

A balance law such as

$$E(t_1) = E(t_0) + \int_{t_0}^{t_1} W(t) dt + \int_{t_0}^{t_1} \mathcal{A}(t) dt$$

or a conservation law such as

$$N(t_1) = N(t_0) + \int_{t_0}^{t_1} \mathcal{J}(t) dt$$

are said to be written in *integral form* because they contain time-integrated fluxes or supplies. These time integrations are necessary for calculating the net amount of quantity that enters through the control surface or is generated in the volume during the time lapse considered.

We can rewrite these laws in a different mathematical form that doesn't show time integrals, but shows time derivatives.

This rewriting is essentially an expression of the **fundamental theorem of calculus**². Consider the case of the conservation law for simplicity. If N is a function of t , and we call $\mathcal{J}$ its derivative with respect to t , then we can also say that N is a **primitive function**³ for $\mathcal{J}$, and the definite integral of $\mathcal{J}$ is given by a difference of N at different arguments. In symbols:

$$\frac{dN(t)}{dt} = \mathcal{J}(t) \quad \Longleftrightarrow \quad \int_{t_0}^{t_1} \mathcal{J}(t) dt = N(t_1) - N(t_0) .$$

But the equation on the right is just a conservation law written in a slightly different way! This means that it can equivalently be expressed by the equation on the left.

Let's interpret this relation from the point of view of the physical measurement of volume content and flux, and for full generality let's consider a balance law.

Change the symbol for the time t_0 into t , and take t_1 to come after a *very short* lapse of time Δt after t :

$$t_0 \text{ becomes } t , \quad t_1 \text{ becomes } t + \Delta t$$

Then the integrated flux can be approximated by

$$\int_t^{t+\Delta t} W(t) dt \approx W(t) \Delta t$$

that is, the flux at time t multiplied by the time lapse. The reason for this approximation is that the flux W is approximately constant during the short time lapse Δt , and equal to its initial value at time t . Similarly for the supply:

$$\int_t^{t+\Delta t} \mathcal{A}(t) dt \approx \mathcal{A}(t) \Delta t .$$

The balance law above can then be approximated as follows:

$$E(t + \Delta t) \approx E(t) + [W(t) + \mathcal{A}(t)] \Delta t .$$

Now move the term $E(t)$ to the left side, and divide the whole expression by Δt :

$$\frac{E(t + \Delta t) - E(t)}{\Delta t} \approx W(t) + \mathcal{A}(t) .$$

Finally, consider smaller and smaller Δt : in the limit the ratio $\frac{E(t+\Delta t)-E(t)}{\Delta t}$ becomes, by definition, a derivative; and the approximation of the integrals becomes an exact equality. So we find:

! Some mathematical conditions must be satisfied to make these steps; otherwise the derivative below must be carefully defined in a generalized sense.

Balance and conservation laws in differential form

A quantity is said to satisfy a **balance law** if the following equality holds for any closed control surface and volume, and any coordinate time t :

$$\frac{d \text{ volume content}(t)}{dt} = \text{influx}(t) + \text{supply}(t) \quad (5.5)$$

Analogously, a quantity is said to satisfy a **conservation law** if the following equality holds for any closed control surface and volume, and any coordinate time t :

$$\frac{d \text{ volume content}(t)}{dt} = \text{influx}(t) \quad (5.6)$$

For instance, the balance and conservation laws written above can be re-expressed as

$$E(t_1) = E(t_0) + \int_{t_0}^{t_1} W(t) dt + \int_{t_0}^{t_1} \mathcal{A}(t) dt \quad \text{or} \quad \frac{dE(t)}{dt} = W(t) + \mathcal{A}(t)$$

$$N(t_1) = N(t_0) + \int_{t_0}^{t_1} \mathcal{J}(t) dt \quad \text{or} \quad \frac{dN(t)}{dt} = \mathcal{J}(t).$$

This differential expression says that the *rate of change of the volume content* of a quantity must equal the sum of influx of that quantity through the control surface and the supply of that quantity in the control volume.

When to use the integral or the differential expression?

First of all, let's remark again that the integral expression and the differential expression of balance and conservation laws are mathematically equivalent (under some conditions that can become important in more advanced applications). So in choosing the one or the other we are not choosing between different physical laws.

But using the one or the other expression as our starting point can be more convenient in some situations and less convenient in others. Some obvious guidelines:

- If the data in a problem include time-integrated fluxes or supplies, then the integral expression is probably more convenient, because it directly uses these integrals.

- If a problem asks for some volume contents at a given time, and gives as data the volume contents at some other time, then the integral expression is probably more convenient.
- If a problem involves fluxes and supplies that are zero, then the differential expression may be more practical, because it says that the volume content is then constant in time: its time derivative is zero.
- If a problem asks for the time dependence of some quantity, then the differential expression is probably more convenient, because it involves a generic time instant.

But keep in mind that these are only guidelines. Some problems may be easily solved with either expression; some problems may require both expressions to be used: one expression for one quantity and the other for another quantity.

Example

As an example of use of the differential expression, let's consider again the [problem with the tennis ball](#) §5.4 p. 134. The physical situation, choice of control surface and volume, and the known information and question are the same as before.

We solved this problem using the integral expression of the balance law for momentum. But it can also be solved using its differential expression:

$$\frac{d\mathbf{P}(t)}{dt} = \mathbf{F}(t) + \mathbf{G}(t)$$

or explicitly in components

$$\begin{cases} \frac{dP_x(t)}{dt} = F_x(t) + G_x(t) \\ \frac{dP_y(t)}{dt} = F_y(t) + G_y(t) \\ \frac{dP_z(t)}{dt} = F_z(t) + G_z(t) \end{cases}$$

The differential expression can be convenient because the value of the influx $\mathbf{F}$ and supply $\mathbf{G}$ are constant, and zero, at every time t . Therefore



the rate of change of momentum at any time t between t_0 and t_1 is

$$\begin{aligned}\frac{d\mathbf{P}(t)}{dt} &= \mathbf{F}(t) + \mathbf{G}(t) \\ &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ N} + \begin{bmatrix} 0 \\ 0 \\ -0.579 \end{bmatrix} \text{ N} \\ &= \begin{bmatrix} 0 \\ 0 \\ -0.579 \end{bmatrix} \text{ N}\end{aligned}$$

We can find the momentum at any time t between and including t_0 and t_1 by an easy integration:

$$\begin{aligned}\mathbf{P}(t_1) &= \mathbf{P}(t_0) + \int_{t_0}^{t_1} \frac{d\mathbf{P}(t)}{dt} dt \\ &= \begin{bmatrix} 0 \\ 1.70 \\ 0.98 \end{bmatrix} \text{ N s} + \int_{t_0}^{t_1} \begin{bmatrix} 0 \\ 0 \\ -0.579 \end{bmatrix} \text{ N} dt \\ &= \begin{bmatrix} 0 \\ 1.70 \\ 0.98 \end{bmatrix} \text{ N s} + \begin{bmatrix} 0 \\ 0 \\ -0.579 \end{bmatrix} \text{ N} \cdot (t_1 - t_0) && \text{(because the integrand is constant)} \\ &= \begin{bmatrix} 0 \\ 1.70 \\ 0.98 \end{bmatrix} \text{ N s} + \begin{bmatrix} 0 \\ 0 \\ -0.579 \end{bmatrix} \text{ N} \cdot 2 \text{ s} \\ &= \begin{bmatrix} 0 \\ 1.70 \\ -0.18 \end{bmatrix} \text{ N s}\end{aligned}$$



Exercise 5.4

Solve the following exercises not just by using intuition, but **by explaining step-by-step how you use a balance or conservation law** to obtain the result:

- What is the relevant time interval, if using the integral expression?
- How do you define the closed control surface and its movement, as well as any subdivisions of the surface?

- What are the values of volume content and of time-integrated flux known to you? Which ones do you want to find?

For the present exercise we assume that **argon**⁴ atoms satisfy a conservation law.

A **distillation column**⁵ in a chemical plant has an **influx** of 2 mol/s of argon atoms constant in time at one inlet, and an **efflux** of argon atoms, constant in time but possibly at a different rate, at an outlet. An amount of 500 mol of argon is measured in the chamber at a given time, and one minute later the amount is measured to be 620 mol. How much is the argon efflux at the outlet, at any time?



§ 5.8 Seven universal balance laws

The reason for introducing the **seven primitive quantities** ^{§3.1 p.67}, which are common to all our main physical theories, is that **each of these quantities obeys a balance law**, and some of them obeys more strictly a conservation law.

What's remarkable about these seven balance laws?

- They are known, so far, to be satisfied by *all* physical phenomena, from subatomic scales to cosmological scales. No exceptions are known.
- They appear implicitly or explicitly in *all* our main physical theories, approximate or not: from Newtonian mechanics to electromagnetism, from general relativity to quantum theory.
- They are valid in any system of coordinates, for any kind of control volumes and surfaces, at any time.
- Each of these balances can be expressed by the *same* mathematical equation in all of these theories.

In other words these seven balances are, as far as we know, **universal** ^{§5.2 p.128}.

These balances are truly the *Seven Wonders of the World*, even more long-lasting than the **traditional "seven wonders"**⁶:



The seven universal balance laws

Balance of matter
Conservation of magnetic flux
Conservation of electric charge
Balance of energy
Balance of momentum
Balance of angular momentum
Balance of entropy

The mathematical expressions of these balances are reported in table 5.1 on page 153.

In some approximate physical theories, such as Newtonian thermomechanics, all these balance laws are, or can easily be taken to be, the [fundamental laws](#) ^{§5.1 p. 127} on which the theory is built. In other theories, only some of these laws are taken as fundamental, while the rest are derived from other fundamental laws; but nevertheless all these balances are still universally satisfied. In General Relativity, for instance, the conservation of matter, electric charge, magnetic flux, and the balance of entropy are taken as fundamental; but the balances of energy, momentum, angular momentum are a *consequence* of the so-called Einstein equations, which are taken as more fundamental.

It is therefore important and useful to learn these universal balances, no matter what kind of specialized theory and physical phenomena you may end up working with in the future. You will apply these balances to any kind of physical phenomenon or engineering or physics field you'll work with: construction of bridges, control of chemical reactions, operation of GPS navigation and satellites, monitoring of nuclear power plants, sending robots to Mars, design of fuel cells, collisions of subatomic particles, cosmology, or who knows what else. Every physical phenomenon involves at least one of these seven balances in its physical description.

The *Einstein equations*, which can deceptively simply be written

$$G = \frac{8\pi\kappa}{c^4}T$$

include the balances of energy, momentum, angular momentum, boost momentum as special consequences.



Concise mathematical form of the universal balances

The seven balances can be expressed very concisely if we use the language of *differential forms*⁷. These are geometric objects that associate a number to any curve, surface, or volume of our choice. The balances for matter, momentum, energy, angular momentum,

electric charge, magnetic flux, entropy then take on these very concise expressions:

$$dN = \mathcal{R} \quad d\Phi = -\mathcal{E} \quad dQ = \mathcal{I} \quad dE = \mathcal{A} \quad d\mathbf{P} = \mathbf{G} \quad d\mathbf{L} = \mathcal{T} \quad dS \geq 0$$

where the symbols on the left of the equations are taken in a four-dimensional sense.

If you want to learn more about differential forms, take a look at the books by Burke 1987; 1995 and Bossavit 1991.

The seven balances for description and prediction

The seven universal balances govern every physical phenomenon. Yet this doesn't mean that all of them are always used explicitly in the description or prediction of physical phenomena.

- For some physical phenomena, all the seven universal balances enter our calculations.
- For some other physical phenomena, some of the seven balances do not appear *explicitly* in our calculations. But they still enter *implicitly* in the way we choose to set up or describe the phenomenon.

For example, we may choose control volumes or control surfaces in such a way that some conservation laws are automatically satisfied. A typical case is the choice of control surface around a given object (*matter*), which guarantees that the law of conservation of matter is automatically satisfied. As another example, sometimes we simplify a physical phenomenon to one spatial dimension only. Think of when we throw a ball vertically in the air, and only consider its height from the ground. In such a case we can make some predictions using only the balance of energy, apparently avoiding the balance of momentum. But in reality, the fact that the ball can be considered as moving vertically is possible because momentum is balanced in the horizontal directions. The balance of momentum is therefore still necessary for this prediction, but it has silently been taken care of.

- For still other physical phenomena, some of the seven balances may not be required because we do not need the kind of physical information they provide. We saw an example of this with the [flat-tyre problem](#) §5.4 p. 134, where we only used the conservation of matter but we weren't interested in what happened to other quantities like energy or momentum, or in how the tyre was moving.



The description of how a common lighter works, thanks to [piezoelectricity](#)⁸, requires more or less all seven universal laws to be explicitly accounted for.

	integral expression	differential expression
matter	$N(t_1) = N(t_0) + \int_{t_0}^{t_1} \mathcal{J}(t) dt + \int_{t_0}^{t_1} \mathcal{R}(t) dt$	$\frac{dN(t)}{dt} = \mathcal{J}(t) + \mathcal{R}(t)$
electric charge	$Q(t_1) = Q(t_0) + \int_{t_0}^{t_1} \mathcal{I}(t) dt$	$\frac{dQ(t)}{dt} = \mathcal{I}(t)$
magnetic flux	$\Phi(t_1) = \Phi(t_0) - \int_{t_0}^{t_1} \mathcal{E}(t) dt$	$\frac{d\Phi(t)}{dt} = -\mathcal{E}(t)$
momentum	$\mathbf{P}(t_1) = \mathbf{P}(t_0) + \int_{t_0}^{t_1} \mathbf{F}(t) dt + \int_{t_0}^{t_1} \mathbf{G}(t) dt$	$\frac{d\mathbf{P}(t)}{dt} = \mathbf{F}(t) + \mathbf{G}(t)$
energy	$E(t_1) = E(t_0) + \int_{t_0}^{t_1} W(t) dt + \int_{t_0}^{t_1} \mathcal{A}(t) dt$	$\frac{dE(t)}{dt} = W(t) + \mathcal{A}(t)$
angular momentum	$\mathbf{L}(t_1) = \mathbf{L}(t_0) + \int_{t_0}^{t_1} \mathbf{M}(t) dt + \int_{t_0}^{t_1} \mathcal{T}(t) dt$	$\frac{d\mathbf{L}(t)}{dt} = \mathbf{M}(t) + \mathcal{T}(t)$
entropy	$S(t_1) \geq S(t_0) + \int_{t_0}^{t_1} \Pi(t) dt$	$\frac{dS(t)}{dt} \geq \Pi(t)$

Table 5.1 The seven universal balance laws. These formulae are valid in Newtonian mechanics, General Relativity, and even quantum theory if their symbols are interpreted as ‘statistical operators’.

Some physical phenomena may be equally predicted using either one particular subset of the seven balances, or a different subset, as we please. For instance, a given problem might be solved using conservation of matter and balance of momentum, or alternatively by using conservation of matter and balance of energy. We shall see examples of all these possibilities in later applications.

But the fact that the seven universal balances govern every physical phenomenon doesn't mean that they can be used alone, by themselves. In the vast majority of cases they need to be augmented by appropriate *constitutive relations*, which we discuss in the next section.

In the next chapters we shall explore and apply the seven universal balance laws in more detail. For each of them we shall recall its mathematical expression, discuss some constitutive relations that are commonly used with it, and examine some example applications.

§ 5.9 Constitutive relations

In the previous sections we studied the mathematical form of balance and conservation laws, and found out that seven balances are of special importance.

From their formulation and from the examples, you noticed that each balance law connects the volume content, flux, supply of one extensive quantity at different times. For instance, the balance law for energy

$$E(t_1) = E(t_0) + \int_{t_0}^{t_1} W(t) dt + \int_{t_0}^{t_1} \mathcal{A}(t) dt$$

connects the volume contents E of energy, the flux W of energy, the supply $\mathcal{A}$ of energy. It does not involve, say, the volume content N of matter, or the flux $\mathbf{F}$ of momentum.

But we know that there must also exist connections among the amounts of *different* quantities. This fact is implicit in many expressions we use everyday, like “the pressure of air” (pressure is *momentum* flux, air is *matter*), or “the energy of the battery” (the battery is made of *matter*, and involves *electric charge*). In previous sections we often made statements such as “the momentum is zero, because there's no matter”.

Physical laws that connect different kinds of quantity are called *constitutive relations*; we briefly [discussed them before](#) ^{§5.2 p. 128}. As was mentioned in that discussion, the term ‘constitutive’ actually refers not to the

characteristic of connecting different quantities, but to the fact that each of these laws typically applies only to specific situations:

Constitutive relation

A **constitutive relation**, also called *constitutive equation*, or *closure equation*, or *constitutive property*, is a physical relationship or that is true only under specific conditions; for example: only for specific physical phenomena, or only on specific scales of space and time, or only for control volumes or surfaces of particular sizes or shapes, or only for specific ranges of measurement precision. Therefore they are often used in specific physical theories.

Constitutive relations express the amazing diversity of physical phenomena that we observe around and within us. For example the fact that a body of water can easily change shape, as opposed to a block of concrete; or that we can store electric energy in a batter but not in a piece of wood. The differences between **states of matter**¹⁰ – solid, liquid, gas, plasma, and there are others – arise from different constitutive relations.

Constitutive relations also express approximations that we use in describing physical phenomena in particular conditions, and therefore mark the difference between specialized or approximate physical theories; for example between Newtonian mechanics, which applies only for low speeds and low energy-mass concentrations (hence weak gravitational fields and small spacetime curvature), and General Relativity, which applies on most if not all known scales, including cosmological ones.

When we read that a new physical phenomenon has been discovered, usually that means that a new *constitutive relation* has been discovered. Depending on the specific scientific field you'll work in, you'll learn some constitutive relations in more detail than others.

Constitutive relations come in a great variety of mathematical forms. Some of them are simple algebraic relations between the volume content, or flux, or supply of one quantity, and the volume content, or flux, or supply of another. Some constitutive relations involve derivatives; some involve integrals.

Many constitutive relations connect volume contents, fluxes, supplies of different primitive quantities, not directly, but through the intermediary of derived or **auxiliary quantities** ^{§3.11 p.86} such as areas and volumes,



Four states of matter, arising from different constitutive relations (image: [Spirit469⁹](#))

temperature, velocity, pressure, polarization, magnetization. We shall see some examples below.

A very powerful characteristic of many constitutive relations is that they connect the volume content of a primitive quantity at a given time with the flux or supply of another primitive quantity *at the same time*. For instance, a constitutive law may allow us to find the flux of matter across some control surface at time t : $\mathcal{J}(t)$, from the content of momentum in a neighbouring control volume at the same time t : $\mathbf{P}(t)$; this example is further discussed below. This characteristic greatly extends the predictive power of balance laws when used together with constitutive relations, as we shall see in chapter 6.

§ 5.10 Examples of constitutive relations

Let us briefly reveal some constitutive equations that have tacitly been used in examples of the previous sections. This is only an overview; we shall study these constitutive relations at length in the next chapters. All these constitutive relations are valid only in “Newtonian approximation”, that is, when speeds are much lower than the speed of light; gravitation is weak, as it is on Earth; and any present electromagnetic fields are enough weak. Note how all relations are only valid for particular kinds of control volumes or surfaces.

Constitutive relation for mass-energy and matter. If a small control volume contains an amount of matter N , then it also contains an amount of *rest mass-energy*

$$m = \rho N$$

where ρ , called *molar mass*, is approximately a constant that depends on the kind of matter. This constitutive relation is the reason why an amount of matter is often quantified in terms of mass.

Newton’s constitutive relation for momentum and matter. One of the most used constitutive relations connects the amount of matter in a small control volume, or the flux of matter through its surface, with the amount of momentum in the volume. It can be stated in several ways. Here are two alternatives: the first valid for a *moving* control volume, the second for a *static* one.

First formulation. Take a small control volume moving with velocity $\mathbf{v}$, with speed much smaller than light’s. If this control volume contains an

amount of matter N and there's *no flux* of matter through its surface, then it also contains, to a very good approximation, an amount of momentum

$$\mathbf{P} = m\mathbf{v}.$$

Second formulation. Take a small *static* cuboid control volume of cubical shape and sides parallel to the coordinate axes. The volume has size V and each of its sides area A . If the fluxes of matter through the sides orthogonal to the three coordinates, in a positive direction, are $[J_x, J_y, J_z]$, then the control volume contains momentum

$$\mathbf{P} = \rho [J_x, J_y, J_z] \frac{V}{A}$$

again only if weak electromagnetic fields are present. This relation is connected to the one [between velocity and the flux of matter](#) §4.15 p. 124.

Constitutive relation for supply of momentum. If a small control volume contains an amount of rest mass-energy m , then it also has a supply of momentum

$$\mathbf{G} = m\mathbf{g}$$

This constitutive relation expresses the *gravitational volume force*. The vector $\mathbf{g}$ is called the **acceleration of free fall**. It expresses the gravitational field, that is, spacetime curvature. It generally depends on time, position, and the motion of the matter or electromagnetic field having mass-energy m , and the coordinate system.

For physical phenomena on, or close to, Earth's surface, in a coordinate system (t, x, y, z) fixed the ground and with z pointing upward, the vector $\mathbf{g}$ can be taken to be approximately constant; it points towards the ground:

$$\mathbf{g} = -g \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \text{with} \quad g := |\mathbf{g}| \approx 9.8 \text{ N/kg} \equiv 9.8 \text{ m/s}^2.$$

In more precise applications in [geodesy](#)¹¹ this vector may depend on latitude, longitude, altitude, and even time, owing to Earth's internal motion. Further away from Earth's, the expression of $\mathbf{g}$ becomes more complex. We shall discuss this further in chapter 10.

Constitutive relation for energy-mass of matter. Take a small control volume on Earth's surface containing an amount of matter N , and such

Old textbooks take this formula as the *definition* of momentum.

$$\begin{aligned} & \sum_{j \neq i} \frac{\mu_j (\mathbf{r}_j - \mathbf{r}_i)}{r_{ij}^3} \left\{ 1 - \frac{2(\beta + \gamma)}{c^2} \sum_{l \neq i} \frac{\mu_l}{r_{il}} - \frac{2\beta - 1}{c^2} \sum_{k \neq j} \frac{\mu_k}{r_{jk}} \right. \\ & + \gamma \left(\frac{\dot{\mathbf{r}}_i}{c} \right)^2 + (1 + \gamma) \left(\frac{\dot{\mathbf{r}}_j}{c} \right)^2 - \frac{2(1 + \gamma)}{c^2} \dot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_j \\ & \left. - \frac{3}{2c^2} \left[\frac{(\mathbf{r}_i - \mathbf{r}_j) \cdot \dot{\mathbf{r}}_i}{r_{ij}} \right]^2 + \frac{1}{2c^2} (\mathbf{r}_j - \mathbf{r}_i) \cdot \dot{\mathbf{r}}_j \right\} \\ & + \frac{1}{c^2} \sum_{j \neq i} \frac{\mu_j}{r_{ij}^3} \left\{ [\mathbf{r}_i - \mathbf{r}_j] \cdot [(2 + 2\gamma) \dot{\mathbf{r}}_i - (1 + 2\gamma) \dot{\mathbf{r}}_j] \right\} (\dot{\mathbf{r}}_i - \dot{\mathbf{r}}_j) \\ & + \frac{3 + 4\gamma}{2c^2} \sum_{j \neq i} \frac{\mu_j \ddot{\mathbf{r}}_j}{r_{ij}} \end{aligned}$$

The expression for $\mathbf{g}$ used at NASA for satellite and spacecraft motion (Moyer 2000).

that there is no flux of matter $\mathcal{J}$ across its surface. In a coordinate system (t, x, y, z) where z points upward, this control volume also contains an amount of energy-mass

$$E = m_{\text{r}} c^2 + U + \frac{1}{2} m_{\text{r}} \boldsymbol{v}^2 + m_{\text{r}} g z .$$

Remember that
 $\boldsymbol{v}^2 \equiv \boldsymbol{v} \cdot \boldsymbol{v} \equiv |\boldsymbol{v}|^2 \equiv v^2$

The first term in this sum is called *rest energy-mass* (a [huge amount](#) §3.6 p.74 of energy in ordinary situations), the third term U is called *internal energy*, the fourth is called *kinetic energy*, the fifth *gravitational potential energy*. The separation between the rest- and internal-energy terms is actually arbitrary. The rest mass-energy m and the internal energy U depend on the amount of matter N .

§ 5.11 Summary of differences between the seven balance laws and constitutive relations

Let’s summarize the most important typical differences between the seven universal balance laws for our seven primitive quantities, and constitutive relations:

Universal balance law	Constitutive relation
Seven	A virtually infinite number
Valid for every phenomenon	Valid for a specific phenomenon
Valid for any control volume and surface	Valid only for control volumes or surfaces of specific size, shape, location
Connects contents, fluxes, supplies of the same quantity	Often connects contents, fluxes, supplies of different quantities
Connects contents, fluxes, supplies at different time instants	Often connects contents, fluxes, supplies at the same time instant
Valid in all modern theories	Valid in specific theories

§ 5.12 Approximations, observation scales, ‘small enough’

The amazing convenience of the seven universal balance laws is that we do not have to worry about the size of the control volumes and surfaces to which we apply them; we do not have to worry about the coordinate system we’re using; we do not have to worry on whether we are applying them on a molecular scale and resolution or on a galactic scale and resolution;

we do not have to worry about the magnitudes of the quantities involved. These laws apply, without approximation, to all situations.

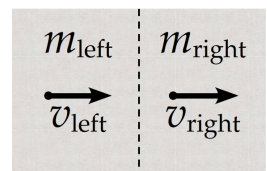
In stating and using constitutive relations, instead, we do have to pay attention to the size and even shape of the control volumes and surfaces to which we apply them; to coordinates; to magnitudes of the quantities involved; and to whether we're studying a phenomenon on molecular or galactic scales and resolutions. And even when we have paid attention to all these aspects, it may still be that a constitutive relation holds only approximately, so our study will have some quantitative error.

This is the reason why in the statement of a constitutive relation there are often qualifying phrases like "only if the volume is small enough", "only for low velocities", "only for weak gravitational fields", "only if the velocity is approximately the same everywhere", "only at large enough resolutions", "quasi-static", and similar. You saw some examples in §5.10; in discussing Newton's constitutive relation for momentum, for instance, we said "take a small control volume". . .

Most important, you'll notice that *such qualifying phrases are often vague*. What's a 'small control volume'? is a volume of 1 cm^3 small enough? or do we need to go down to 1 mm^3 ? or maybe 1 m^3 is already enough? And what about a velocity being 'the same everywhere'? is it acceptable if the velocity in one part of the control volume is 1.00 m/s and in another 1.01 m/s ? or must we go down to 1.000 m/s and 1.001 m/s ? And what's 'quasi-static'? is a motion at 1 mm/s quasi-static? could be 1 m/s be accepted as quasi-static?

Unfortunately *no general answer can be given to such questions*. This is why the qualifying phrases use vague words like "small enough". The answer depends, among other factors, on **the numerical precision that we require in the final answer**, and on **the relative magnitudes of all quantities involved**. A velocity, or a variability in the velocity, equal to 1 m/s may be considered very low in some circumstances, and very high in others.

Let's see a very simple example. Suppose that there is some matter in a control volume, and we need to know the total momentum contained in the volume. Only one horizontal coordinate is relevant. We want to use Newton's constitutive relation for momentum $P = mv$. The left half of the volume contains matter with total mass-energy of 1.00 kg and everywhere a velocity of 0.100 m/s . The right half contains matter with total mass-energy of 1.00 kg and everywhere a velocity of 0.103 m/s . We could apply Newton's relation to each half and applying the extensivity



property of momentum, obtaining

$$\begin{aligned} P &= m_{\text{left}} v_{\text{left}} + m_{\text{right}} v_{\text{right}} \\ &= (1.00 \times 0.100 + 1.00 \times 0.103) \text{ kg m/s} = 0.203 \text{ N s} . \end{aligned}$$

But suppose we only need to know the momentum to *two* significant digit, that is, it's enough to know whether the momentum is between 0.195 N s and 0.205 N s. In this case the velocity v may be considered 'approximately the same' in the whole control volume, with value $v = 0.10 \text{ m/s}$. The total mass in the volume is $m = 2.0 \text{ kg}$ and we can apply Newton's constitutive relation without dividing the control volume:

$$P = mv = (2.0 \times 0.10) \text{ kg m/s} = 0.20 \text{ N s} .$$

The example above was extremely simple. In real applications it can be quite difficult to assess whether the conditions for applying a constitutive relation – say, whether some quantities or volumes are “small enough” – are satisfied for the precision required. In such situations one can explore a test case, first with a less approximate analysis, for instance using many small control volumes and more precise constitutive relations; and then with a more approximate analysis, for instance using just a couple of control volumes and less precise constitutive relations. If the more approximate analysis agrees with the less approximate one to the number of significant digits required, then we know that the more approximate analysis can be used in other similar cases.

In many engineering fields it is known, from experience, what expressions like ‘small enough’ mean in numerical terms – even if this meaning is often left implicit. So if you will work in any specific field you will learn which approximations can be considered acceptable. But it's also good to periodically re-check that such experience is still valid and does not lead to the inappropriate use of some constitutive relations.

§ 5.13 Newton's laws

Many books present, mainly out of tradition, a view of physical laws based on Newton's three axioms, or laws of motion, stated in his *Philosophiæ naturalis principia mathematica*. It is therefore convenient to get acquainted with their statements. In the present section we examine Newton's laws and discuss their limitations as a foundations of today's physical theories.

The statements below are from the translation by Cajori (Newton 1974); the original Latin, from the third edition of Newton's *Principia* (Newton 1726b), is reported on the side. We must keep in mind that these statements are inextricably connected with the rest of definitions and reasoning given in the *Principia*, and with the knowledge of the time in which they were written. This makes it difficult, or maybe even meaningless, to fully interpret them. See for instance the analysis by Smith 2024.

First law

LAW I. Every body continues in its state of rest, or of uniform motion in a right line, unless it is compelled to change that state by forces impressed upon it.

LEX I. Corpus omne perseverare in statu suo quiescendi vel movendi uniformiter in directum, nisi quatenus illud a viribus impressis cogitur statum suum mutare.

This is usually called the ‘law of inertia’, and it was accepted already sometime before Newton. Today this law is often viewed as a consequence of the second law below, together with Newton’s definition of momentum: if there are no forces, then there is no change in ‘motion’, and if ‘motion’ is mass times velocity, then the velocity is constant, in magnitude and direction. But there are also differing views, which see it as an independent statement about the properties of space or of absolute motion, or about the differences between “real” and “fictitious” forces. Eddington (1958) made a little fun of the first law, rephrasing it roughly this way: *Every body continues in its state of rest or uniform motion in a straight line, except when it doesn’t.*

From a 21st-century point of view, centred on the relativity of motion and coordinate systems and frames, these debates about the first law lose some of their physical importance.

Second law

LAW II. The change of motion is proportional to the motive force impressed; and is made in the direction of the right line in which that force is impressed.

LEX II. Mutationem motus proportionalem esse vi motrici impressæ, & fieri secundum lineam rectam qua vis illa imprimitur.

This law will be further discussed in chapter 10. It is sometimes mathematically reported as $\mathbf{F} = m\mathbf{a}$, but Newton never wrote this formula, which moreover may not fully represent the second law.

With some liberty we may interpret this law as the balance of momentum, but with two important warnings.

First, Newton did not make any explicit distinction between surface forces and volume forces. The absence of this important distinction caused some difficulties in applying Newton's laws to bodies capable of deformation, like fluids. These difficulties were solved when the Bernoullis and especially Euler in the 1700s introduced more clearly the concept of control surface and surface force, eventually developed in full generality by Cauchy in the 1800s. Modern mechanical engineering and fluid mechanics would have been impossible without this conceptual distinction.

Second, in the *Principia* Newton defines 'quantity of motion' as "arising from the velocity and quantity of matter conjointly", that is, as ' $m\mathbf{v}$ ', so in this context the second law is specific to matter. But we know today that the exact expression for matter is different, and that electromagnetic fields have momentum as well.

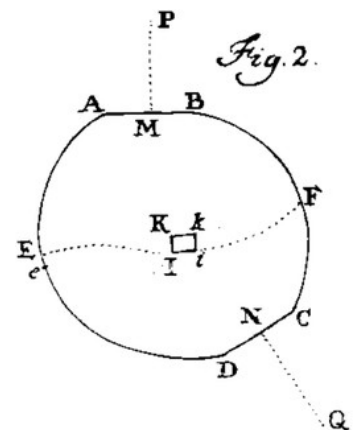
Third law

LAW III. To every action there is always opposed an equal reaction: or, the mutual actions of two bodies upon each other are always equal, and directed to contrary parts.

This is usually called the law of *action and reaction*. This law is only valid for *surface* forces, that is, for momentum flux; but not for *volume* forces, that is, momentum supply. For volume forces this law is generally not true, or only valid in special approximations and coordinate choices. We already mentioned that this law can be viewed as the [principle of symmetry of flux for momentum](#) §4.11 p. 117. Therefore an analogous law is true for the fluxes of all other quantities, not only of momentum.

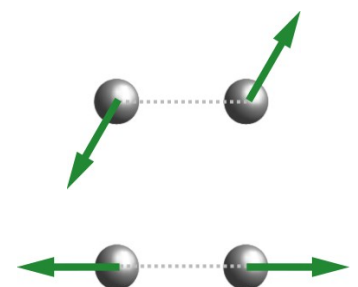
In the 1900s literature there was a discussion on whether the third law could allow pairs of *volume* forces on two bodies as in the top side figure: equal and opposite but with a direction that doesn't necessarily align with a line connecting the two bodies. Or whether the third law would only allow for a direction as depicted in the bottom side figure.

Today we know that this discussion is somewhat pointless: as already said, the third law is in general not true for supplies of momentum. Moreover in a curved spacetime it becomes unclear what a 'line' connecting two bodies is. For *surface* forces – momentum flux – this situation doesn't arise, because the flux concerns the surface as a whole, so there is no "application point".



Imaginary surface $EliF$ in a fluid, conceived by Euler to describe internal pressure. From Euler 1761.

LEX III. Actioni contrariam semper & æqualem esse reactionem: sive corporum duorum actiones in se mutuo semper esse æquales & in partes contrarias dirigi.



Further limitations

Besides the shortcomings mentioned in the comments above, there are further limitations that make Newton's laws insufficient today as a general basis to understand physical phenomena.

One basic limitation is that they do not cover all balances necessary to describe most mechanical phenomena. If we want to study, for example: the oceans, the atmosphere, a car engine, the stability of a bridge (all year round), aeroplane flight, the behaviour of materials, and similar examples, then besides Newton's laws we need the balance of energy, of angular momentum, and of matter; all of which are *independent* of Newton's laws. The need for a separate balance of angular momentum, for instance, was definitively recognized by the Bernoullis and Euler in the 1600–1700s in trying to study the behaviour of fluids. The recognition of the balance of energy as a separate law led to modern thermomechanics. General relativity later showed that a further set of three balance equations must be taken into account in considering relativistic mechanical phenomena. Therefore Newton's laws account for less than half of the universal equations necessary in mechanics. And we're not considering applications involving electromagnetics.

If you're curious about the discovery of the balance of angular momentum, take a look at Truesdell 1968a.

Another unfortunate limitation, not of Newton's laws per se but of the way they are often presented and taught, is their mathematical vagueness. You can find Newton's second law variously written as

$$\mathbf{F} = m\mathbf{a}, \quad \frac{d m \mathbf{v}}{dt} = \mathbf{F}, \quad m \frac{d \mathbf{v}}{dt} = \mathbf{F}, \quad \frac{d \mathbf{P}}{dt} = \mathbf{F}, \quad \frac{\partial \rho \mathbf{v}}{\partial t} = -\nabla \cdot \mathbf{T} + \rho \mathbf{g},$$

and other variations, some of which are more general than others. The reason of this variety is that equations like the ones above come from the combinations of one law – the balance of momentum – with different constitutive relations.

Sometimes one hears the statement “Just use Newton's laws!” or “Just use Newton's second law!”. Seeing the limitations above, the ‘just’ in that statement is a little ridiculous. First, one needs more laws than Newton's; and second, it's always unclear what precise formula that statement is referring to. So that statement is not very helpful – they could just as well say “Just use physics!”

One more, obvious limitation of Newton's laws is that they are about ‘bodies’, that is, *matter*. The idea of electromagnetic field was unknown to Newton and until the 1800s. The combination of mechanical and electromagnetic phenomena leads to conceptual difficulties with Newton's

laws. For instance, if a charged body experiences an electromagnetic force or 'action', then *what* is experiencing the 'reaction'?

If you translate '*force exerted on. . .*' and similar expressions to '*momentum influx into. . .*' or '*momentum supply into. . .*', paying attention on whether they're surface or volume forces, then you'll be able to frame any physical problems in terms of control volumes, control surfaces, and balances.

URLs for chapter 5

1. https://pglpm.github.io/7wonders/media/volume_moving.gif
2. https://encyclopediaofmath.org/wiki/Newton-Leibniz_formula
3. https://encyclopediaofmath.org/wiki/Primitive_function
4. <https://pubchem.ncbi.nlm.nih.gov/element/Argon>
5. <https://encyclopedia.che.engin.umich.edu/distillation-columns/>
6. <https://education.nationalgeographic.org/resource/seven-wonders-ancient-world/>
7. http://encyclopediaofmath.org/index.php?title=Differential_form
8. <https://www.aps.org/archives/publications/apsnews/201403/physicshistory.cfm>
9. https://commons.wikimedia.org/wiki/images/File:Four_Fundamental_States_of_Matter.png
10. <http://hyperphysics.phy-astr.gsu.edu/hbase/Chemical/chemeas.html#c4>
11. <https://oceanservice.noaa.gov/geodesy>

Inference, prediction, simulation

[Marco Polo:] "... You take delight not in a city's seven or seventy wonders, but in the answer it gives to a question of yours."

[Kublai Khan:] "Or the question it asks you, forcing you to answer, like Thebes through the mouth of the Sphinx."

I. Calvino 1979

§ 6.1 General inference of physical quantities

Whole books can be filled with discussion on how physical laws are used to get insight on physical phenomena; and vice versa, on how the exploration of physical phenomena can lead to new physical laws or modifications of existing ones. It can even lead to the invention of new mathematics to express new physical laws.

In this chapter we shall focus on how physical laws can be used to find information about physical situations, or predict the outcomes of an evolving physical situation. These tasks are at the heart of many technologies and are the starting point for the invention of new ones.

In general terms, the use of physical laws is one of *inference*: to find the values of some physical quantities, given known values of other quantities. Both the quantities that we know, and the one we want to find, may be of different kinds and may refer to several different locations and times. We may also have an *infinite* number of quantities that we know, for example the values of the amount of matter $N(t)$ in a volume for *all* times t between 0 s and 3600 s. Or we may need to find an infinite number of quantities, for instance the momentum $\mathbf{P}(t)$ of a rocket for *all* times t between 0 s and 3600 s.

The universal balance laws and constitutive equations can both be used for such purposes. We saw some [examples](#) ^{§5.6 p. 140} in our introductory discussion about balance laws. For instance, given the initial momentum

of a tennis ball, and the momentum influx and supply to it during a time lapse of 2 s, we found its momentum at the end of the time lapse.

Constitutive relations can also be used for inference. If we know the velocity $\boldsymbol{v}$ of a small control volume, and the amount m of rest mass-energy in it (all at a given coordinate time t), then we can find the momentum $\boldsymbol{P}$ in the volume by means of the constitutive relation

$$\boldsymbol{P} = m\boldsymbol{v} .$$

Or we might know the rest mass-energy and the momentum, and need to find the velocity. This could also be done by means of the same constitutive relation:

$$\boldsymbol{v} = \frac{\boldsymbol{P}}{m} .$$

Note an important difference, though, between the example with the balance law and the one with the constitutive relation:

Necessity of validation of constitutive relations

If we want to use a *balance law* to draw a physical inference, we may use it in any kinds of circumstances.

If we want to use a *constitutive relation* to draw a physical inference, we must first **make certain that it applies** to the specific phenomenon and control volumes and surfaces involved.

For instance, in the example above we would not be allowed to use the constitutive relation $\boldsymbol{P} = m\boldsymbol{v}$ if the speed involved were close to the speed of light; or if the control volume contained not matter but electromagnetic field; or if it were visibly deforming and therefore have an ambiguous velocity.

Use of multiple balances or constitutive relations

We can obviously also combine different balance laws or constitutive relations to arrive at the desired information.

 To be completed. As examples, please see exercises 5.3 and 5.4 in the [exercise text](#)¹, checking the provided solutions if necessary.

§ 6.2 Prediction and forecast

The term *prediction* has many related everyday meanings. Here we shall use it in the specific sense of *finding physical information about a future time*,

given physical information at an earlier time or times. Another and perhaps better term is *forecast*.

For example we may need to predict where on Mars a probe will land, 200 days from now, given its current launch position and velocity, and given trajectory corrections that will be performed later. Often these kinds of prediction are made in order to choose appropriate initial conditions that allow us to reach a desired final condition.

When it comes to prediction we notice a critical difference between the universal balance laws and constitutive relations: *most constitutive relations, used alone, do not allow us to make predictions*. This is because most, if not all, of them connect quantities at the same coordinate time. We must therefore always use at least one or more balance laws whenever we want to make predictions.

But we have also seen that balance laws only connect amounts of the same quantity at different times. Therefore *if we need to predict the value of one quantity later, given the value of a different quantity now, we must perform use both balance laws and constitutive relations*. Each kind of law by itself would be useless for this kind of prediction.

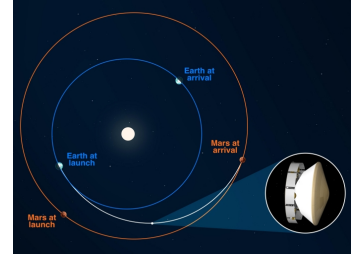
Example: oscillating reaction

The combination of balance laws and constitutive relations can have amazing predictive capabilities. Let's see a simple but not too unrealistic example.

Suppose in a control volume we have two substances called a and b . Let's say that the control volume corresponds to a sealed container, so the net influxes $\mathcal{J}_a$ and $\mathcal{J}_b$ of the two substances are always zero; we directly omit them in the following. We know that the two substances obey balance laws:

$$\begin{aligned} \frac{dN_a(t)}{dt} &= \mathcal{R}_a(t) & \text{or} & & N_a(t_1) &= N_a(t_0) + \int_{t_0}^{t_1} \mathcal{R}_a(t) dt, \\ \frac{dN_b(t)}{dt} &= \mathcal{R}_b(t) & \text{or} & & N_b(t_1) &= N_b(t_0) + \int_{t_0}^{t_1} \mathcal{R}_b(t) dt. \end{aligned}$$

From the integral expression it is clear that if we know $N_a(t_0)$ and we want to predict $N_a(t_1)$ at the later time, then we need to know how $\mathcal{R}_a(t)$ depends on time, for *all* times between t_0 and t_1 . That's a lot of information. Similarly for the b substance.



Sending probes, such as the [Perseverance rover](#)², to other planets requires a careful prediction of the planet's and the probe's trajectories (image: [NASA](#)³)

Now suppose that in these specific physical conditions two constitutive relations are known to be valid:

$$\mathcal{R}_b(t) = \lambda N_a(t), \quad \mathcal{R}_a(t) = -\lambda N_b(t),$$

for some positive physical constant λ . These constitutive relations express chemical reactions. The first constitutive relation says that the supply of substance b is always proportional to the volume content of substance a . The second relation says that the sink (negative supply) of substance a is always proportional to the volume content of substance b . Intuitively, a large amount of a favours the production of b , and a large amount of b disfavours the production of a .

Combine these two constitutive relations with the balance laws in their differential expression: take each $\mathcal{R}$ and replace its expression in the balance laws, so that all $\mathcal{R}$ s disappear (we're omitting the time argument, for brevity):

$$\frac{dN_a}{dt} = -\lambda N_b, \quad \frac{dN_b}{dt} = \lambda N_a. \quad (6.1)$$

This is called a *system of coupled, first-order ordinary differential equations*⁴. It is 'coupled' because each equation contains both $N_a(t)$ and $N_b(t)$. We can also substitute $N_a(t)$ from the second equation into the derivative of the first, and vice versa. We find

$$\frac{d^2 N_a}{dt^2} = -\lambda^2 \frac{dN_a}{dt}, \quad \frac{d^2 N_b}{dt^2} = -\lambda^2 \frac{dN_b}{dt},$$

which is a set of uncoupled *second-order ordinary differential equations*⁵. Each one is the equation for *simple harmonic motion*⁶.

Something amazing has actually happened. It turns out that the differential equations above are satisfied if N_a has this specific time dependence:

$$N_a(t) = C \cos(\lambda t) + D \sin(\lambda t) \quad (6.2)$$

where C, D are two constants. If we substitute this expression in the system (6.1), and thereafter in the constitutive relations, we finally find

$$\begin{aligned} N_a(t) &= C \cos(\lambda t) + D \sin(\lambda t), & \mathcal{R}_a &= \lambda D \cos(\lambda t) - \lambda C \sin(\lambda t), \\ N_b(t) &= -D \cos(\lambda t) + C \sin(\lambda t), & \mathcal{R}_b &= \lambda C \cos(\lambda t) + \lambda D \sin(\lambda t). \end{aligned} \quad (6.3)$$

The two constants C, D can be found if we know $N_a(t_0)$ and $N_b(t_0)$ at the initial time t_0 .

Let's summarize what all this means. If we know:



The *Belousov-Zhabotinsky reaction*⁷ (above) and the *Briggs-Rauscher reaction*⁸ are two kinds of real oscillating chemical reactions that have the main features of our simplified example; see the videos and interactive applets in the links above (image: *Rochester Institute of Technology*⁹).

- that N_a and N_b satisfy balance laws,
- that the net influxes are zero,
- that the constitutive relations $\mathcal{R}_b = \lambda N_a$ and $\mathcal{R}_a = -\lambda N_b$ are valid,
- the values $N_a(t_0)$, $N_b(t_0)$ at time t_0 ,

then we can predict the values of the volume contents and of the supplies *at all times*, or at least all times at which the conditions above are true!



Exercise 6.1

1. Find the final solutions (6.3), by substituting the expression (6.2) in the equations (6.1) to find $N_b(t)$, and then finding the supplies $\mathcal{R}$ from the constitutive relations.
2. What are physical dimensions of λ ?
3. Find the concrete numerical expression of the solution (6.3) in the case where

$$\lambda = 6.283 \text{ s}^{-1}, \quad t_0 = 0 \text{ s}, \quad N_a(t_0) = 2 \text{ mol}, \quad N_b(t_0) = 3 \text{ mol},$$

and plot the graph of $N_a(t)$ against time t between $t = 0 \text{ s}$ and $t = 5 \text{ s}$.

4. Try to explain the time behaviour of $N_a(t)$, from the intuitive explanation of the constitutive relation given in the example above.
5. Is there any generation of antimatter in this imaginary example?

§ 6.3 Numerical time integration and simulations

The example with the imaginary chemical reaction of the previous section shows the powerful predictive powers that can be achieved by combining balance laws and constitutive relations. In that example we were able to find the whole time-dependence of the contents and production rates of the two substances, and to express this time dependence with an *analytical formula*.

An analytical formula is a combination of **elementary functions**¹⁰, like ‘cos’ and ‘sin’, and sometimes also of some **special functions**¹¹, in which we can plug the time t and read out the result almost immediately, with the help of a calculator. Having an analytical formula is extremely useful, because it often allows us to immediately grasp important properties of the physical phenomenon. For instance, if a volume content is given by

$N(t) = C \cos(\lambda t)$, then we automatically know that its values can only change between $-C$ and C , owing to the properties of the cosine function.

Unfortunately, however, the *vast majority* of combinations of balance laws and constitutive relations lead to time-dependencies that *cannot be expressed by analytical formulae*. For instance, if for the oscillating reaction we had to use constitutive relations such as

$$\mathcal{R}_b(t) = \lambda \sin[N_a(t)/\text{mol}], \quad \mathcal{R}_a(t) = -\lambda \cos[N_b(t)/\text{mol}],$$

(with a new λ having different physical dimensions), then an analytical solution is hardly available. In these cases we can't simply plug in the time t and quickly calculate what the result is. How can we predict the physical behaviour then?

We must resort to **numerical time integration**, a computation that in some situations can be very time- and energy-consuming. It is often also called **simulation**, because it essentially reproduces, step by step, the evolution of the physical phenomenon.

Numerical time integration is essential for present-day engineering and physical applications and research. But it also offers a point of view from which we can see quite clearly the roles that diverse physical laws have in physical phenomena; thus it has pedagogical value. In the next sections we shall study its basics.

Finite-difference approximation

Recall again the integral and differential expressions of a balance law. In the following discussion we shall use the symbols for energy, but the reasoning is valid for any other quantity:

$$E(t_1) = E(t_0) + \int_{t_0}^{t_1} W(t) dt + \int_{t_0}^{t_1} \mathcal{A}(t) dt \qquad \frac{dE(t)}{dt} = W(t) + \mathcal{A}(t)$$

integral expression differential expression

This is a conservation law if $\mathcal{A}(t) = 0$ in all circumstances.

From either of these expressions we can find an approximate mathematical procedure to make predictions about a short instant of time ahead. Let's see how to do this from the integral expression.

Suppose that the lapse of time Δt between t_0 and t_1 ,

$$\Delta t = t_1 - t_0$$

is extremely short. So short that the flux $W(t)$ and the supply $\mathcal{A}(t)$ don't change appreciably during this time lapse. We can take their values as approximately constant in time, equal to their initial values:

$$W(t) \approx W(t_0) \quad \mathcal{A}(t) \approx \mathcal{A}(t_0) \quad \text{for all } t \text{ between } t_0 \text{ and } t_1 .$$

If they do change appreciably, then we take an even shorter Δt .

Since they are approximately constant, their integrals can also be approximated as follows:

$$\begin{aligned} \int_{t_0}^{t_1} W(t) dt &\approx \int_{t_0}^{t_1} W(t_0) dt = W(t_0) (t_1 - t_0) = W(t_0) \Delta t \\ \int_{t_0}^{t_1} \mathcal{A}(t) dt &\approx \int_{t_0}^{t_1} \mathcal{A}(t_0) dt = \mathcal{A}(t_0) (t_1 - t_0) = \mathcal{A}(t_0) \Delta t \end{aligned}$$

Recall that

$$\int_a^b \text{const } dx = \text{const} \cdot (b - a)$$

Now let's replace these approximations in the balance law, also taking $t_1 = t_0 + \Delta t$. We find:

$$\begin{aligned} E(t_0 + \Delta t) &\approx E(t_0) + W(t_0) \Delta t + \mathcal{A}(t_0) \Delta t \\ &\stackrel{=E(t_1)}{\approx} \int_{t_0}^{t_1} W(t) dt \approx \int_{t_0}^{t_1} \mathcal{A}(t) dt \end{aligned}$$

Factoring Δt , and replacing the symbol ' t_0 ' with ' t ' (since we don't need to distinguish initial and final time), we obtain:



Finite-difference approximation

$$E(t + \Delta t) \approx E(t) + [W(t) + \mathcal{A}(t)] \Delta t \quad (6.4)$$

is a **finite-difference approximation** of a balance law.

This formula says that if we know the value of the volume content $E(t)$, the total influx $W(t)$, and the total supply $\mathcal{A}(t)$ at time t_0 , then we can *approximately* predict the value of the volume content $E(t_0 + \Delta t)$ a short lapse Δt later. This basic idea is the same as the one behind **Euler's method**¹².

Example

Suppose that the total amount of energy in a given control volume at time $t = 20$ s is $E(t) = 500$ J. At that same time there's a net influx of $W(t) = -30$ J/s, and the supply is zero, $\mathcal{A}(t) = 0$ J/s.

Then the amount of energy in the control volume 0.01 s later, at time $t + \Delta t = 20 \text{ s} + 0.01 \text{ s} = 20.01 \text{ s}$, is approximately

$$\begin{aligned} E(t + \Delta t) &\approx E(t) + [W(t) + \mathcal{A}(t)] \Delta t \\ E(20.01 \text{ s}) &\approx E(20 \text{ s}) + [W(20 \text{ s}) + \mathcal{A}(20 \text{ s})] \cdot 0.01 \text{ s} \\ &\approx 500 \text{ J} + [-30 \text{ J/s} + 0 \text{ J/s}] \cdot 0.01 \text{ s} \\ &\approx 499.7 \text{ J}. \end{aligned}$$



Exercise 6.2

1. At time $t = 0 \text{ s}$ the amount of oxygen in a control volume is 0 mol, and at that instant there is an influx of 8 mol/s. Assume that oxygen satisfies a conservation law, and calculate its amount in the control volume at time $t' = 0.01 \text{ s}$.
2. Try to obtain the finite-difference approximation starting from the differential expression of the balance law instead:

$$\frac{dE(t)}{dt} = W(t) + \mathcal{A}(t).$$

Use the fact that the derivative at a time t can be approximately calculated as

$$\frac{dE(t)}{dt} \approx \frac{E(t + \Delta t) - E(t)}{\Delta t}.$$

Vector quantities

The finite-difference approximation is also valid for the balance law of a vector quantity like momentum. We only have to remember that a vector equation is a collection of three equations, one per vector component:

$$\begin{aligned} \mathbf{P}(t + \Delta t) &\approx \mathbf{P}(t) + [\mathbf{F}(t) + \mathbf{G}(t)] \Delta t \\ \text{or } \begin{cases} P_x(t + \Delta t) \approx P_x(t) + [F_x(t) + G_x(t)] \Delta t \\ P_y(t + \Delta t) \approx P_y(t) + [F_y(t) + G_y(t)] \Delta t \\ P_z(t + \Delta t) \approx P_z(t) + [F_z(t) + G_z(t)] \Delta t \end{cases} & (6.5) \end{aligned}$$



Exercise 6.3

- Let's follow the flight of a tennis ball by choosing a control volume coinciding with the ball, as we did in previous similar examples.

At time $t_0 = 0$ s the amount of momentum and the supply of momentum in the tennis ball are

$$\mathbf{P}(t_0) = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix} \text{ N s} \quad \mathbf{G}(t_0) = \begin{bmatrix} 0 \\ 0 \\ -0.579 \end{bmatrix} \text{ N}.$$

The total influx of momentum is zero. Calculate the momentum within the control volume 0.01 s later.

- The tennis ball has a mass-energy $m = 0.059$ kg. Assume the [constitutive relation for momentum](#) ^{§5.10 p.156}:

$$\mathbf{P} = m \mathbf{v}$$

What was the velocity of the tennis ball at time t_0 ? How much is it 0.01 s later?



§ 6.4 Iteration: the special role of balance laws

The finite-difference approximation (6.4) can be used iteratively: once we have the volume content $E(t + \Delta t)$ at time $t + \Delta t$, we can use it to find the value at a slightly later time $t + \Delta t + \Delta t$, and so on:

$$\begin{aligned} E(t + \Delta t) &\approx E(t) + [W(t) + \mathcal{A}(t)] \Delta t \\ E(t + 2\Delta t) &\approx E(t + \Delta t) + [W(t + \Delta t) + \mathcal{A}(t + \Delta t)] \Delta t \\ E(t + 3\Delta t) &\approx E(t + 2\Delta t) + [W(t + 2\Delta t) + \mathcal{A}(t + 2\Delta t)] \Delta t \\ &\dots \end{aligned} \tag{6.6}$$

And an analogous iteration could be used for a vector quantity such as momentum.

This iteration is what we more precisely mean by 'numerical time integration':

```
p.vx = p.vx + dt * p.fx / this.mass;
p.vy = p.vy + dt * (p.fy / this.mass + t

// integrate position
p.x = p.x + dt * p.vx;
p.y = p.y + dt * p.vy;
```

Code snippet¹³ from the [Particle System webpage](#)¹⁴, iterating an evolution equation for momentum.



Numerical time integration

The numerical calculation of one or more quantities at successive time steps, with algorithms similar to that illustrated above, is called **numerical time integration**, often simply shortened to *integration*.

In some circumstances this procedure is also called **simulation**, especially when we visually follow, for instance by means of graphs or animated pictures, the time evolution of some of the quantities.

It is apparent that each of the lines in the iteration (6.6) above is an just application of the balance law in its approximate form, at each time step. This is again a consequence of the fact that balance laws relate different times. The iterative expression above makes it even clearer that *balance laws are the physical laws that 'drive' physical systems forward in time*.



Exercise 6.4

Try to write a script, in your preferred programming language, that implements the time-stepped evolution algorithm (6.6) and evolves the energy content E in a control volume.

Assume:

- the initial time is $t_0 = 0$ s,
- the energy content at the initial time is $E(t_0) = 3.0$ J,
- the energy influx is constant in time: $W(t) = 0.5$ J/s,
- the energy supply is zero: $\mathcal{A} = 0$ J/s
- total duration of simulation is 300 s
- timestep is $\Delta t = 0.1$ s

The script should output the energy content at the final time.

§ 6.5 Iteration: the roles of boundary conditions and constitutive relations

In order to perform the numerical time integration (6.6), we need to know the new values of influx W and supply $\mathcal{A}$ at the initial time and at *all* subsequent time steps. The finite-difference equation updates only the volume content E , not the influx or the supply.

From where can we get the values of influx and supply at later times? There are two possibilities:

- They are *acquired* at every time step.
- They are *calculated*, by means of constitutive relations, from the volume contents of physical quantities which are in turn predicted by means of numerical time integration.

Let's discuss them in turn.

The first possibility occurs when we actually *control* the influxes or supplies; or when we *monitor* them; or when we simply know them in advance out of acquaintance with previous instances of the same phenomenon.

For example, in a reaction chamber we may control the influx and efflux of different substances through valves and similar devices. In a power grid control centre the effluxes of electric charge and energy – the power demand – are continuously monitored. In the case of the tennis ball in the last exercise, it is known that the influx of momentum is zero if the object is moving in vacuum, or negligible when moving through air.

The set of values known for all future times thanks to control, monitoring, or early knowledge has a special name:



Old power-grid control room in Frøland, Norway (image: [Kraftmuseet¹⁵](#)).



Boundary conditions

The quantities that are set or known in advance *at every time* for the prediction or simulation of a physical phenomenon are called **boundary conditions**, and their values *boundary values*. Boundary conditions are typically fluxes or supplies.

A common and useful case is when we know that some flux or some supply is zero for all relevant times.



Exercise 6.5

1. Modify your script from Exercise 6.4, now assuming that the energy influx depends on time as follows:

$$W(t) = C \exp[-2 \sin(\lambda t)] \quad \text{with} \quad C = 10 \text{ J/s} \quad \lambda = 0.157 \text{ s}^{-1}.$$

2. Try to add to your script the possibility of plotting the energy content E as it evolves in time.
3. In Exercise 6.3 you calculated the evolution of the tennis ball's momentum – all its three components – for one timestep of 0.01 s.

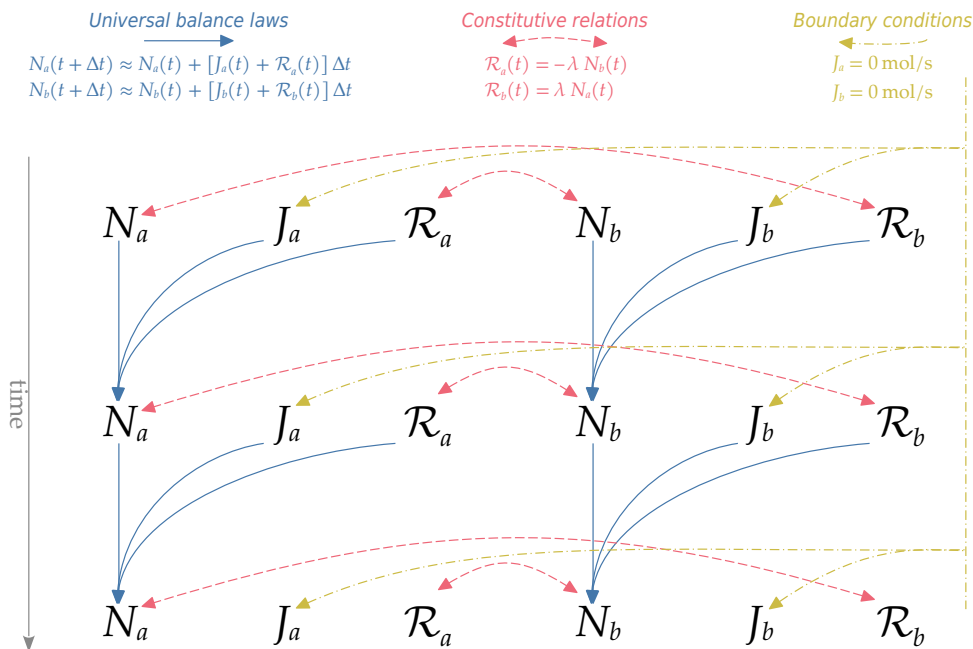
Try to write a script that implements the time-stepped evolution algorithm (6.6) and evolves all three components of the tennis ball's momentum. Assume:

- the initial time is $t_0 = 0$ s,
- the momentum at the initial time is $\mathbf{P}(t_0) = [3, 0, 3]$ N s,
- the momentum influx is zero at all times,
- the momentum supply is constant in time: $\mathbf{G} = [0, 0, -0.579]$ N,
- total duration of simulation is 2 s
- timestep is $\Delta t = 0.01$ s

Try to plot the values of the three momentum components of the tennis ball against time.

The second possibility is achieved by means of *constitutive relations*, which, as we know well by now, relate contents, fluxes, supplies of different physical quantities at the same time instant. Let's see by means of an example how they are used in the iteration for time integration.

The role of balance laws, constitutive relations, and boundary conditions in numerical time-integration could be visually represented as follows, in the case of the oscillating-reaction example:



Example: oscillating reaction again

Let's consider again the [oscillating reaction](#) ^{§6.2 p. 168}, which we studied with analytical methods. This time we shall approach it by numerical time integration.

We have two balance laws for matter that drive the physical system forward in time. Their iteration is therefore

$$\begin{aligned}
 N_a(t + \Delta t) &\approx N_a(t) + [\mathcal{J}_a(t) + \mathcal{R}_a(t)] \Delta t \\
 N_b(t + \Delta t) &\approx N_b(t) + [\mathcal{J}_b(t) + \mathcal{R}_b(t)] \Delta t \\
 N_a(t + 2\Delta t) &\approx N_a(t + \Delta t) + [\mathcal{J}_a(t + \Delta t) + \mathcal{R}_a(t + \Delta t)] \Delta t \\
 N_b(t + 2\Delta t) &\approx N_b(t + \Delta t) + [\mathcal{J}_b(t + \Delta t) + \mathcal{R}_b(t + \Delta t)] \Delta t \\
 &\dots \\
 &\dots
 \end{aligned} \tag{6.7}$$

We apply the *boundary conditions* from that example: the two influxes $\mathcal{J}_a(t)$, $\mathcal{J}_b(t)$ are always zero. For simplicity we shall therefore simply omit them.

Now we use the constitutive relations between the contents and the supply: $\mathcal{R}_b(t) = \lambda N_a(t)$, $\mathcal{R}_a(t) = -\lambda N_b(t)$. At every time step, after calculating the volume contents N_a , N_b from the balance laws, we can use their new values to calculate the supplies $\mathcal{R}_b$, $\mathcal{R}_a$ at that same time step. These

supplies can then be used for the next time step:

$$\begin{aligned}
 \mathcal{R}_a(t) &= -\lambda N_b(t) \\
 \mathcal{R}_b(t) &= \lambda N_a(t) \\
 N_a(t + \Delta t) &\approx N_a(t) + \mathcal{R}_a(t) \Delta t \\
 N_b(t + \Delta t) &\approx N_b(t) + \mathcal{R}_b(t) \Delta t \\
 \mathcal{R}_a(t + \Delta t) &= -\lambda N_b(t + \Delta t) \\
 \mathcal{R}_b(t + \Delta t) &= \lambda N_a(t + \Delta t) \\
 N_a(t + 2\Delta t) &\approx N_a(t + \Delta t) + \mathcal{R}_a(t + \Delta t) \Delta t \\
 N_b(t + 2\Delta t) &\approx N_b(t + \Delta t) + \mathcal{R}_b(t + \Delta t) \Delta t \\
 &\dots \\
 &\dots
 \end{aligned} \tag{6.8}$$

In the iteration above, note how after specifying the initial $N_a(t)$, $N_b(t)$ we don't need to worry about anything else: the iteration can proceed indefinitely without any new input from us.

§ 6.6 Computer code for numerical time integration

The iteration at the core of numerical time integration lends itself well to calculation by a computer. Here is an example script that implements the numerical time integration of the oscillating reaction and of the tennis ball from a previous exercise. Recall the [remarks about computer code](#) ^{§0.4 p. 18} used in the present notes.

Script for oscillating reaction

The script shown in table 6.1 on page 182 is a realization of the iteration for the [oscillating reaction](#) ^{§6.5 p. 179}. The script has the following main blocks, each one with a clear physical meaning:

Line 4: constants. The values of known physical constants relevant to the physical laws are given here, so that we don't need to write decimal numbers all over the script.

Lines 7–9: initial conditions. The initial values of the quantities to be time-integrated are given here. They will be used to start the iteration loop, but may also be necessary in other parts of the script, for instance in setting up the boundary conditions.

Lines 12–13: boundary conditions. Here are given the values of fluxes, supplies, and sometimes even volume contents that are known beforehand to be constant throughout the iteration. In some cases the time dependence of some quantities may be not constant, but still known beforehand. In such cases this known dependence is given directly in the iteration loop.

Lines 16–17: time-iteration parameters. The chosen duration of the simulation and the time step are given here.

Lines 30–47: iteration. This is the heart of the physical evolution of the physical system.

Lines 32–33: the *constitutive relations* first allow us to calculate, at the present time, the values necessary for using the balance laws.

Lines 36–38: then the *balance laws* allow us to calculate the volume contents at the next time instant; the time is updated accordingly.

§ 6.7 Numerical time integration of position

In many applications we are interested in how the time-varying position $\mathbf{r}(t)$ of a small control volume, coinciding with an object (matter) or with part of an object, changes in time. Such a control volume must be small – compared to the spatial dimensions of the physical phenomenon under study – because otherwise it wouldn't make sense to represent its position with just one vector.

The numerical time integration that we discussed for the volume contents of physical quantities can be formulated in an similar way to the position vector $\mathbf{r}$ of a small control volume.

The [coordinate velocity](#) ^{§2.8 p. 62} $\mathbf{v}(t)$ of such a control volume at time t is the time derivative of its position:

$$\mathbf{v}(t) = \frac{d\mathbf{r}(t)}{dt}.$$

For a small time step Δt we can approximate this derivative with

$$\mathbf{v}(t) \approx \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

```

1 %% Numerical time integration of imaginary chemical reaction
2
3 %% Constants
4 lambda = 6.283; % s-1: conversion rate
5
6 %% Initial conditions
7 t = 0; % s: time
8 Na = 2; % mol: volume content substance a
9 Nb = 3; % mol: volume content substance b
10
11 %% Boundary conditions
12 Ja = 0; % mol/s: influx substance a
13 Jb = 0; % mol/s: influx substance b
14
15 %% Time-iteration parameters
16 t1 = 5; % s: duration of time integration
17 dt = 0.0001; % s: time step
18
19 %% Plotting
20 dtplot = t1/360; % time interval between plots
21 tplot = dtplot; % time for next plot
22 figure
23 plot(t, Na, 'o', 'color', '#CCBB44'); hold on
24 plot(t, Nb, 'v', 'color', '#EE6677')
25 xlabel('time {\it t}/s'); ylabel('amount {\it N}/mol')
26 axis tight; xlim([0, t1]); grid on
27
28 %% Numerical time integration
29 while t < t1
30     %% constitutive relations
31     Ra = -lambda * Nb;
32     Rb = lambda * Na;
33
34     %% step forward in time with balance laws
35     t = t + dt;
36     Na = Na + (Ja + Ra) * dt;
37     Nb = Nb + (Jb + Rb) * dt;
38
39     %% plot
40     if t > tplot
41         plot(t, Na, 'o', 'color', '#CCBB44')
42         plot(t, Nb, 'v', 'color', '#EE6677')
43         pause(0)
44         tplot = tplot + dtplot;
45     end
46 end

```

Download Octave version
[oscillatingreaction.m](#)¹⁶

Download Python version
[oscillatingreaction.py](#)¹⁷

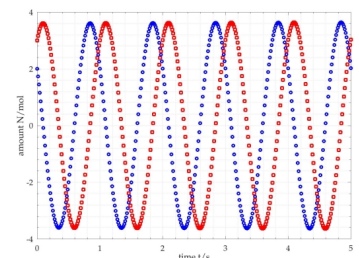


Table 6.1 Script for the oscillating reaction

Multiply by Δt and keep on the right side only the terms that refer to time t . We obtain the following approximate formula:



Finite-difference approximation for position

$$\mathbf{r}(t + \Delta t) \approx \mathbf{r}(t) + \mathbf{v}(t) \Delta t \quad \text{or} \quad \begin{cases} x(t + \Delta t) \approx x(t) + v_x(t) \Delta t \\ y(t + \Delta t) \approx y(t) + v_y(t) \Delta t \\ z(t + \Delta t) \approx z(t) + v_z(t) \Delta t \end{cases} \quad (6.9)$$

This formula says that if we know the position $\mathbf{r}(t)$ and the velocity $\mathbf{v}(t)$ at time t , then we can approximately predict the position $\mathbf{r}(t + \Delta t)$ a short lapse Δt later.

This formula can also be iterated as we did with a balance law. This way we can numerically time-integrate the position $\mathbf{r}(t)$ and therefore keep track of and predict the motion of a small control volume.

Connection with the balance of matter

If the small control volume having position $\mathbf{r}(t)$ contains matter, then we know that there's a [connection](#) ^{§4.15 p. 124} between its velocity $\mathbf{v}(t)$ at time t and the fluxes of that matter through three *static* surfaces orthogonal to the coordinate axes:

$$[v_x, v_y, v_z] = [J_x, J_y, J_z] \frac{V}{AN} .$$

The appearance of the matter content N and flux $\mathcal{J}$ suggests that there may be a connection between the finite-difference formulae for position and for matter balance:

$$\begin{aligned} \mathbf{r}(t + \Delta t) &\approx \mathbf{r}(t) + \mathbf{v}(t) \Delta t \\ N(t + \Delta t) &\approx N(t) + \mathcal{J}(t) \Delta t . \end{aligned}$$

Indeed such a connection exists; but its discussion is somewhat involved, as can be suspected from the fact that the first relation above is vectorial, while the second is scalar. In these notes you won't need to make use of this mathematical connection, but it may be useful to keep in mind that such a connection exists.

In physics simulations where the finite-difference formula for the position of a control volume is used, you typically won't see the finite-difference formula for the matter content of that volume; and vice versa.

This will be a consequence of the connection above: the evolution equation for position is often a “proxy” for the balance of matter.

§ 6.8 Applicability of numerical time integration

Keep in mind that the finite-difference approximation (6.4) that we derived is not the only possible one. The key step in our derivation was the approximation of the integrals appearing in the integral expression of the balance law. Other approximation approaches are possible, and they lead to finite-difference approximations of slightly different forms. Entire books are devoted to these approximation schemes, and if you’ll work in some physics or engineering fields you’ll learn several of them in more detail. Some of these alternative forms lead much better approximations and numerical predictions; but they are more complicated and therefore we don’t discuss them here.

We have illustrated the numerical-evolution procedure with energy (6.6) and amount of matter (6.8). But obviously this procedure can be applied to any quantities that satisfy balance laws.

The time-stepping scheme discussed in the previous sections is very crude and quickly leads to increasing numerical errors. See for instance the examples on the [Mass-Spring Model webpage](#)¹⁸ of *Physics Simulation in Visual Computing*¹⁹. More refined and complex schemes are used for concrete applications.

Yet, the simple time-stepping scheme that we have explored remains the core of most numerical-evolution procedures, and allows us to understand how they essentially work.

URLs for chapter 6

1. <https://github.com/pglpm/7wonders/raw/master/seven-wonders-exercises.pdf>
2. <https://mars.nasa.gov/mars2020/>
3. <https://www.jpl.nasa.gov/news/nasas-perseverance-rover-is-midway-to-mars>
4. <https://mathworld.wolfram.com/First-OrderOrdinaryDifferentialEquation.html>
5. <https://mathworld.wolfram.com/Second-OrderOrdinaryDifferentialEquation.html>
6. <https://mathworld.wolfram.com/SimpleHarmonicMotion.html>
7. <https://ccl.northwestern.edu/netlogo/models/B-ZReaction>
8. <https://chemistrytalk.org/briggs-rauscher-reaction>
9. <https://www.rit.edu/spotlights/chemical-waves-belousov-zhabotinsky-bz-reaction>
10. <https://mathworld.wolfram.com/ElementaryFunction.html>
11. <https://mathworld.wolfram.com/SpecialFunction.html>
12. https://encyclopediaofmath.org/wiki/Euler_method
13. https://github.com/InteractiveComputerGraphics/physics-simulation/blob/5bfb1022ac8b055bf52b197da22b0c2e4122d5f8/examples/particle_system.html#L181
14. https://interactivecomputergraphics.github.io/physics-simulation/examples/particle_system.html
15. <https://digitaltmuseum.no/nvim>
16. <https://pglpm.github.io/7wonders/scripts/oscillatingreaction.m>
17. <https://pglpm.github.io/7wonders/scripts/oscillatingreaction.py>
18. https://interactivecomputergraphics.github.io/physics-simulation/examples/spring_plot.html
19. <https://interactivecomputergraphics.github.io/physics-simulation/>

Balance & conservation of matter

[T]he inertia of energy does not obviate the necessity for assuming the conservation of matter. *Matter* is to be interpreted as number of molecules, therefore, and not as inertia.

C. Eckart 1940

§ 7.1 Formulation and generalities



Balance and conservation of matter

Volume content: N Flux: $\mathcal{J}$ Supply: $\mathcal{R}$

$$N(t_1) = N(t_0) + \int_{t_0}^{t_1} \mathcal{J}(t) dt + \int_{t_0}^{t_1} \mathcal{R}(t) dt \qquad \frac{dN(t)}{dt} = \mathcal{J}(t) + \mathcal{R}(t)$$

integral expression differential expression

(7.1)

The supply $\mathcal{R}$ is called **rate of formation**, or **rate of consumption** if negative. The negative supply $-\mathcal{R}$ is also called **activity** in applications involving radioactivity. The supply is identically zero, $\mathcal{R}(t) \equiv 0$, in case of conservation.

Conservation of matter is a physical law that we intuitively take for granted and use continuously in our life. The very notion of ‘object’ – including living objects – is possible thanks to this fundamental regularity of nature: we can speak of objects because they exist for some time and we can follow them as they move in space; these are typical consequences of a conservation law..

Despite our everyday acquaintance with it, this law is tricky in several respects:

- How can we quantify amounts of matter?
- How many kinds of matter are there?

- Does matter satisfy a balance law, or a conservation law instead?

The answer to each of these questions depends on the scale on which we observe matter, and on the particular physics or engineering discipline.

§ 7.2 Quantification of amounts of matter

There are three ways in which amounts of matter are usually quantified: by simple counting, in terms of ‘amount of substance’, and in terms of mass-energy.

Count. As our observation resolution and scale become finer and finer, matter appears to exist in “bundles” or entities: molecules, atoms, subatomic particles, and so on. In some disciplines these entities are therefore simply *counted*: we can say for example that a control volume contains 41 protons, or 2 electrons, or 35 atoms of oxygen, or 485 molecules of H_2O (water), and so on.

Quantified this way, the volume content N is a quantity of dimension one, with no unit (or more precisely the unit is the number 1). The flux $\mathcal{J}$ and supply $\mathcal{R}$ are quantities of dimension 1/time with unit *inverse second*, s^{-1} .

Amount of substance. Especially in chemistry and when large amounts of matter are involved, we can count the entities of matter in multiples of a large amount called a *mole*. The number per mole is given by the **Avogadro constant**:

$$N_A := 6.022\,140\,76 \times 10^{23} \text{ mol}^{-1} \quad (\text{exactly}) . \quad (7.2)$$

So one mole of protons means exactly $6.022\,140\,76 \times 10^{23}$ protons; one mole of H_2O molecules means exactly $6.022\,140\,76 \times 10^{23}$ H_2O molecules; and so on.



Always specify the entities

It’s important to specify which entities we are quantifying. For example, one oxygen *molecule*, chemical formula O_2 , is composed of two oxygen *atoms*. Therefore one mole of oxygen molecules is not the same as one mole of oxygen atoms; for every mole of oxygen molecules we have two moles of atoms.

Quantified this way, the volume content N is a quantity of dimension amount of substance, with unit *mole* (mol). The flux $\mathcal{J}$ and supply $\mathcal{R}$ have dimension amount of substance/time with unit *mole per second*, mol/s.

Mass. If a control volume contains matter, then it also always contains a non-zero amount of mass-energy. Yet, **matter and mass-energy are not the same thing**. In fact, two control volumes that contain exactly the *same* amount of the same kind of matter – say, 1000 molecules of H_2O – usually contain *different* amounts of mass-energy.

But it's also true that in many important applications the same amount of the same kind of matter has *approximately* the same amount of mass-energy: the difference could be beyond the 10th significant digit. For example, two control volumes, each with exactly the same number of silicon-28 atoms, could contain different mass-energies of

$$1.000\,000\,000\,00\text{ kg} \quad \text{vs} \quad 1.000\,000\,000\,18\text{ kg}.$$

Both mass-energies can therefore be considered equal to 1.000 000 kg to six or seven significant digits, which is high enough precision in many applications. [Later on](#) ^{§7.4 p.192} we shall formulate a constitutive relation based on this fact.

For this reason an amount of matter in many disciplines is quantified in terms of *amount of mass-energy*, with dimension mass and unit *kilogram*, kg. Thus we may say

“this control volume contains 1 kg of H_2O molecules”

by which we mean something like this:

“this control volume contains such an amount of H_2O molecules as to have mass-energy of 1 kg, within a precision of less than 9 significant digits.”

This kind of quantification is useful because it tells us at once both the amount of matter and the amount of mass-energy, even if in an approximate way.

The approximately constant amount of mass-energy associated with a given amount of matter depends on the kind of matter itself. For example, one mole of hydrogen molecules (H_2) has around 0.002 kg mass-energy, whereas one mole of oxygen molecules (O_2) has around 0.032 kg.

“although an exact relationship between mass and amount of substance is not fundamentally challenged by the technologies used to weigh chemicals (which will subsequently be improved), a point will be reached when the equivalent masses of molecular bond energies can no longer be considered negligible.”

Davis & Milton 2014

§ 7.3 Categorization of kinds of matter

Also the categorization of matter into different kinds, each of which may satisfy a balance or a conservation law, depends on the particular discipline, application, and scale of observation. We can distinguish four main categorizations, which are obviously not mutually exclusive:

High-energy particle physics. In the [Standard Model](#)¹ of high-energy particle physics, which tries to describe sub-nuclear scales, we generally speak of different kinds of *fields*. What we call ‘matter’ can be taken to be the set of *fermion*² fields. The way fermion fields are grouped together into different kinds of ‘matter’ is somewhat arbitrary: they could be considered as just one, or twelve, or more.

In particular it’s possible to group them into two kinds. One kind consists of *quarks*³, which make up protons and neutrons among other particles, and is characterized by a *baryon number*⁴. The other consists of *leptons*⁵, like the electron, and is characterized by a *lepton number*⁶. Each of these two kinds of matter satisfies a *conservation law for all practical purposes*.

If we consider these two kinds of matter as just one kind, with baryonic and anti-leptonic counting as ‘positive’, and leptonic and anti-baryonic counting as ‘negative’, then this kind should be conserved even beyond the Standard Model.

In any case, the theories we have for these scales of length and energy are still incomplete or tentative; and they also need the use of probability theory by means of quantum theory.

Baryons and fermions. In applications on length scales of atoms and of their nuclei, and often in disciplines like astrophysics, we can consider matter as consisting of two kinds (compare with the previous categorization): *baryonic*⁷, which comprises protons, neutrons, and their anti-particles; and leptonic, which comprises electrons and their anti-particles positrons. Either kind essentially obeys a *conservation law*, even in phenomena involving *radioactivity*⁸ and nuclear *fission*⁹ and *fusion*¹⁰.

With this kind of categorization, an amount of matter can be quantified by counting, or in moles, or in mass-energy terms. There’s one convenient feature: owing to the definition of the mole, we have

“The anomalies cancel if we take the difference of the baryon and lepton currents, with the resulting anomaly equations

$$\partial_\mu (\mathcal{J}_B^\mu - \mathcal{J}_L^\mu) = 0$$

[...] Here we see that, because of the anomaly, baryon number is in fact not conserved in the Standard Model, although $B - L$ is.” Donoghue et al. [2022](#)

that 1.0 mol of baryonic matter has a mass-energy of approximately 0.0010 kg. That is:

$$\begin{array}{ll} 1.0 \text{ kg of baryonic matter} & \text{correspond to} \\ 1.0 \times 10^3 \text{ mol of baryonic matter} & \text{correspond to} \\ 6.0 \times 10^{26} \text{ baryons} & \end{array} \quad (7.3)$$

So if we're told that a control volume contains 2.3 kg baryonic matter, then we know that it approximately contains 2.3×10^3 mol of matter, or 1.4×10^{27} baryons.

By contrast, 1.0 mol of leptons – essentially electrons – has a much lower mass-energy of approximately 0.000 000 55 kg. For this reason in disciplines like astrophysics ‘matter’ is often simply identified with ‘baryons’.

‘every quantity is referred to a unit of “matter,” i.e., to a unit of baryonic charge.’

Zel’dovich & Novikov 1996

Materials, substances, elements. In everyday phenomena that do not involve nuclear energy or radioactive decay, we often consider different kinds of matter such as wood, air, water, iron, skin, and so on. We call them different **materials**¹¹ or **mixtures**¹² or **solutions**¹³ or **substances**¹⁴, or other terms which often have very specific meanings.

These different kinds obey *balance* laws. For instance, a closed control volume at some time t_0 could contain oxygen and hydrogen gases, and at some later time t_1 it could contain just water – without the oxygen, hydrogen, water ever leaving or entering through the control surface. These three substances must therefore have non-zero supplies. As another example, a closed control volume at some time t_0 could contain air, wood, and a slight amount of other materials like **sulphur**¹⁵ and **phosphorus**¹⁶; and at some later time t_1 it could contain just air and ash – without the wood or ash leaving or entering through the volume’s surface. They must therefore have non-zero supplies. That’s what happens in a common **match-head reaction**¹⁷.

In such type of phenomena, which we can call *chemical reactions* in a general sense, amounts of some kinds of matter appear while amounts of other kinds disappear. The supplies and sinks in these phenomena are connected by precise constitutive relations, which are the subject of chemistry and of **stoichiometry**¹⁸.

We know of course that despite such “appearances” and “disappearances” there is something that is conserved: most of the **chemical elements**¹⁹. If we described matter not in terms of wood, air, and so on,



but in terms of amounts of atoms of hydrogen, helium, lithium, and so on, then these different kinds of matter would obey conservation laws.

Depending on the application and discipline, amounts of these different kinds of matter are quantified in moles or in terms of mass-energy; both ways of quantification have advantages and disadvantages.

“Just matter” In some applications we do not distinguish between different kinds of matter; all that counts is whether in some control volume there is or there isn’t matter. In this case matter obeys a *conservation* law.

In such applications matter is typically quantified in terms of mass-energy, since this provides mathematically more convenient connections with momentum and energy, as we shall see later. But just like in the case of the baryon-fermion categorization, in this case we can practically identify ‘matter’ with baryonic matter, and therefore we have the approximate equivalence (7.3) above.



Exercise 7.1

1. Approximately how many baryons does your body have? Use formula (7.3).
2. At a particular time, a party balloon contains 0.012 kg of helium atoms. One helium atom has two baryons: two protons and two neutrons. How many moles of helium atoms does the balloon approximately contain?



§ 7.4 Constitutive relation between matter and mass-energy

As mentioned in §7.2, a very important constitutive relation connects the amount of matter and the amount of mass-energy in a control volume. According to our present physical knowledge, if a control volume contains an amount of matter N then it also always contains an amount of mass-energy m . The opposite is not true: a control volume can contain mass-energy and yet no matter.

The amount of mass-energy of matter is typically huge when measured in joules. For instance, a control volume with $N = 1$ mol of water (which

occupies roughly $2 \times 10^{-5} \text{ m}^3$ in everyday conditions) has approximately $m = 0.018 \text{ kg}$ or $1.6 \times 10^{15} \text{ J}$ of mass-energy.

The exact amount of mass-energy mainly depends on the kind of matter contained in the control volume, and to a very minor degree also on the matter's temperature and flux, that is, motion, within the volume. In many circumstances the dependence on temperature and motion is so small as to be safely neglected. So we can simply consider the amount of mass-energy at low temperature and if there was no motion; we call this amount **rest mass-energy**.



Mass-energy of matter

If a control volume contains an amount N of matter, then it also contains an amount of mass-energy m proportional to N :

$$m = \rho N$$

where the proportionality factor ρ , called **molar mass**, depends on the kind of matter and can be taken as practically constant if the matter's velocity is small compared to the speed of light and temperatures are not too high.

In many applications the amount of matter in the volume is often directly reported in terms of its mass-energy; that is, ρN is reported instead of N .

Precise values of molar masses for various materials can be found in published tables. The molar mass is, to a good approximation (say two significant digits), numerically equal to the number of baryons – protons and neutrons – in a given amount of matter, divided by that amount; this is again an expression of formula (7.3).

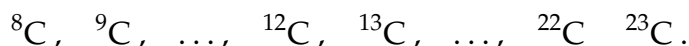
Here are a couple of examples, starting from the case of simple elements.

Isotopes and average molar mass

What is the mass-energy of one **carbon**²⁰ atom? We cannot give a definite answer, because what we call 'carbon atom' comes in different varieties called **isotopes**.

All these varieties have the same number of protons: 6. This is called the **atomic number**²¹ and determines their identical chemical properties; this is why we call all of them 'carbon'. But they differ in their number of neutrons, which can range from 2 to 17. In the case of carbon there

are therefore sixteen known isotopes. The total number of protons and neutrons of an isotope is called the **nucleon number or mass number**²², and is usually shown as a left superscript of the chemical symbol:



Each isotope, having a different number of baryons, has a different mass-energy. The mass-energy of known isotopes can be found in special tables, like the *Atomic Weights and Isotopic Compositions for All Elements*²⁴ table at NIST²⁵.

Owing to the relation (7.3) between amounts and mass-energies of baryons, the mass-energy of one mole of an isotope is roughly equal to its nucleon number in grams. Therefore the carbon isotopes above have molar masses ranging from 0.008 kg/mol to 0.23 kg/mol. More precise values can be found in isotope tables, like the one mentioned above, or the *IUPAC Periodic Table of the Elements and Isotopes*²⁶. The molar mass of an isotope is usually reported as a multiple, called **relative molar mass**²⁷, of $1.000\,000\,001 \times 10^{-3}$ kg/mol.

The isotopes of an element are not equally common in nature: in a sample of an element they occur in different amounts; many of them decay into isotopes of other elements. In the case of carbon, around 99 % of atoms are ${}^{12}\text{C}$ isotopes and 1 % ${}^{13}\text{C}$ isotopes. A chemical element is assigned a molar mass that's the *average* of the molar masses of all its isotopes, weighed by their natural occurrences. This average is the value usually reported as *atomic weight* in periodic tables, always in terms of atomic mass units. In the case of carbon we find an average molar mass of 0.013 003 kg/mol.

Molar mass of other materials

We can find the molar mass of any material in a straightforward way, once we know:

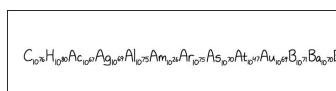
- the chemical composition of the material, that is, how many atoms of each element are in a given amount of material;
- the average molar mass of those atoms.

Let's see an example. What's the mass-energy of 1 mol of air? Air is a **mixture of several substances**²⁸. Those in the largest amounts are:

- **nitrogen**²⁹, N_2 , around 78 %;
- **oxygen**³⁰, O_2 , around 21 %;
- **argon**³¹, Ar, around 1 %.

6	C	8	8.037 643(20)
		9	9.031 0372(23)
		10	10.016 853 31(42)
		11	11.011 4336(10)
		12	12.000 000 00(00)
		13	13.003 354 835 07(23)
		14	14.003 241 9884(40)
		15	15.010 599 26(86)
		16	16.014 7013(38)
		17	17.022 577(19)
		18	18.026 751(32)
		19	19.034 80(11)
		20	20.040 32(26)
		21	21.049 00(43#)
		22	22.057 53(26)
		23	23.0689(11#)

List of known carbon isotopes with their nucleon number and atomic mass. From NIST²³.



THE APPROXIMATE CHEMICAL FORMULA FOR THE UNIVERSE.

<https://xkcd.com/3200/>

Air usually also contains water vapour in amounts that can be very variable depending on the geographical location and period of the year; for simplicity let's imagine very dry conditions with no water vapour.

So 1 mol of air approximately consists of

$$2 \times 0.78 \text{ mol} = 1.56 \text{ mol of nitrogen atoms}$$

$$2 \times 0.21 \text{ mol} = 0.42 \text{ mol of oxygen atoms}$$

$$1 \times 0.01 \text{ mol} = 0.01 \text{ mol of argon atoms}$$

A look at a [periodic table](#)³² shows these average molar masses for the atoms of these three elements:

$$\rho_{\text{N}} \approx 0.014 \text{ kg/mol}, \quad \rho_{\text{O}} \approx 0.016 \text{ kg/mol}, \quad \rho_{\text{Ar}} \approx 0.040 \text{ kg/mol}.$$

For air we therefore find

$$\begin{aligned} \rho_{\text{air}} &\approx (1.56 \times 0.014 + 0.42 \times 0.016 + 0.01 \times 0.040) \text{ kg/mol} \\ &\approx 0.029 \text{ kg/mol}, \end{aligned}$$

and a control volume containing 1 mol of air (and no water vapour) approximately contains 0.029 kg mass-energy.



Exercise 7.2

At a particular time, a party balloon contains 0.012 kg of helium atoms. A minute later, the same balloon contains 0.010 kg of helium atoms. The [molar mass for helium](#)³³ is $\rho_{\text{He}} = 4.0026 \times 10^{-3} \text{ kg/mol}$. How many moles of helium have been lost during the one-minute time lapse?



§ 7.5 Radioactive decay

For radioactive substances, for which conservation does not hold for individual kinds of matter, we still have a balance law where the supply $\mathcal{R}$ has a specific constitutive equation:

Matter supply in radioactive decay

A control volume, containing an amount $N(t)$ of a radioactive substance at time t , has also a (negative) supply

$$\mathcal{R}(t) = -\lambda N(t) \quad (7.4)$$

where λ is positive and called the **decay constant** of that radioactive substance. The supply with changed sign, $-\mathcal{R}$, is usually called **activity**³⁴.



The ISO warning symbol for radioactive material or ionizing radiation.

Law of radioactive decay

Consider a control volume containing a radioactive material. The balance of matter obviously applies to the amount of material N in this control volume:

$$\frac{dN(t)}{dt} = \mathcal{J}(t) + \mathcal{R}(t).$$

If no material is leaving or entering the volume through the surface, the influx is always zero: $\mathcal{J}(t) \equiv 0$. The supply $\mathcal{R}(t)$ is given by the constitutive relation (7.4). The balance of matter therefore becomes

$$\frac{dN(t)}{dt} = -\lambda N(t). \quad (7.5)$$

This expression is called the **law of radioactive decay**. It is a differential equation with $N(t)$ as the unknown function.

Analytical time integration of the law of radioactive decay

Suppose that at time $t_0 = 0$ s we have an amount $N(t_0)$ of radioactive material in the control volume, and we want to know how much material will be there at a generic time t ; this time could be in the future or in the past. To find this amount $N(t)$ we need to solve the law of radioactive decay (7.5).

This differential equation is easy to solve. Let's try, as a solution, a function with the expression $N(t) = A e^{Bt}$, where A and B are constants:

$$\frac{d \overbrace{A e^{Bt}}^{N(t)}}{dt} = -\lambda \overbrace{A e^{Bt}}^{N(t)} \quad \implies \quad B A e^{Bt} = -\lambda A e^{Bt}$$

which is satisfied if $B = -\lambda$. We find A by setting the initial amount at $t_0 = 0$ s:

$$N(t_0) = A e^{-\lambda t_0} = A e^{-\lambda \cdot (0 \text{ s})} = A .$$

Recall that $e^0 = 1$.

The amount of radioactive material at time t is therefore given by

$$N(t) = e^{-\lambda t} N(t_0) . \quad (7.6)$$

Numerical time integration of the law of radioactive decay

In any situation where the law of radioactive decay applies, it is most convenient and quick to use the analytic expression found above for the time dependence of the volume content $N(t)$. But as an instructive exercise let's write a script that numerically integrates that law in the way we learned in §6.3.

Take the following concrete scenario. We have a control volume containing 1×10^{-12} mol of the carbon radioactive isotope ^{14}C at time $t_0 = 0$ yr. The decay constant for this isotope is $\lambda = 0.000\,122 \text{ yr}^{-1}$. We are interested in monitoring the amount of ^{14}C over 20 000 yr.

There is no official 'yr' (year) unit; informally let's set $1 \text{ yr} = 31 \times 10^6 \text{ s}$.

We're going to examine the script for such a task soon; but try to write it on your own first, before continuing:



Exercise 7.3

Try to write a script for the numerical time integration of an amount $N(t)$ of radioactive material in a control volume. Follow the scheme and example from §6.5 and §6.6; note that the present problem is much simpler!

For the present problem:

- The balance of matter $\frac{dN}{dt} = \mathcal{J} + \mathcal{R}$, in its [finite-difference approximation](#) ^{§6.3 p.173}, drives the system forward in time.
- We have the [boundary condition](#) ^{§6.5 p.176} of zero flux: $\mathcal{J} \equiv 0 \text{ mol/s}$.
- The constitutive relation (7.4) for radioactive decay allows us to determine all necessary quantities for performing a new time step. Take $\lambda = 0.000\,122 \text{ yr}^{-1}$.
- As initial values take

$$N(t_0) = 10^{-12} \text{ mol} \quad \text{at} \quad t_0 = 0 \text{ yr} .$$

- Integrate until time $t_1 = 20\,000 \text{ yr}$ using a timestep $\Delta t = 1 \text{ yr}$ or 0.1 yr .

The time iteration looks as follows:

$$\begin{aligned}
 &\mathcal{R}(t) = -\lambda N(t) \\
 &N(t + \Delta t) \approx N(t) + [\mathcal{J} + \mathcal{R}(t)] \Delta t \\
 &\mathcal{R}(t + \Delta t) = -\lambda N(t + \Delta t) \\
 &N(t + 2\Delta t) \approx N(t + \Delta t) + [\mathcal{J} + \mathcal{R}(t + \Delta t)] \Delta t \\
 &\dots
 \end{aligned} \tag{7.7}$$

We note again the important roles of the two physical laws involved:

- The balance of matter, written as a finite-difference approximation, allows us to determine the amount N at the next time step.
- In order to use the balance of matter to step forward from t to $t + \Delta t$ we also need the values of content influx $\mathcal{J}$, supply $\mathcal{R}$ at time t . It's the constitutive relation (7.4) that allow us to find $\mathcal{R}(t)$ from $N(t)$.

The influx $\mathcal{J}$ is instead given as a constant boundary condition: it is always zero.

A script implementing this numerical time integration is shown in table 7.1 on page 199. Recall the [remarks about computer code](#) §0.4 p. 18 used in the present notes. Light-grey code lines are for drawing the plot of $N(t)$ against time t ; they are of no importance for the physical simulation. The resulting plot is shown at the bottom of the table.

Let's examine the structure of the script, starting from the while-loop; the lines for plotting, which are physically irrelevant, are omitted:

Lines 26–40: iteration. At each time step the supply of matter is calculated from the amount of matter in **line 28**, using the constitutive relation for radioactive decay. Then the matter content is calculated in **line 32**, using the approximate form of the matter balance.

Lines 14–15: time-iteration parameters. We time-integrate the system until time $t_1 = 20\,000$ yr, with a timestep $\Delta t = 1$ yr.

Line 11: boundary condition. We assume that the flux of matter is zero.

Lines 7–8: initial conditions. We start the time integration from $t_0 = 0$ yr, with the initial value $N(t_0) = 1 \times 10^{-12}$ mol.

Line 4: constant. Our equations use the decay constant $\lambda = 0.000\,122$ yr⁻¹.

Table 7.1 Script for radioactive decay

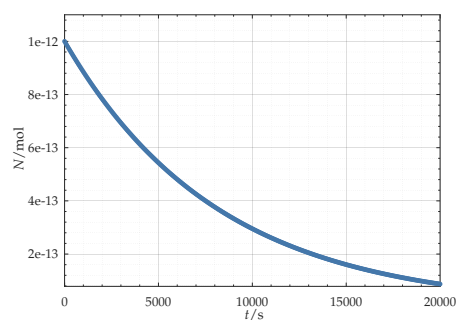
```

1 %% Time integration of radioactive decay
2
3 %% Constants
4 lambda = 0.000122; % 1/yr: decay constant for C14
5
6 %% Initial conditions
7 t = 0; % s: time
8 N = 1e-12; % mol: amount of C14
9
10 %% Boundary conditions
11 J = 0; % mol/s: matter influx of C14
12
13 %% Parameters for time loop
14 t1 = 20000; % yr: final time
15 dt = 1; % yr: time step
16
17 %% Plotting
18 dtplot = t1/360; % time interval between plots
19 tplot = dtplot; % time for next plot
20 figure
21 plot(t, N, 'o', 'color', '#4477AA')
22 xlabel('\it t/s'); ylabel('\it N/mol')
23 axis('tight'); xlim([0, t1]); grid on; hold on
24
25 %% Numerical time integration
26 while t < t1
27     %% constitutive relations
28     R = -lambda * N;
29
30     %% step forward in time with balance law
31     t = t + dt;
32     N = N + (J + R) * dt;
33
34     %% plot
35     if t > tplot
36         plot(t, N, 'o', 'color', '#4477AA')
37         pause(0)
38         tplot = tplot + dtplot;
39     end
40 end

```

Download Octave version
[radioactive_decay.m](#)³⁵

Download Python version
[radioactive_decay.py](#)³⁶



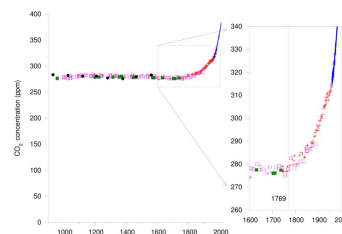
Application to climate

The balance of matter together with the constitutive relation (7.4) for radioactive decay are very useful in climate research.

On Earth's surface and atmosphere we can assume a law of conservation of matter for each of the **stable isotopes**³⁷ of the chemical elements, for instance the **two stable carbon isotopes**³⁸ and the **three stable oxygen isotopes**³⁹. We can therefore follow these isotopes as they flow between different physical systems, like the atmosphere, the oceans, and the biosphere, especially plants – in practice we are using huge control volumes.

For the radioactive isotopes, the **law of radioactive decay** §7.5 p.195 applies, and from it we can deduce the age of different materials and objects like ice or wood.

This is how we are able to say that human usage of fossil fuel is an important factor in the increase of carbon dioxide (CO₂) in the atmosphere during the past 200 years or so. Take a look at the more detailed explanations given by the **Global Monitoring Laboratory**⁴⁰.



Volume content of CO₂ in the atmosphere in the years 1000–2000. By the law of matter conservation, there must have been a net influx of CO₂ into the atmosphere in these years. A vertical line is drawn at year 1769, when James Watt patented his steam engine (from MacKay 2008 p. 6)



Exercise 7.4

You measure the relative amounts of carbon within an air pocket near the surface of a block of ice, and find that one mole of air in this pocket contains 1×10^{-12} mol of the radioactive isotope ^{14}C . You then make an analogous measurement within an air pocket deeper in the ice, and find a relative amount of 2×10^{-13} mol of ^{14}C there. The smaller amount in the same volume size indicates that some of the original ^{14}C has disappeared from radioactive decay. How older is the deeper section of ice?

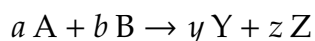
Find out the age t_{ice} of the deeper section by using either the analytical or the numerical time integration used in the present section. (*Hint: if you use numerical time integration, you can terminate the while-loop when the desired amount $N(t) = 2 \times 10^{-13}$ mol has been reached, and then read t_{ice} .)*

§ 7.6 Constitutive relations for rates of formation: stoichiometry

If in a control volume there are different kinds of matter, then their supplies or *rates of formation* satisfy particular constitutive relations. These relations express the fact that, even if different kinds of matter can appear or

disappear, the chemical elements which constitute them are nevertheless conserved.

The disappearance and appearance of different kinds of matter in a control volume can be called a chemical reaction, and is usually expressed by means of a **stoichiometric equation**, which looks as follows:

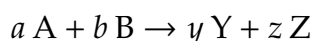


where $A, \dots, Z$ stand for chemical formulae of some substances; and $a, \dots, z$ for numeric coefficients. The substances on the left are called **reactants** and those on the right **products**. The coefficients $-a, -b, y, z$, with negative sign for the reactants, are called **stoichiometric numbers**. This equation expresses the fact that for every amount a of substance A that disappears, also an amount b of B disappears (hence the negative sign for their stoichiometric numbers), and amounts y, z of substances Y, Z appear. A chemical reaction may even involve just one reactant and one product, or more than two.

If no other substances appear in the control volume besides those written in the stoichiometric equation, then the corresponding reaction is called **elementary**. Chemical reactions are usually **composite**, consisting of many elementary reactions that occur simultaneously or in steps.

Rate of conversion

If an *elementary* reaction occurs in a control volume among some substances, say A, B, Y, Z , with stoichiometric equation



then the supplies of the substances satisfy the constitutive relations

$$-\frac{1}{a}\mathcal{R}_A(t) = -\frac{1}{b}\mathcal{R}_B(t) = \frac{1}{y}\mathcal{R}_Y(t) = \frac{1}{z}\mathcal{R}_Z(t) =: \dot{\xi}(t) \quad (7.8)$$

or analogous relations if more or less reactants or products are involved. Note the negative sign for the reactants.

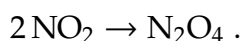
The quantity $\dot{\xi}(t)$ is called the **rate of conversion** for the elementary reaction, and depends on other quantities such as temperature, the contents of the substances, the volume, momentum fluxes, and others. We shall discuss some of these dependencies shortly.

This constitutive relation can also be written with the time-integrated supplies:

$$-\frac{1}{a} \int_{t_0}^{t_1} \mathcal{R}_A(t) dt = \dots = \frac{1}{z} \int_{t_0}^{t_1} \mathcal{R}_Z(t) dt = \int_{t_0}^{t_1} \xi(t) dt \quad (7.9)$$

and the quantity $\int_{t_0}^{t_1} \xi(t) dt$ is called the **extent of reaction**.

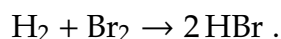
Let's see a simple example of this constitutive relation. Take the elementary reaction where **nitrogen dioxide**⁴¹ NO_2 converts into **dinitrogen tetroxide**⁴² N_2O_4 :



This reaction has only one reactant, NO_2 , with stoichiometric number -2 ; and only one product, N_2O_4 , with stoichiometric number $+1$. Their supplies then satisfy the constitutive relation

$$-\frac{1}{2} \mathcal{R}_{\text{NO}_2}(t) = \mathcal{R}_{\text{N}_2\text{O}_4}(t) .$$

As another example take the reaction between hydrogen and **bromine**⁴³:



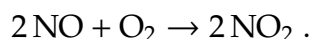
This reaction is not elementary, but in some situations it may be considered approximately so. The supplies of the three substances satisfy the constitutive relation

$$-\mathcal{R}_{\text{H}_2}(t) = -\mathcal{R}_{\text{Br}_2}(t) = \frac{1}{2} \mathcal{R}_{\text{HBr}}(t) .$$



Exercise 7.5

Write down the constitutive relation for the three supplies in the approximately elementary reaction



§ 7.7 Constitutive relations for rates of formation: reaction rates

The constitutive relation discussed in the previous section relates the supplies of different kinds of matter, but does not give an explicit expression

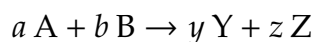
for them in terms of other quantities. Finding constitutive relations between supplies and other quantities like the volume contents of the substances, temperature, volume, forces, and so on, is the goal of *chemical kinetics*. Such relations can have extremely diverse mathematical expressions.

Here we discuss one of the simplest: an example of constitutive relation that applies for some elementary reactions, and that connects together: the rate of conversion $\dot{\xi}$, and therefore the supplies of all reactants and products of the reaction; their volume contents; the size of the control volume; and the temperature in the control volume. It's actually the combination of two constitutive relations: the first is an *empirical rate law*, and the second is called *Arrhenius equation*.



Empirical rate law

Consider an approximately elementary reaction occurring in a control volume of size V , say with stoichiometric equation



and rate of conversion

$$\dot{\xi}(t) := -\frac{1}{a}\mathcal{R}_A(t) = -\frac{1}{b}\mathcal{R}_B(t) = \frac{1}{y}\mathcal{R}_Y(t) = \frac{1}{z}\mathcal{R}_Z(t).$$

This reaction obeys an *empirical rate law* with *Arrhenius equation* if the rate of conversion satisfies the equation

$$\dot{\xi} = V C \cdot \left(\frac{N_A}{V}\right)^\alpha \cdot \left(\frac{N_B}{V}\right)^\beta \quad (7.10)$$

where:

- the exponents α, β are called **partial orders**;
- C is a factor that depends on the temperature and possibly on other quantities;

An analogous formula holds if there are more or fewer reactants.

In this expression, the volume, volume contents, and the factor C may depend on time t . This expression is only valid if the conditions are uniform throughout the control volume, which is usually the case for a small enough control volume.

**Partial orders need not be related to stoichiometric numbers**

The partial orders α, β of the rate law (7.10) are in principle unrelated to the stoichiometric numbers a, b of the stoichiometric equation. In some situations they are equal, in others not.

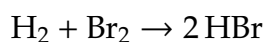
**Chemistry notation for N/V**

In the chemistry literature, the ratio $\frac{N_A}{V}$ between the volume content of a substance A and the volume size, called **amount-of-substance concentration**, is often denoted by square brackets: $[A]$.

In the present sections we try to keep fractions like $\frac{N_A}{V}$ together, even if they lead to expression where the volume V could be simplified, so that you can easily locate concentrations $[A]$ in case you're also studying chemistry.

We shall discuss the factor C that appears in eq. (7.10) more in detail later, after we study the balance of energy-mass.

It's important to keep in mind that the constitutive relation above is *only an example*. Rates of conversion may satisfy constitutive relations that are mathematically quite different. For instance, the reaction



has the rate law

$$\dot{\xi} = V C \frac{N_{\text{H}_2}}{V} \sqrt{\frac{N_{\text{Br}_2}}{V}} \left(1 + \gamma \frac{N_{\text{HBr}}}{N_{\text{Br}_2}} \right)$$

where γ is a constant. Note that the volume content of the *product*, HBr, also appears in the constitutive law above. Other examples exist, involving sums of powers or other mathematical expressions.

§ 7.8 Constitutive relations for rates of formation: examples

As examples of the constitutive relations discussed in the previous sections, let's study two opposite reactions involving **nitrogen dioxide**⁴⁵ (NO_2) and **dinitrogen tetroxide**⁴⁶ (N_2O_4):

(1) in a so-called **dimerization**⁴⁷ reaction, NO_2 turns into N_2O_4 :



(2) in a so-called **dissociation**⁴⁸ reaction, N_2O_4 turns into NO_2 :



We shall refer to these two reactions by (1) and (2).

Dimerization $2\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$

The stoichiometric equation for reaction (1) indicates that the NO_2 content in a control volume, N_{NO_2} , decreases, whereas the N_2O_4 content $N_{\text{N}_2\text{O}_4}$ increases. Let's find the supplies $\mathcal{R}_{1,\text{NO}_2}$ and $\mathcal{R}_{1,\text{N}_2\text{O}_4}$ provided by this reaction for the two substances.

This reaction involves only one reactant, with stoichiometric number -2 , and one product, with stoichiometric number 1 . Therefore the supplies satisfy the constitutive relation

$$-\frac{1}{2}\mathcal{R}_{1,\text{NO}_2}(t) = \mathcal{R}_{1,\text{N}_2\text{O}_4}(t) = \dot{\xi}_1(t). \quad (7.13)$$

The rate of conversion $\dot{\xi}_1$ can be described by a constitutive relation like eq. (7.10), with exponent $\alpha = 2$:

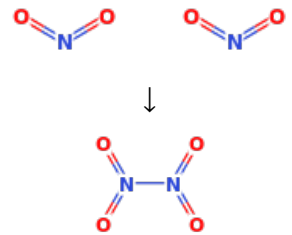
$$\dot{\xi}_1 = V C_1 \cdot \left(\frac{N_{\text{NO}_2}}{V} \right)^2 \quad (7.14)$$

with $C_1 \approx 4.2 \times 10^5 \text{ m}^3/(\text{mol s})$ at a temperature of 330 K;

a temperature of 330 K is around 57°C .



Left: NO_2 ; right: N_2O_4 (from Educador Brasil Escola⁴⁴)



Combining the two equations above, we find the supplies

$$\begin{aligned}\mathcal{R}_{1,\text{NO}_2} &= -2 V C_1 \cdot \left(\frac{N_{\text{NO}_2}}{V} \right)^2, \\ \mathcal{R}_{1,\text{N}_2\text{O}_4} &= V C_1 \cdot \left(\frac{N_{\text{NO}_2}}{V} \right)^2.\end{aligned}\quad (7.15)$$

Keep in mind that in many concrete applications the factor C_1 and the volume V might vary with time; but in the following we shall consider them constant, to keep things simple.



Exercise 7.6

1. In eq. (7.14), verify that the units of the factor C_1 make sense.
2. In eq. (7.15), verify that the supply $\mathcal{R}_{1,\text{NO}_2}$ is always negative, and $\mathcal{R}_{1,\text{N}_2\text{O}_4}$ always positive. Do these signs make sense in this chemical reaction?
3. If we decrease the volume V (keeping everything else constant), will the supply magnitudes $|\mathcal{R}_{1,\text{NO}_2}|$ and $|\mathcal{R}_{1,\text{N}_2\text{O}_4}|$ increase or decrease?

Dissociation $\text{N}_2\text{O}_4 \rightarrow 2\text{NO}_2$

The stoichiometric equation for reaction (2) indicates that the N_2O_4 content $N_{\text{N}_2\text{O}_4}$, decreases, whereas the NO_2 content N_{NO_2} increases. Let's find the supplies $\mathcal{R}_{2,\text{NO}_2}$ and $\mathcal{R}_{2,\text{N}_2\text{O}_4}$ provided by this reaction.

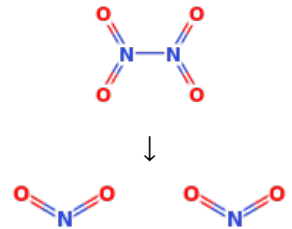
We have only one reactant, with stoichiometric number -1 , and one product, with stoichiometric number 2 . The supplies satisfy the constitutive relation

$$-\mathcal{R}_{2,\text{N}_2\text{O}_4}(t) = \frac{1}{2} \mathcal{R}_{2,\text{NO}_2}(t) = \dot{\xi}_2(t). \quad (7.16)$$

The rate of conversion $\dot{\xi}_2$ can be described by a constitutive relation similar to eq. (7.10) but with exponent $\alpha = 1$:

$$\dot{\xi}_2 = V C_2 \cdot \frac{N_{\text{N}_2\text{O}_4}}{V} \quad (7.17)$$

with $C_2 \approx 1.6 \times 10^7 \text{ s}^{-1}$ at a temperature of 330 K.



Combining the two equations above, we find the supplies

$$\begin{aligned}\mathcal{R}_{2,\text{NO}_2} &= 2 V C_2 \cdot \frac{N_{\text{N}_2\text{O}_4}}{V}, \\ \mathcal{R}_{2,\text{N}_2\text{O}_4} &= -V C_2 \cdot \frac{N_{\text{N}_2\text{O}_4}}{V}.\end{aligned}\tag{7.18}$$

Also in this expression the factor C_2 might vary with time.



Exercise 7.7

1. In eq. (7.17), verify that the units of the factor C_2 make sense; note that they are different from those of the previous factor C_1 .
2. In eq. (7.18), verify that the supply $\mathcal{R}_{2,\text{NO}_2}$ is always positive, and $\mathcal{R}_{2,\text{N}_2\text{O}_4}$ always negative. Do these signs make sense in this reaction?
3. If we decrease the volume V (keeping everything else constant), will the supply magnitudes $|\mathcal{R}_{2,\text{NO}_2}|$ and $|\mathcal{R}_{2,\text{N}_2\text{O}_4}|$ increase or decrease?

§ 7.9 Analytical time integration of amount of matter in a chemical reaction

In applications involving chemical reactions, our goal may be to predict the amount of a particular substance in a control volume after a given amount of time. The balance of matter and the constitutive relations discussed in the previous sections allow us to make such predictions – provided they apply to the specific situation under study. In simple situations such predictions can be made [6.2](#)analytically.

As an example, let's suppose we have a control volume with some amounts of nitrogen dioxide NO_2 and dinitrogen tetroxide N_2O_4 , and:

- *only* the dissociation reaction $\text{N}_2\text{O}_4 \rightarrow 2\text{NO}_2$ takes place in the volume;
- there are no matter fluxes into or out of the control volume;
- the volume size V and temperature T are constant in time.

It's unrealistic to assume that only the dissociation reaction is taking place, but under some conditions it might be the prevalent one.

If at time $t_0 = 0$ s there are amounts $N_{\text{NO}_2}(t_0)$ and $N_{\text{N}_2\text{O}_4}(t)$ of the two substances, what will be the amounts at time t_1 ?

The physical laws that we must apply to this control volume are the balance of matter as well as the constitutive relations (7.18) between supplies and volume contents, for both substances:

$$\begin{aligned}
 \frac{dN_{\text{NO}_2}(t)}{dt} &= \mathcal{J}_{\text{NO}_2}(t) + \mathcal{R}_{2,\text{NO}_2}(t) && \text{matter balance for NO}_2 \\
 \frac{dN_{\text{N}_2\text{O}_4}(t)}{dt} &= \mathcal{J}_{\text{N}_2\text{O}_4}(t) + \mathcal{R}_{2,\text{N}_2\text{O}_4}(t) && \text{matter balance for N}_2\text{O}_4 \\
 \mathcal{R}_{2,\text{NO}_2}(t) &= 2 V C_2 \frac{N_{\text{N}_2\text{O}_4}(t)}{V} && \text{const. relation for NO}_2 \text{ supply} \\
 \mathcal{R}_{2,\text{N}_2\text{O}_4}(t) &= -V C_2 \frac{N_{\text{N}_2\text{O}_4}(t)}{V} && \text{const. relation for N}_2\text{O}_4 \text{ supply} \\
 \mathcal{J}_{\text{NO}_2}(t) &\equiv 0 \text{ mol/s} , \quad \mathcal{J}_{\text{N}_2\text{O}_4}(t) \equiv 0 \text{ mol/s} && \text{no fluxes}
 \end{aligned}$$

Substituting the constitutive relations and no-flux conditions in the matter balances, and simplifying with some simple algebra, we arrive at this system of differential equations:

$$\begin{aligned}
 \frac{dN_{\text{NO}_2}(t)}{dt} &= 2 C_2 N_{\text{N}_2\text{O}_4}(t) \\
 \frac{dN_{\text{N}_2\text{O}_4}(t)}{dt} &= -C_2 N_{\text{N}_2\text{O}_4}(t)
 \end{aligned} \tag{7.19}$$

where $N_{\text{NO}_2}(t)$ and $N_{\text{N}_2\text{O}_4}(t)$ are the unknown functions; all other quantities are constant and their values are supposed to be known.

Let's try to analytically solve this system of differential equations.

In the first equation both amounts $N_{\text{NO}_2}(t)$ and $N_{\text{N}_2\text{O}_4}(t)$ appear, whereas in the second, only $N_{\text{N}_2\text{O}_4}(t)$ does. Therefore focus first on the second equation. It's a **first-order homogeneous linear differential equation**⁴⁹ with constant coefficients, which can be solved analytically. The general solution is

$$N_{\text{N}_2\text{O}_4}(t) = B e^{-C_2 t} .$$

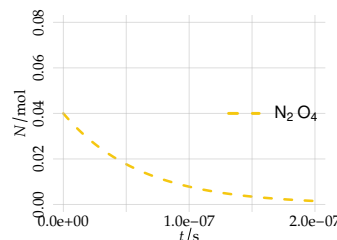
With an undetermined constant B . This constant can be found by imposing the initial condition at $t_0 = 0$ s:

$$B e^{-C_2 \cdot 0 \text{ s}} = N_{\text{N}_2\text{O}_4}(t_0) \quad \Rightarrow \quad B = N_{\text{N}_2\text{O}_4}(t_0) . \quad \text{Recall that } e^0 = 1 .$$

We finally arrive at the formula for the explicit time-dependence of the amount $N_{\text{N}_2\text{O}_4}(t)$ of dinitrogen tetroxide:

$$N_{\text{N}_2\text{O}_4}(t) = e^{-C_2 t} N_{\text{N}_2\text{O}_4}(t_0) . \tag{7.20}$$

The side plot shows how the amount would change in 2×10^{-7} s from an initial amount $N_{\text{N}_2\text{O}_4}(t_0) = 0.04$ mol (recall that the present value of C_2 , from eq. (7.17) is for a temperature of 330 K). We see that at $t = 2 \times 10^{-7}$ s there is almost no N_2O_4 left; but remember that our current example is unrealistic, because the dissociation reaction is not the only one that occurs in reality.



Now that we have the explicit time dependence for $N_{\text{N}_2\text{O}_4}(t)$, we can also find the one for $N_{\text{NO}_2}(t)$ from the first of eqs (7.19). All we have to do is integrate its left and right sides with respect to time between $t_0 = 0$ s and a generic time t :

$$\begin{aligned} \int_{t_0}^t \frac{dN_{\text{NO}_2}(t)}{dt} dt &= 2 C_2 \int_{t_0}^t N_{\text{N}_2\text{O}_4}(t) dt \\ &= 2 C_2 \int_{t_0}^t e^{-C_2 t} dt \cdot N_{\text{N}_2\text{O}_4}(t_0) \end{aligned}$$

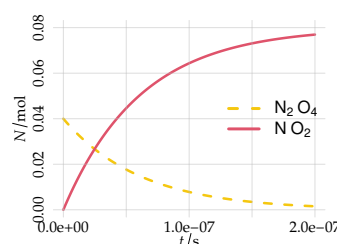
Recall that $\int_a^b e^{Ct} dt = \frac{1}{C} (e^{Cb} - e^{Ca})$

$$\Rightarrow N_{\text{NO}_2}(t) - N_{\text{NO}_2}(t_0) = -2 \frac{C_2}{C_2} (e^{-C_2 t} - 1) N_{\text{N}_2\text{O}_4}(t_0)$$

from which we finally find the time dependence of the amount of nitrogen dioxide:

$$N_{\text{NO}_2}(t) = N_{\text{NO}_2}(t_0) + 2 (1 - e^{-C_2 t}) N_{\text{N}_2\text{O}_4}(t_0). \quad (7.21)$$

The side figure shows how this amount changes in 2×10^{-7} s from a zero initial amount $N_{\text{NO}_2}(t_0) = 0$ mol. The previously plotted amount for N_2O_4 is also shown. Consistently with the dissociation reaction under discussion, the amount of NO_2 increases with time. At $t = 2 \times 10^{-7}$ s, when almost all 0.04 mol of N_2O_4 has vanished, an amount of almost 0.08 mol of NO_2 has appeared: as expected, twice as much.



Exercise 7.8

- Using the time dependence (7.20) for N_2O_4 , prove that the time dependence (7.21) for NO_2 can be written as follows:

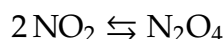
$$N_{\text{NO}_2}(t) = N_{\text{NO}_2}(t_0) + 2 [N_{\text{N}_2\text{O}_4}(t_0) - N_{\text{N}_2\text{O}_4}(t)] .$$

- Intuitively explain the mathematical expression above in terms of the dissociation reaction.

3. Plot the time dependencies (7.20), (7.21) for other values of the factor C_2 . If we decrease this factor, is NO_2 produced faster or more slowly?
4. Find the differential equations, analogous to (7.19), for the dimerization reaction $2\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$. Try to solve them.

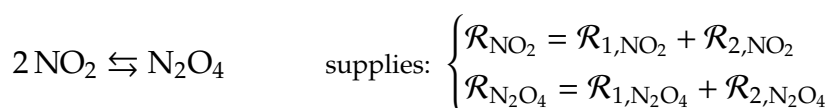
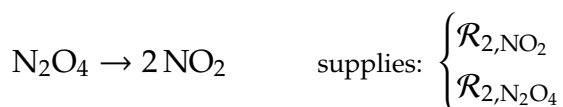
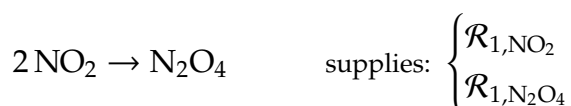
§ 7.10 Simultaneous reactions

In reality neither the dimerization nor the dissociation reactions discussed so far occurs alone. They occur together – in fact, together with still other reactions. A more realistic analysis of the time dependence of NO_2 and N_2O_4 amounts must therefore consider both reactions at once; we write



To indicate that they occur simultaneously

When several reactions occur simultaneously between a substance A and other substances, the supply $\mathcal{R}_A$ may have a complicated constitutive relation, quite different from those of the individual reactions in which substance A participates. In many cases, however, we can consider the supply to be *the sum* of the supplies for the individual reactions. For instance, in the case of the dimerization and association reactions of the previous section we have



This is a constitutive relation:



Supply for simultaneous reactions

Under particular conditions, if a substance A simultaneously participates in several chemical reactions $1, 2, \dots$ having supplies $\mathcal{R}_{1,A}(t), \mathcal{R}_{2,A}(t), \dots$,

then its supply $\mathcal{R}_A(t)$ for all reactions at once is given by the sum

$$\mathcal{R}_A(t) = \mathcal{R}_{1,A}(t) + \mathcal{R}_{2,A}(t) + \cdots \quad (7.22)$$

You might wonder: the fact that the supply for both reactions is the sum of the supplies for the individual reactions, is it a consequence of the [property of extensivity](#) §3.2 p. 69? The answer to this intelligent question is not clear-cut. We might say that the two reactions occur in many distinct microscopic sub-volumes, and this is the reason why the supplies add up. But in this picture we must also be aware that there might be microscopic *fluxes* of the substances between the various sub-volumes. If these microscopic fluxes are important and we neglect them, the effect from a macroscopic perspective is that the supplies don't add up anymore.

Let's see an example of this constitutive relation for the dimerization-dissociation reaction $2\text{NO}_2 \rightleftharpoons \text{N}_2\text{O}_4$. According to the supplies (7.15) and (7.18) for the individual reactions, the supplies for the joint reaction are

$$\begin{aligned} \mathcal{R}_{\text{NO}_2} &= -2 V C_1 \cdot \left(\frac{N_{\text{NO}_2}}{V} \right)^2 + 2 V C_2 \cdot \frac{N_{\text{N}_2\text{O}_4}}{V} \\ \mathcal{R}_{\text{N}_2\text{O}_4} &= V C_1 \left(\frac{N_{\text{NO}_2}}{V} \right)^2 - V C_2 \frac{N_{\text{N}_2\text{O}_4}}{V} \end{aligned} \quad (7.23)$$

with $C_1 \approx 4.2 \times 10^5 \text{ m}^3/(\text{mol s})$, $C_2 \approx 1.6 \times 10^7 \text{ s}^{-1}$ at 330 K.

In the equations above, the supplies, the volume contents, and possibly also the volume and the factors C_1, C_2 depend on time.

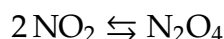


Exercise 7.9

Following the example from §7.9, combine the balances of matter for NO_2 and N_2O_4 , with the constitutive relations above for the supplies in the simultaneous reaction $2\text{NO}_2 \rightleftharpoons \text{N}_2\text{O}_4$; and write down the two differential equations that describe the time evolution of the volume contents $N_{\text{NO}_2}(t)$, $N_{\text{N}_2\text{O}_4}(t)$.

§ 7.11 Numerical time integration for amount of matter in a chemical reaction

For the simultaneous dimerization-dissociation reaction



if we combine the balances of matter for NO_2 and N_2O_4 with the constitutive relations (7.23) we arrive at the system of differential equations

$$\begin{aligned}\frac{dN_{\text{NO}_2}(t)}{dt} &= -2 \frac{C_1}{V} N_{\text{NO}_2}(t)^2 + 2 C_2 N_{\text{N}_2\text{O}_4}(t) \\ \frac{dN_{\text{N}_2\text{O}_4}(t)}{dt} &= \frac{C_1}{V} N_{\text{NO}_2}(t)^2 - C_2 N_{\text{N}_2\text{O}_4}(t)\end{aligned}\tag{7.24}$$

where $N_{\text{NO}_2}(t)$ and $N_{\text{N}_2\text{O}_4}(t)$ are the unknown functions.

This system is more complicated than the one studied in §7.9. Both unknown functions appear in both equations, and one of them is squared. This is generally called a **Riccati**⁵⁰ system of differential equations, and can actually be solved analytically. It may nevertheless be convenient to numerically integrate the time evolution of this joint reaction, as we learned in §6.3 and as we did, as an instructive example, with **radioactive decay** ▶ §7.5 p. 197.

Consider the following scenario. A control volume of constant size $V = 0.001 \text{ m}^3$ (one litre) contains 0.04 mol of N_2O_4 and no NO_2 at the initial time $t_0 = 0 \text{ s}$. The temperature is constant at $T = 330 \text{ K}$. The two reactions $2\text{NO}_2 \rightleftharpoons \text{N}_2\text{O}_4$ occur; the rates of formation of NO_2 and N_2O_4 are given by eqs (7.23). All influxes of matter are zero. We want to know the amounts of NO_2 and N_2O_4 a very short time $t_1 = 0.01 \text{ s}$ later.

Before discussing the script for this time integration, try to write it on your own:



Exercise 7.10

Try to write a script for the numerical time integration of the chemical reaction described above. Follow the scheme and example from §6.5 and §6.6; the present problem is similar, it only has slightly more complicated supplies.

For the present problem:

- The balances of matter for NO_2 and N_2O_4 , in their finite-difference approximations, drive the system forward in time.
- We have the boundary conditions of zero fluxes.
- The constitutive relations (7.23) allow us to determine the supplies of the two substances, necessary for performing a new time step.
- Set $V = 0.001 \text{ m}^3$.
- As initial values take

$$N_{\text{NO}_2}(t_0) = 0.00 \text{ mol} \quad N_{\text{N}_2\text{O}_4}(t_0) = 0.04 \text{ mol} \quad \text{at} \quad t_0 = 0 \text{ s} .$$

- Integrate until time $t_1 = 0.01 \text{ s}$ using a timestep $\Delta t = 1 \times 10^{-8} \text{ s}$.

The time iteration is very similar to the one for the [oscillating reaction](#) ^{eq. (6.8) p. 180}, except that in the present case both volume contents $N_{\text{NO}_2}(t)$, $N_{\text{N}_2\text{O}_4}(t)$ allow us to calculate both supplies $\mathcal{R}_{\text{NO}_2}(t)$, $\mathcal{R}_{\text{N}_2\text{O}_4}(t)$ at every time step t .

A script implementing the numerical time integration from $t_0 = 0 \text{ s}$ to $t_1 = 0.01 \text{ s}$ is shown in table 7.2 on page 214 and the following page. The structure of the time-stepping loop is the usual one. Note in particular how the supplies $\mathcal{R}_{\text{NO}_2}(t)$, $\mathcal{R}_{\text{N}_2\text{O}_4}(t)$ are calculated (lines 46–51) before the actual timestepping, in two stages:

Lines 38–40: First the rates of conversion $\dot{\xi}_1$, $\dot{\xi}_2$ are calculated using eqs (7.14), (7.17); they are called `xidot1`, `xidot2` in the script.

Lines 41–43: Then the rates of formation $\mathcal{R}_{\text{NO}_2}$, $\mathcal{R}_{\text{N}_2\text{O}_4}$ are calculated using eqs (7.13), (7.16), added together as in eq. (7.22).

This calculation could have been done in other equivalent ways. For instance we could have used directly formulae (7.23).

Table 7.2 Script for $2\text{NO}_2 \rightleftharpoons \text{N}_2\text{O}_4$ reaction

```

1  %% Time integration of dimerization-dissociation reaction
2  %% Coordinates (t)
3
4  %% Constants
5  V = 0.001; % m^3: volume
6  C1 = 4.2e5; % m^3/(mol*s): factor 2NO2 -> N2O4
7  C2 = 1.6e7; % 1/s: factor N2O4 -> 2NO2
8
9  %% Initial conditions
10 t = 0; % s: time
11 N_NO2 = 0.00; % mol: amount NO2
12 N_N2O4 = 0.04; % mol: amount N2O4
13
14 %% Boundary conditions
15 J_NO2 = 0; % mol/s: influx of NO2
16 J_N2O4 = 0; % mol/s: influx of N2O4
17
18 %% Time-iteration parameters
19 %% uncomment these for longer time integration
20 t1 = 0.01; % s: final time
21 dt = 1e-8; % s: time step
22 %% uncomment these for shorter time integration
23 % t1 = 2e-7; % s: final time
24 % dt = 1e-10; % s: time step
25
26 %% Plotting
27 dtplot = t1/60; % time interval between plots
28 tplot = dtplot; % time for next plot
29 figure; plot(t, N_NO2, 'v', 'color', '#EE6677')
30 hold on; plot(t, N_N2O4, 'x', 'color', '#CCBB44')
31 legend('NO_2', 'N_2O_4')
32 legend('fontsize', 24, 'box', 'off', 'autoupdate', 'off')
33 xlabel('{\it t}/s'); ylabel('{\it N}/mol')
34 axis('tight'); xlim([0, t1]); grid on
35
36 %% Numerical time integration
37 while t < t1
38     %% constitutive relations: rates of conversion
39     xidot1 = C1 * (N_NO2)^2 / V;
40     xidot2 = C2 * N_N2O4;
41     %% constitutive relations: rates of formation (supplies)
42     R_NO2 = -2 * xidot1 + 2 * xidot2;
43     R_N2O4 = xidot1 - xidot2;
44

```

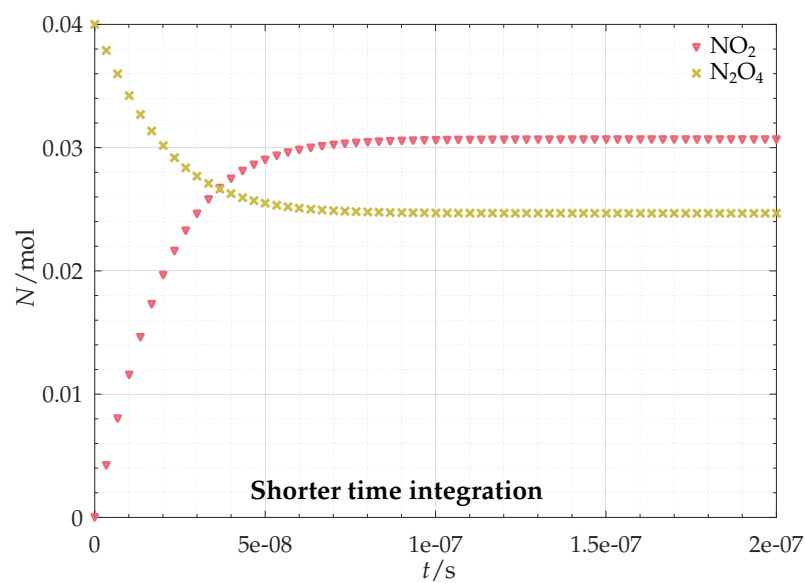
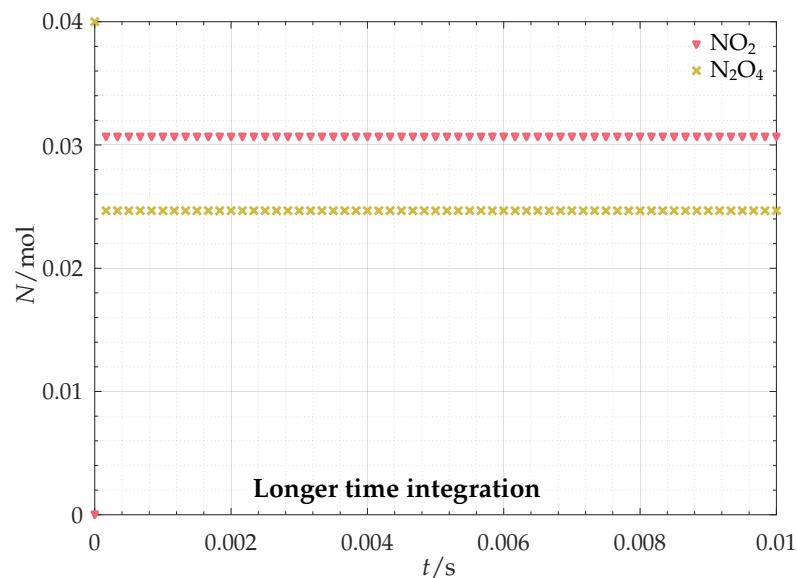
Download Octave version
[N02_N204_reaction.m](#)⁵¹

Download Python version
[N02_N204_reaction.py](#)⁵²

```

45 %% step forward in time with balance laws
46 t = t + dt;
47 N_N02 = N_N02 + (J_N02 + R_N02) * dt;
48 N_N204 = N_N204 + (J_N204 + R_N204) * dt;
49
50 %% plot
51 if t > tplot
52     plot(t, N_N02, 'v', 'color', '#EE6677')
53     plot(t, N_N204, 'x', 'color', '#CCBB44')
54     pause(0)
55     tplot = tplot + dtplot;
56 end
57 end

```



Equilibrium

The plot of the volume contents calculated by numerical time integration is shown as the first plot at the bottom of the table, labelled **Longer time integration**. NO₂ amounts are represented by **red triangles**, and N₂O₄ amounts by **yellow crosses**.

We notice something peculiar: although we started with 0.04 mol of N₂O₄ and no NO₂, the plot seems to show constant amounts 0.025 mol of N₂O₄ and 0.030 mol of NO₂! Is there an error somewhere in the script? A closer look reveals that the very initial amounts at $t = 0$ s are actually as prescribed in the script: look for the markers at the upper and lower left corners of the plot area. This means that at the very beginning of the time integration there's an extremely rapid change in the amounts of N₂O₄ and NO₂.

To explore this rapid change, let's do a shorter time integration with final time $t_1 = 2 \times 10^{-7}$ s and timestep $\Delta t = 10^{-10}$ s (see lines 22–24 in the script). The resulting plot, labelled **Shorter time integration**, is shown below the first. We observe that the volume contents of NO₂ and N₂O₄ start from their initially prescribed values and reach different, stable values within 2×10^{-7} s. This time scale – around 10^{-7} s – is *not* reliable, because on such short scales our constitutive relations cannot be trusted anymore. The real time scale of the change seems to be somewhere between 10^{-7} s and 10^{-4} s (Samukov et al. 2025 §1). But the first, longer time integration is reliable, and it shows that equilibrium is quickly reached in the control volume.

Equilibrium in a chemical reaction is the condition where the volume contents of reactants and products do not change in time; that is, in our case,

$$\frac{dN_{\text{NO}_2}(t)}{dt} = \frac{dN_{\text{N}_2\text{O}_4}(t)}{dt} = 0 \text{ mol/s}.$$

Usually a condition of zero-influx, $\mathcal{J} = 0$ mol/s, is also implicitly understood. Substituting the equilibrium condition in eqs (7.19) for the time evolution of the amounts, we obtain a system of two equations (no more differential equations) for two unknowns: N_{NO_2} and $N_{\text{N}_2\text{O}_4}$:

$$\begin{aligned} -2 \frac{C_1}{V} (N_{\text{NO}_2})^2 + 2 C_2 N_{\text{N}_2\text{O}_4} &= 0 \text{ mol/s} \\ \frac{C_1}{V} (N_{\text{NO}_2})^2 - C_2 N_{\text{N}_2\text{O}_4} &= 0 \text{ mol/s} \end{aligned}$$



Exercise 7.11

1. Solve the system above of two equations in N_{NO_2} and $N_{\text{N}_2\text{O}_4}$, and verify that the solution agrees with the result of the simulation. (The values of the constants in the equations can be read from lines 5–7 in the script.)
2. Perform the longer time integration several times, trying out different values for the volume V (line 5). How do the equilibrium amounts of NO_2 and N_2O_4 change? Does each increase or decrease with an increase or decrease of temperature or volume? (Warning: try very small changes at first, modifying the volume by around 0.0005 m^3 .)

§ 7.12 Rigid-body and particle mechanics

In many applications, the law of conservation of matter can be used in such a subtle way that we almost don't notice that we're actually using it.

This happens when we choose closed control surfaces through which there is no flux of matter: $\mathcal{J}(t) = 0 \text{ mol/s}$. For example, in studying solid objects, we choose control surfaces that tightly “wrap” and follow the object; think of what we did in the tennis-ball examples and exercises of the chapter on [physical laws](#) ▶ §5.6 p. 140. With this choice, the amount of matter N within the control surfaces doesn't change in time, thanks to the law of conservation of matter. So this law is hidden in the fact that we're taking that amount of matter as constant.

We shall see an example of this procedure in the [numerical time integration of the motion of a tennis ball](#) ▶ §10.10 p. 255. At each timestep the mass-energy m of the tennis ball is assumed to be constant in time. But as explained above, [this mass is proportional to the amount of matter](#) ▶ §7.4 p. 193: $m = \rho N$. So this assumption is implicitly guaranteed by the law of conservation of matter.

§ 7.13 Nozzle flow

The law of conservation of matter in its explicit form is at the heart of fluid-dynamic problems. Consider the flow of a fluid (liquid or gas) for instance through a pipe or through a jet engine. When we say that a flow is **steady** we mean that the volume contents and the fluxes taken for whatever

control volumes and surfaces do not change in time (though they may change in space). This condition can be viewed as a constitutive relation.

Consider a control volume, for instance the one indicated in **light blue** in the side picture. The amount of fluid N in this volume is constant in time: $\frac{dN(t)}{dt} = 0$. By the law of conservation of matter, the total influx must then be zero:

$$0 = \frac{dN(t)}{dt} = \mathcal{J}(t)$$

and it is given by the influxes through three surfaces: the side surface, the one at the top, and the one at the bottom. The influx through the side surface is zero. Let us denote the influx through the top surface by $\mathcal{J}_1$, and the influx through the bottom surface by $\mathcal{J}_2$. We must therefore have

$$0 = \mathcal{J}(t) = \mathcal{J}_1(t) + \mathcal{J}_2(t) \quad \implies \quad \mathcal{J}_1(t) = -\mathcal{J}_2(t)$$

that is, the influx through the top surface must equal the efflux through the bottom one. This is a very powerful deduction: consider that it's valid at different sections of a jet engine, even if the flow of the fluid is turbulent; it's valid in even more general situations.

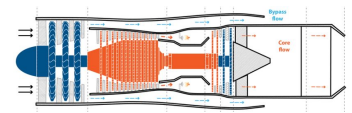
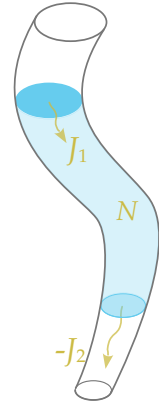
If the surfaces are small enough, we can also use the **connection between flux and velocity** ^{§4.15 p. 124}:

$$v_1 = \frac{\mathcal{J}_1/A_1}{N/V} \quad v_2 = \frac{-\mathcal{J}_2/A_2}{N/V}$$

where A_1, A_2 are the areas of the top and bottom surfaces, and v_1, v_2 the *downward* velocities through them (hence the minus sign for the bottom surface). From $\mathcal{J}_1 = -\mathcal{J}_2$ we then find this important relationship:

$$v_1 A_1 = v_2 A_2$$

that is, if the area through which the flux occur decreases, then the velocity of the fluid through it increases, and vice versa. This is what we often observe in water running from our taps. We can feel with our fingers that the water stream is slightly faster at the bottom, right before it hits the basin; and at this point the stream usually also thinner than at the top (the presence of an aerator or of turbulence can mask this effect).



(image from JetX⁵³)



The thickness and velocity variations of tap water are a consequence of the law of conservation of matter

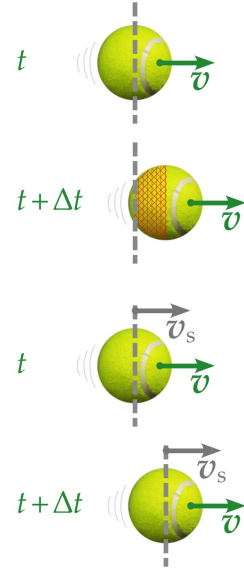
§ 7.14 Matter flux through a moving control surface

As with all fluxes, the flux of matter through a control surface at a coordinate time t depends on the instantaneous motion of the surface at that same time. Take for example a tennis ball moving to the right with velocity $\mathbf{v}$ at time t , as in the side figures, and imagine a vertical control surface of given area, depicted by the dashed grey line, passing through the centre of the ball. A time $t + \Delta t$ later the ball will have moved a little to the right.

If the control surface is at rest at time t , then at time $t + \Delta t$ it will still be at the same position. Part of the tennis ball, indicated by the **hatched red area**, will therefore go through it during the time interval Δt ; in other words, there is a non-zero flux $\mathcal{J}$ of matter through the surface at time t .

If the control surface is instead moving with velocity $\mathbf{v}_s = \mathbf{v}$ at time t , then at time $t + \Delta t$ it will still be at the centre of the tennis ball. No matter has passed through it, and the matter flux is zero. There are obviously also intermediate situations, depending on the magnitude and direction of the velocity $\mathbf{v}$ of matter and $\mathbf{v}_s$ of the surface.

It is convenient to have a formula that yields the matter flux $\mathcal{J}$ if we know these velocities and some other details:



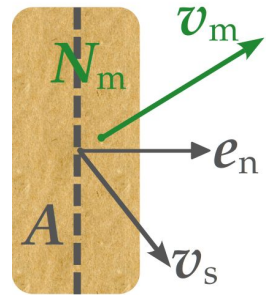
Matter flux through a moving control surface

Consider a control surface at time t , having area A and normal unit vector $\mathbf{e}_n$, oriented like the surface crossing direction. The surface at time t has velocity $\mathbf{v}_s$.

If in a thin volume V_m containing the surface there is an amount of matter N_m having velocity $\mathbf{v}_m$, then the flux through the surface at time t is

$$\mathcal{J} = A \mathbf{e}_n \cdot (\mathbf{v}_m - \mathbf{v}_s) \frac{N_m}{V_m}. \quad (7.25)$$

This formula is only valid if the control surface is flat, so that the normal $\mathbf{e}_n$ is well-defined; if matter is uniform throughout the volume; and if the velocity of matter $\mathbf{v}_m$ is the same all around the surface. These conditions are usually satisfied if the control surface is *small* enough. The velocities must be smaller than the speed of light, and gravitational fields must be weak.



The relation above connects the matter flux $\mathcal{J}$, the component of matter velocity normal to the control-surface $\mathbf{e}_n \cdot \mathbf{v}_m$, and the normal component

of the surface velocity $\mathbf{e}_n \cdot \mathbf{v}_s$. It therefore allows us to find any one of these quantities given the other two.



Exercise 7.12

1. Verify that the formula above makes sense:

- (i) How much is the flux if neither matter nor the surface are moving, that is if $\mathbf{v}_m = \mathbf{v}_s = 0$ m/s?
- (ii) How much is the flux if matter and surface are moving with the same velocity, that is $\mathbf{v}_m = \mathbf{v}_s$?
- (iii) How much is the flux if the surface is not moving, $\mathbf{v}_s = 0$ m/s, and matter is moving parallel to the surface, that is $\mathbf{e}_n \cdot \mathbf{v}_m = 0$?
- (iv) How much is the flux if there is no matter around the surface, that is $N_m = 0$ mol?
- (v) How much is the flux if matter is not moving, $\mathbf{v}_m = 0$ m/s, and the surface is moving parallel to itself, that is $\mathbf{e}_n \cdot \mathbf{v}_s = 0$?

2. Take a vertical coordinate, say z , directed upward. Through a horizontal control surface of area 7.5 m^2 and vertical normal $\mathbf{e}_n$ directed downward, there is a (downward) matter flux $\mathcal{J} = 80 \times 10^3$ mol/s. The surface is moving upward with velocity $\mathbf{v}_s = 300$ m/s. The ratio between amount of matter and volume at the surface is $N_m/V_m = 25$ mol/m³. How much is the component of the matter velocity along the normal, $\mathbf{e}_n \cdot \mathbf{v}_m$?



Matter flux for any speeds and gravitational fields

A formula very similar to (7.25) holds more generally, also in general relativity, provided the surface is small enough:

$$\mathcal{J} = A \mathbf{e}_n \cdot \left(\mathbf{l} - \mathbf{v}_s \frac{N_m}{V_m} \right),$$

where $\mathbf{l} := [j_x, j_y, j_z]$ are the fluxes, divided by coordinate area, across three small surfaces parallel to the yz , zx , and xy coordinate planes.

URLs for chapter 7

1. <http://hyperphysics.phy-astr.gsu.edu/hbase/Forces/particleint.html>
2. <http://hyperphysics.phy-astr.gsu.edu/hbase/Particles/spinc.html#c2>
3. <http://hyperphysics.phy-astr.gsu.edu/hbase/particles/quark.html>
4. <http://hyperphysics.phy-astr.gsu.edu/hbase/Particles/parint.html#c2>
5. <http://hyperphysics.phy-astr.gsu.edu/hbase/particles/lepton.html>
6. <http://hyperphysics.phy-astr.gsu.edu/hbase/Particles/parint.html#c3>
7. <http://hyperphysics.phy-astr.gsu.edu/hbase/particles/hadron.html#c6>
8. <http://hyperphysics.phy-astr.gsu.edu/hbase/Nuclear/radact.html>
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Conservation of magnetic flux

Some writers regard electromagnetism as a well-understood and even as a closed field of study. Maxwell's equations can be accepted and all that is needed is to find appropriate solutions. This, however, is an onlooker's view, practitioners know that there are many fundamental problems to be investigated.

P. Hammond 1988

§ 8.1 Formulation and generalities



Conservation of magnetic flux

Flux: Φ Circulation: $-\mathcal{E}$

$$\Phi(t_1) = \Phi(t_0) - \int_{t_0}^{t_1} \mathcal{E}(t) dt \qquad \frac{d\Phi(t)}{dt} = -\mathcal{E}(t) \qquad (8.1)$$

integral expression differential expression

✂ To be written in a later version

Conservation of electric charge

Furthermore, is not the intuitive notion of conservation of charge, for example, quite independent of measurements of length and time? We view the conservation of charge and magnetic flux as independent of ideas like inertial frames, rigid rods, absolute or uniform time, Lorentz transformations, Galilean transformations, etc., and hence as deserving an independent mathematical expression.

C. A. Truesdell & R. A. Toupin 1960

§ 9.1 Formulation and generalities



Conservation of electric charge

Volume content: Q Flux: I

$$Q(t_1) = Q(t_0) + \int_{t_0}^{t_1} I(t) \, dt \qquad \frac{dQ(t)}{dt} = I(t) \qquad (9.1)$$

integral expression differential expression

✂ To be written in a later version

Balance of momentum

LEX II.

Mutationem motus proportionalem esse vi motrici impressæ, & fieri secundum lineam rectam qua vis illa imprimitur.

I. Newton 1726a

LAW II. *The change of motion is proportional to the motive force impressed, and is made in the direction of the straight line in which that force is impressed.*

§ 10.1 Formulation and generalities



Balance of momentum

Volume content: $\mathbf{P}$ Flux: $\mathbf{F}$ Supply: $\mathbf{G}$

$$\mathbf{P}(t_1) = \mathbf{P}(t_0) + \int_{t_0}^{t_1} \mathbf{F}(t) dt + \int_{t_0}^{t_1} \mathbf{G}(t) dt \qquad \frac{d\mathbf{P}(t)}{dt} = \mathbf{F}(t) + \mathbf{G}(t)$$

integral expression differential expression

(10.1)

Among the seven universal balances, the balance of momentum is probably the most used in applications where motion or stability are important. Newton's famous "second law" is included in this balance as a special case, [provided we disregard](#) ^{§5.13 p.160} Newton's specific association between momentum, mass, and velocity, and lack of the important distinction between surface forces and volume forces.

The importance of the balance of momentum comes to a great extent from two kinds of constitutive relations:

- a small number of constitutive relations that connect the volume content $\mathbf{P}$ of momentum with the motion and flux of matter and of the electromagnetic field;
- an amazingly wide variety of constitutive relations that connect, in the most diverse ways, the flux $\mathbf{F}$ of momentum with many properties of matter, like its extension, deformation, motion, temperature, and the simultaneous presence of charge and electromagnetic field.

Through this balance we can therefore describe and predict motions. But this balance is also essential in the opposite problem: when we need to study things that do not and should not move, like a building or a bridge. In this case the balance is used to study which momentum fluxes $\mathbf{F}$ and supply $\mathbf{G}$ are necessary to ensure that the volume content $\mathbf{P}$ of momentum is constantly zero.



Amounts of momentum depend on the coordinate system

Remember that the amount of momentum $\mathbf{P}$ in a control volume, the flux $\mathbf{F}$ through a control surface, and the supply $\mathbf{G}$ in a control volume **all depend on the specific coordinate system** (t, x, y, z) . If we use a different coordinate system, these amounts will be different *for the same volume and surface*. For example, a tennis ball can contain a huge amount of momentum with respect to one coordinate system, and zero momentum with respect to another! The supply of momentum, or gravitational-inertial force, can also be zero or non-zero, in the same physical situation, depending on the coordinate system.



Momentum balance keeps buildings from collapsing (photo: Bryggen, Bergen, from [UNESCO World Heritage Centre](#)¹)

In the rest of this chapter we shall explore several common constitutive relations that involve momentum, and some of their applications together with the balance law for momentum.

§ 10.2 Newton's approximate relation between momentum and matter

One of the most important constitutive relations is the one connecting momentum with the flux of matter, briefly mentioned in §3.7.

Choose a coordinate system (t, x, y, z) . Consider a control volume containing an amount of matter N , and where electromagnetic fields are negligible. From the [constitutive relation between matter and mass-energy](#) ^{§7.4 p. 192} we know that this volume also contains an amount of mass-energy $m = \rho N$. If this control volume is small enough, the matter in it has also a unique velocity $\mathbf{v}$, possibly zero, [related to its flux](#) ^{§4.15 p. 124}. The **Newtonian formula for the momentum of matter** then says that this control volume also contains an amount of momentum $\mathbf{P}$ that is proportional to the mass-energy and the velocity:

Newtonian constitutive relation for momentum

If a control volume contains an amount of matter N having rest mass-energy m and velocity $\mathbf{v}$, then it also contains an amount of momentum

$$\mathbf{P} = m\mathbf{v} \quad \text{or in components} \quad \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = m \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} \quad (10.2)$$

This relation is only valid if the matter's velocity $\mathbf{v}$ is the same throughout the volume. This is usually the case if the volume is *small* enough.

You are probably familiar with the equality above; old textbooks present it as the definition of momentum. Today we know that this equality cannot be taken as a definition of momentum: it only holds for matter but not, for instance, for the electromagnetic field. And even for matter it is only approximate: it's only valid when the speed v is small compared to the speed of light, only in weak gravitational fields, and only when changes in energy-mass are small compared to the total energy-mass.

In [particle physics](#)² the following constitutive relation between momentum, mass-energy, and velocity of matter is ordinarily used instead:

$$\mathbf{P} = \frac{m\mathbf{v}}{\sqrt{1 - v^2/c^2}} \approx m\mathbf{v} + \frac{1}{2c^2}v^2 m\mathbf{v} + \dots$$

where c is the speed of light. This relation is also valid at high speeds, but only with weak gravitational fields and only in particular coordinates.

In astrophysics, in order to describe and predict the behaviour of stars and planets, a more general expression is used:

$$\mathbf{P} = m_r \mathbf{v} + \frac{1}{c^2} \left(\frac{1}{2}v^2 + \frac{U}{V} - 2g \right) m_r \mathbf{v} + \frac{1}{c^2} \Phi + \dots$$

where m_r is the rest mass-energy, the term U represents internal energy, the term g comes from gravitational effects, and the term Φ represents fluxes of heat and of mechanical energy, also involving pressure. We shall say more about these quantities in [chapter 11](#). One important aspect of this formula is that *momentum $\mathbf{P}$ and velocity of matter $\mathbf{v}$ are not exactly parallel*. Relations among momentum, matter, velocity of this kind are also used for our Earth in some applications of geodesy.

If a control volume contains not only matter but also electric charge and electromagnetic fields, then we must use constitutive relations that connect

the momentum content with all those quantities. Such constitutive relations can be quite complex. In general they cannot even be written as a *sum* of some expression involving only velocity and mass, plus some expression involving only electromagnetic fields. For example, the constitutive relation for the momentum content of a control volume containing inviscid fluid matter, electric charge, and electromagnetic fields may look as follows:

$$\begin{aligned} & [\rho_0(c^2 + \epsilon_i) + \phi] \gamma^2 \frac{\mathbf{v}}{c} + \epsilon_0 \mathbf{E} \times \mathbf{B} + \frac{1}{c\alpha} \mathbf{P} \times \left(\frac{\mathbf{v}}{c} \times \mathbf{P} \right) \\ & + \frac{1}{c\beta} \mathbf{M} \times \left(\frac{\mathbf{v}}{c} \times \mathbf{M} \right) - \frac{1}{c} \mathbf{B} \times \left(\frac{\mathbf{v}}{c} \times \mathbf{M} \right) \\ & + \frac{1}{c^2} \mathbf{E} \times \mathbf{M}, \end{aligned}$$

From Pfeifer et al. 2007. Velocity is $\mathbf{v}$, mass density is ρ_0 . Symbols $\mathbf{P}$, $\mathbf{M}$ are related to electric charge; $\mathbf{E}$, $\mathbf{B}$ to electromagnetic field.

This applies, for instance, for a control volume containing water or glass traversed by a laser beam, as occurs in applications like water-assisted laser inscription. This inseparability means that we cannot really speak of “momentum of matter” or “momentum of the electromagnetic field”: the momentum content is related to the whole.

§ 10.3 Momentum supply: gravity and inertial forces

According to General Relativity, there can be creation of momentum in a control volume, that is, there can be a momentum supply $\mathbf{G}$, which we also call a *volume force or body force* ^{§4.10 p. 114}. This supply depends on two aspects: our choice of coordinate system, and the nearby presence of large amounts of energy-mass, momentum, and of their fluxes. General Relativity also makes it clear that it doesn’t make sense to distinguish between these two aspects in a small control volume: if we find a non-zero supply of momentum $\mathbf{G}$ in a small region of space, it can be because of our coordinate system, or because of the nearby presence of energy-mass or momentum. In fact, by changing our coordinate system we can always make the supply of momentum to be zero *in a small region*, though not everywhere.

Examples of these kinds of supplies are the *gravitational force*, the *centrifugal force*, and other forces called *inertial*. For example, when we’re travelling in a car that speeds up or slows down, we feel a horizontal force pushing us against our seat or pulling us away from it; we feel this force in all our body. This force is an example of supply of momentum.



The supply of momentum appearing in a slowed-down car is deadly real.

How is the supply of momentum $\mathbf{G}$ in a control volume related to other quantities?

When we consider physical phenomena that happen close to a planet's surface and involve spatial extensions that are small compared to the planet's size, and we choose a coordinate system fixed with the planet's ground, then we have a simple constitutive relation for the supply of momentum:

Gravitational force near a planet's surface

Take a coordinate system (t, x, y, z) fixed with the ground, with z as vertical upward coordinate. A control volume containing an amount of mass-energy m also has a supply of downward momentum, the **gravitational force**, given by

$$\mathbf{G} = m \mathbf{g}, \quad \text{with} \quad \mathbf{g} = -g \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad (10.3)$$

where $\mathbf{g}$, the **acceleration of free fall**, depends on the planet. On Earth it is taken to have the standard magnitude

$$g = 9.806\,65 \text{ N/kg} \equiv 9.806\,65 \text{ m/s}^2, \quad (10.4)$$

although it actually depends slightly on location and time.

This constitutive relation is used in most of everyday applications of mechanical and building engineering and fluid mechanics.

Further away from a planet's surface, an approximate expression for the supply of momentum is given by **Newton's law of gravitation**. Consider a small control volume a at position $\mathbf{r}_a$, containing an amount of mass-energy m_a ; and a planet b with centre at position $\mathbf{r}_b$, containing an amount of mass-energy m_b . We suppose that other amounts of mass-energy are negligible or far away. Then the control volume a has a supply of momentum approximately given by

$$\mathbf{G}_a = \kappa \frac{m_a m_b}{|\mathbf{r}_a - \mathbf{r}_b|^3} (\mathbf{r}_b - \mathbf{r}_a),$$

where $\kappa \approx 6.673\,84 \times 10^{-11} \text{ N m}^2/\text{kg}^2$ is the *gravitational constant*, more often denoted ' G ' (a symbol we don't use here to avoid confusion with momentum supply).

Newton's law of gravitation is not used in present-day technologies that must account for gravitational forces on planetary scale, because it is too approximate for the precision of modern measuring instruments, also in [geodesy](#)³ (Müller et al. 2008). But it is useful for pedagogical purposes and for quick, approximate checks.

At NASA's [Jet Propulsion Laboratory](#)⁴, the following expression for gravitational force is used for spacecraft, satellite, and planetary motions:

$$\begin{aligned} & \sum_{j \neq i} \frac{\mu_j (\mathbf{r}_j - \mathbf{r}_i)}{r_{ij}^3} \left\{ 1 - \frac{2(\beta + \gamma)}{c^2} \sum_{l \neq i} \frac{\mu_l}{r_{il}} - \frac{2\beta - 1}{c^2} \sum_{k \neq j} \frac{\mu_k}{r_{jk}} \right. \\ & + \gamma \left(\frac{\dot{s}_i}{c} \right)^2 + (1 + \gamma) \left(\frac{\dot{s}_j}{c} \right)^2 - \frac{2(1 + \gamma)}{c^2} \dot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_j \\ & \left. - \frac{3}{2c^2} \left[\frac{(\mathbf{r}_i - \mathbf{r}_j) \cdot \dot{\mathbf{r}}_j}{r_{ij}} \right]^2 + \frac{1}{2c^2} (\mathbf{r}_j - \mathbf{r}_i) \cdot \ddot{\mathbf{r}}_j \right\} \\ & + \frac{1}{c^2} \sum_{j \neq i} \frac{\mu_j}{r_{ij}^3} \left\{ [\mathbf{r}_i - \mathbf{r}_j] \cdot [(2 + 2\gamma) \dot{\mathbf{r}}_i - (1 + 2\gamma) \dot{\mathbf{r}}_j] \right\} (\dot{\mathbf{r}}_i - \dot{\mathbf{r}}_j) \\ & + \frac{3 + 4\gamma}{2c^2} \sum_{j \neq i} \frac{\mu_j \ddot{\mathbf{r}}_j}{r_{ij}} \end{aligned} \quad \begin{array}{l} \text{from Moyer 2000 §4.4.1} \\ \text{eq. (4.26)} \end{array}$$

The [International Earth Rotation and Reference Systems Service \(IERS\)](#)⁵ uses the following expression, to be added to Newton's:

$$\begin{aligned} & \frac{GM_E}{c^2 r^3} \left\{ \left[2(\beta + \gamma) \frac{GM_E}{r} - \gamma \vec{r} \cdot \vec{r} \right] \vec{r} + 2(1 + \gamma) (\vec{r} \cdot \vec{r}) \vec{r} \right\} + \\ & (1 + \gamma) \frac{GM_E}{c^2 r^3} \left[\frac{3}{r^2} (\vec{r} \times \vec{r}) (\vec{r} \cdot \vec{J}) + (\vec{r} \times \vec{J}) \right] + \\ & \left\{ (1 + 2\gamma) \left[\vec{R} \times \left(\frac{-GM_S \vec{R}}{c^2 R^3} \right) \right] \times \vec{r} \right\} \end{aligned} \quad \begin{array}{l} \text{from Petit \& Klioner 2010} \\ \text{§10.3 eq. (10.12)} \end{array}$$

These more precise expressions show interesting features: for example, the force of gravity (besides the centrifugal force) depends on how fast a planet rotates; this is represented by the terms $\dot{s}_i$ and $\vec{J}$.

You may have noticed a peculiar coincidence about body forces like the gravitational and the inertial ones. When *we* feel these *body* forces, we also always feel some accompanying *contact* forces. Most of the time we feel the force of gravity in our body, and also the contact force of the ground pressing upward against the soles of our feet. Sitting in an accelerating car,

we feel the force pushing our body against the seat, but also the contact force from the seat pushing against our back. When we don't feel any body force and momentarily feel weightless, for instance when we reach the peak while going back and forth in a swing or if we fall from a height, then we don't feel any contact forces either.

The reason for this concurrence of the two kinds of force is the balance of momentum, together with a peculiarity of each person's coordinate system. *With respect to your own coordinate system, you are always at rest:* your momentum content $\mathbf{P}_{\text{you}}$ is always zero. Therefore $d\mathbf{P}_{\text{you}}/dt = 0 \text{ N}$ always. The balance of momentum for your control volume then becomes

$$0 \text{ N} = \frac{d\mathbf{P}_{\text{you}}}{dt} = \mathbf{F}_{\text{you}} + \mathbf{G}_{\text{you}} \quad \implies \quad \mathbf{F}_{\text{you}} = -\mathbf{G}_{\text{you}} .$$

So whenever you feel a body force $\mathbf{G}_{\text{you}}$, you must also feel somewhere a contact force $\mathbf{F}_{\text{you}}$ that is equal and opposite to the body force. In fact our feeling of gravity comes from our "force-receptors", [mechanoreceptors](#)⁶ in our skin and bones. And these actually measure momentum *fluxes*, that is, contact forces, among different parts of our bodies. They really measure " $\mathbf{F}_{\text{you}}$ ", which you interpret as " $\mathbf{G}_{\text{you}}$ ".

§ 10.4 Statics

There are situations where, in some coordinate system:

- the momentum in one or more control volumes and in their sub-volumes is *zero* at all times;
- the fluxes and supplies of momentum (forces) are *constant* in time.

In such situations, for each control volume we have

$$\begin{aligned} \mathbf{P}(t) &= \mathbf{0} \text{ N s} , & \frac{d\mathbf{P}}{dt} &\equiv \mathbf{0} \text{ N} , \\ \mathbf{F}(t) &\text{ constant} , & \mathbf{G}(t) &\text{ constant} . \end{aligned} \tag{10.5}$$

Such a situation is called **static**. Since contents, fluxes, supplies are constant in time, the time-dependence ' (t) ' is often omitted.

Two typical tasks in static situations are: to find what are the momentum fluxes (forces) $\mathbf{F}$ and supplies $\mathbf{G}$ that make situation static; or to design momentum fluxes and supplies such that the physical condition will be and remain static. This is the domain of the discipline of **statics**.

Let's see a simple example, approached in four different scenarios.

Scenario 1. Initial question and information

An object, say a book, is resting on a table. Which momentum fluxes occur in such a static situation? Let's see if we can answer this question with the information given, and what happens as additional pieces of information are provided.

Choose a coordinate system (see side figure) and a control volume coinciding with the object. We collect the physical laws that are relevant for this control volume, together with all information we have about the physical situation. Obviously we have the balance of momentum, and we know that the situation is static; therefore what we know is expressed mathematically as

$$\begin{cases} \frac{d\mathbf{P}}{dt} = \mathbf{F} + \mathbf{G} & \text{momentum balance} \\ \mathbf{P} = \mathbf{0} \text{ N s} & \text{static condition} \\ \frac{d\mathbf{P}}{dt} = \mathbf{0} \text{ N} & \text{static condition} \end{cases}$$

If this is all information we have, then it is *not* possible to determine the total influx $\mathbf{F}$, or any influxes through parts of the control surface, such as the one between the object and the table. From the equations above we easily find $\mathbf{F} = -\mathbf{G}$, but since the supply $\mathbf{G}$ is not given, the total flux $\mathbf{F}$ is unknown too.



Scenario 2. Additional information: mass-energy and a constitutive relation

Now suppose we're given an additional piece of information: the object has mass-energy $m = 0.20 \text{ kg}$. We can combine this information with a constitutive relation: the momentum supply for matter close to Earth's surface, $\mathbf{G} = m \mathbf{g}$, with $\mathbf{g} \approx -[0, 0, 9.8] \text{ N}$. The information we have now can thus be expressed by

$$\begin{cases} \frac{d\mathbf{P}}{dt} = \mathbf{F} + \mathbf{G} & \text{momentum balance} \\ \mathbf{P} = \mathbf{0} \text{ N s} & \text{static condition} \\ \frac{d\mathbf{P}}{dt} = \mathbf{0} \text{ N} & \text{static condition} \\ \mathbf{G} = m \mathbf{g} \approx -[0, 0, 2.0] \text{ N} & \text{constitutive relation} \end{cases}$$

With this information we are now able to find the *total* flux $\mathbf{F}$. This is intuitively clear from a quick look at the equations above, but it can also be seen from a more formal mathematical analysis.

The set of physical laws and static conditions above can be seen as a linear system: *four* vector equations, and *four* vector unknowns: $d\mathbf{P}/dt$, $\mathbf{F}$, $\mathbf{G}$, $\mathbf{P}$. We find

$$\mathbf{F} = -\mathbf{G} \approx [0, 0, 2.0] \text{ N} .$$

If the problem was to find the *total* flux $\mathbf{F}$, then it's solved.

The equation $\mathbf{P} = \mathbf{0} \text{ N s}$ and the “unknown” $\mathbf{P}$ are trivial, but let's count them in for formality's sake.

Scenario 3. Different task: particular flux

Now let's consider a different task. We don't want to find the total flux of momentum (force) to the object, but more specifically the flux of momentum *from the table to the object through the bottom surface*, that is, the force exerted from the table to the object.

Since we're asking about the influx through part of our closed control surface, we must invoke the principle of [extensivity](#) §3.2 p. 69, and write the total flux $\mathbf{F}$ as the sum of fluxes through the different surfaces, for instance:

$$\mathbf{F} = \mathbf{F}_{\text{table} \rightsquigarrow \text{book}} + \mathbf{F}_{\text{other}} .$$

This sum could be even more precise, considering for instance the flux through the lateral sides of the object and the flux from the top. But the statement of the problem doesn't give us much information on how this separation should be done.

The information we have now is the same as before, plus the equation for extensivity:

$$\left\{ \begin{array}{ll} \frac{d\mathbf{P}}{dt} = \mathbf{F} + \mathbf{G} & \text{momentum balance} \\ \mathbf{P} = \mathbf{0} \text{ N s} & \text{static condition} \\ \frac{d\mathbf{P}}{dt} = \mathbf{0} \text{ N} & \text{static condition} \\ \mathbf{G} = m \mathbf{g} \approx -[0, 0, 2.0] \text{ N} & \text{constitutive relation} \\ \mathbf{F} = \mathbf{F}_{\text{table} \rightsquigarrow \text{book}} + \mathbf{F}_{\text{other}} & \text{extensivity} \end{array} \right.$$

and our unknown is $\mathbf{F}_{\text{table} \rightsquigarrow \text{book}}$. Clearly we do *not* have enough information to solve the problem. In terms of an equation system we have five vector equations in six vector unknowns: $d\mathbf{P}/dt$, $\mathbf{F}$, $\mathbf{G}$, $\mathbf{F}_{\text{table} \rightsquigarrow \text{book}}$, $\mathbf{F}_{\text{other}}$, $\mathbf{P}$. All we can say is that

$$\mathbf{F}_{\text{table} \rightsquigarrow \text{book}} \approx -[0, 0, 2.0] \text{ N} - \mathbf{F}_{\text{other}} .$$

Scenario 4. Additional information: momentum flux from air

Finally let's suppose that the task is as before, finding the force from the table to the object, but we're given an additional piece of information: *the other influx of momentum to consider is from the air surrounding the object to the object itself, but this influx is negligible*^{§10.6 p.244}. Mathematically we can express this with two equations:

$$\begin{aligned}\mathbf{F} &= \mathbf{F}_{\text{table} \rightsquigarrow \text{book}} + \mathbf{F}_{\text{air} \rightsquigarrow \text{book}} \\ \mathbf{F}_{\text{air} \rightsquigarrow \text{book}} &\approx \mathbf{0} \text{ N} .\end{aligned}$$

The information we have in this last set-up is mathematically expressed by these equations:

$$\left\{ \begin{array}{ll} \frac{d\mathbf{P}}{dt} = \mathbf{F} + \mathbf{G} & \text{momentum balance} \\ \mathbf{P} = \mathbf{0} \text{ N s} & \text{static condition} \\ \frac{d\mathbf{P}}{dt} = \mathbf{0} \text{ N} & \text{static condition} \\ \mathbf{G} = m \mathbf{g} \approx -[0, 0, 2.0] \text{ N} & \text{constitutive relation} \\ \mathbf{F} = \mathbf{F}_{\text{table} \rightsquigarrow \text{book}} + \mathbf{F}_{\text{air} \rightsquigarrow \text{book}} & \text{extensivity} \\ \mathbf{F}_{\text{air} \rightsquigarrow \text{book}} \approx \mathbf{0} \text{ N} . & \text{extra information} \end{array} \right.$$

and our unknown is still $\mathbf{F}_{\text{table} \rightsquigarrow \text{book}}$. It's obvious that we can now find the answer; indeed we have *six* vector equations in *six* vector unknowns: $d\mathbf{P}/dt$, $\mathbf{F}$, $\mathbf{G}$, $\mathbf{F}_{\text{table} \rightsquigarrow \text{book}}$, $\mathbf{F}_{\text{air} \rightsquigarrow \text{book}}$, $\mathbf{P}$. The answer is

$$\begin{aligned}\mathbf{F}_{\text{table} \rightsquigarrow \text{book}} &= -\mathbf{G} - \mathbf{F}_{\text{air} \rightsquigarrow \text{book}} \\ &\approx +[0, 0, 2.0] \text{ N} - [0, 0, 0] \text{ N} = [0, 0, 2.0] \text{ N} .\end{aligned}$$

Why the above scenarios?

The problem in each of the four scenarios above was quite simple, easily solvable, or easily recognized as unsolvable. If it had been a concrete application, we wouldn't have needed to go through all the formal steps of writing down physical laws and extra information, counting equations and unknowns, and so on. So why all the hullabaloo?

Concrete applications, in static and non-static problems, can be vastly more complex than the one just discussed; so complex that we don't have any clear intuition and we easily get lost among the information given and the tasks to solve. In complex applications we therefore need to use a more

systematic and formal procedure. Note some of the most important steps involved in such a procedure:

- **Choose appropriate control volumes.**
- **List balances and constitutive relations.**
- **List the information given, expressing it mathematically.**
- **Identify the need for principles such as extensivity and symmetry of fluxes, and use them.**
- **Identify the unknowns and the equations.**

These steps are only roughly in chronological order. It may happen that to complete a step you need to take a look at subsequent steps; it may also happen that some step leads you to re-examine and modify a previous step.

The detailed analyses of the four scenarios above are example applications of this systematic procedure. The fact that they are simple helps you understand intuitively the systematic steps.

§ 10.5 Statics and the choice of control volumes

In the problem of statics discussed in the previous section it was obvious how to choose the control volume. More complex statics problems may require the choice of several control volumes. But there are no clear-cut rule for such choice. Different choices of number, shapes, locations of control volumes can all lead to the solution of the problem – and obviously the solution will be the same. Other choices may prove to be unfruitful.

Let's see three different analysis of the same statics problem, with three different choices of control volume. Before continuing, try to solve the problem on your own; then compare your approach with the three approaches explained below. Did you use one of them? or maybe a new one?



Exercise 10.1

Two identical books rest on a table, on top of each other. Each book has mass-energy 0.2 kg. The momentum flux from the air to each book is negligible.

Find the flux of momentum between the table and the bottom book, using any appropriate physical laws.



First approach. Bottom book as control volume

Denote the top book with '1' and the bottom book with '2'.

Choose control volume. Since our task is to find the momentum flux from table to book 2, let's try to analyse the situation with one control volume only, coinciding with book 2; see margin figure. The momentum content, influx, supply for this control volume are denoted $\mathbf{P}_2, \mathbf{F}_2, \mathbf{G}_2$.

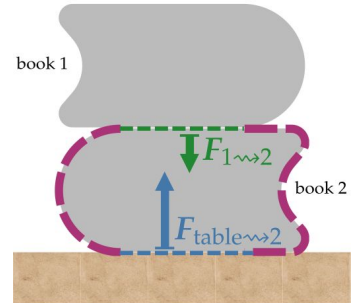
Physical laws. From the information given, two physical laws seem relevant for this control volume: the balance of momentum and the constitutive relation for momentum supply (force) from gravity.

Other information. The text implies that we have a static configuration, so the momentum content and its time derivative are zero. We are also given the mass-energy of the books, which can be used in the constitutive relation for the gravitational force. Finally, the text implies that our closed control surface should be divided into several pieces, and several momentum fluxes should be considered. We have:

- The surface (dashed green thin top line in the figure) separating book 1 and book 2, with influx $\mathbf{F}_{1 \rightsquigarrow 2}$.
- The surface (dashed blue thin bottom line) separating table and book 2, with influx $\mathbf{F}_{\text{table} \rightsquigarrow 2}$.
- The remaining surface (dashed red thick side curves), with a negligible influx $\mathbf{F}_{\text{air} \rightsquigarrow 2}$.

Extensivity for flux. Since we have influxes through several surfaces, we must use the principle of extensivity to find the total momentum influx:

$$\mathbf{F}_2 = \mathbf{F}_{1 \rightsquigarrow 2} + \mathbf{F}_{\text{table} \rightsquigarrow 2} + \mathbf{F}_{\text{air} \rightsquigarrow 2}.$$



(Supplies and some fluxes are not shown.)

Equations and unknowns. All the information above can be collected in the following *five* vector equations:

$$\left\{ \begin{array}{ll} \frac{d\mathbf{P}_2}{dt} = \mathbf{F}_2 + \mathbf{G}_2 & \text{momentum balance} \\ \frac{d\mathbf{P}_2}{dt} = \mathbf{0} \text{ N} & \text{static condition} \\ \mathbf{G}_2 = m_2 \mathbf{g} \approx -[0, 0, 2.0] \text{ N} & \text{constitutive relation} \\ \mathbf{F}_2 = \mathbf{F}_{1 \rightsquigarrow 2} + \mathbf{F}_{\text{table} \rightsquigarrow 2} + \mathbf{F}_{\text{air} \rightsquigarrow 2} & \text{extensivity influx} \\ \mathbf{F}_{\text{air} \rightsquigarrow 2} \approx \mathbf{0} \text{ N} & \text{extra information} \end{array} \right.$$

with *six* vector unknowns: $d\mathbf{P}_2/dt$, $\mathbf{F}_2$, $\mathbf{G}_2$, $\mathbf{F}_{1 \rightsquigarrow 2}$, $\mathbf{F}_{\text{air} \rightsquigarrow 2}$, $\mathbf{F}_{\text{table} \rightsquigarrow 2}$. The unknown that interests us is the last one.

It's clear that the system of equations is under-determined. Indeed either $\mathbf{F}_{\text{table} \rightsquigarrow 2}$ or $\mathbf{F}_{1 \rightsquigarrow 2}$ could be chosen as a **free variable**⁷. We arrive at

$$\mathbf{F}_{\text{table} \rightsquigarrow 2} + \mathbf{F}_{1 \rightsquigarrow 2} \approx [0, 0, 2.0] \text{ N}$$

where both fluxes on the left are unknown.

In order to solve our original task we must therefore consider additional or alternative control volumes.

The analysis was correct but insufficient

Note that the analysis above is physically correct; there is nothing wrong with it from a physics point of view. Its only problem is that it's insufficient for arriving at the information we want.



Exercise 10.2

Ordering the unknowns as follows:

$$d\mathbf{P}_2/dt, \quad \mathbf{F}_2, \quad \mathbf{G}_2, \quad \mathbf{F}_{1 \rightsquigarrow 2}, \quad \mathbf{F}_{\text{air} \rightsquigarrow 2}$$

verify that the **augmented matrix**⁸ corresponding to the system of equations above is this:

$$\left(\begin{array}{cccccc|c} 1 & -1 & -1 & 0 & 0 & 0 & \mathbf{0} \\ 1 & 0 & 0 & 0 & 0 & 0 & \mathbf{0} \\ 0 & 0 & 1 & 0 & 0 & 0 & m_2 \mathbf{g} \\ 0 & 1 & 0 & -1 & -1 & -1 & \mathbf{0} \\ 0 & 0 & 0 & 0 & 0 & 1 & \mathbf{0} \end{array} \right)$$

Second approach. Both books as control volumes

Choose control volume. In our previous tentative analysis, the momentum flux from book 1 to book 2 appeared to be relevant. Let us therefore try adding one more control volume, coinciding with book 1. The boundaries of this new control volume and of the initial one have one part in common (dashed green thin line). The momentum content, influx, supply for the additional control volume are denoted $\mathbf{P}_1, \mathbf{F}_1, \mathbf{G}_1$.

Physical laws. We now have two control volumes, so for each one we must use the balance of momentum and the constitutive relation for momentum supply (force) from gravity.

Other information. The information listed previously for book 2 applies also in the present analysis. We must add the information about book 1. The static configuration also holds for book 1, so the time-derivative of momentum for this control volume is zero. We are also given the mass-energy of book 1, identical to that of book 2, to be used for the gravitational force. The flux from air is also negligible for this book. It seems that the closed control surface for book 1 should be divided into two parts, with corresponding influxes:

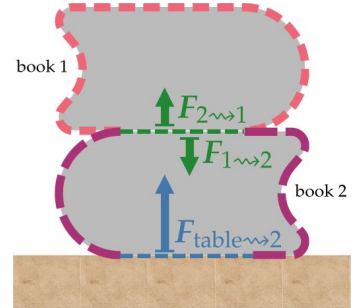
- The surface (dashed green thin line) separating book 1 and book 2, with influx $\mathbf{F}_{2 \rightsquigarrow 1}$; note that this is the *influx* into the new control volume.
- The remaining surface (dashed light-red thick curves), with a negligible influx $\mathbf{F}_{\text{air} \rightsquigarrow 1}$.

Flux extensivity and symmetry. The use of extensivity for the initial control surface still applies. We must apply extensivity also for the new closed control surface:

$$\mathbf{F}_1 = \mathbf{F}_{2 \rightsquigarrow 1} + \mathbf{F}_{\text{air} \rightsquigarrow 1}.$$

Moreover, the two control volumes have part of their boundaries in common. The principle of [symmetry of flux](#) ^{§4.6 p.106} (Newton's third law in this case) therefore applies there:

$$\mathbf{F}_{2 \rightsquigarrow 1} = -\mathbf{F}_{1 \rightsquigarrow 2}.$$



(Supplies and some fluxes are not shown.)

Equations and unknowns. We arrive at 11 vector equations:

$$\left\{ \begin{array}{ll} \frac{d\mathbf{P}_1}{dt} = \mathbf{F}_1 + \mathbf{G}_1 & \text{momentum balance book 1} \\ \frac{d\mathbf{P}_2}{dt} = \mathbf{F}_2 + \mathbf{G}_2 & \text{momentum balance book 2} \\ \frac{d\mathbf{P}_1}{dt} = \mathbf{0} \text{ N} & \text{static condition book 1} \\ \frac{d\mathbf{P}_2}{dt} = \mathbf{0} \text{ N} & \text{static condition book 2} \\ \mathbf{G}_1 = m_1 \mathbf{g} \approx -[0, 0, 2.0] \text{ N} & \text{constitutive relation book 1} \\ \mathbf{G}_2 = m_2 \mathbf{g} \approx -[0, 0, 2.0] \text{ N} & \text{constitutive relation book 2} \\ \mathbf{F}_1 = \mathbf{F}_{2 \rightsquigarrow 1} + \mathbf{F}_{\text{air} \rightsquigarrow 1} & \text{extensivity influx book 1} \\ \mathbf{F}_2 = \mathbf{F}_{1 \rightsquigarrow 2} + \mathbf{F}_{\text{table} \rightsquigarrow 2} + \mathbf{F}_{\text{air} \rightsquigarrow 2} & \text{extensivity influx book 2} \\ \mathbf{F}_{2 \rightsquigarrow 1} = -\mathbf{F}_{1 \rightsquigarrow 2} & \text{flux symmetry} \\ \mathbf{F}_{\text{air} \rightsquigarrow 1} \approx \mathbf{0} \text{ N} & \text{extra information} \\ \mathbf{F}_{\text{air} \rightsquigarrow 2} \approx \mathbf{0} \text{ N} & \text{extra information} \end{array} \right.$$

with 11 vector unknowns:

$$d\mathbf{P}_1/dt, \mathbf{F}_1, \mathbf{G}_1, d\mathbf{P}_2/dt, \mathbf{F}_2, \mathbf{G}_2, \mathbf{F}_{2 \rightsquigarrow 1}, \mathbf{F}_{\text{air} \rightsquigarrow 1}, \mathbf{F}_{1 \rightsquigarrow 2}, \mathbf{F}_{\text{air} \rightsquigarrow 2}, \mathbf{F}_{\text{table} \rightsquigarrow 2} .$$

The last one being our target.

Now we can arrive at a unique solution. Solving the system above, for instance by the method of substitution, we find in particular

$$\begin{aligned} \mathbf{F}_{1 \rightsquigarrow 2} &= \mathbf{G}_1 \approx -[0, 0, 2.0] \text{ N} \\ \mathbf{F}_{\text{table} \rightsquigarrow 2} &= -\mathbf{G}_1 - \mathbf{F}_{1 \rightsquigarrow 2} \end{aligned}$$

and the final answer to our task:

$$\mathbf{F}_{\text{table} \rightsquigarrow 2} \approx [0, 0, 4.0] \text{ N} .$$

This answer was probably obvious to you from the beginning, after a quick intuitive analysis. But let's emphasize again: **our focus in the example above is not on the answer, but on the *procedure*.**

Third approach. Both books as a single control volume

The last analysis led to many equations because we were considering two control volumes – and therefore balances and constitutive relations for each. Is it possible to solve the problem with fewer equations, analysing it in terms of a single control volume? Let's try such an approach.

Choose control volume. It seems important to take into account book 1, so let's choose one control volume that coincides with *both* books, as illustrated in the margin figure. The momentum content, influx, supply for this alternative control volume are denoted $\mathbf{P}, \mathbf{F}, \mathbf{G}$.

Physical laws. We now have one volume, so we need to use the balance of momentum only once.

We also need the constitutive relation for the momentum supply: $\mathbf{G} = m\mathbf{g}$. Note, however, that strictly speaking this constitutive relation cannot be used as-is in the present case: *we need to use it together with the principle of extensivity*, as explained below.

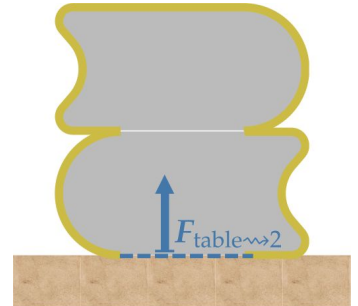
Other information. The configuration is static, so the time-derivative of momentum for this control volume is zero. We are given the mass-energies of the two books separately; as discussed below we need to use extensivity for them. Our present closed control surface should be divided into two parts, with corresponding influxes:

- The surface (**dashed blue thin bottom line**) separating table and our control volume, with influx $\mathbf{F}_{\text{table} \rightsquigarrow 2}$ as before.
- The remaining surface (**solid yellow curve**) with a negligible influx $\mathbf{F}_{\text{air} \rightsquigarrow \text{books}}$.

Note that *we don't need to consider any control surface or momentum flux between the two books* (the surface depicted in **dashed green** in the previous figures)! This is the consequence of using a single control volume including both books.

Extensivity for flux and for supply. We have influxes through two surfaces, we must use the principle of extensivity:

$$\mathbf{F} = \mathbf{F}_{\text{table} \rightsquigarrow 2} + \mathbf{F}_{\text{air} \rightsquigarrow \text{books}}.$$



(Supplies and some fluxes are not shown.)

We also need to use this principle to calculate the total supply, as mentioned above. The reason is that the ‘ m ’ in the constitutive relation for the gravitational force refers to the *total* mass-energy in the control volume, and problem’s text doesn’t give this directly. The principle of extensivity must be used in two possible alternative ways:

- *Extensivity of mass-energy*: the total mass-energy m in the control volume is the sum of the mass-energies in two sub-volumes: $m = m_1 + m_2$. Once we have the total mass-energy, we can use it in the constitutive relation for the momentum supply.
- *Extensivity of momentum supply*: the total momentum supply $\mathbf{G}$ in the control volume is the sum of the supplies in two sub-volumes: $\mathbf{G} = \mathbf{G}_1 + \mathbf{G}_2$. The constitutive relation for momentum supply can be used for each of the two supplies separately.

These two ways are obviously equivalent; you can choose whichever you prefer. What’s important here is to note that the principle of extensivity was used, in one way or another.

Equations and unknowns. We arrive at *seven* vector equations:

$$\left\{ \begin{array}{ll} \frac{d\mathbf{P}}{dt} = \mathbf{F} + \mathbf{G} & \text{momentum balance} \\ \frac{d\mathbf{P}}{dt} = \mathbf{0} \text{ N} & \text{static condition} \\ \mathbf{G}_1 = m_1 \mathbf{g} \approx -[0, 0, 2.0] \text{ N} & \text{constitutive relation book 1} \\ \mathbf{G}_2 = m_2 \mathbf{g} \approx -[0, 0, 2.0] \text{ N} & \text{constitutive relation book 2} \\ \mathbf{G} = \mathbf{G}_1 + \mathbf{G}_2 & \text{extensivity supply} \\ \mathbf{F} = \mathbf{F}_{\text{air} \rightsquigarrow \text{books}} + \mathbf{F}_{\text{table} \rightsquigarrow 2} & \text{extensivity influx} \\ \mathbf{F}_{\text{air} \rightsquigarrow \text{books}} \approx \mathbf{0} \text{ N} & \text{extra information} \end{array} \right.$$

with *seven* vector unknowns: $d\mathbf{P}/dt$, $\mathbf{F}$, $\mathbf{G}$, $\mathbf{G}_1$, $\mathbf{G}_2$, $\mathbf{F}_{\text{air} \rightsquigarrow \text{books}}$, $\mathbf{F}_{\text{table} \rightsquigarrow 2}$.

All unknowns can therefore be found. In particular we find, as before, $\mathbf{F}_{\text{table} \rightsquigarrow 2} \approx [0, 0, 4.0] \text{ N}$.



Exercise 10.3

Solve the linear system of equations above, using your favourite method.

Summary

Comparing the three approaches above, you notice some important aspects of the solution of a physical problem:

- The physical problem doesn't always clearly indicate how control volumes and surfaces should be chosen. Often you have great freedom of choice.
- Different choices of control volumes can lead to the solution.
- Some choices may *not* lead to the solution; so you have to alter your initial choice.
- Depending on the choice of control volumes, some fluxes or supplies may have to be considered or may be neglected.



Exercise 10.4

Try solving the two-book problem of this section by using *two* control volumes chosen as follows:

- one coinciding with both books, as in the [third approach](#) ^{§10.5 p. 242};
- one containing part of the table under the two books, or the whole table if you prefer.

For the procedure, work by analogy with the [second approach](#) ^{§10.5 p. 240}.

Do you obtain a system of equations where *all* unknowns can be found?

Can you still manage to find the unknown of interest, $F_{\text{table} \rightarrow 2}$?

§ 10.6 Atmospheric pressure

There is a continuous flux of momentum across any surface that has air on at least one of its sides. Not only any surface where air is in contact with objects, but also any imaginary surface, say in the sky over an open field, with air on both sides. This flux is called *atmospheric pressure*⁹ and has some important constitutive properties:

- It is always *compressive* ^{§4.12 p. 118}, that is, its direction is always orthogonal to the surface, and its orientation is the same as the crossing orientation.
- Its magnitude, for a surface of 1 m^2 at sea level, is approximately equal to 10^5 N (corresponding to a weight of about 10 000 kg!).

- Its magnitude slowly decreases with altitude; that is, a 1 m^2 surface 1 km above the ground will have an atmospheric pressure (momentum flux) lower than a 1 m^2 surface close to the ground.

In examining momentum fluxes for objects on Earth we often neglect the presence of air and atmospheric pressure. We did so in the previous example about a book on a table. The reason is that the *total* momentum flux from atmospheric pressure around an object is usually approximately zero. This happens on account of three factors: (a) there's always at least a thin layer of air around an object; even an object lying on a table or on the ground has a thin layer of air underneath; (b) the atmospheric-pressure momentum flux is always orthogonal to every small piece of surface; (c) its magnitude is approximately the same at the same altitude. Together, these factors lead to a zero total when we sum up all the atmospheric-pressure fluxes across the different parts of a surface that encloses an object.

Indeed we suddenly feel the heavy presence of atmospheric pressure when the first or the last of those factors doesn't hold anymore.

For example you may sometime have made the mistake of laying a wide, smooth slab of glass upon another, only to find out that you couldn't easily separate them anymore: the two slabs were sort of "stuck" together. The reason is that the glass surface is so smooth that effectively there's no space for a thin layer of air between the two glass slabs. So air is exerting a contact force on the upper side of the top slab and on the lower side of the bottom slab, pressing the two against each other. As mentioned, for a 1 m^2 slab this momentum flux is equivalent to a weight of 10 000 kg! You can't separate the slabs, not because they're "glued" together, but because they're pressed against each other by a net force of 10 tonnes on each side. The only thing you can do is to try to let air again between the two slabs.

The same phenomenon occurs, in a more useful way, with a suction cup: the atmospheric pressure on its two sides is very different, and therefore the cup receives a net influx of momentum pointing towards the wall. This, in turn, leads to an influx of upward momentum owing to [friction](#) ^{§10.27 p. 297}, which compensates for the momentum supply from gravity. Thus the cup can stay in place without falling, and it can even support some object.

In some cases the magnitude of the atmospheric pressure is different because of altitude, and this can be exploited for flying, as we discuss below. But in order to do that, try to solve this exercise first:





Exercise 10.5

In this exercise we apply a reasoning very similar to that of Exercise 10.1. Consider a body of air at rest. Imagine a closed control surface which encloses some of this air. This is analogous to the [example with the book on the table](#) ▶ §10.4 p. 233; now instead of a book we simply have a volume of air, and instead of a table we have air also underneath the volume. For definiteness let's say that the volume is a cuboid with small height h and horizontal base area A , so the volume is $h A$.

Air is matter, and therefore this control volume has a constant supply of downward momentum owing to gravity. On Earth's surface we can say (this is a constitutive relation) that a volume $h A$ of air has a momentum supply with z -component approximately equal to

$$G_z = -h \cdot A \cdot 11.8 \text{ N/m}^3.$$

Assume that the momentum flux on the lateral surface of the volume is zero.

1. Apply the momentum balance, and extensivity of momentum, to find the **difference in magnitude** between the momentum influx through the bottom of the cuboid and the momentum influx through the top of the cuboid.
2. Pressure is defined as momentum flux *divided by area*. How much is the pressure difference between the bottom and top of the cuboid of air?
3. Try to generalize and write an approximate formula of how pressure changes with altitude.

❗ The results obtained in this exercise are approximate, because the supply of momentum $\mathbf{G}$ actually changes with altitude as well, and depends on other quantities such as temperature. But they do have the correct order of magnitude.

§ 10.7 Statics: cable cars

Consider a cable car suspended on a horizontal cable, and not moving. Which momentum fluxes occur in this situation?

This problem has some similarities with the previous one about the book on a table. Choose a control volume containing the cable car.

- This is a static problem. The total momentum in the control volume is zero and constant in time, because the cable car isn't moving.
- There's a constant momentum supply because of gravity; let's say that in this case it's

$$\mathbf{G}(t) = -[0, 0, 8 \times 10^4] \text{ N} \quad (\text{constant}).$$

- By the balance of momentum, there must then be an influx of upward-momentum of the same magnitude as the supply:

$$\begin{aligned} \mathbf{F}(t) &= \underbrace{\frac{d\mathbf{P}(t)}{dt}}_{[0,0,0] \text{ because constant}} - \underbrace{\mathbf{G}(t)}_{[0,0,-8 \times 10^4] \text{ N}} \\ &= [0, 0, 8 \times 10^4] \text{ N} \quad (\text{constant}). \end{aligned}$$

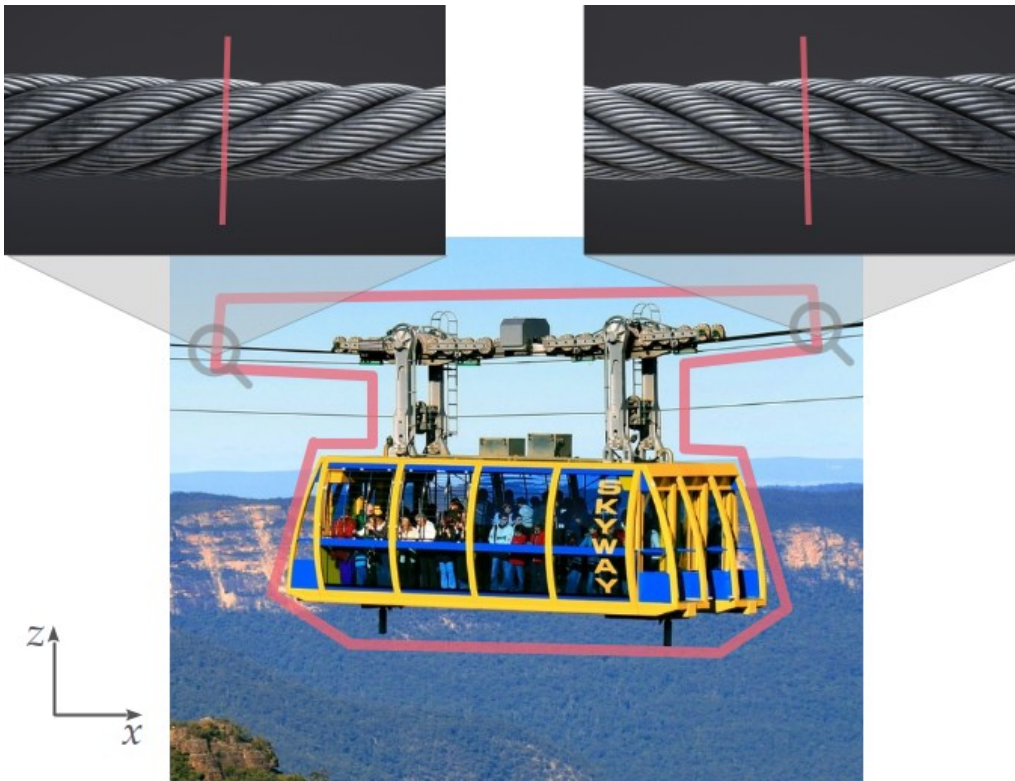
- Also in this case the influx through the parts of the control surface in contact with air is practically zero:

$$\mathbf{F}_{\text{air} \rightsquigarrow \text{car}}(t) \approx \mathbf{0} \text{ N}.$$

All the influx of momentum must therefore come through the cable. But there are some interesting aspects on how this happens. In this case it's interesting to choose a static closed control surface that wraps the car and part of the cables, "cutting" the cables in imagination, as illustrated by the **red polygon** in the picture below:



Scenic Skyway¹⁰ cable car, Australia



This picture also depicts, as zoom-ins, the two parts of the imaginary control surface that “cut” the cable (for simplicity imagine that there’s only one cable). The total influx $\mathbf{F}$ must happen through these two small regions of the surface, in the cable. An interesting question is: *How much is the momentum influx through each one?* Call these two momentum influxes $\mathbf{F}_{\text{left}}$ through surface cutting the cable on the left; and $\mathbf{F}_{\text{right}}$ through surface cutting the cable on the right. Since all these fluxes are constant in time, let’s drop the argument ‘(t)’; and let’s drop the y -components, which are all zero.

We know that

$$\mathbf{F}_{\text{left}} + \mathbf{F}_{\text{right}} = \mathbf{F} = \begin{bmatrix} 0 \\ 8 \times 10^4 \end{bmatrix} \text{ N}$$

from this equation we find

$$(\mathbf{F}_{\text{left}})_x = -(\mathbf{F}_{\text{right}})_x \quad (\mathbf{F}_{\text{left}})_z = 8 \times 10^4 \text{ N} - (\mathbf{F}_{\text{right}})_z$$

where ‘ $(\dots)_x$ ’ denotes the x -component, and similarly for the z -component. But this doesn’t tell us how much $\mathbf{F}_{\text{left}}$ and $\mathbf{F}_{\text{right}}$ are individually. For example we could have

$$\mathbf{F}_{\text{left}} \stackrel{?}{=} \begin{bmatrix} 5 \\ -2 \times 10^4 \end{bmatrix} \text{ N} \quad \mathbf{F}_{\text{right}} \stackrel{?}{=} \begin{bmatrix} -5 \\ 20 \times 10^4 \end{bmatrix} \text{ N}$$

and the total influx would be the correct one.

We shall see later that the *balance of angular momentum* applied to this problem leads to a further constraint: $(\mathbf{F}_{\text{left}})_z = (\mathbf{F}_{\text{right}})_z$. So we can conclude that

$$(\mathbf{F}_{\text{left}})_z = (\mathbf{F}_{\text{right}})_z = 4 \times 10^4 \text{ N}$$

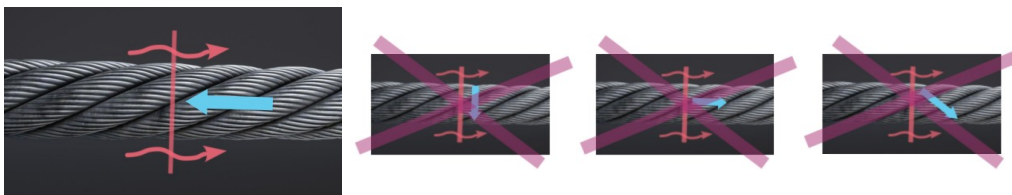
that is, each half of the cable is “taking half of the weight”, as intuitively expected.

The last part of our mystery, about the x -components, can be solved thanks to an additional constitutive property:

Momentum flux allowed in cables and ropes

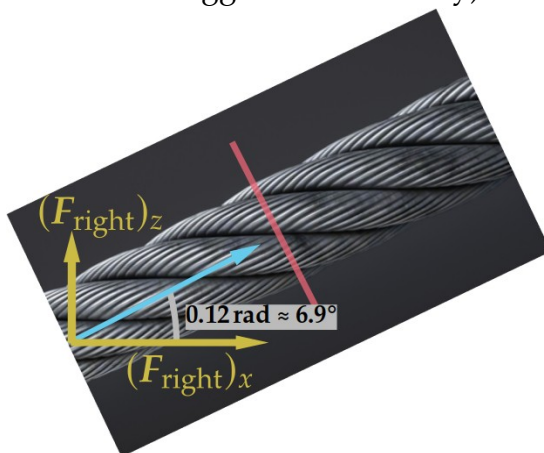
A cable, rope, or similar object can (approximately) only transmit *tensile momentum flux* ^{§4.12 p.119}, aligned along its axis.

Pictorially this means that only the momentum flux represented in the left picture below – tension – is physically possible:



the other three (from left to right: shear, pressure, mixed) are not.

We therefore need to know the directions of the axes of the two parts of the cable. Let's suppose that each part has an inclination of $0.12 \text{ rad} \approx 6.9^\circ$, but in opposite directions. This must then also be the angle between the influx $\mathbf{F}_{\text{right}}$ and its horizontal component $(\mathbf{F}_{\text{right}})_x$, as in the following picture (the angle has been exaggerated for clarity):



From trigonometry we find

$$(\mathbf{F}_{\text{right}})_x = (\mathbf{F}_{\text{right}})_z / \tan(0.12 \text{ rad}) = 4 \times 10^4 \text{ N} / 0.12 \approx 3.3 \times 10^5 \text{ N}$$

and therefore $(\mathbf{F}_{\text{left}})_x \approx -3.3 \times 10^5 \text{ N}$:

$$\mathbf{F}_{\text{left}} \approx \begin{bmatrix} -3.3 \times 10^5 \\ 4 \times 10^4 \end{bmatrix} \text{ N} \quad \mathbf{F}_{\text{right}} \approx \begin{bmatrix} 3.3 \times 10^5 \\ 4 \times 10^4 \end{bmatrix} \text{ N}$$

The magnitude of these tensions is also approximately $|\mathbf{F}_{\text{left}}| = |\mathbf{F}_{\text{right}}| \approx 3 \times 10^5 \text{ N}$ or an equivalent weight of 35 tonnes (this result seems in the correct order of magnitude, comparing with the data in Brownjohn 1998).

In solving the cable-car problem above we used the balance of momentum, but note and keep in mind how that balance alone wasn't enough. We also had to use:

- the balance of angular momentum
- constitutive relations regarding:
 - the momentum supply
 - the amount of momentum flux between air and the cable car
 - the kind of momentum flux allowed in a cable

§ 10.8 Airborne flight

Airborne flight is a conceptually extremely simple application of momentum balance. In terms of momentum, the objective of flight is to compensate or over-compensate the constant downward-momentum supply $\mathbf{G}$ that an object, not touching any other solid object, receives because of gravity. One way to compensate this supply is by creating a *net* influx of upward-momentum, or equivalently a *net* efflux of downward-momentum, between the object and the air or atmosphere that surrounds it. This is airborne flight.

The ways in which such a momentum flux between object and air is realized can be very different:

Buoyancy In the preceding section we saw that the magnitude of the compressive momentum flux – atmospheric pressure – between air and any object *decreases* with height. So if the object is large enough compared to its weight, it automatically receives a net influx of upward momentum that can balance its weight. A calculation shows that this net upward force is proportional to the volume of the object, the mass per volume of the surrounding air (or atmosphere or other fluid), and the gravitational acceleration; this formula is called ‘**Archimedes’s principle**’¹². Flight by buoyancy can therefore be achieved by making the object large, light, or both. This is the principle upon which helium-filled party balloons, hot-air balloons, airships, and also submarines, are based.



Airship *Airlander* from Hybrid Air Vehicles¹¹

Soaring Air is not always at rest (with respect to the Earth’s surface). Owing to energy flow in the atmosphere and Earth’s rotation, bodies of air – air currents – can be moving in different directions, including upward. An upward-moving body of air contains a net upward momentum. If an object stops the vertical motion of this air, for example deflecting it horizontally, then by the balance of momentum there must be a flux of upward momentum from the body of air to the object. This upward momentum can compensate the gravity supply of downward momentum, and therefore the object can float or rise. This mechanism is called **soaring or gliding**¹⁴ and is used by birds and gliders; it’s also the mechanism that lifts light objects like paper or feathers.



A hang-glider (image from Jæren Luftsportsklubb¹³)

Propelled flight We can try to create a flow of downward momentum from an object to the surrounding air, even if air is initially at rest. As a result, a body of air surrounding the object will acquire a net, partially downward movement. By the **symmetry of flux** §4.6 p.106 this also means that momentum with an upward component flows from the air to the object, and this component can compensate the object’s weight. This is achieved by birds by flapping their wings, and by aeroplanes and helicopters through horizontally moving wings or blades.

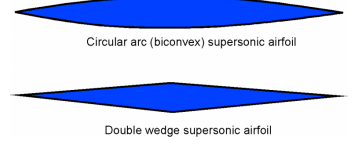


It's not the difference in air pressure

Some texts say that wings can sustain an aeroplane because their cross-section has an asymmetric shape, slightly more bulged upward than downward, leading to a difference in the air pressure between the top and bottom of the wing. This is not true. In fact, aeroplanes with symmetric wings fly as well.

the popular theory of lift generation found in many textbooks is completely wrong! The upper surface doesn't have to be longer than the lower surface to generate lift. The lift occurs because the airfoil turns the flow of air and both the lower and upper surface contribute to the turning.

Wing geometry, NASA Glenn Research Center¹⁵



Sections of two kinds of wings for supersonic flight (Leishman 2024 §26).

§ 10.9 Analytical time integration of position and momentum

In many situations we want to track the position $n(t)$ of a small control volume containing an amount of matter, together with its momentum content. Think about tracking the motion of a tennis ball for instance.

We must therefore use the balance of momentum together with the relation between position and velocity (which is often a [proxy for the balance of matter](#) ▶ §6.7 p. 183):

$$\begin{aligned} \frac{d\mathbf{P}(t)}{dt} &= \mathbf{F}(t) + \mathbf{G}(t) & \text{or} & & \mathbf{P}(t_1) &= \mathbf{P}(t_0) + \int_{t_0}^{t_1} \mathbf{F}(t) dt + \int_{t_0}^{t_1} \mathbf{G}(t) dt, \\ \frac{d\mathbf{r}(t)}{dt} &= \mathbf{v}(t) & \text{or} & & \mathbf{r}(t_1) &= \mathbf{r}(t_0) + \int_{t_0}^{t_1} \mathbf{v}(t) dt. \end{aligned}$$

Even if we have the influx $\mathbf{F}$ and supply $\mathbf{G}$ of momentum are given as [boundary conditions](#) ▶ §6.5 p. 176, and therefore we can find the momentum at future times, we would still need to specify the velocity $\mathbf{v}(t)$ at all times in order to find the future position $\mathbf{r}$ at future times.

Thanks to the constitutive relation $\mathbf{P} = m\mathbf{v}$, knowledge of the momentum at a later timestep allows us to know the velocity at that timestep, or vice versa:

$$\mathbf{v}(t) = \mathbf{P}(t)/m.$$

In simple cases this allows us to find the time dependence of momentum $\mathbf{P}(t)$ and position $\mathbf{r}(t)$ for all time by means of [analytical methods](#) ▶ §6.2 p. 168.

Example: motion of tennis ball

Let's see a simple example with a flying tennis ball. As usual we consider a control volume coinciding with the tennis ball, and which can be localized just by a position vector $\mathbf{r}(t)$ in some coordinate system (t, x, y, z) . Suppose the mass-energy of the ball is $m = 0.06$ kg. Suppose that we know the following boundary conditions:

- Momentum flux is zero: $\mathbf{F}(t) \equiv 0$ N.
- Momentum supply, owing to gravity, is $\mathbf{G}(t) = m\mathbf{g} \equiv -mg \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, constant in time.

Insert these two boundary conditions into the integral expression of the balance of momentum, and solve the simple integrals. If the initial time is t_0 and we call the final time t_1 simply t , that is $t_1 = t$, we find

$$\mathbf{P}(t) = \mathbf{P}(t_0) + m\mathbf{g}(t - t_0) .$$

The initial momentum is $\mathbf{P}(t_0)$, and thanks to the Newtonian relation between momentum and velocity we also find the initial velocity:

$$\mathbf{v}(t_0) = \mathbf{P}(t_0)/m .$$

With the same relation we can also find the velocity at any time t :

$$\begin{aligned} \mathbf{v}(t) &= \mathbf{P}(t)/m = \mathbf{P}(t_0)/m + m\mathbf{g}(t - t_0)/m \\ &= \mathbf{v}(t_0) + \mathbf{g}(t - t_0) \end{aligned}$$

Note that this is simply a linear function ' $C + D t$ ' of the variable t .

Finally replace the expression for $\mathbf{v}(t)$ above in the integral formula for the position $\mathbf{r}(t)$, and integrate. The result is

$$\mathbf{r}(t) = \mathbf{r}(t_0) + \mathbf{v}(t_0)(t - t_0) + \frac{1}{2}\mathbf{g}(t - t_0)^2 .$$

Let's put together the time dependence for momentum and position that we've found:

$$\begin{aligned} \mathbf{P}(t) &= m\mathbf{v}(t_0) + m\mathbf{g}(t - t_0) , \\ \mathbf{r}(t) &= \mathbf{r}(t_0) + \mathbf{v}(t_0)(t - t_0) + \frac{1}{2}\mathbf{g}(t - t_0)^2 . \end{aligned} \tag{10.6}$$

Or, writing explicitly all components (remember that $\mathbf{g} := -g [0, 0, 1]$):

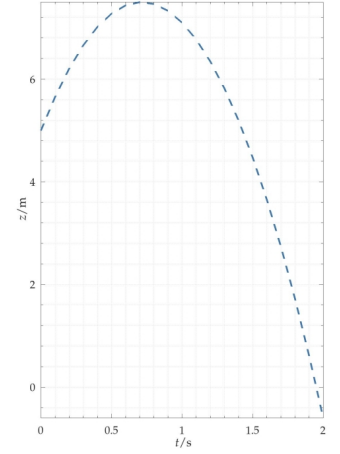
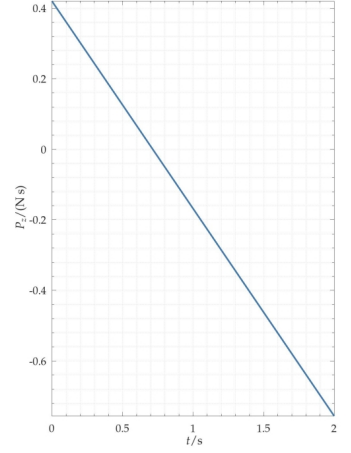
$$\begin{bmatrix} P_x(t) \\ P_y(t) \\ P_z(t) \end{bmatrix} = m \begin{bmatrix} v_x(t_0) \\ v_y(t_0) \\ v_z(t_0) \end{bmatrix} - mg \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} (t - t_0),$$

$$\begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} x(t_0) \\ y(t_0) \\ z(t_0) \end{bmatrix} + \begin{bmatrix} v_x(t_0) \\ v_y(t_0) \\ v_z(t_0) \end{bmatrix} (t - t_0) - \frac{1}{2}g \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} (t - t_0)^2,$$

Therefore if we know the initial position $\mathbf{r}(t_0)$, and the initial velocity $\mathbf{v}(t_0)$ or equivalently the initial momentum $\mathbf{P}(t_0)$, then we can predict the position and momentum for all future times. In particular:

- The x - and y -components of momentum are constant.
- The z -component of momentum decreases linearly with time.
- The x and y position coordinates are constant.
- The z coordinate is a *quadratic* function of time.

Note again how the analytic method manages to tell us beforehand about important features of the time behaviour, as [remarked previously](#) §6.3 p.171. In the present example it tells us for instance that there are no periodic oscillations, and that the z -component of position and momentum can increase or decrease indefinitely.



Plots of $P_z(t)$ and $z(t)$ vs t with initial conditions $t_0 = 0$ s, $P_z(t_0) = 0.42$ N s, $z(t_0) = 5$ m.



Exercise 10.6

In the example just discussed, the dependence of $\mathbf{r}(t)$ and $\mathbf{P}(t)$ on time shown in formula (10.6) was found starting from the integral expression of the balance of momentum and of the relation between position and velocity.

Try to prove again that time dependence, but starting instead from the *differential* expressions of the balance law of momentum and of the relation between position and velocity:

$$\frac{d\mathbf{P}(t)}{dt} = \mathbf{F}(t) + \mathbf{G}(t), \quad \frac{d\mathbf{r}(t)}{dt} = \mathbf{v}(t)$$

and taking into consideration the constitutive relation and the initial conditions:

$$\mathbf{P}(t) = m\mathbf{v}(t), \quad \mathbf{r}(t_0) = \text{known}, \quad \mathbf{v}(t_0) = \text{known}.$$

If you're unsure on how to proceed, follow a strategy similar to the example of the [oscillating chemical reaction](#) ^{§6.2 p. 169}: assume that the position vector has the following quadratic dependence on time:

$$\mathbf{r}(t) = B + C t + D t^2,$$

where B, C, D are generic coefficients. Calculate its time derivative and substitute it in the various equations above. Find out what the coefficients B, C, D should be so that the equations are identically satisfied.

§ 10.10 Numerical time integration of position and momentum

The previous example with the tennis ball was simple enough for the analytic method to be available. But as soon as we introduce more constitutive relations between momentum $\mathbf{P}$ and velocity $\mathbf{v}$ that are more precise than the Newtonian one, or new constitutive relations that connect momentum flux and supply to position or velocity, then finding an analytical solution might be mathematically impossible. We shall therefore have to use [numerical time-integration](#) ^{§6.3 p. 171} methods instead.

Let us study the tennis-ball example with numerical time integration. We already discussed in chapter 6 the [finite-difference approximation of the balance law for momentum](#) ^{eq. (6.5) p. 174} and [that for position](#) ^{eq. (6.9) p. 183}. Thanks to the constitutive relation $\mathbf{P}(t) = m\mathbf{v}(t)$, knowledge of the momentum at every timestep allows us to calculate the velocity at that same timestep, or vice versa. The time iteration therefore looks as follows:

$$\begin{aligned}
 & \mathbf{v}(t_0) = \mathbf{P}(t_0)/m \\
 & \mathbf{r}(t_0 + \Delta t) \approx \mathbf{r}(t_0) + \mathbf{v}(t_0) \Delta t \\
 & \mathbf{P}(t_0 + \Delta t) \approx \mathbf{P}(t_0) + [\mathbf{F}(t_0) + \mathbf{G}(t_0)] \Delta t \\
 & \mathbf{v}(t_0 + \Delta t) = \mathbf{P}(t_0 + \Delta t)/m \\
 & \mathbf{r}(t_0 + 2\Delta t) \approx \mathbf{r}(t_0 + \Delta t) + \mathbf{v}(t_0 + \Delta t) \Delta t \\
 & \mathbf{P}(t_0 + 2\Delta t) \approx \mathbf{P}(t_0 + \Delta t) + [\mathbf{F}(t_0 + \Delta t) + \mathbf{G}(t_0 + \Delta t)] \Delta t \\
 & \dots
 \end{aligned} \tag{10.7}$$

These are vector equations; each one corresponds to three components. Note again how the constitutive relations, in this case Newton's formula for momentum, connect quantities at the same time instant; whereas the balance laws drive the physical phenomenon forward in time (the update of position is a proxy of the balance of matter).

A script implementing the numerical time integration based on the iteration above is shown in table 10.1 on page 258. Recall the [remarks about computer code](#) ^{§0.4 p. 18} used in the present notes. Grey lines draw the plots of $\mathbf{P}_z(t)$ and $z(t)$ against time t .

Let's examine the structure of this script, starting from the bottom, and omitting the lines for plotting, which are physically irrelevant:

Lines 32–49: iteration. At each time step the velocity is calculated from the momentum in **line 35**, using Newton's constitutive relation. Then the momentum content and the position vector at the next timestep are calculated in **lines 39–40**, using the approximate forms of the momentum balance and of the relation between position and velocity.

Lines 17–19: time-iteration parameters. We decide to time-integrate the system until time $t_1 = 2$ s, with a timestep $\Delta t = 0.01$ s.

Lines 13–15: boundary conditions. We assume that the flux of momentum is zero, and the supply of momentum is the gravitational force, constant in time.

Lines 8–11: initial conditions. We decide to start the time integration from $t_0 = 0$ s, with the initial values $\mathbf{r}(t_0) = [0, 0, 5]$ m and $\mathbf{P} = [0, 0, 0.42]$ N s.

Lines 4–6: constants. Our equations use the constant mass-energy of the tennis ball $m = 0.06$ kg, and the magnitude of free-fall acceleration $g = 9.8$ N/kg.



Exercise 10.7

Modify and play around with the script for the tennis ball of table 10.1 on page 258, changing the various parameters in the script and changing the physical situation a little. For example you can try the following:

- Change the mass m , so that the script describes not a tennis ball, but a golf ball or a bowling ball. How does this modify the time behaviour of the vertical position and of the vertical momentum?
- Change the initial momentum $\mathbf{P}$. What happens if the object

initially has a downward momentum? What if there is a horizontal component of momentum? Note that you can plot the x -component of momentum or position by changing 'P(3)' into 'P(1)' and 'r(3)' into 'r(1)'.

- Change the boundary condition for the momentum influx. You can approximately represent the effect of wind by using a horizontal influx of momentum, constant in time; for instance

$$\mathbf{F}(t) = [2, 0, 0] \text{ N}.$$

How does this affect the horizontal and the vertical motion of the object?

Table 10.1 Script for the example of a flying tennis ball

```

1 %% Numerical simulation of object motion with gravity
2 %% Coordinates (t, x, y, z)
3
4 %% Constants
5 m = 0.06; % kg: tennis ball's mass-energy
6 g = 9.8; % N/kg: gravitational acceleration
7
8 %% Initial conditions
9 t = 0; % s: time
10 r = [0, 0, 5]; % m: position vector
11 P = [0, 0, 0.42]; % N s: momentum vector
12
13 %% Boundary conditions
14 F = [0, 0, 0]; % N: momentum influx
15 G = -m * g * [0, 0, 1]; % N: momentum supply
16
17 %% Time-iteration parameters
18 t1 = 2; % s: final time
19 dt = 0.01; % s: time step
20
21 %% Plotting
22 dtplot = t1/360; % time interval between plots
23 tplot = dtplot; % time for next plot
24 figure
25 subplot(2, 1, 1); plot(t, P(3), '^', 'color', '#4477AA')
26 xlabel('time {\it t}/s'); ylabel('z-momentum {\it P_z}/(N s)')
27 axis('tight'); xlim([0, t1]); grid on; hold on
28 subplot(2, 1, 2); plot(t, r(3), 'o', 'color', '#EE6677')
29 xlabel('time {\it t}/s'); ylabel('coordinate {\it z}/m')
30 axis('tight'); xlim([0, t1]); grid on; hold on
31
32 %% Numerical time integration
33 while t < t1
34
35     %% constitutive relations
36     v = P / m;
37
38     %% step forward in time with balance laws
39     t = t + dt;
40     P = P + (F + G) * dt;
41     r = r + v * dt;
42
43     %% plot
44     if t > tplot
45         subplot(2, 1, 1); plot(t, P(3), '^', 'color', '#4477AA')
46         subplot(2, 1, 2); plot(t, r(3), 'o', 'color', '#EE6677')
47         pause(0)
48         tplot = tplot + dtplot;
49     end
50 end

```

Download Octave version
[tennisball_rP.m](#)¹⁶

Download Python version
[tennisball_rP.py](#)¹⁷

Note: In Octave, vectors are simply represented by square brackets, as in [lines 10–11](#), similar to the maths notation. The 1st component of a vector **P** is retrieved by means of round brackets: '**P(1)**'.

§ 10.11 Constitutive relations for momentum flux: materials science]

In everyday situations we can experimentally observe that the constitutive relation between *volume content* of momentum $\mathbf{P}$ and matter is essentially the Newtonian one, when electromagnetic fields can be neglected.

When it comes to *flux* of momentum, that is, *surface force* through a control surface, we instead observe an astounding, seemingly endless variety of constitutive relations which connect momentum flux to matter content and flux, and to the shape and dimensions of the control volumes and surfaces involved. These constitutive relations reflect the spectacular variety of physical behaviour offered by objects and materials that surround us and that make up our bodies.

Consider for instance what happens if you hold and squeeze different kinds of materials between your fingers; the ensuing behaviour is called the **response** of the material. A block of metal or hard wood may not deform at all as you squeeze it; it is a *rigid* response. A piece of rubber may deform and return to its initial shape as you release it; it has an *elastic* response. A piece of plastic or wax may deform and thereafter keep the deformed shape; it has a *plastic* response. With water, it's even impossible to hold it between your fingers! We have *liquid* and *gaseous* materials, which in turn can also have *elastic* (in a slightly different sense than the solid case) or *viscous* responses, or many other kinds of behaviour.

The exploration of these physical behaviours and responses, and of their constitutive relations, goes under the umbrella field of **materials science**¹⁸. The study of these constitutive relations via time integration – typically numerical – has two complementary goals. We perform it to check whether a candidate constitutive relation really reflects the physical behaviour that we experimentally observe. And when we know that a constitutive relation is valid, we may use it to predict physical behaviour in situations where experimentation would be too difficult or costly.

In the following sections we shall study several extremely simplified examples of constitutive relations between the momentum flux through a closed control surface and several other quantities, in particular the dimension and shape of the corresponding control volume. The goal is to simply give you a glimpse of how this kind of studies can be performed, and of the physical responses that we can observe.

§ 10.12 Momentum flux for a simplified piece of material

Simplifying assumptions

To start with, we consider a piece of material in the following simplified situation:

- Momentum flux only occurs through two separate parts of the control surfaces, usually diametrically opposite.
- The shape and dimensions of the closed control surface can be summarized by the distance between the two parts.
- There is no flux of matter through the control surface.
- The matter within the closed control surface has negligible mass-energy, and therefore the momentum content is practically zero at all times.

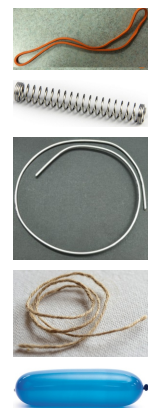
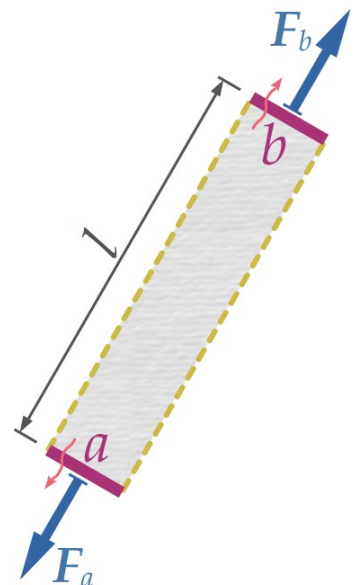
These simplifications are illustrated in the side figure. The closed control surface is represented by the rectangle with dashed yellow and solid red lines. Keep in mind that the control volume does not need to have the elongated shape shown in the figure; it could be round or have a helical shape instead. The matter within the control volume is represented by the greyish fill.

The momentum fluxes, that is, surface forces, only occur at the two partial surfaces represented by the solid red lines, which we may call a and b (or '1' and '2', or other similar symbols). The relevant geometrical dimension of the control volume is the distance between the two partial control surfaces; it may be denoted l .

If the partial surfaces a and b are enough small, we may identify their positions with position vectors $\mathbf{r}_a$ and $\mathbf{r}_b$ (not indicated in the figure). In this case we have $l = |\mathbf{r}_a - \mathbf{r}_b|$.

The momentum effluxes through the partial surfaces are $\mathbf{F}_a$ and $\mathbf{F}_b$, represented by the blue arrows. We focus on the effluxes, rather than the influxes, to make the discussion in the following sections easier. In the figure, the momentum effluxes are depicted as compressive ^{§4.12 p.118}, but note that they might be tensile ^{§4.12 p.119} instead.

With the simplified setup just described we can approximately represent the response of objects like an elastic band, a spring, a piece of wire, a piece of string, and even a balloon or a small cylinder containing a gaseous substance.



Consequences of the simplifications

The simplification of having momentum fluxes through two separate partial control surfaces, together with the simplification of negligible mass-energy content and momentum content, lead to two important conditions that must be satisfied by the momentum fluxes. These conditions are *consequences of the balances of momentum and of angular momentum* §12.5 p.379:

C1 The fluxes $\mathbf{F}_a$ and $\mathbf{F}_b$ must have equal magnitudes and directions, and opposite orientations, that is,

$$\mathbf{F}_a = -\mathbf{F}_b .$$

C2 If the partial control surfaces a and b are enough small, then the fluxes $\mathbf{F}_a$ and $\mathbf{F}_b$ must each have the same direction as the line connecting the two partial surfaces:

$$\mathbf{F}_a , \mathbf{F}_b , \mathbf{r}_a - \mathbf{r}_b \text{ are parallel}$$

this condition can equivalently be written as follows:

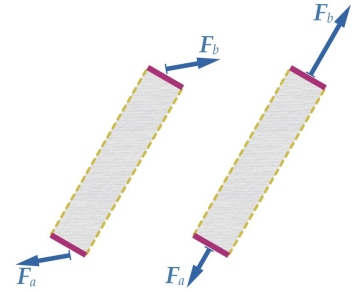
$$\pm \frac{\mathbf{F}_a}{|\mathbf{F}_a|} = \pm \frac{\mathbf{F}_b}{|\mathbf{F}_b|} = \pm \frac{\mathbf{r}_a - \mathbf{r}_b}{|\mathbf{r}_a - \mathbf{r}_b|} .$$

The difference between these two conditions is illustrated in the two side figures. In the left figure the effluxes $\mathbf{F}_a$ and $\mathbf{F}_b$ satisfy condition C1 but break condition C2. In the right figure the effluxes satisfy condition C2 but break condition C1.

In words, these two conditions say that the material in this control volume is a *perfect conductor of momentum and of angular momentum*. The material, owing to its negligible mass-energy, cannot “store” momentum or angular momentum. Therefore any influx of either on one side is instantaneously transferred, unaltered, to the other side as an efflux, and vice versa.

Note that the conditions above are not satisfied any longer, as soon as we consider the more realistic case of a control volume containing some amount of mass-energy. The two fluxes $\mathbf{F}_a$ and $\mathbf{F}_b$ can then have different directions, orientations, and magnitudes.

An important consequence of the two conditions above, is that the direction – but not the magnitude or orientation – of either efflux $\mathbf{F}_a$ or $\mathbf{F}_b$ is completely determined by the positions $\mathbf{r}_a, \mathbf{r}_b$ of the surfaces a, b :



it must be parallel to their difference. Any constitutive relation for the effluxes must therefore have the following mathematical expression:

$$\mathbf{F}_a = -\mathbf{F}_b = f(\dots) \frac{\mathbf{r}_a - \mathbf{r}_b}{l} \quad (10.8)$$

where $l := |\mathbf{r}_a - \mathbf{r}_b|$ and $f(\dots)$ is a *scalar* function that may depend on other physical quantities, such as the length l . This function determines the magnitude and orientation of the vectors $\mathbf{F}_a$ and $\mathbf{F}_b$.

 The condition $\mathbf{F}_a = -\mathbf{F}_b$ is not Newton's third law

Condition C1 is **not** an expression of the principle of symmetry of momentum flux ([Newton's third law](#) ▶ §4.11 p.117), despite looking mathematically similar to that principle. It cannot be, because that principle expresses a property of the flux through *one* surface; the condition above is instead a relation between the fluxes through *two* different surfaces. You shall now prove that this condition is a *consequence of the balance of momentum*.



Exercise 10.8

Prove that condition C1 follows from the balance of momentum. Proceed by taking into consideration that:

- If the mass-energy content in the control volume is negligible, then the momentum content $\mathbf{P}$ is always zero, and its derivative is also always zero:

$$\frac{d\mathbf{P}}{dt} = 0 \text{ N}.$$

- Likewise for the supply:

$$\mathbf{G} = 0 \text{ N}.$$

- By the principle of symmetry of fluxes, the total momentum influx is equal to

$$\mathbf{F} = -\mathbf{F}_a - \mathbf{F}_b.$$

Combine the three formulae above with the [balance of momentum](#) ▶ eq. (10.1) p.227 and derive condition C1.

In chapter 12 we shall prove how condition C2 follows from the balance of angular momentum.



Impossibility of perfect conductors of momentum

In physical situations where the speed of light becomes relevant, such as relativistic conditions or the presence of strong electromagnetic fields, we cannot make the simplifying assumptions above because they break other physical laws (the balance of [boost momentum](#) ▶ [chapter 13 p.395](#) which can be seen as a generalization of the balance of angular momentum). In such situations we cannot treat pieces of matter as rigid or as perfect conductors of momentum. This impossibility became quite clear while researchers tried to model the propagation of light in material such as water or glass: contradictory results kept on appearing, until they realized that they were inadvertently making the simplifying assumptions above (Pfeifer et al. [2007](#)).

§ 10.13 Hookean materials

In the previous sections we discussed a simplified physical situation to study how momentum fluxes, that is, [surface forces](#) ▶ [§4.11 p.115](#), can be connected with geometric quantities in a piece of some kind of material.

We also found that if the mass-energy of the piece of material is negligible, and if there are just two main momentum effluxes $\mathbf{F}_a$ and $\mathbf{F}_b$ across two small surfaces a and b , then these effluxes must satisfy constitutive relations having mathematical form

$$\mathbf{F}_a = -\mathbf{F}_b = f(\dots) \frac{\mathbf{r}_a - \mathbf{r}_b}{l},$$

where $\mathbf{r}_a, \mathbf{r}_b$ are the positions of the small surfaces, which may of course change with time. The function $f(\dots)$ therefore characterizes how the magnitude and orientation of the surface forces depend on other quantities.

We are all familiar with elastic springs, and similar objects which approximately agree with our simplified setup:

- one of their dimensions, which we may call their *length*, is somehow singled out, for instance because more extended, with respect to the other two;
- if we try to modify their length along that particular dimension, they experience (and exert) a surface force.

These objects often show two additional characteristics:

- the surface forces are proportional to the modification of the object's length;
- these objects return to a given length when we don't apply surface forces.



These two characteristics are reflected in a particular mathematical form of the constitutive function $f(\dots)$:

Hookean material

A **Hookean material** is characterized by momentum effluxes related to its length $l := |\mathbf{r}_a - \mathbf{r}_b|$ by the constitutive relation

$$\mathbf{F}_a = -\mathbf{F}_b = -k(l - l_n) \frac{\mathbf{r}_a - \mathbf{r}_b}{l}, \quad (10.9)$$

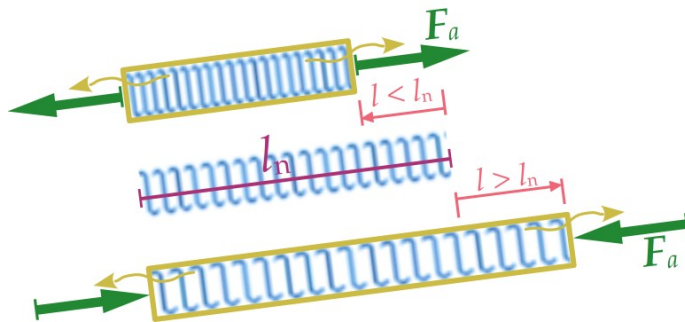
called **Hooke's law**, where l_n and k are positive constants.

The constant l_n is the *natural length* or *unstretched length* of the piece of material; k is the **elastic constant** of the material. The variable difference $l - l_n$ is called the *elongation* of the material; if negative, it means that the material has shortened.

Note the signs of the surface forces *exerted* (momentum effluxes) and *experienced* (momentum influxes) by the piece of material:

- When $l > l_n$, the surface forces are *tensile*: the material experiences a force that stretches it, and exerts a force directed towards itself.
- When $l < l_n$, the surface forces are *compressive*: the material experiences a force that compresses it, and exerts a force directed away from itself.
- When $l = l_n$, the material does not experience or exert any surface force.

The figure below illustrates these situations for the surface force $\mathbf{F}_a$ *exerted* (momentum efflux) by the piece of material.



In some physical situations the natural length of the piece of material is negligible compared to the amounts by which it is typically stretched,

so we can put $l_n = 0$ m. Hooke's law then takes on the very simple form

$$\mathbf{F}_a = -\mathbf{F}_b = -k(\mathbf{r}_a - \mathbf{r}_b).$$

Note that in this case the material always exerts a tension, never a pressure.

The elastic constant k determines the *stiffness* of the Hookean material, or how hard it is to stretch or compress it. Take a particular magnitude of the elongation $|l - l_n|$; let's say 0.01 m. If $k = 2$ N/m, then the force corresponding to that elongation has magnitude 0.02 N; if the elastic constant is ten times larger, $k = 20$ N/m, then the force corresponding to that elongation is 0.2 N instead: also ten times larger. In other words, for larger values of k we need larger forces to obtain a given amount of elongation.

The elastic constant k can *never be negative*. This is a consequence of the *balance of entropy* which we'll discuss in chapter (15.1).



Exercise 10.9

1. Verify that the elastic constant k has physical dimension force/length.
2. What if the elastic constant k were negative? Examine Hooke's law (10.9), and try to deduce what would happen if you stretched out such a bizarre kind of material starting from its natural length.
3. Hooke's law is usually stated for the surface forces *exerted* by the material; in other words, for the *effluxes* rather than for the *influxes*.

Denote by $\mathbf{F}_{a,\text{in}}$ the influx through the surface a . Write Hooke's law for $\mathbf{F}_{a,\text{in}}$ rather than for $\mathbf{F}_a$.

§ 10.14 Hookean spring and harmonic oscillator

The zero-mass simplification that we're making to study the momentum fluxes of materials has a drawback: we cannot study the motion of such a piece of material *alone*. By saying that its mass-energy is negligible, we're effectively saying that its momentum is too small to be measured. We're thus forfeiting exactly the quantity allows us to study its motion.

This is apparent from the fact that the balance of momentum for this unrealistic physical system *degenerates* ^{§10.12 p. 260} to the equation $\mathbf{F}_a(t) = -\mathbf{F}_b(t)$, which connects two quantities at the *same* time instant t : we have lost the expression of the balance that "drives" the system forward in time.

One way to bypass this situation and get a glimpse at the motion that the material under study leads to, is to connect the control volume of the material with two other control volumes containing matter with non-zero mass-energy. For example, to study the behaviour of a massless spring we attach two small bodies with non-zero masses at its ends. We can thus track the momenta and movement of these auxiliary bodies and, indirectly, the motion and the effects of the momentum fluxes of the material.

In this section we study this three-body physical system step by step, in order to study its motion later, by numeric and if possible analytical methods.



Exercise 10.10

Before continuing, try to do on your own the analysis of two small bodies connected by a piece of Hookean material. Consider in particular:

- How would you set up a coordinate system and closed control surfaces to describe this physical phenomenon? how many control volumes?
- How many applications of the balance of momentum would you need to make?
- What would be the volume contents, fluxes, and supplies in these balances?
- What would be the relevant constitutive relations?
- What would be the initial conditions and the boundary conditions?

Consider two bodies of matter, call them a and b , of small extension compared to the distance between them. Attach them at the two ends of a piece of Hookean material, such as a small spring. For simplicity, in this analysis we take the natural length of the piece to be negligible: $l_n \approx 0$ m.

Choose a coordinate system (t, x, z) at rest with the Earth's surface, with z pointing upward. The only momentum supply, if considered, is given by the constitutive equation for [gravitational force](#) §10.3 p. 230.

In terms of control surfaces, we have three of them: one tightly enveloping body a , one body b , and a third enveloping the piece of Hookean material. Let us analyse each one in turn, writing down its relevant balances. Note that all quantities below, except the masses and the elastic constant k , depend on the time t .

Body a : The control volume for body a is small and can be characterized simply by its position vector $\mathbf{r}_a$. The mass-energy in this control volume is m_a .

By construction, [conservation of matter automatically holds](#) ^{§7.12 p. 217} for this control volume. It remains, hidden, in the relation between the position and velocity: $\mathbf{v}_a(t) = d\mathbf{r}_a(t)/dt$.

For the amount of momentum in this control volume, use the [Newtonian constitutive equation for momentum](#) ^{§10.2 p. 228} $\mathbf{P}_a = m_a \mathbf{v}_a$.

Assume that the only momentum flux across the closed control surface occurs on the small portion of surface in common with the control volume of the Hookean material; this portion is represented by the **thick red lines** in the side picture below. Across that small surface there is an **influx** $\mathbf{F}_a$. We are using this notation for the fluxes: ' $\mathbf{F}_a$ ' is the influx in body a coming from the Hookean material, the "spring"; analogously for $\mathbf{F}_b$ and body b .

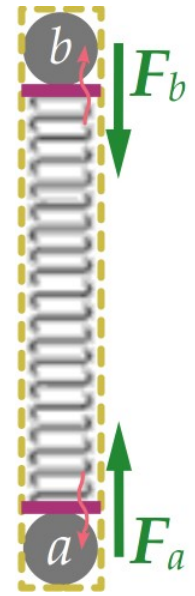
If we consider gravity effects, we can use the constitutive equation for gravitational force $\mathbf{G}_a = -m_a g [0, 1]$.

For this control volume we have therefore the following momentum balance and constitutive relations:

momentum balance	$\frac{d\mathbf{P}_a(t)}{dt} = \mathbf{F}_a(t) + \mathbf{G}_a(t)$
velocity	$\frac{d\mathbf{r}_a(t)}{dt} = \mathbf{v}_a(t)$
constit. relation	$\mathbf{P}_a(t) = m_a \mathbf{v}_a(t)$
constit. relation	$\mathbf{G}_a(t) = -m_a g [0, 1]$

Body b : An analogous analysis can be made for the control volume of body b . It has position $\mathbf{r}_b$, total mass-energy m_b , total momentum $\mathbf{P}_b = m_b \mathbf{v}_b$.

The only momentum flux occurs across the portion of control surface in common with the control volume of the Hookean material; the **influx** there is $\mathbf{F}_b$.



The control volumes are indicated by **dashed yellow lines**. The surfaces in common between the Hookean material and the two bodies are indicated by two **thick dark-red lines**. The fluxes $\mathbf{F}_a$, $\mathbf{F}_b$ occur across these surfaces, but their vectors are placed on the side in the illustration for clarity. The velocities $\mathbf{v}_a$, $\mathbf{v}_b$, and momentum supplies $\mathbf{G}_a$, $\mathbf{G}_b$ are not shown.

For this control volume we have the following momentum balance and constitutive relations:

momentum balance	$\frac{d\mathbf{P}_b(t)}{dt} = \mathbf{F}_b(t) + \mathbf{G}_b(t)$
velocity	$\frac{d\mathbf{r}_b(t)}{dt} = \mathbf{v}_b(t)$
constit. relation	$\mathbf{P}_b(t) = m_b \mathbf{v}_b(t)$
constit. relation	$\mathbf{G}_b(t) = -m_b g [0, 1]$

Spring: The control volume for the piece of Hookean material has two negligible dimensions and can be represented by a long, narrow tube extending between the extremities at $\mathbf{r}_a$ and $\mathbf{r}_b$. The two surfaces at these extremities are in common with the control volumes for bodies a and b . The amount of matter and matter flux in this control volume are considered to be zero. The same is true for the mass-energy and the momentum content and supply.

At the surface where the piece is in contact with body a there is an **influx** of momentum equal to $-\mathbf{F}_a$. The minus sign comes from the [symmetry of flux](#) §4.6 p. 106, because $\mathbf{F}_a$ is the influx into body a .

Analogously, at the surface of contact with body b , there is a momentum influx equal to $-\mathbf{F}_b$.

The balance of momentum for this control volume degenerates into the equation

$$\mathbf{F}_a = -\mathbf{F}_b,$$

as already discussed. Note that the *total* influx of momentum into this control volume is also zero.

The Hookean constitutive relation (10.9) says that the momentum **efflux** across the surface of contact with body a is

$$\mathbf{F}_a = -k (\mathbf{r}_a - \mathbf{r}_b).$$

The momentum balance and constitutive relations for the control volume of the Hookean material are therefore:

momentum balance	$\mathbf{F}_a = -\mathbf{F}_b$
simplification	$\mathbf{P}(t) = 0$
simplification	$\mathbf{G}(t) = 0$
Hooke constit. relation	$\mathbf{F}_a(t) = -k [\mathbf{r}_a(t) - \mathbf{r}_b(t)]$

The setup for the description of this physical system is thus complete. This setup was maybe intuitively clear, but I'd like you to stop for a moment and note all the subtle steps and details that it involves. We often *cannot* reason *intuitively* in the analysis of more complex physical systems, and need to spell out their description step by step, to avoid neglecting important details. It is therefore a good exercise to do this kind of analysis for a simpler system, as we just did.

In this analysis, note how some control volumes have parts of their control surfaces in common. These common surfaces are important because they connect the fluxes of momentum and other quantities between the various control volumes, by means of the principle of symmetry of flux. In modelling extended solids and fluids we often use a *grid* of control volumes, so that the surface of one control volume has regions in common with the surrounding control volumes.

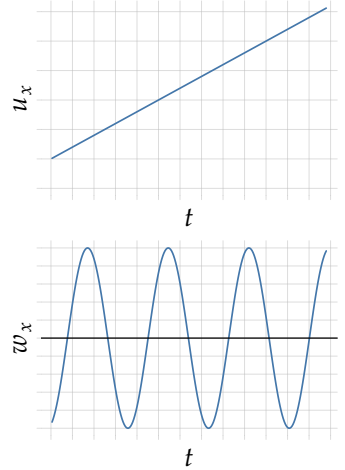
The set of equations above is usually solved analytically by introducing two alternative vector variables:

$$\mathbf{u} := \frac{m_a \mathbf{r}_a + m_b \mathbf{r}_b}{m_a + m_b} \quad \mathbf{w} := \mathbf{r}_a - \mathbf{r}_b$$

the first, $\mathbf{u}$, is called the *centre of mass* of this physical system. In terms of these variables the whole set of equations above reduces to these two linear differential equations with constant coefficients (each equation is a set of three, one for each component):

$$\frac{d^2 \mathbf{u}(t)}{dt^2} = 0 \quad \frac{d^2 \mathbf{w}(t)}{dt^2} + k \frac{m_a + m_b}{m_a m_b} \mathbf{w}(t) = 0$$

that can be solved analytically.



Every component of $\mathbf{u}(t)$ is a linear function of time, and every component of $\mathbf{w}(t)$ has a harmonic dependence on time with period $\frac{1}{2\pi} \sqrt{\frac{m_a m_b}{k(m_a + m_b)}}$. The slopes and intercepts of $\mathbf{u}(t)$ and the amplitudes and phases of $\mathbf{w}(t)$ depend on the initial conditions.



Exercise 10.11

Try to refresh your knowledge of linear differential equations with constant coefficients, and find the general solution for the equations above.

Then express this solution in terms of the positions $\mathbf{r}_a$ and $\mathbf{r}_b$.

§ 10.15 A strategy for numerical time integration.

The concept of physical state

The present section is somewhat peculiar because we shall discuss two very different topics, one practical and one conceptual:

- A step-by-step strategy to write a script for numerical time integration for physical systems, starting from the physical analysis of the system.
- The concepts of **physical state** and **state variables** of a physical system.

The reason for a joint section is that the scripting strategy is very helpful for understanding the concept of physical state and state variables; and vice versa this concept is key to understanding some steps of the scripting strategy.

The physical system of two masses connected by a Hookean spring can be solved, and its behaviour predicted, analytically. But finding analytical solutions becomes more difficult or even impossible as soon as we consider [non-Hookean springs](#) ^{› §10.16 p. 280} and more general constitutive relations, or more masses. We then resort to numerical time-integration methods to predict the behaviour of such systems.

The [essential idea of such numerical methods](#) ^{› §6.3 p. 171} was introduced in an earlier chapter. Let us now apply it to the masses-and-spring system. If we write an algorithm that numerically simulates a Hookean spring, it is then very easy to generalize it to a non-Hookean spring and to even more complex constitutive relations.

We wrote our [first time-integration algorithm](#) ^{› §6.8 p. 184} by proceeding intuitively, without following any particular scheme. Our present physical system involves more quantities and more constitutive equations. It is therefore convenient to find a more systematic way to build a simulation algorithm. Let's see the main steps using the present physical system as an example.

In trying to reach a systematic way we shall also realize again the importance of the universal balance laws.

Overview of the relevant equations

First we must have a clear list of the balance laws and constitutive relations that apply to the physical system. In our case [§10.14 p. 265](#) they are

$\frac{d\mathbf{P}_a(t)}{dt} = \mathbf{F}_a(t) + \mathbf{G}_a(t)$	momentum balance for body a
$\frac{d\mathbf{P}_b(t)}{dt} = \mathbf{F}_b(t) + \mathbf{G}_b(t)$	momentum balance for body b
$\frac{d\mathbf{r}_a(t)}{dt} = \mathbf{v}_a(t)$	velocity of body a
$\frac{d\mathbf{r}_b(t)}{dt} = \mathbf{v}_b(t)$	velocity of body b
$\mathbf{F}_b(t) = -\mathbf{F}_a(t)$	momentum balance for spring
$\mathbf{P}_a(t) = m_a \mathbf{v}_a(t)$	Newtonian momentum body a
$\mathbf{P}_b(t) = m_b \mathbf{v}_b(t)$	Newtonian momentum body b
$\mathbf{G}_a(t) = -m_a g [0, 1]$	momentum supply body a
$\mathbf{G}_b(t) = -m_b g [0, 1]$	momentum supply body b
$\mathbf{F}_a(t) = -k [\mathbf{r}_a(t) - \mathbf{r}_b(t)]$	Hooke's law

where each vector equation represents two equations: one for component y , one for z . We could have written the balance laws in integral, rather than differential, form. The latter form is simply more compact. Recall the peculiarity about the momentum balance for the spring: since the spring is assumed to always have zero momentum, the balance simplified to the equation $\mathbf{F}_b(t) = -\mathbf{F}_a(t)$, without a time derivative.

In order to do numerical time integration, we approximate the balance laws and the velocities with [finite-difference approximations](#) [§6.3 p. 173](#). Our

set of equations is therefore rewritten as follows:

$\mathbf{P}_a(t + \Delta t) \approx \mathbf{P}_a(t) + [\mathbf{F}_a(t) + \mathbf{G}_a(t)] \Delta t$	momentum balance for body a
$\mathbf{P}_b(t + \Delta t) \approx \mathbf{P}_b(t) + [\mathbf{F}_b(t) + \mathbf{G}_b(t)] \Delta t$	momentum balance for body b
$\mathbf{r}_a(t + \Delta t) \approx \mathbf{r}_a(t) + \mathbf{v}_a(t) \Delta t$	velocity of body a
$\mathbf{r}_b(t + \Delta t) \approx \mathbf{r}_b(t) + \mathbf{v}_b(t) \Delta t$	velocity of body b
$\mathbf{F}_b(t) = -\mathbf{F}_a(t)$	momentum balance for spring
$\mathbf{P}_a(t) = m_a \mathbf{v}_a(t)$	Newtonian momentum body a
$\mathbf{P}_b(t) = m_b \mathbf{v}_b(t)$	Newtonian momentum body b
$\mathbf{G}_a(t) = -m_a g [0, 1]$	momentum supply body a
$\mathbf{G}_b(t) = -m_b g [0, 1]$	momentum supply body b
$\mathbf{F}_a(t) = -k [\mathbf{r}_a(t) - \mathbf{r}_b(t)]$	Hooke's law

This is now our starting point. Let's see a reasoned set of steps to write a script that implements these equations.

A strategy for writing numerical time-integration algorithms

Our strategy can be divided into seven systematic steps, which we reason out now. The script can be written along, as we follow them. After each step, take a look at the example script of table 10.2 on page 278, and locate the lines where that step was implemented.

0. Find any constants appearing in the equations. Constitutive relations typically contain constant quantities, that is, quantities that don't change in time. These constants must be known in order to simulate the system. They are therefore **declared at the beginning of the script**. In our case the constants are the masses m_a , m_b , the gravitational acceleration g , and the elastic constant k . Other examples could be fixed lengths, areas, volumes.

1. Find which equations drive the system forward in time. Some equations must tell us the values of some quantities at the next time point $t + \Delta t$, given quantities at the present time point t . They must therefore be written in the **core of the time-stepping loop**, together with the update of the time variable t .

These driving equations easy to identify: ‘ $t + \Delta t$ ’ appears in them. In our case they are the balances for the momenta $\mathbf{P}_a$, $\mathbf{P}_b$, and the velocity equations for the two bodies, having two coordinate components each.

The fact that the forward-driving equations are fundamental balances is not a coincidence: as mentioned several times already, *the universal balances are important because they relate later times to earlier times*. This fact becomes especially evident when we write a simulation algorithm.

In our case there’s also one more balance, the momentum balance for the spring, but owing to the special assumptions about the spring (being massless), it simplifies to a same-time equation.

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Physical state and state variables

2. Choose the state variables for the physical system. In the time loop, the driving equations calculate later quantities from a particular set of present quantities. We must therefore make sure that the values of all necessary present quantities are defined before the core lines, and also that they are known before the time loop begins. In our case we need 8 vector quantities, underlined below, with two components each, for a total of 16 numbers:

$$\begin{aligned}\mathbf{P}_a(t + \Delta t) &\approx \underbrace{\mathbf{P}_a(t)}_{\text{needed}} + [\underbrace{\mathbf{F}_a(t)}_{\text{needed}} + \mathbf{G}_a(t)] \Delta t \\ \mathbf{P}_b(t + \Delta t) &\approx \underbrace{\mathbf{P}_b(t)}_{\text{needed}} + [\underbrace{\mathbf{F}_b(t)}_{\text{needed}} + \mathbf{G}_b(t)] \Delta t \\ \mathbf{r}_a(t + \Delta t) &\approx \underbrace{\mathbf{r}_a(t)}_{\text{needed}} + \underbrace{\mathbf{v}_a(t)}_{\text{needed}} \Delta t \\ \mathbf{r}_b(t + \Delta t) &\approx \underbrace{\mathbf{r}_b(t)}_{\text{needed}} + \underbrace{\mathbf{v}_b(t)}_{\text{needed}} \Delta t\end{aligned}$$

The supplies $\mathbf{G}_a$, $\mathbf{G}_b$ are given as boundary conditions, so we don’t need to worry about them.

The eight quantities above are not all independent, owing to the constitutive relations that relate some of them. For instance, if we assign the velocity $\mathbf{v}_a$, then the momentum $\mathbf{P}_a$ is determined by the constitutive relation $\mathbf{P}_a = m_a \mathbf{v}_a$; and if we assign the positions $\mathbf{r}_a$, $\mathbf{r}_b$, then the force $\mathbf{F}_a$ is determined by the constitutive relation $\mathbf{F}_a = -k(\mathbf{r}_a - \mathbf{r}_b)$.

Our task now is to find a *minimum* set of available quantities from which all others can be determined via constitutive relations. As a rule of thumb, the minimum number of quantities is given by

$$(\text{number needed in driving equations}) - (\text{number of same-time constitutive relations})$$

considering all vector components. In our case the same-time relations are four, with two components each:

$$\begin{aligned}\mathbf{F}_b(t) &= -\mathbf{F}_a(t) \\ \mathbf{P}_a(t) &= m_a \mathbf{v}_a(t) \quad \mathbf{P}_b(t) = m_b \mathbf{v}_b(t) \\ \mathbf{F}_a(t) &= -k [\mathbf{r}_a(t) - \mathbf{r}_b(t)]\end{aligned}$$

so we should find a minimum set of $(2 \times 8) - (2 \times 4) = 8$ quantities.

The choice is not unique. Convince yourself that this minimal set could be used:

$$\mathbf{r}_a, \mathbf{v}_a, \mathbf{r}_b, \mathbf{v}_b$$

or alternatively this:

$$\mathbf{r}_a, \mathbf{P}_a, \mathbf{F}_a, \mathbf{v}_b$$

Usually we prefer a minimal set that consists of easily observable or measurable quantities, like positions, velocities, temperatures. In the present case let's agree to use $\mathbf{r}_a, \mathbf{v}_a, \mathbf{r}_b, \mathbf{v}_b$. Each of these has two components: $x_a, z_a, x_b, \dots$ and so on, for a total of eight.

These are called *state variables* of the physical system:



State and state variables of a physical system

The **physical state** of a physical system is the minimal amount of information needed to drive the system from one coordinate-time instant to the next.

The state is usually encoded in a minimal set of time-dependent quantities, called **state variables**. The choice of state variables is often not unique. In this case we say that different sets of state variables *represent the same state*.

The values of the state variables at the beginning of a numerical time integration are called **initial conditions**.

Note the subtle difference between *state* and *state variables*. In our current example, suppose for a moment that $m_a = m_b = 2$ kg. The set

$$(\mathbf{r}_a, \mathbf{v}_a, \mathbf{r}_b, \mathbf{v}_b)$$

and the set

$$(\mathbf{r}_a, \mathbf{P}_a, \mathbf{r}_b, \mathbf{P}_b)$$

are *different* state variables. But the values

$$\mathbf{r}_a = [1, 3] \text{ m}, \quad \mathbf{v}_a = [0, 2] \text{ m/s}, \quad \mathbf{r}_b = [7, 5] \text{ m}, \quad \mathbf{v}_b = [-1, 0] \text{ m/s}$$

and

$$\mathbf{r}_a = [1, 3] \text{ m}, \quad \mathbf{P}_a = [0, 4] \text{ N s}, \quad \mathbf{r}_b = [7, 5] \text{ m}, \quad \mathbf{P}_b = [-2, 0] \text{ N s}$$

represent the *same state*. This is because the values of the first set of state variables correspond to the values of the second set of state variables: the information they provide is equivalent.

We have chosen the state variables of our system to be $(\mathbf{r}_a, \mathbf{v}_a, \mathbf{r}_b, \mathbf{v}_b)$. The values of the state variables at the initial time, that is, the initial conditions, need to be **declared before the time-stepping loop**, together with the value of the initial time.

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3. Determine the boundary conditions. The time dependence of some fluxes, supplies, and volume contents may be known beforehand. These are what we call boundary conditions. We declare them before the time iteration. In our case the gravity supplies of the two bodies, $\mathbf{G}_a(t) = -m_a g [0, 1]$ and $\mathbf{G}_b(t) = -m_b g [0, 1]$ are constant, because the masses, the gravitational acceleration, and the unit vector $[0, 1]$ are constant.

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4. From the state variables, determine the quantities necessary for forward-driving. Recall that the forward-driving equations require, at each time step, the 2×8 quantities

$$\mathbf{P}_a(t), \mathbf{F}_a(t), \mathbf{P}_b(t), \mathbf{F}_b(t), \mathbf{r}_a(t), \mathbf{v}_a(t), \mathbf{r}_b(t), \mathbf{v}_b(t)$$

We must therefore find these from our state variables, at each new time step, by using any necessary constitutive relations. In the present case:

$$\mathbf{P}_a(t) = m_a \mathbf{v}_a(t)$$

$$\mathbf{F}_a(t) = -k [\mathbf{r}_a(t) - \mathbf{r}_b(t)]$$

$$\mathbf{P}_b(t) = m_b \mathbf{v}_b(t)$$

$$\mathbf{F}_b(t) = -\mathbf{F}_a(t)$$

$$\mathbf{r}_a(t) \quad \text{given}$$

$$\mathbf{v}_a(t) \quad \text{given}$$

$$\mathbf{r}_b(t) \quad \text{given}$$

$$\mathbf{v}_b(t) \quad \text{given}$$

Note that to find the force $\mathbf{F}_b$ from the state variables we needed to use two equations.

In the code, the formulae that calculate the quantities necessary to the forward-driving lines from the state variables **need to be written within the time-stepping loop, right before the forward-driving lines.**

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5. Find the new values of the state variables from the time-updated quantities. The time-stepping loop of our code is now able to determine quantities at the next time step $t + \Delta t$, given the state at the present time t . The quantities that we obtain at the next time step are

$$\mathbf{P}_a(t + \Delta t), \mathbf{P}_b(t + \Delta t), \mathbf{r}_a(t + \Delta t), \mathbf{r}_b(t + \Delta t)$$

We now want to be able to start the next loop iteration. The next iteration needs the new value of the state variables $(\mathbf{r}_a, \mathbf{v}_a, \mathbf{r}_b, \mathbf{v}_b)$; what we have is $\mathbf{P}_a, \mathbf{P}_b, \mathbf{r}_a, \mathbf{r}_b$. We need therefore to **convert the updated quantities back to the state variables, within the time loop.** This is again done by using same-time constitutive equations. In our case we use

$$\begin{aligned} \mathbf{r}_a(t + \Delta t) & \text{ given} \\ \mathbf{r}_b(t + \Delta t) & \text{ given} \\ \mathbf{v}_a(t + \Delta t) &= \mathbf{P}_a(t)/m_a \\ \mathbf{v}_b(t + \Delta t) &= \mathbf{P}_b(t)/m_b \end{aligned}$$

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6. Set the time-iteration parameters. The discrete timestep Δt is a constant in our scripts. In more advanced techniques the timestep may be *adaptive*, that is, it may change at every iteration, depending on particular conditions. This can lead to smaller numerical errors.

For how long should our simulation run? This obviously depends on the application. We may want to run it for a given amount of time. Or we may want to stop it as soon as some condition is met; for instance, if the position or speed of a body reach particular values. The possibilities are endless.

We set these conditions before the time-loop.

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After these seven steps, the essential part of our numerical-time-integration script is ready. See table 10.2 on the next page. The script will probably need additional lines to perform tasks that depend on the specific problem. For example we may want to store the numerical values of some quantities at different times, for later analysis; or plot some of them as the simulation evolves. The implementation of these tasks depend on the

programming language, and we therefore cannot discuss them in detail here.

It is also easy to generalize the script in various ways. For instance, we can use the more general version of Hooke's law with a non-zero natural length:

$$\mathbf{F}_a(t) = -k(l - l_n) \frac{\mathbf{r}_a - \mathbf{r}_b}{l}.$$

An example script that also implements plotting can be downloaded as an [Octave version](#) `hooke_spring_simple.m`¹⁹ and a [Python version](#) `hooke_spring_simple.py`²⁰.

Table 10.2 Example script for numerical time integration of spring & bodies system.

```

%%% 2D simulation of two bodies connected by Hookean material
%% Coordinates (t, x, z)

%% 0. Constants
ma = 2; % kg: mass of object a
mb = 2; % kg: mass of object b
g = 9.8; % N/kg: gravitational acceleration
k = 5; % N/m: spring constant

%% 2. State variables and initial conditions
t = 0; % s: initial time
ra = [-3, 0]; % m: initial position of object a
rb = [3, 0]; % m: initial position of object b
va = [0, 10]; % m/s: initial velocity of object a
vb = [0, 0]; % m/s: initial velocity of object b

%% 3. Boundary conditions
Ga = -ma * g * [0, 1]; % N: gravity supply in object a
Gb = -mb * g * [0, 1]; % N: gravity supply in object b

%% 6. Parameters for time integration
t1 = 5; % s: final time
dt = 0.0001; % s: time step

%% Numerical time integration
while t < t1

    %% 4. Calculate forward-driving quantities from state
    Pa = ma * va;
    Pb = mb * vb;
    Fa = -k * (ra - rb);
    Fb = -Fa; % balance of momentum for Hookean material

    %% 1. Balances drive forward in time
    t = t + dt;
    Pa = Pa + (Fa + Ga) * dt;
    Pb = Pb + (Fb + Gb) * dt;
    ra = ra + va * dt;
    rb = rb + vb * dt;

    %% 5. Calculate new state from forward-driven quantities
    va = Pa / ma;
    vb = Pb / mb;

end

```

Download Octave version
[hooke_spring_simple.m](#)²¹

Download Python version
[hooke_spring_simple.py](#)²²



Exercise 10.12

1. Examine our [previous script for the oscillating reaction](#) ^{§6.1 p. 182}, and pinpoint where the seven building steps above are implemented.
2. Which quantities constitute the *state variables* in that script?
3. Implement the pseudo-code of table 10.2 on page 278 in your preferred programming language.

Try to write your script so that some values of the state variables ($\mathbf{r}_a, \mathbf{r}_b, \mathbf{v}_a, \mathbf{v}_b$), and their corresponding t values are plotted or saved at regular intervals.

(If you had difficulties writing your script, for the following exercises you can download and use [hooke_spring.m](#) (Octave)²³ or [hooke_spring.py](#) (Python)²⁴ as a starting point.)

4. Run your script with the following values:

$$\begin{aligned}
 t_0 &= 0 \text{ s} & t_1 &= 10 \text{ s} & \Delta t &= 0.0001 \text{ s} \\
 m_a &= m_b = 2 \text{ kg} & k &= 5 \text{ N/m} \\
 \mathbf{r}_a(t_0) &= [-3, 0] \text{ m} & \mathbf{r}_b(t_0) &= [3, 0] \text{ m} \\
 \mathbf{v}_a(t_0) &= [0, 0] \text{ m/s} & \mathbf{v}_b(t_0) &= [0, 0] \text{ m/s} \\
 \mathbf{G}_a(t_0) &= [0, 0] \text{ N} & \mathbf{G}_b(t_0) &= [0, 0] \text{ N}
 \end{aligned}$$

- Plot the x -coordinate of body a against time t . What kind of time dependence do you observe? can you explain it intuitively?
 - Now plot the trajectories of the two bodies, that is, z_a against x_a , and z_b against x_b . What do you observe? can you explain it intuitively?
 - Plot, against time t , the x - and z -components of the *total* momentum $\mathbf{P}_a + \mathbf{P}_b$ for the system composed by the two bodies and the spring. How do these component change? Why?
5. Run the script with the same parameter as before but the following initial velocity values:

$$\mathbf{v}_a(t_0) = [1, 1] \text{ m/s} \quad \mathbf{v}_b(t_0) = [-1, -1] \text{ m/s}$$

- Plot again the x -coordinate of body a against time t . Is the time dependence different from the previous simulation? How do the trajectories of the two bodies look like this time?

- Plot again the components of the total momentum against time. How do they change? Why have they the same time dependence as before?

6. Run the script with the following initial velocity values:

$$\mathbf{v}_a(t_0) = [2, 2] \text{ m/s} \quad \mathbf{v}_b(t_0) = [1, 1] \text{ m/s}$$

- Plot the trajectories of the two bodies. What do you observe this time?
 - How do the components of the total momentum differ from the previous simulation? Try to explain why.
7. Play with all the parameter values and initial conditions, and see what happens. Before simulating, try to intuitively predict what the behaviour of the system will be.

§ 10.16 Non-Hookean and more general materials

Hooke's law ^{§10.13 p. 263} is just a very simple example of how the magnitude and orientation of the momentum influx in our simplified piece of material can depend on other quantities:

$$\mathbf{F}_a = -\mathbf{F}_b = f(\dots) \frac{\mathbf{r}_a - \mathbf{r}_b}{l}.$$

But we can easily find materials and objects that deviate from Hooke's law. We can try to model their behaviour by choosing alternative expressions for the constitutive function $f(\dots)$. We discuss some examples below.

Small mathematical changes in the constitutive function $f(\dots)$ can lead to different, interesting new physical behaviours; take for instance a look at the plots in Exercise 10.13 below. Alternative mathematical expressions of this constitutive function may be difficult or impossible to be approached with analytical methods, but can easily be time-integrated numerically.

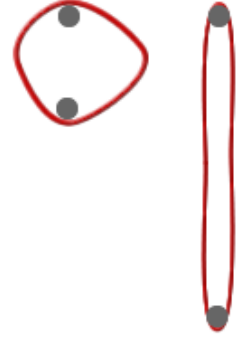
It's therefore important that you *play* with the scripts by changing the constitutive function and observing what happens. In some of these explorations your script may break: it means that you've hit upon some extreme physical situation, which need careful numerical handling. Just

start anew and try different values of the physical parameters, or of the time step, or some less drastic changes to the constitutive function.

Rubber bands and strings

Items like rubber bands and strings deviate from Hooke's law in a notable way: they exert a purely tensile surface force, and they only exert it if they are stretched beyond their natural, relaxed length. If the distance between their extremities is shorter than their natural length, then no forces are exerted – no momentum fluxes occur.

This kind of behaviour can be modelled, at least qualitatively, by a constitutive equation like the following:



Constitutive equation for rubber bands and strings

Suppose that the two extremities of a piece of material, such as a rubber band, have position vectors $\mathbf{r}_a$ and $\mathbf{r}_b$. The natural length of the piece is l_n . Then the momentum effluxes at the extremities $\mathbf{r}_a$ and $\mathbf{r}_b$ are approximately given by

$$\mathbf{F}_a = \begin{cases} 0, & \text{if } l \leq l_n \\ -k(l - l_n) \frac{\mathbf{r}_a - \mathbf{r}_b}{l}, & \text{if } l > l_n \end{cases} \quad (10.10)$$

with $l := |\mathbf{r}_a - \mathbf{r}_b|$.

A string can be approximately modelled by such a constitutive equation with a large constant k .

Pairwise forces

By using peculiar functions $f(\dots)$, we can also model control volumes that actually contain electromagnetic fields. These fields transport momentum almost instantaneously from some body of matter to another. We call them *pairwise forces*. An example is the force of the so-called *Lennard-Jones potential*:

$$\mathbf{F}_a = \frac{\epsilon}{l} \left[12 \left(\frac{\sigma}{l} \right)^{12} - 6 \left(\frac{\sigma}{l} \right)^6 \right] \frac{\mathbf{r}_a - \mathbf{r}_b}{l} \quad (10.11)$$

Which is often used to model the flux of momentum among molecules of some fluids and solids.



Exercise 10.13

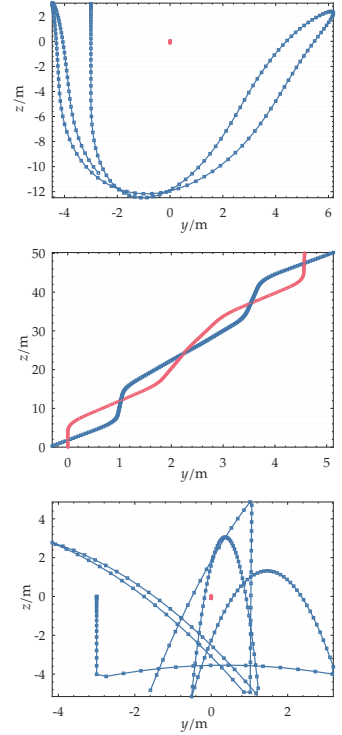
(You can download `hooke_spring.m` (Octave)²⁵ or `hooke_spring.py` (Python)²⁶ as a starting point.)

1. Modify the script for the Hookean material in table 10.2 on page 278 so as to implement the non-Hookean constitutive relation (10.10). This constitutive relation can be implemented with an if-statement. If you need some help, download the script rubberband (Octave: `rubberband.m`²⁷; Python: `rubberband.py`²⁸), which implements this kind of constitutive relation.
2. Then numerically time-integrate the system with the following four sets of parameters and initial values. Find out which of these correspond to the trajectories shown in the three plots on the margin:

Set 1: $t_0 = 0 \text{ s}$ $t_1 = 10 \text{ s}$ $\Delta t = 0.0001 \text{ s}$
 $m_a = 0.1 \text{ kg}$ $m_b = 0.1 \text{ kg}$ $k = 0.5 \text{ N/m}$ $l_n = 0.5 \text{ m}$
 $\mathbf{r}_a(t_0) = [-0.3, 0.3] \text{ m}$ $\mathbf{r}_b(t_0) = [0, 0] \text{ m}$
 $\mathbf{v}_a(t_0) = [1, 5] \text{ m/s}$ $\mathbf{v}_b(t_0) = [0, 5] \text{ m/s}$
 $\mathbf{G}_a(t_0) = m_a \cdot [0, 0] \text{ N/kg}$ $\mathbf{G}_b(t_0) = [0, 0] \text{ N}$

Set 2: $t_0 = 0 \text{ s}$ $t_1 = 10 \text{ s}$ $\Delta t = 0.0001 \text{ s}$
 $m_a = 2 \text{ kg}$ $m_b = 2 \text{ kg}$ $k = 5 \text{ N/m}$ $l_n = 5 \text{ m}$
 $\mathbf{r}_a(t_0) = [-3, 0] \text{ m}$ $\mathbf{r}_b(t_0) = [3, 0] \text{ m}$
 $\mathbf{v}_a(t_0) = [0, 0] \text{ m/s}$ $\mathbf{v}_b(t_0) = [0, 0] \text{ m/s}$
 $\mathbf{G}_a(t_0) = [0, 0] \text{ N}$ $\mathbf{G}_b(t_0) = [0, 0] \text{ N}$

Set 3: $t_0 = 0 \text{ s}$ $t_1 = 10 \text{ s}$ $\Delta t = 0.0001 \text{ s}$
 $m_a = 1 \text{ kg}$ $m_b = 5000 \text{ kg}$ $k = 5000 \text{ N/m}$ $l_n = 5 \text{ m}$
 $\mathbf{r}_a(t_0) = [-3, 0] \text{ m}$ $\mathbf{r}_b(t_0) = [0, 0] \text{ m}$
 $\mathbf{v}_a(t_0) = [0, 0] \text{ m/s}$ $\mathbf{v}_b(t_0) = [0, 0] \text{ m/s}$
 $\mathbf{G}_a(t_0) = -m_a g \cdot [0, 1]$ $\mathbf{G}_b(t_0) = [0, 0] \text{ N}$



Examples of trajectories of two masses connected by a non-Hookean spring, for different values of parameters and initial conditions. Mass *a* in blue, mass *b* in red.

Set 4: $t_0 = 0 \text{ s}$ $t_1 = 10 \text{ s}$ $\Delta t = 0.0001 \text{ s}$
 $m_a = 1 \text{ kg}$ $m_b = 5000 \text{ kg}$ $k = 5 \text{ N/m}$ $l_n = 5 \text{ m}$
 $\mathbf{r}_a(t_0) = [-3, 3] \text{ m}$ $\mathbf{r}_b(t_0) = [0, 0] \text{ m}$
 $\mathbf{v}_a(t_0) = [0, 0] \text{ m/s}$ $\mathbf{v}_b(t_0) = [0, 0] \text{ m/s}$
 $\mathbf{G}_a(t_0) = m_a g \cdot [0, 1]$ $\mathbf{G}_b(t_0) = [0, 0] \text{ N}$

Play with other parameters and initial values!

§ 10.17 Momentum fluxes in a gas

Imagine a rigid box at rest containing an amount of gas, and vacuum outside the box. Let gravity be negligible. It is common knowledge that if we open one side of the box, the gas within gets out. The gas's behaviour is to be contrasted with that of a solid: if the box contained, say, a brick, then the brick would simply stay where it is upon opening one side of the box.

This peculiar behaviour of gases gives us an idea of what kind of momentum fluxes must occur at the boundaries of a control volume containing some gas.



Exercise 10.14

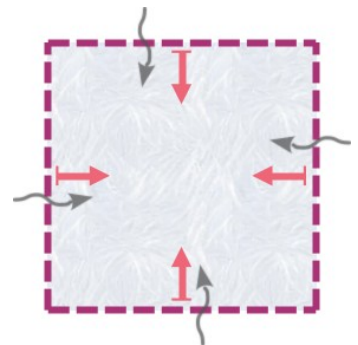
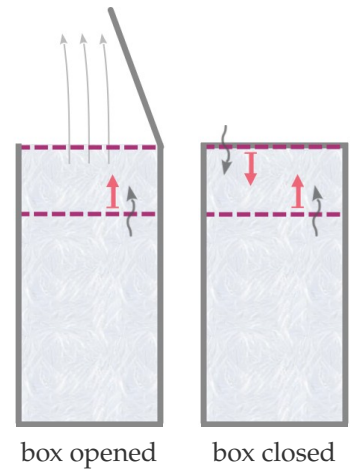
Before continuing to read, try to figure out by yourself how the momentum fluxes in a gas should be; at least formulate some hypotheses. Remember that you must first choose control surfaces. For instance, think of what kind of momentum flux there could be through control surfaces like the **dashed red lines** in the side figure: (a) at the opened side; (b) in between the opened side and the middle of the box, parallel to the opened side.



Let us try to infer what are the momentum fluxes through the two horizontal control surfaces depicted in the previous illustration.

- The portion of gas between the two control surfaces gets out as soon as we open the top side of the box. This portion of gas must therefore be receiving upward-oriented momentum. This momentum cannot come from the gravitational momentum supply, because the latter is oriented downward. It must therefore be the result of a momentum flux. This flux cannot be through the control surface separating the gas from the vacuum, and it seems implausible that it is a flux through the lateral contact surface with the box. Intuitively we understand that it is a momentum flux coming from the gas beneath the lower control surface; see margin figure on the left.
- As long as the top side of the box is closed, however, the portion of gas between the two control surfaces stays at rest. The flux of upward-oriented momentum coming from the gas beneath the lower control surface must therefore be compensated by a flux of downward-oriented momentum that comes from the top side of the box; see margin figure on the right.

Repeating this kind of reasoning with horizontal control surfaces further within the box, and then with vertical control surfaces, we arrive at the following conclusion: through any small portion of a closed control surface that delimits some gas *at rest*, there is an influx of inward-oriented momentum, as illustrated in the margin figure. The total momentum within the control volume doesn't change because the influxes from opposite parts of the closed control surface cancel each other.



Internal pressure, pressure vector, stress vector

The compressive momentum flux that occurs through any control surface that we choose within a gas, is called the **internal pressure** of the gas. More precisely, the internal pressure is this momentum flux divided by the area through which it occurs:

For a control surface of area A where the momentum flux $\mathbf{F}$ is uniform throughout, and through which *there is no matter flux*, we call **pressure vector** or **stress vector** $\mathbf{p}$ the momentum flux divided by the area: $\mathbf{p} := \mathbf{F}/A$.

If the momentum flux $\mathbf{F}$, and therefore the pressure vector, are orthogonal to the surface and point *away from the surface*, then we call **pressure**

the magnitude of the pressure vector, and usually denote it p :

$$p := |\boldsymbol{p}| \equiv \frac{|\boldsymbol{F}|}{A} \quad (10.12)$$

The name *tension* is used instead of *pressure* if the momentum flux points towards the surface. Compare our discussion about [compressive and tensile momentum fluxes](#) ▶ §4.12 p. 118.

Pressure has physical dimensions of force per area, with units N/m^2 ; this unit also called *pascal* (Pa).

! 'Pressure' has many different meanings

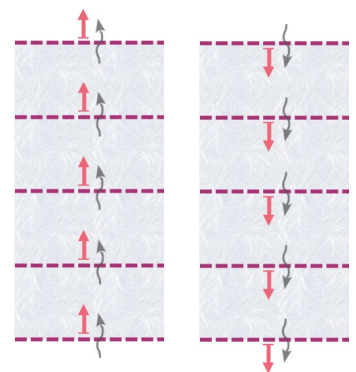
Be aware that the term 'pressure' is used in many different and sometimes even opposite ways in the physics literature. Some texts use 'pressure' to indicate not just the magnitude of $\boldsymbol{F}/A$, but the full vector; so for them 'pressure' is not just a number, but a set of three vector components. Some texts use 'pressure' to denote the pressure vector; so for them a pressure need not be orthogonal to a control surface. Some texts use 'pressure' also to indicate 'tension', or vice versa.

So when you read a text that uses or discusses 'pressure', make sure to get correctly in which sense the text is using this word.

In these notes we shall sometimes use 'pressure' in a slightly different or more general meaning, which should be clear from the context.

If we consider a collection of parallel control surfaces within a gas, we see that there is a flux of momentum – pressure – across all of them. This momentum has the same orientation as the chosen crossing direction. In the margin figures, for instance, if we cross the horizontal control surfaces (**dashed red lines**) from bottom to top, we'll measure a flux of upward momentum, as in the left figure. By flux symmetry, if we choose the opposite crossing direction, from top to bottom, then we'll measure a flux of downward momentum, as in the right figure. These are just two different aspects of the same situation.

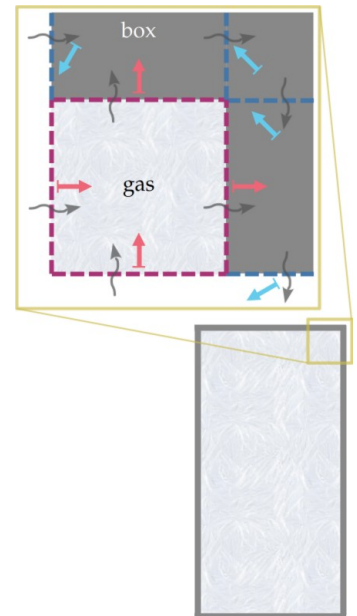
From a molecular point of view, this pressure momentum flux mainly comes from the [momentum transported](#) ▶ §10.28 p. 302 by the molecules that make up the gas. Some molecules are moving with an upward velocity, and therefore each of them has an upward momentum; this upward



momentum travels upward, transported by the molecule. Other molecules are moving with a downward velocity, and therefore each of them has a downward momentum, which is also transported downward.

At the surface where the gas is in contact with its containing box, the momentum gets transferred to the box: we say that the gas presses against its container. An interesting question then arises. If the box is also at rest, the momentum contained in each part of it must be zero and remain zero. Where does the momentum received from the gas go?

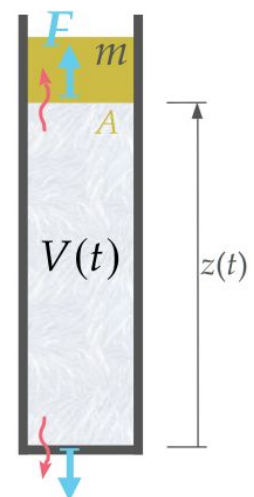
The answer is that an opposite and slightly more complex circulation of momentum takes place through walls of the box. The momentum that one side of the box receives from the gas is transported to the opposite side, where it is given back to the gas. By the symmetry of flux, this circulation can also be visualized in the opposite direction. The flux of momentum along the walls of the box is therefore partially *tensile* ^{§4.12 p. 119} and partially *shearing* ^{§4.12 p. 119}. The margin picture gives a *qualitative* illustration of the momentum fluxes (*light-blue arrows*) that occur through various control surfaces (*dashed dark-blue lines*) at the corner of a box that contains an amount of gas and is surrounded by a vacuum. From a molecular point of view, the flux of momentum through the box occurs because of transport, not by molecules, but by the electromagnetic field that coexists with the box's molecules.



§ 10.18 Constitutive relation for momentum flux of an ideal gas

The *simplifying assumptions* ^{§10.12 p. 260} that we made to study pieces of material like rubber bands, small springs, or strings, can also be made in circumstances involving gaseous substances.

Consider for instance the setup illustrated in the margin figure. A chamber is filled with a gaseous substance, represented by the light-bluish filling. One end of the chamber is movable; for instance it could be a freely movable piston, represented by the *yellow rectangle*. Note that we imagine the chamber walls to be indefinitely long, so that the piston never gets out. The chamber has one main relevant length $z(t)$, possibly variable with time. There is a flux of momentum (surface force) from the gas, and we focus on the fluxes (*blue arrows*) through two main surfaces: to the movable piston, and to the bottom of the container. If the section of the container has area A , then the container has variable volume $V(t) = A z(t)$.



The chamber walls should be imagined indefinitely high: the piston never gets out.

If the piston has mass m , the mass of the gas in the container may be negligible compared to it. This condition is especially true if the gas is very rarefied, that is, if the ratio N/V of the gas amount N divided by the volume is very small.

We then have the same assumptions discussed in §10.12, and the same consequences. In particular the gas may be considered to have zero momentum at all times, even if it's expanding or contracting; and the force $\mathbf{F}$ that it exerts on the piston must be directed in the z direction and have magnitude

$$F = f(\dots)$$

that may depend on the length $z(t)$, and therefore on the volume $V(t)$, and also on other quantities.

Pressure of an ideal gas

Fluids are characterized by the fact that $f(\dots)$ may also depend on the *rate of change* of the volume, that is, the time-derivative $dV(t)/dt$. Dependence on other, non-geometric quantities is also common, for instance on the [temperature](#) §3.12 p. 86 of the fluid.

We now discuss a constitutive relation that connects:

- the flux of momentum F , or more precisely the pressure $p := F/A$;
- the amount N of gas in the container;
- the temperature T of the gas;
- the volume V of the container;
- the rate of change of the volume, $dV(t)/dt$.

This constitutive relation can be used for many gases especially when they are rarefied: low N/V . A gas for which such a constitutive relation can be applied is said to behave as an **ideal gas**. We shall discuss ideal gases further in chapter 11.



Ideal-gas law with viscosity

If a closed control surface encloses a volume V with an amount N of an ideal gas, at uniform temperature T , and the momentum flux is orthogonal to the surface and uniform throughout, and there is no matter flux through the surface, then the pressure p is given by

$$p = \frac{RNT}{V} - \mu \frac{1}{V} \frac{dV}{dt} \quad (10.13)$$

where $\frac{dV}{dt}$ is the rate of change of volume; μ is called the **viscosity coefficient**, which is always positive, has physical dimension force/area, and may also depend on temperature; and R is the **molar gas constant**²⁹, having universal value

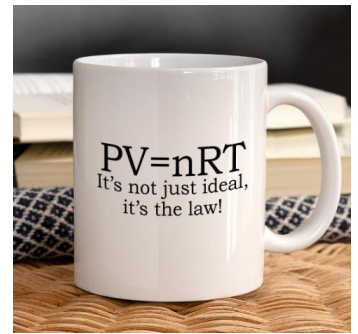
$$R = 8.314\,462\,618\,153\,24 \text{ N m/(mol} \cdot \text{K)} \quad (\text{exactly})$$

The equation above is called the **ideal-gas law with viscosity**.

In the expression above for the pressure you may recognize the famous “ $pV = NRT$ ” formula, called the **ideal-gas law**. The fact that the viscosity coefficient is positive is a consequence of the *balance of entropy*, which we’ll discuss in chapter 15.

! The classic ideal-gas formula is valid only at rest

The famous formula “ $pV = NRT$ ” is strictly speaking valid only when the ideal gas is fully at rest, that is, when its volume is not changing with time: $dV/dt = 0 \text{ m}^3/\text{s}$. If the ideal gas is expanding or contracting, then additional terms enter that formula, as the constitutive relation (10.13) above shows. In many cases, however, these additional terms are extremely small and may therefore be neglected.



§ 10.19 Simulation of an ideal gas & piston

Let us return to our simple setup with a gas in a container with a movable piston. Let’s assume that the gas behaves like an ideal gas. The volume is connected to the length z and section area A of the container by $V = Az$. From the ideal-gas law (10.13), the magnitude of the surface force F on the piston is

$$F = Ap = A \frac{RNT}{V} - A\mu \frac{1}{V} \frac{dV}{dt} = \frac{RNT}{z} - \mu \frac{A}{z} v_z. \quad (10.14)$$

Suppose that the temperature T is constant and the viscosity term is negligible, $\mu \approx 0 \text{ N/m}^2$. Compare the constitutive relation for the force, that is, momentum efflux, by a Hookean spring and that by an ideal gas:

$$\text{Hooke's law: } F = -\text{constant} \cdot (z - l_n)$$

$$\text{ideal-gas law: } F = \frac{\text{constant}}{z}$$

There are several interesting differences between the two:

- The force from Hooke's law can be positive (compressive), when $z < l_n$, or negative (tensile), when $z > l_n$.
- The force from the ideal-gas law can only be compressive.
- For large z , the force from Hooke's law is negative (tensile).
- For large z , the force from the ideal-gas law is positive (compressive).
- As z increases, the force from Hooke's law becomes stronger.
- As z increases, the force from the ideal-gas law becomes weaker.

These differences have important consequences for the motion of an object on which such a force is exerted. As we saw in our previous [analytic](#) §10.14 p. 265 and [numeric](#) §10.15 p. 269 analyses, the Hookean force gives rise to oscillatory motion around a position of equilibrium; if the object is initially at this position with zero velocity, it stays there. The ideal-gas force instead will indefinitely keep on pushing the object, unless some other force counteracts it; it does not have an equilibrium position.

But notwithstanding these important differences, the forces are both given by a function of the distance z . It is therefore very easy to adapt our previous simulations of a Hookean mass-and-spring system to the simulation of a piston-and-ideal-gas system. We shall do this now; but first try on your own:



Exercise 10.15

1. Start from the simulation script for the spring & bodies system in table 10.2, analysed in §10.15, and modify it as follows:
 - Eliminate one spatial dimension, keeping only the z -coordinate. Feel free to rename the position ' $\mathbf{r}$ ' to just ' z '.
 - Eliminate object b . Feel free to eliminate all ' a ' subscripts (so ' $\mathbf{P}_a$ ' becomes just ' $\mathbf{P}$ ', and so on).
 - Consider the b -end of the spring to be attached to the origin $z = 0$. What are the state variables of this simplified system?
 - Use the following constants, initial conditions, and time-integration parameters:

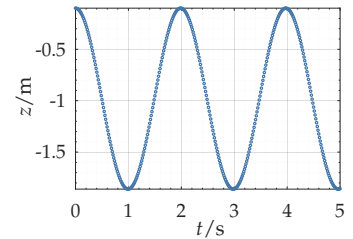
$$\text{constants: } m = 20 \text{ kg} \quad k = 200 \text{ N/m}$$

$$\text{initial conditions: } z = -0.1 \text{ m} \quad v = 0 \text{ m/s}$$

$$\text{time-integration parameters: } t_1 = 5 \text{ s} \quad \Delta t = 0.000\,01 \text{ s}$$

- Simulate and plot the position of object a versus time. Verify that you obtain a plot like the one in the margin.

(If you had difficulties modifying the script, for the following exercise you can download and use `hooke_spring_exercise.m` (Octave)³⁰ or `hooke_spring_exercise.py` (Python)³¹ as a starting point.)



Now we want to replace the Hookean material that connects object a to the origin, with an “ideal-gas material” (and no air outside of the ideal gas’s container). This new material provides a surface force described by the constitutive relation (10.14), but let’s say without the μ -term:

2. Which physical laws need to be changed in the script? Where in the while-loop should the modification occur?
3. Modify the previous script accordingly.
4. Use the following constants and initial conditions:

$$\text{constants: } R = 8.314 \text{ N m/(K mol)} \quad N = 0.02 \text{ mol} \quad T = 298.15 \text{ K}$$

$$\text{initial conditions: } z = +0.1 \text{ m} \quad v = 0 \text{ m/s}$$

5. Before starting the simulation: what kind of behaviour do you expect to observe for the piston’s position z ? For instance, will the piston be pushed upward indefinitely? Or maybe it will stop at some position?
6. Simulate and plot the piston’s position z versus time. Were your expectations correct?

An example script for the simulation of the ideal gas & piston according to the previous exercise is shown in table 10.4 on page 314, with the resulting plot at the margin.

In § 10.14 we were able to find an analytic expression for the motion of an object under the force of a Hookean material: the result is a harmonic function, that is, a sine or cosine function with some specific frequency and phase. In the case of the motion of a piston under the force of an ideal gas, *no explicit analytic expression exists*. So our only way to find this motion is by numerical time integration, as done in the script above.

In chapter 11 we shall consider what happens if the temperature is not constant but changes with time.

Table 10.3 Script for ideal gas & piston

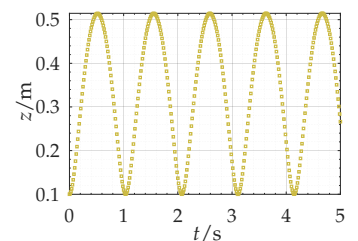
```

1 %% Simulation of ideal gas & piston
2 %% Coordinates (t, z)
3
4 %% Constants
5 m = 20;           % kg: mass piston
6 g = 9.8;          % N/kg: gravitational acceleration
7 R = 8.31446261815; % N m/(K mol): molar gas constant
8 N = 0.02;         % mol: amount of ideal gas
9 T = 298.15;       % K: temperature of ideal gas
10
11 %% Initial conditions, state: (z, v)
12 t = 0;           % s: time
13 z = 0.1;         % m: position piston
14 v = 0;           % m/s: velocity piston
15
16 %% Boundary conditions
17 G = -m * g;      % N: gravity supply piston
18
19 %% Parameters for time loop
20 t1 = 5;          % s: final time
21 dt = 0.00001;    % s: time step
22
23 %% Plotting
24 dtplot = t1/360; % time interval between plots
25 tplot = dtplot;  % time for next plot
26 clf
27 plot(t, z, 's', 'color', '#CCBB44')
28 xlabel('\it t/s'); ylabel('\it z/m')
29 hold on; axis('tight'); xlim([0, t1]); grid on
30
31 %% Numerical time integration
32 while t < t1
33     %% constitutive relations
34     P = m * v;
35     F = R * N * T / z;
36
37     %% balance laws
38     t = t + dt;
39     P = P + (F + G) * dt;
40     z = z + v * dt;
41
42     %% constitutive relations: find new state variables
43     v = P / m;
44
45     %% plot
46     if t > tplot
47         plot(t, z, 's', 'color', '#CCBB44')
48         pause(0)
49         tplot = tplot + dtplot;
50     end
51 end

```

Download Octave version
[idealgas_exercise.m](#)³²

Download Python version
[idealgas_exercise.py](#)³³



Under the force of the ideal gas, the piston has an oscillatory motion. This is not a sinusoidal curve, unlike the case of the Hookean spring.

§ 10.20 Chemical reaction with variable volume

✂ to be written

§ 10.21 Beyond the no-mass simplification

The physical phenomena that we explored in the previous sections involve two small bodies attached to the ends of a piece of material, for instance two tennis balls connected by a spring. The two bodies are assumed to always stay attached to the piece of material. So you may wonder: what if the two bodies are just touching the extremities of the material, without being “glued” to them? If we compress a spring by means of two tennis balls, these will be pushed apart, and at some point they will lose contact with the spring and continue moving. Otherwise pinball machines and other devices that rely on spring launchers wouldn’t work.

Our simplified setup cannot handle this kind of situation: as soon as the two bodies are not in contact with the piece of material, we must again face the problem of its masslessness and its lack of momentum. The only way is therefore to remove the zero-mass simplification, and treat the material as having mass and momentum. This realistic step, however, leads to a host of complications.

If the material can now accumulate momentum, then $d\mathbf{P}/dt$ is non-zero, and therefore the influxes $\mathbf{F}_a$ and $\mathbf{F}_b$ don’t need to satisfy condition C1 (page 261): $\mathbf{F}_a = -\mathbf{F}_b$.

This in turn means that the influxes $\mathbf{F}_a$, $\mathbf{F}_b$ cannot have the same dependence on the geometric properties, apart from a sign. The constitutive function $f(\dots)$ of formula (10.8), page 262, must be different for $\mathbf{F}_a$ and $\mathbf{F}_b$, and must depend on more geometric quantities than just the length l .

Condition C2 is no longer satisfied either, because angular momentum can be accumulated. The piece of material can in fact rotate, and therefore we must take care of the *balance of angular momentum* for its control volume.

For this reason we shall study this more realistic setup when we discuss the balance of angular momentum in chapter 12.

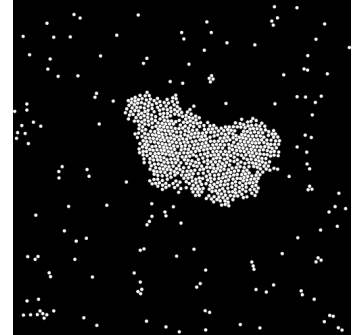


Wooden pinball machine. Spring launcher on the bottom right (image: Thebigkegg³⁴)

§ 10.22 Many-body systems

We can obviously consider more than two objects connected by a number of springs or other pairwise forces. The analytical description of such a system becomes quickly intractable. For numerical time integration, however, we only need to add additional lines of code to compute the additional momentum fluxes and timestep the momentum for the individual bodies; this can be done by appropriate for-loops for the pairwise forces and for the momenta.

This way we can numerically simulate complex physical phenomena like a planetary system, or a collection of molecules – thus entering the field of [molecular-dynamics simulation](#)³⁶. You can see an example of the latter kind of simulation, using the Lennard-Jones pairwise force of formula (10.11), at the [Interactive Molecular Dynamics webpage](#)³⁷.



Snapshot of a molecular-dynamics simulation from [Interactive Molecular Dynamics](#)³⁵

Velocities at contact surfaces: slip conditions

✂ To be written

§ 10.23 Contact and impact forces

✂ To be written

§ 10.24 Motion from forces? or forces from motion?

For the discussion in the next sections it's useful to quickly review two main ways in which we have used the balance of momentum so far.

Consider the general differential expression of the balance of momentum

$$\frac{d\mathbf{P}(t)}{dt} = \mathbf{F}(t) + \mathbf{G}(t)$$

and to simplify the discussion assume that the momentum supply $\mathbf{G}$ is known; for instance it's the gravitational supply ' $m\mathbf{g}$ '. This balance therefore connects the rate of change of momentum $d\mathbf{P}/dt$ in a control volume, and the total momentum flux or surface force $\mathbf{F}$ through the boundary of the control volume.

One simple use of this balance law – and of any other balance law, as we discussed in chapter 5 – can therefore be to find the force, given the rate of change of momentum:

$$\left. \begin{array}{l} \text{balance of momentum} \\ \text{rate of change of momentum } \frac{d\mathbf{P}}{dt} \end{array} \right\} \longrightarrow \text{momentum flux } \mathbf{F}$$

If we take a panoramic look back at the initial sections of the present chapter, where we discussed statics problems, we see that this is indeed the way we used the balance of momentum there: the motion, including the special case of no motion, was known; and from it we calculated the values of some unknown forces.

Another simple use of the balance law is the reverse: to find the rate of change of momentum, given the surface force:

$$\left. \begin{array}{l} \text{balance of momentum} \\ \text{momentum flux } \mathbf{F} \end{array} \right\} \longrightarrow \text{rate of change of momentum } \frac{d\mathbf{P}}{dt}$$

Another panoramic look back at the subsequent sections of the present chapter, where we discussed constitutive relations for the momentum flux in several materials like elastic springs or ideal gases, shows that this is the way we used the balance of momentum there: the forces were known or calculated thanks to constitutive equations; and from the forces we were able to find how the momentum should change. This is also the way the momentum balance is used in simulations.

In any physical analysis it's very important to recognize which of the two uses above of the balance of momentum is the necessary one, or the one adopted in the analysis. Otherwise we may end up trying to find the change in momentum by using formulae for forces that require that we *already* know the change in momentum!

§ 10.25 Normal to a flat control surface

Recall that a [control surface](#) ^{§4.3 p. 99} can have arbitrary shape and dimension, as long as we can choose a crossing direction through it.

When we have a *flat* control surface, we can also define a special vector in relation to it:

Unit normal to a flat control surface

The **unit normal** vector to a *flat* control surface, usually denoted

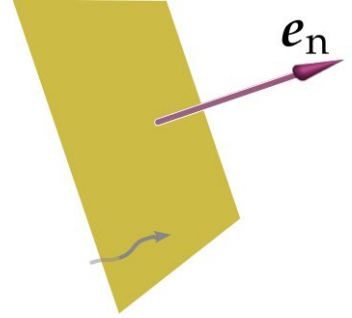
$$\mathbf{e}_n = \begin{bmatrix} e_{n,x} \\ e_{n,y} \\ e_{n,z} \end{bmatrix},$$

is a *unique* vector that is perpendicular to the surface and has unit (dimensionless) magnitude:

$$|\mathbf{e}_n| = \sqrt{e_{n,x}^2 + e_{n,y}^2 + e_{n,z}^2} = 1.$$

This vector not have any specific “application point”.

A unit normal defines a crossing direction through the control surface, and vice versa.



Note the difference between a crossing direction and a unit normal. Any control surface, even a curved one, can be assigned a crossing direction. But only a *flat* surface can be assigned a unit normal. Take a sphere for example: it can be given the crossing direction ‘from outside to inside’; but which vector could be its normal? in which direction would such a unique vector point?

The best we can do with a spherical surface is to divide it into extremely small sub-surfaces, so small that each can be considered practically flat. To each such sub-surface we can then assign a unique unit normal vector. But the unit normal vectors of different sub-surfaces will point in different directions.

§ 10.26 Normal and tangential surface forces

In studying the flux of momentum, that is the surface force, at the surface of contact between two bodies of matter, it is often useful to imagine this flux as the sum of two fluxes.

Normal- and tangential-momentum flux (surface force)

Consider a control surface that is either flat or enough small that a unique unit normal vector $\mathbf{e}_n$ to it is well-defined. In the margin figure such a control surface is represented by the solid grey line; a crossing direction is also shown by the wiggly arrows.

Suppose there is a momentum flux $\mathbf{F}$ through this control surface, in the given crossing direction (solid blue arrow in the figure, representing a compressive flux). We can decompose this momentum flux as a sum

$$\mathbf{F} = \mathbf{F}_n + \mathbf{F}_t$$

of:

- a flux of momentum $\mathbf{F}_n$ which is *orthogonal* to the surface (long-dashed purple arrow). It is called the **normal component** of the force and is given by

$$\mathbf{F}_n := (\mathbf{F} \cdot \mathbf{e}_n) \mathbf{e}_n . \quad (10.15)$$

- A flux of momentum $\mathbf{F}_t$ which is *tangential* or, said otherwise, parallel to the surface (short-dashed red arrow). It is called the **tangential component** of the force and is given by

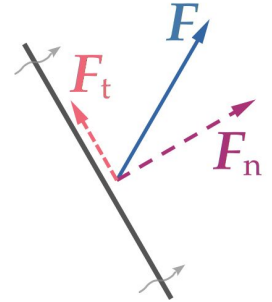
$$\mathbf{F}_t := \mathbf{F} - (\mathbf{F} \cdot \mathbf{e}_n) \mathbf{e}_n \equiv \mathbf{F} - \mathbf{F}_n . \quad (10.16)$$

The margin figure above shows the decomposition for a surface force of a **compressive** ^{§4.12 p. 118} kind (the flowing momentum points in the same direction as the flow). The figure on the margin here shows instead a similar decomposition for a surface force of a **tensile** ^{§4.12 p. 119} kind (the flowing momentum points in a direction opposite to the flow).

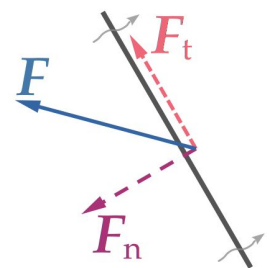
Different approaches for normal components of force

In some applications the normal component of a surface force is modelled in ways like those described in §10.23. This means that in such applications the normal component is calculated by means of a constitutive relation, and can thereafter be used to calculate how momentum changes.

In other applications the motion of matter – and therefore the change in momentum – in a direction normal to a surface is known beforehand, and



(normal $\mathbf{e}_n$ not depicted)



decomposition for tensile $\mathbf{F}$

the normal component of a surface force is calculated as a consequence of this.

As discussed in the previous section, be careful and aware of whether one or the other approach is used.

§ 10.27 Friction

Many physical phenomena involve **contact** between two bulks of matter that lean on each other without moving, or move with different velocities. Think of a house of cards, or a stone pushed across the floor, or a child going down a slide, or air flowing past an aeroplane wing. The contact forces involved in such phenomena partially prevent two kinds of movement: that the two bulk of matters move *into* each other, and that they move laterally with respect to each other. The normal component of the contact force prevents the first kind of motion, and the tangential component prevents the second. The tangential component is generically called **friction**, or **drag** when the surface separates solid and fluid matter.

There are many constitutive relations for contact forces involving friction. These relations are peculiar because they do not relate a whole momentum flux with other quantities such as some length (think of Hooke's law) or volume (think of the ideal-gas law) of a control volume. Instead, they relate the *magnitude* and *direction* of a momentum flux, or equivalently its *tangential* and *normal* components. In other words, these constitutive relations allow us to calculate one geometrical aspect of the surface force from another aspect, for instance one component from the other. The other component must therefore be found by other means.

We discuss here two constitutive relations of this kind which are common, although there are many applications in which they cannot be used. They pertain to friction under two different circumstances:

static friction: when matter on the two sides of the surface is not in relative motion, that is, it has the same velocity – usually zero in the coordinate system used;

kinetic or dynamic friction: when matter on the two sides has different velocities.

Static friction

When we want to push or drag an object such as a table or large box across the floor, it's a common experience that exerting a small horizontal force,



image: peterjroberts³⁸



*Five men pushing a stone*³⁹ by Jacques Stella

for instance pushing with a finger, does not move the object. We need to exert a minimal amount of force to set it into motion; and usually this minimal horizontal force is the larger, the heavier is the object.

Since the object does not move as we apply a small horizontal force, it means that its momentum components parallel to the ground are not changing. From the balance of momentum, this means that the total force parallel to the ground must be zero. Neglecting air pressure, this total horizontal force is the sum of two surface forces: the horizontal one coming from us, and the *tangential component* of the force coming from the ground to the object.

Let's put this mathematically. Denote these momentum components parallel to the ground with $\mathbf{P}_t$, the horizontal force coming from us with $\mathbf{F}_{us}$, and the tangential component of the force coming from the ground by $\mathbf{F}_t$. We observe that $d\mathbf{P}_t/dt = 0$ N. From the balance of momentum, this means that

$$0 \text{ N} = \frac{d\mathbf{P}_t}{dt} = \mathbf{F}_{us} + \mathbf{F}_t \quad \implies \quad \mathbf{F}_t = -\mathbf{F}_{us} .$$

Therefore the tangential component of the force from the ground must be equal and opposite to the one coming from us.

There is one peculiarity: we can increase or decrease our force $\mathbf{F}_{us}$ a little – it's changing with time – and the object still does not move. This means that the tangential component for the force from the ground continually “adapts”, so to speak, to our force.

All this occurs only as long as the force $\mathbf{F}_{us}$ from us does not exceed a certain threshold; then the object starts to move. This threshold force is higher, the heavier the object is, and obviously depends on the kind of matter on the two sides of the contact surface: it may be easier to set the object into motion if it rests on ice than if it rests on concrete.

These observations can be condensed into

Amontons's law for static friction

In some situations, the tangential component $\mathbf{F}_t$ of the contact force between matter on two sides of a control surface, when the matter is not in relative motion, is given by

$$\mathbf{F}_t = -\mathbf{F}_{\text{other},t} \quad \text{as long as} \quad |\mathbf{F}_{\text{other},t}| \leq \mu_s |\mathbf{F}_n| \quad (10.17)$$

where

$$\mathbf{F}_{\text{other},t} := \left(\begin{array}{c} \text{tangential component of the sum} \\ \text{of all other momentum fluxes and supplies} \end{array} \right) ,$$

$\mathbf{F}_n$ is the normal component of the force, and μ_s is called **static-friction coefficient** or, better, **static-friction factor**

The tangential, frictional component is often denoted F_s .

The static-friction factor depends on the kind (wood, metal, concrete, etc.) and condition (smooth, rough, dry, wet, etc.) of matter on the two sides of the control surface.

Amontons's law for static friction is not a constitutive relation, although in some cases it can be used as one. Consider a body of matter as a control volume. If static friction occurs only on one part of the surface of this control volume, then it might be possible to apply Amontons's law by first calculating the total of the tangential forces on the other partial surfaces – by means of constitutive equations – and then setting static friction equal to minus this total. But if static friction occurs on several separate contact surfaces, call them '1', '2', '3', and so on, then it may become impossible to find all these static-friction forces by means of Amontons's law: the problem becomes indeterminate. We would have to find several unknowns $\mathbf{F}_{1,t}$, $\mathbf{F}_{2,t}$, $\mathbf{F}_{3,t}$ from only one equation

$$\mathbf{F}_{1,t} + \mathbf{F}_{2,t} + \mathbf{F}_{3,t} + \cdots = -\mathbf{F}_{\text{other},t} .$$

Kinetic friction

Continuing with the previous example, it's also a common experience that once we manage to set the object into motion, we need to keep applying some horizontal force if we want it to move with a constant velocity. The force we need to apply is larger, the heavier the object is. But sometimes this force can be smaller than the force we initially needed to apply to set the object into motion. We can also increase the velocity if we apply a larger force for a short time.

These observations indicate that also in this case the contact force from the ground to the object must have a tangential component, with the following characteristics:

- it points in a direction opposite to the relative velocity between the object and the ground;
- it has a constant magnitude, independent on the force we apply;
- it is larger if the object exerts a larger normal force on the ground;
- it depends on the kind and condition of matter on the two sides.

More simply we can say that the contact force must have a *fixed inclination* with respect to the surface, and this inclination points directly away from the velocity

These observations can be condensed into

Amontons's law for kinetic friction

Consider a contact surface between two bodies of matter, the one on side a having velocity $\mathbf{v}_a$, parallel to the surface, and the one on side b having a *different* velocity $\mathbf{v}_b$, parallel to the surface: $\mathbf{v}_a \neq \mathbf{v}_b$.

In some situations of this kind the tangential $\mathbf{F}_t$ and normal $\mathbf{F}_n$ of the contact force on matter on the a side are related by

$$\mathbf{F}_t = -\mu_k |\mathbf{F}_n| \frac{\mathbf{v}}{|\mathbf{v}|} \quad (10.18)$$

where $\mathbf{F}_n$ is the normal component of the force, μ_k is the **kinetic-friction coefficient** or, better, **kinetic-friction factor**, and

$$\mathbf{v} := \mathbf{v}_a - \mathbf{v}_b$$

is the *relative velocity* of matter on side a with respect to matter on side b .

The tangential, frictional component is often denoted $\mathbf{F}_k$.

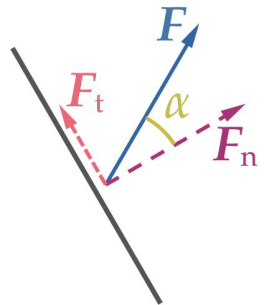
The kinetic-friction factor depends on the kind and condition of matter on the two sides of the control surface.

The $-\mathbf{v}/|\mathbf{v}|$ factor in the law is simply a unit vector that points in the direction opposite to the relative velocity $\mathbf{v}$. So it mathematically expresses the fact that the friction component is always opposite to the relative velocity.

Amonton's law for kinetic friction effectively says that the contact force $\mathbf{F}$ must form a *constant* angle α with the unit normal to the surface, such that

$$\tan \alpha \equiv \mu_k = \frac{|\mathbf{F}_t|}{|\mathbf{F}_n|}.$$

The static- and kinetic-friction coefficients μ_s and μ_k need not have the same value; in many cases the kinetic-friction coefficient is smaller than the static-friction coefficient.





Dependence of kinetic friction on the area of the contact surface

You notice that the area A of the contact surface does *not* appear in eq. (10.18) for kinetic friction. One says that kinetic friction is independent of the size of the contact surface. This statement may appear as counter-intuitive, and is in fact somewhat imprecise.

If we divide the normal force $\mathbf{F}_n$ on the contact surface by the surface's area A , we obtain the normal pressure on the surface, $p_n := \mathbf{F}_n/A$. Then Amontons's law for kinetic friction can be rewritten as follows:

$$\mathbf{F}_t = -\mu_k A |p_n| \frac{\mathbf{v}}{|\mathbf{v}|}.$$

Written this way we can actually say that kinetic friction is proportional to the area of the contact surface and to the normal pressure on it.

An analogous comment can be made for the law (10.17) for static friction.



There is only one force

Keep in mind that at the contact surface *there is only one flux of momentum, only one force*. Amontons's laws say that the *magnitude* and *inclination* of this force with respect to the surface change, depending on the conditions of the surface, the state of motion on its two sides, and the presence of forces on other surfaces of the bodies in contact.

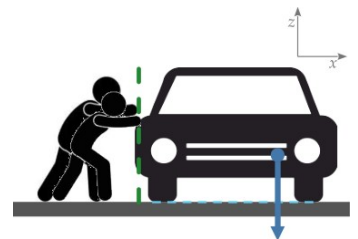
The division into 'normal' and 'tangential' components is only imaginary: we make it in order to simplify our analysis of contact forces.



Exercise 10.16

1. Find the physical dimensions and units of the coefficients of static and kinetic friction from formulae (10.17) and (10.18).
2. Two persons are pushing a parked car. The car rests on a thin layer of ice that formed on the tarmac. Each person is exerting a horizontal force of 600 N. The car has mass 1200 kg, and the acceleration of gravity is 9.8 N/kg. The coefficient of static friction between the wheel's rubber and ice is $\mu_s = 0.1$.

Will the two persons manage to move the car?



3. Consider the following Octave/MATLAB function to calculate the friction in a numerical simulation:

```

1 function Ft = friction(Fothert, Fn, v, mu_s, mu_k)
2   if norm(v) == 0 % relative velocity is zero: static friction
3
4       if norm(Fothert) < mu_s * norm(Fn)
5           Ft = -Fothert;
6       else
7           Ft = 0;
8       end
9
10  else % relative velocity is zero: kinetic friction
11      Ft = -mu_k * norm(Fn) * v / norm(v);
12  end
13
14  end

```

`norm(x)` is $|x|$

Check that this function correctly covers and represents both formulae (10.17) and (10.18).

§ 10.28 Constitutive relations for momentum flux in the presence of matter flux

In the previous sections we have studied different constitutive relations and applications for momentum flux through a control surface under a particular condition: there was *no flux of matter* through the surface. This condition can often be satisfied in applications involving solid materials or in static situations.

In applications involving fluids in motion, however, it can be very difficult to choose control surfaces that satisfy a no-matter-flux condition. Fluids can undergo very complex stretching and bending motions – think of how a large drop of milk on a cup of coffee gets deformed as it's stirred with a teaspoon. This is why in the study of fluid motion we typically choose imaginary control surfaces, static or in motion, which are independent of the motion of the fluid. The fluid therefore flows through them.

For a control surface through which there is a flux of matter, there is a more general constitutive relation for the contact force or momentum flux. The momentum flux can be seen as consisting of two contributions:



- A contact force that occurs simply because matter is present, even if no matter is flowing through the control surface.
- A contact force that occurs because matter is flowing through the control surface. If this matter has momentum, then in passing through the surface it brings its momentum from one side to the other. There is therefore a momentum flux, a contact force.

Let us see how these two contributions are expressed in mathematical form. What we have is not just a constitutive relation, but a *family* of constitutive relations which have a common mathematical form.



Momentum flux in the presence of matter flux

Fix a coordinate system (t, x, y, z) , and consider an imaginary control surface with a given crossing direction, and satisfying the following conditions:

- electromagnetic fields are negligible, velocities and gravitational fields are small enough;
- there's a matter flux $\mathcal{J}$, possibly zero, through the surface;
- a thin control volume containing the surface has *uniform* matter content N_m and momentum content $\mathbf{P}_m$;
- conditions are uniform over the surface; this is true for instance if the surface is small enough.

Then the flux of momentum or contact force $\mathbf{F}$ through the surface can be written, in many circumstances, as the sum of two terms:

$$\mathbf{F} = \mathcal{F} + \mathcal{J} \frac{\mathbf{P}_m}{N_m} \quad (10.19a)$$

The term $\mathcal{F}$ is called **stress vector** or **traction** or **molecular momentum flux**, among other names. It can be present even if there is no matter flux, $\mathcal{J} = 0$ mol/s, through the surface. It obeys further constitutive relations that depend on the kind and state of matter, like those we studied in the previous sections.

The term $\mathcal{J} \mathbf{P}_m / N_m$ is called **convective momentum flux** or **momentum transfer**, among other names. It is zero if there is no matter flux, $\mathcal{J} = 0$ mol/s. If the control surface has area A and the thin volume around it has volume V_m , then using the formula for matter

flux (7.25) and Newton's formula for momentum (10.2), we can rewrite the convective momentum flux as follows:

$$\mathcal{J} \frac{\mathbf{P}_m}{N_m} \equiv A \mathbf{e}_n \cdot (\mathbf{v}_m - \mathbf{v}_s) \frac{m_m}{V_m} \mathbf{v}_m.$$

The constitutive formula for momentum flux can then be rewritten

$$\mathbf{F} = \mathcal{F} + A \mathbf{e}_n \cdot (\mathbf{v}_m - \mathbf{v}_s) \frac{m_m}{V_m} \mathbf{v}_m \quad (10.19b)$$

In the applications of momentum flux studied in the previous sections, there was never a matter flux through the control surfaces of interest. All forces considered there had therefore only the stress vector contribution $\mathcal{F}$. The constitutive relations that we explored were, effectively, constitutive relations for the stress vector $\mathcal{F}$. The contact forces that we experience on our skin – for example when we pull or push an object with our hands, or the pressure and friction from the ground on the soles of our feet – consist only of a stress vector, because no matter flows through the control surface that is our skin.

In the next section we shall explore situations where matter flux occurs. The convective momentum flux will therefore appear and be quite relevant. There are also hybrid situations, where some control surfaces enclosing a control volume do not have a matter flux, whereas other do.



Exercise 10.17

1. In formula (10.19a), verify that the convective-flux term $\mathcal{J} \frac{\mathbf{P}_m}{N_m}$ has the correct dimensions of force.
2. In formula (10.19b), verify that the convective-flux term also has the correct dimensions of force.
3. Fix a coordinate system (t, x, y, z) at rest with the Earth's surface, with z upwards. Consider a horizontal control surface of area 7.5 m^2 , with a *downward* crossing direction; this is therefore also the direction of its normal $\mathbf{e}_n$.

At some time point, the control surface has positive vertical velocity 300 m/s . Matter, say some gases, is crossing the surface downwards, with vertical velocity -130 m/s . The mass of the gases divided

by volume is $m_m/V_m = 4.3 \text{ kg/m}^3$. How much is the convective momentum flux through the control surface, including its sign?

§ 10.29 Stress vector and convective momentum flux: observation scales

✂ to be written

§ 10.30 Rockets

The flight mechanisms discussed in §10.8 rely on the presence of air or some kind of atmosphere around the object, with which a momentum exchange through a surface can occur. This is not possible in outer space; and even if air is surrounding an object, that kind of momentum flux may not be enough to lift it anyway.

Another way of producing an influx of upward momentum is to release matter having downward momentum, creating a *convective momentum flux*. This is the momentum-flux mechanism employed by rocket engines, for which the convective momentum flux should have a large magnitude. The constitutive relation (10.19b), rewritten here:

$$\mathbf{F} = \mathcal{F} + A \mathbf{e}_n \cdot (\mathbf{v}_m - \mathbf{v}_s) \frac{m_m}{V_m} \mathbf{v}_m ,$$

is called the *rocket thrust equation*⁴⁰ in applications with rockets and jet engines. We see that the flux can be increased by increasing the area A of the surface, or the matter velocity $\mathbf{v}$, or the mass-to-volume ratio $\frac{m_m}{V_m}$ around the surface. Typically the focus is on the first two factors. Rockets that must achieve high velocities, for instance in order to leave Earth, typically have *nozzles*⁴¹ with large outlet areas. The released matter consists of gases produced by combustion, which produces large pressure and velocity; and this velocity is increased even more by means of a specific nozzle design.

Simplified simulation of a rocket's motion

Let us study and simulate, in a simplified setting, the motion of a rocket like the [Saturn V](#)⁴³; see side figure. We follow our [general strategy for numerical time integration](#) ^{§10.15 p.269}.

Choose a coordinate system (t, z) with just one spatial coordinate z , vertical and directed upward, with origin fixed on the Earth's surface. All vectors below have therefore just one component. Note that rockets like Saturn V typically only have a purely vertical motion at the very beginning; their motion gradually becomes horizontal and goes into an orbital motion around the Earth.

We consider a control volume bounded by two imaginary control surfaces:

- One surface, represented by the **solid yellow line**, wraps the rocket's frame, except the nozzle at the bottom. There is no matter flux through this surface.
- The other imaginary surface, represented by the **dotted red line**, is at the rocket's nozzle. This surface is static with respect to the rocket, but there is a matter flux through it: the exhaust gases ejected from the rocket.

For both surfaces we consider an *inward* crossing direction, so that the fluxes through them are influxes.

We associate a position vector $\mathbf{r}(t)$, the altitude, with the rocket's control volume, let's say with its lowest point, at the nozzle. The control volume and its two bounding surfaces are moving with velocity $\mathbf{v}(t) = d\mathbf{r}(t)/dt$.



Saturn V SA-513 rocket (image: [Gunter's Space Page](#)⁴²).

Which balances do we have to consider for this control volume?

Matter balance: The control volume contains many different kinds of matter: metal, plastic, astronauts, fuel, and so on. For our purposes we can approximately consider just two kinds. Let's call a the matter that always stay within the control volume; and b the matter, consisting mainly of fuel and oxidizing agents, that is transformed into gas by combustion and expelled at the nozzle.

For the first kind of matter no explicit balance is necessary, because we know there is no flux or supply and its amount is constant. For the second kind we need to keep track of its amount $N_b(t)$, which depends on time owing to a non-zero flux $\mathcal{J}$ at the nozzle. If we

wanted to simulate and keep track of the combustion process, we would need to distinguish among more kinds of matter and their balances, in the form of chemical reactions.

Momentum balance: We are interested in the motion of the rocket, so the *balance of momentum* is necessary for the total momentum $\mathbf{P}$ in the control volume enclosing the rocket. This control volume does not include the gases expelled out of the nozzle.

We are considering two kinds of matter, which as an approximation we may imagine to occupy separate control-subvolumes. The principle of extensivity can therefore be used to calculate their total momentum.

The contact forces for the rocket's surface (solid yellow line) and for the surface at the nozzle (dotted red line) are different, so we shall use the principle of extensivity to calculate the total momentum flux $\mathbf{F}$. The total momentum supply $\mathbf{G}$ is the gravitational force.

Energy and other balances: In a more realistic scenario we might be interested in keeping track of the rocket's energy content and of the energy released. But let's keep things simple and assume that the balance of energy is unimportant to us – hoping that it won't be necessary for our simulation.

Which constitutive relations do we need?

Matter and momentum content: We can use Newton's constitutive equation to relate the total momentum $\mathbf{P}$ and the amounts of matter. The two kinds of matter a and b have different [molar masses](#) ^{§7.4 p.193}.

The amount of matter a is constant, so we can associate with it a constant mass m_a and a momentum $m_a \mathbf{v}(t)$, where $\mathbf{v}$ is the velocity of the rocket. The second kind of matter has constant molar mass

$$\rho = \frac{m_b}{N_b},$$

so its momentum is $\rho N_b(t) \mathbf{v}(t)$. Note that using just one velocity $\mathbf{v}$ is an approximation: the matter within the control volume, especially the one involved in combustion, does not have the same velocity everywhere.

By extensivity the total momentum is

$$\mathbf{P}(t) = [m_a + \rho N_b(t)] \mathbf{v}(t).$$

Matter flux At the rocket's surface there is no flux of matter. At the nozzle there is a flux $\mathcal{J}$ of exhaust gases, matter b ; the relevant constitutive relation is eq. (7.25). Note that for this constitutive equation we must momentarily imagine a *thin control volume around the nozzle*, and this is *not* the whole control volume of the rocket!

Denote by V_{noz} the size of this thin control volume; and by $N_{b,\text{noz}}$ the amount of matter b , the exhaust gases, contained in it. This is not the whole amount N_b contained in the rocket: $N_{b,\text{noz}} \neq N_b$. What's important in the constitutive relation is the molar density of this matter in this thin control volume, which we can denote by δ :

$$\delta := \frac{N_{b,\text{noz}}}{V_{\text{noz}}}.$$

This density can in principle change with time, but for simplicity we assume it is constant.

If the surface area of the nozzle control surface is A , and its normal, directed upward, is $\mathbf{e}_n$, and the velocity of the exhaust gases is $\mathbf{v}_b$, then the constitutive relation for the matter flux $\mathcal{J}$ can be written

$$\mathcal{J} = A \mathbf{e}_n \cdot (\mathbf{v}_b - \mathbf{v}) \delta.$$

In general the matter flux $\mathcal{J}$ can change with time – for instance it is zero before the engines are turned on and after the fuel is used up – and it can depend on other quantities such as pressure and temperature; in fact it can be controlled by the rocket's pilots. For simplicity let's assume that it is constant, and therefore belongs to the boundary conditions of our simulation.

The velocity of matter ' $\mathbf{v}_m$ ' and of the control surface ' $\mathbf{v}_s$ ' in eq. (7.25) are called ' $\mathbf{v}_b$ ' and ' $\mathbf{v}$ ' in the present case.

Momentum flux: By extensivity, the total momentum flux into the control volume is given by the sum of the flux $\mathbf{F}_{\text{atm}}$ into the rocket's control surface, and the flux $\mathbf{F}_{\text{noz}}$ at the nozzle.

At the rocket's surface there is force from the atmosphere, which varies with the altitude of the rocket, as air becomes more rarefied. There is also the drag force from the air, which depends on the rocket's velocity and on the altitude. For simplicity let's assume – quite unrealistically – that the drag force is negligible. Recall that the net force from [atmospheric pressure](#) §10.6 p.244 on an object is approximately zero because it acts from all sides. In the present case, however, it does not act on the nozzle surface, because there are

exhaust gases there instead of air. There is therefore a net amount of force from the atmosphere, directed downward and acting on an area equal to the nozzle surface. We assume this force to be constant; it is therefore given in the boundary conditions.

For the momentum flux $\mathbf{F}_{\text{noz}}$ at the nozzle we must use the general constitutive relation (10.19), or rocket thrust equation:

$$\mathbf{F}_{\text{noz}}(t) = \mathcal{F} + A \mathbf{e}_n \cdot [\mathbf{v}_b(t) - \mathbf{v}(t)] \frac{m_{b,\text{noz}}}{V_{b,\text{noz}}} \mathbf{v}_b(t) ,$$

where $m_{b,\text{noz}}$ is the mass of the matter $N_{b,\text{noz}}$ in the thin control volume around the nozzle.

The stress vector $\mathcal{F}$ expresses the [internal pressure](#) ^{§10.17 p. 284} p of the exhaust gases, exerted on the nozzle area A ; it is a positive influx of momentum having magnitude

$$|\mathcal{F}| = A p .$$

The pressure p generally depends on their temperature and density, and possibly on other quantities, as we'll discuss in chapter 11. For the present case let's assume that this pressure is constant; it is therefore given in the boundary conditions.

The convective momentum flux can be rewritten as follows by multiplying and dividing by $N_{b,\text{noz}}$:

$$\begin{aligned} & A \mathbf{e}_n \cdot [\mathbf{v}_b(t) - \mathbf{v}(t)] \frac{m_{b,\text{noz}}}{V_{b,\text{noz}}} \mathbf{v}_b(t) \\ &= A \mathbf{e}_n \cdot [\mathbf{v}_b(t) - \mathbf{v}(t)] \overbrace{\left(\frac{N_{b,\text{noz}}}{V_{b,\text{noz}}} \right)}^{\delta} \overbrace{\left(\frac{m_b}{N_{b,\text{noz}}} \right)}^{\rho} \mathbf{v}_b(t) \\ &= \mathcal{J} \rho \mathbf{v}_b(t) \end{aligned}$$

The total momentum influx into the rocket's control volume is therefore

$$\begin{aligned} \mathbf{F}(t) &= \mathbf{F}_{\text{atm}} + \mathbf{F}_{\text{noz}} \\ &= \mathbf{F}_{\text{atm}} + \mathcal{F} + \mathcal{J} \rho \mathbf{v}_b(t) . \end{aligned}$$

Momentum supply: The rocket receives the Earth's gravitational force. For the beginning of the flight, when the rocket is still close to the Earth's we could use the simplified [constitutive relation](#) ^{§10.3 p. 231}

$\mathbf{G} = m\mathbf{g}$, with constant $g \approx 9.8 \text{ kg/N}$. More precision could be obtained using Newton's formula for gravitation. In the Saturn V case, the first stage of the flight reached an altitude of around 65 000 km (Orloff 2024), where the acceleration of free fall becomes 9.6 kg/N. So let's use the simplified expression with constant g .

The reasoning is similar to the one for momentum content: we must keep track of two kinds of matter, applying the principle of extensivity. The formula we obtain is

$$\mathbf{G} = [m_a + \rho N_b(t)] \mathbf{g} .$$

This momentum supply is not constant, because the mass changes with time.

Overview of the relevant equations

Let's collect the balance laws and constitutive relations just discussed for the present physical system. Since all vectors have only a z -component, we simply write this component instead of using boldface notation. In particular:

$$\mathbf{r}(t) \equiv z(t) , \quad \mathbf{e}_n \equiv +1 , \quad \mathbf{g} \equiv -g , \quad \mathcal{F} \equiv A p , \quad \mathbf{e}_n \cdot \mathbf{v} = 1 \cdot v = v .$$

Our equations become, making all time dependencies explicit:

$\frac{dP(t)}{dt} = F(t) + G(t)$	momentum balance
$\frac{dN_b(t)}{dt} = \mathcal{J}$	momentum balance for matter b
$\frac{dz(t)}{dt} = v(t)$	velocity of rocket's body
$P(t) = [m_a + \rho N_b(t)] v(t)$	momentum of rocket's body
$\mathcal{J} = A [v_b(t) - v(t)] \delta = \text{const.}$	flux of matter b
$F(t) = F_{\text{atm}} + A p + \mathcal{J} \rho v_b(t)$	Momentum flux
$G(t) = -[m_a + \rho N_b(t)] g$	Momentum supply of rocket's body

The first three balances will occur in our simulation in their finite-difference approximations:

$$\begin{aligned} P(t + \Delta t) &\approx P(t) + [F(t) + G(t)] \Delta t \\ N_b(t + \Delta t) &\approx N_b(t) + \mathcal{J} \Delta t \\ z(t + \Delta t) &\approx z(t) + v(t) \Delta t \end{aligned} \quad (10.20)$$

State variables

The next step is to find the **state variables** ^{§10.15 p.274} that can be used to repeat the time steps of the rocket's simulation. Try first to find them by yourself:



Exercise 10.18

Review §10.15 if necessary. Find a set of state variables from which the time-varying quantities in the time-stepping loop can be calculated at each time step.

Two obvious state variables are the altitude z and velocity v of the rocket. From these we can time-step the altitude itself.

Another natural state variable is the amount of fuel matter N_b . Since the fuel matter flux $\mathcal{J}$ is constant, with this state variable we can time-step the balance of matter.

From the velocity v and amount of matter N_b we can also calculate the rocket's momentum P , by means of its constitutive relation. From the amount of matter N_b we can calculate the gravity force G as well by means of its constitutive relation.

What's left is the force F . A look at its constitutive relation shows that the only time-varying term is the velocity of the exhaust gases v_b . Do we need this quantity as a state variable? Actually not. This velocity appears the constitutive relation for the matter flux, in which all other quantities are constant except for the rocket velocity v , which we have counted in as a state variable. We can therefore calculate v_b from this constitutive relation as follows:

$$v_b(t) = \frac{J}{A \delta} + v(t) .$$

In the simulation, this expression can either be directly substituted in the expression for $F(t)$, which becomes

$$F(t) = F_{\text{atm}} + A p + \mathcal{J} \rho \left[\frac{J}{A \delta} + v(t) \right] ,$$

or it can be computed separately, and then the result substituted in $F(t)$. We shall follow this latter approach.

Now all quantities needed for the time step are accounted for. Our state variables are (z, v, N_b) .

After the update (10.20) of P, N_b, z , we already have the state variables N_b and z for the new timestep. Only v is missing, but it's easily obtained from the new values of P and N_b by means of the constitutive relation for momentum:

$$v(t + \Delta t) = P(t + \Delta t) / [m_a + \rho N_b(t + \Delta t)] .$$

The body of the time-integration loop in our simulation looks therefore as follows; the dependencies between the time-dependent quantities before the time update are indicated by arrows:

$$\begin{aligned}
 & z(t), \quad v(t), \quad N_b(t) \\
 & P(t) = [m_a + \rho N_b(t)] v(t) \\
 & v_b(t) = \frac{J}{A \delta} + v(t) \\
 & F(t) = F_{\text{atm}} + A p + \mathcal{J} \rho v_b(t) \\
 & G(t) = -[m_a + \rho N_b(t)] g \\
 & P(t + \Delta t) \approx P(t) + [F(t) + G(t)] \Delta t \\
 & N_b(t + \Delta t) \approx N_b(t) + \mathcal{J} \Delta t \\
 & z(t + \Delta t) \approx z(t) + v(t) \Delta t \\
 & v(t + \Delta t) = P(t + \Delta t) / [m_a + \rho N_b(t + \Delta t)]
 \end{aligned} \tag{10.21}$$

Constants; initial and boundary conditions

In order to choose somewhat realistic values for the constants and the initial and boundary conditions of our simulation, we use values available for the flight of the Saturn V SA-506, from the [Apollo 11](#)⁴⁴ mission which

brought humans onto the Moon for the first time. Here are the chosen values, together with their sources:

$m_a = 7.9 \times 10^5 \text{ kg}$	rocket mass without 1st-stage fuel (Orloff 2024)
$\rho = 0.22 \text{ kg/mol}$	molar mass of fuel (Sutton & Biblarz 2017 Tables 5-4, 5-5)
$\delta = 0.5 \text{ mol/m}^3$	molar density of exhaust gases (Oefelein & Yang 1993 Table 2)
$\mathcal{J} = -6 \times 10^4 \text{ mol/s}$	matter (exhaust gases) influx at nozzle (Oefelein & Yang 1993 Table 2)
$A = 50 \text{ mol/m}^3$	total area of nozzles (NASA 1972 §III p. 3.7)
$p = 5 \times 10^4 \text{ N/m}^2$	exhaust-gas pressure (Oefelein & Yang 1993 Table 2)
$N_b(t_0) = 9.8 \times 10^6 \text{ mol}$	amount of 1st-stage fuel (Orloff 2024)
$t_1 = 150 \text{ s}$	time to 1st-stage engine cutoff (Orloff 2024)
$g = 9.8 \text{ N/kg}$	free-fall acceleration
$p_{\text{atm}} = 1 \times 10^5 \text{ N/m}^2$	atmospheric pressure (sea level)

The script implementing our simulation is shown in table [10.4](#) on page [314](#)

✂ To be continued

Table 10.4 Script for simplified rocket simulation

```

1  %% Numerical simulation of rocket propulsion
2  %% Coordinates (t, z)
3
4  %% Constants
5  m = 7.9e5;    % kg: rocket mass without fuel
6  rho = 0.22;   % kg/mol: fuel molar mass
7  delta = 0.5; % mol/m^3: fuel molar density
8  A = 50;       % m^2: nozzle area
9  g = 9.8;      % N/kg: free-fall acceleration
10
11 %% conditions. State: (z, v, Nb)
12 t = 0;        % s: time
13 z = 0;        % m: altitude of control volume
14 v = 0;        % m/s: velocity of control volume
15 Nb = 9.8e6;   % mol: amount of fuel
16
17 %% Boundary conditions
18 J = -6e4;      % mol/s: matter influx at nozzle
19 patm = 1e5;    % N/m^2: atmospheric pressure
20 p = 5e4;       % N/m^2: pressure at nozzle
21 Fatm = -A * patm; % N: force on rocket surface
22 Fstress = A * p; % N: stress vector at nozzle
23
24 %% Time-iteration parameters
25 t1 = 150;      % s: final time
26 dt = 0.0001;  % s: time step
27
28 %% Plotting
29 dtplot = t1/360; % time interval between plots
30 tplot = dtplot;  % time for next plot
31 clf
32 subplot(2, 1, 1); plot(t, z, 'o', 'color', '#4477AA')
33 xlabel('time {\it t}/s'); ylabel('{\it z}/m')
34 axis('tight'); xlim([0, t1]); grid on; hold on
35 subplot(2, 1, 2); plot(t, v, '^', 'color', '#EE6677')
36 xlabel('{\it t}/s'); ylabel('{\it v} / (m/s)')
37 axis('tight'); xlim([0, t1]); grid on; hold on
38
39 %% Numerical time integration
40 while t < t1 && Nb > 0 % '&&' means 'and'
41     %% constitutive relations
42     P = (m + rho * Nb) * v;
43     vb = v + J / (A * delta); % needed to calculate F
44     F = Fatm + Fstress + J * rho * vb;
45     G = -(m + rho * Nb) * g;
46

```

Download Octave version
[rocket_simplified.m](#)⁴⁵

Download Python version
[rocket_simplified.py](#)⁴⁶

```
47 %% step forward in time with balance laws
48 t = t + dt;
49 Nb = Nb + J * dt;
50 P = P + (F + G) * dt;
51 z = z + v * dt;
52
53 %% constitutive relations: find new state variables
54 v = P / (m + rho * Nb);
55
56 %% plot
57 if t > tplot
58     subplot(2, 1, 1); plot(t, z, 'o', 'color', '#4477AA')
59     subplot(2, 1, 2); plot(t, v, '^', 'color', '#EE6677')
60     pause(0)
61     tplot = tplot + dtplot;
62 end
63 end
```

URLs for chapter 10

1. <https://whc.unesco.org/en/list/59>
2. <http://hyperphysics.phy-astr.gsu.edu/hbase/Relativ/relmom.html>
3. <https://oceanservice.noaa.gov/geodesy>
4. <https://www.jpl.nasa.gov>
5. <https://www.iers.org>
6. <https://openbooks.lib.msu.edu/neuroscience/chapter/touch-the-skin/>
7. <https://textbooks.math.gatech.edu/ila/parametric-form.html>
8. <https://textbooks.math.gatech.edu/ila/row-reduction.html>
9. <https://www.noaa.gov/jetstream/atmosphere/air-pressure>
10. <https://www.scenicworld.com.au/experience/scenic-skyway>
11. <https://www.hybridairvehicles.com/>
12. <http://hyperphysics.phy-astr.gsu.edu/hbase/pbuoy.html#arch>
13. <https://www.jaerenluftsport.no>
14. <https://www.birdful.org/how-is-soaring-different-from-flying-birds/>
15. <https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/wing-geometry/>
16. https://pglpm.github.io/7wonders/scripts/tennisball_rP.m
17. https://pglpm.github.io/7wonders/scripts/tennisball_rP.py
18. <https://mse.engineering.cmu.edu/about/what-is-mse.html>
19. https://pglpm.github.io/7wonders/scripts/hooke_spring_simple.m
20. https://pglpm.github.io/7wonders/scripts/hooke_spring_simple.py
21. https://pglpm.github.io/7wonders/scripts/hooke_spring_simple.m
22. https://pglpm.github.io/7wonders/scripts/hooke_spring_simple.py
23. https://pglpm.github.io/7wonders/scripts/hooke_spring.m
24. https://pglpm.github.io/7wonders/scripts/hooke_spring.py
25. https://pglpm.github.io/7wonders/scripts/hooke_spring.m
26. https://pglpm.github.io/7wonders/scripts/hooke_spring.py
27. <https://pglpm.github.io/7wonders/scripts/rubberband.m>
28. <https://pglpm.github.io/7wonders/scripts/rubberband.py>
29. <https://doi.org/10.1351/goldbook.G02579>
30. https://pglpm.github.io/7wonders/scripts/hooke_spring_exercise.m
31. https://pglpm.github.io/7wonders/scripts/hooke_spring_exercise.py
32. https://pglpm.github.io/7wonders/scripts/idealgas_exercise.m
33. https://pglpm.github.io/7wonders/scripts/idealgas_exercise.py
34. <https://pin.it/5rsbVkSff>
35. <https://physics.weber.edu/schroeder/md>
36. <https://doi.org/10.1351/goldbook.MT06969>
37. <https://physics.weber.edu/schroeder/md/>
38. <https://www.flickr.com/photos/92798732@N00/4457615801>
39. <https://www.metmuseum.org/art/collection/search/339776>
40. <https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/rocket-thrust/>
41. <https://www.grc.nasa.gov/WWW/k-12/airplane/nozzled.html>
42. https://space.skyrocket.de/doc_lau_det/saturn-5_2st.htm
43. <https://www.smithsonianmag.com/air-space-magazine/we-built-saturn-v-180-964759/>
44. <https://www.nasa.gov/mission/apollo-11/>

- 45. https://pglpm.github.io/7wonders/scripts/rocket_simplified.m
- 46. https://pglpm.github.io/7wonders/scripts/rocket_simplified.py

Balance of energy-mass

I turned the page. The answer was, for the wind-up toy,
 “Energy makes it go.” And for the boy on the bicycle,
 “Energy makes it go.” For everything, “*Energy* makes it go.”
 Now that doesn’t *mean* anything. Suppose it’s “Wakalixes.”
 That’s the general principle: “Wakalixes makes it go.”
 There’s no knowledge coming in. The child doesn’t learn
 anything; it’s just a *word*! [. . .]

It’s also not even true that “energy makes it go,” because if
 it stops, you could say, “energy makes it stop” just as well.

R. P. Feynman 1989

§ 11.1 Formulation and generalities



Balance of energy-mass

Volume content: E Flux: W Supply: $\mathcal{A}$

$$E(t_1) = E(t_0) + \underbrace{\int_{t_0}^{t_1} W(t) dt + \int_{t_0}^{t_1} \mathcal{A}(t) dt}_{\text{integral expression}} \quad \underbrace{\frac{dE(t)}{dt} = W(t) + \mathcal{A}(t)}_{\text{differential expression}} \quad (11.1)$$

The balance of energy-mass is extremely important in physical phenomena that underlie many modern (post-**industrial revolution**¹) technologies. It must often be explicitly accounted for, together with the balance of momentum; and it is often the main governing balance, when the balance of momentum can be neglected. It’s the relevant balance when we put thick clothes on in order to keep warm, or when we watch a video or do computations on a laptop. The balance of energy-mass is also the reason why some things *don’t* happen. For instance you never see rigid objects resting on some static surface, like a pen on a table, suddenly move by

themselves. This is a consequence of the balance of energy-mass together with the balance of entropy which we'll study in chapter 15.

In discussing the [balance of momentum](#) §10.1 p. 227 we saw that there were just a few constitutive relations for its volume content $\mathbf{P}$, and a plethora of constitutive relations for its flux $\mathbf{F}$. In the case of energy-mass there is a great variety of constitutive relations that connect both its volume content E and its flux W to all other quantities: matter, momentum, angular momentum, the metric, electric charge, electromagnetic fields, and [auxiliary quantities](#) §3.11 p. 86 like temperature. These constitutive relations are therefore an important focus of [materials science](#) §10.11 p. 259.

In these notes we shall first restrict our focus on general constitutive relations that can be applied in physical phenomena that involve matter but not electric charge or the electromagnetic field. Then we shall restrict our focus even further, on constitutive relations for specific kinds of matter in particular states, for example ideal gases. Often we shall also simplify the discussion by considering physical phenomena where only one spatial dimension is relevant.

The balance of energy-mass is therefore also very important for numerical time integration, because its constitutive relations allow us to find fluxes and supplies of other quantities, necessary for stepping them forward in time.

§ 11.2 Definitions of total energy

In the physics literature we can read about energies and energy fluxes having all sorts of names – ‘internal’, ‘kinetic’, ‘potential’, ‘elastic’, ‘electromagnetic’, and others. We can also read about different kinds of energy balances, which can be related to one another through particular sequences of mathematical steps.

One confusing aspect of such variety is that it becomes unclear what is, and what is not, to be counted as “energy content”. One example is ‘gravitational potential energy’. Some texts define it as “[energy an object possesses because of its position in a gravitational field](#)”², that is, as a form of energy *content*. Other texts, especially in fluid dynamics or materials science, do not include any gravitational terms in the energy content; related terms appear instead as energy *fluxes* or *supplies*. Another example is ‘kinetic energy’, which is sometimes included as part of the total energy content, and sometimes not, for instance in texts on fluid dynamics.

$$\frac{\partial}{\partial t} \left(\frac{1}{2} \rho v^2 + \rho \hat{u} \right) + \rho (\mathbf{v} \cdot \mathbf{g})$$

rate of increase of energy per unit volume rate of work done on fluid per unit volume by external forces

Snippets from a formula in *Introductory Transport Phenomena* (Bird et al. 2015). This text does not consider gravitational potential energy to be part of the total energy. Compare the expression above on the right with eq. (11.3) on page 326 below.

One can show that all these points of view have mathematically the same consequences. Is the definition of ‘total energy content’ arbitrary, then? Yes it is, and relativity theory gives a clear understanding of the origin of this arbitrariness.

Relativity theory shows that *the definition of “total energy” depends on the choice of a reference velocity and a reference clock*. From a spacetime perspective, this is the choice of a four-dimensional vector. The choice of velocity & clock can also be different at each spacetime point. Furthermore, this choice is *independent from the choice of a coordinate system*. That is, once we have chosen coordinates (t, x, y, z) , we can still choose arbitrary reference velocities and reference clocks to define ‘total energy’.

Given a coordinate system (t, x, y, z) , one of the following three choices is usually implicitly made:

- E1 At each point in space and time we may choose a zero coordinate velocity – that is, we don’t move with respect to the coordinates (x, y, z) – and we may measure time according to coordinate time t .

We shall call **total energy** or **coordinate energy** the energy defined with respect to this velocity & clock.

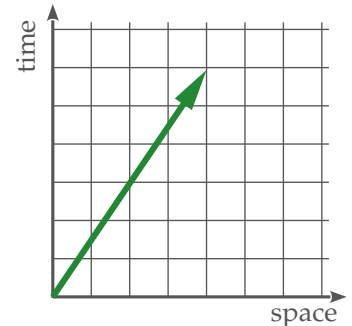
- E2 We may choose a zero coordinate velocity, as in the previous choice; but instead of measuring time according to coordinate time t , we measure it according to the *proper time* ^{§2.1 p.45} of a static observer.

Energy defined with respect to this velocity & clock is similar to total energy defined above, except that it lacks a term. We call this missing term **gravitational potential energy**.

- E3 If matter is present, we may choose the matter’s velocity (which is *related to the matter flux* ^{§4.15 p.124}), and we may measure time according to the proper time of a clock moving with that body of matter.

Energy defined with respect to this velocity & clock is called **internal energy**. It is similar to the energy of the preceding definition, except that it lacks a term. We call this missing term **kinetic energy**. This also means that internal energy lacks *two* terms with respect to the total energy of the first definition: the ‘gravitational potential energy’ term, and the ‘kinetic energy’ term.

This definition of energy is often adopted in the description of fluids. Note that this energy *cannot* be defined if matter is not present, because there’s no velocity of matter to take as reference.



A reference ‘velocity & clock’ is simply a vector with four components in four-dimensional spacetime

If we denote by U the ‘internal energy’ of choice E3, by E_k the ‘kinetic energy’ that makes the difference between choices E2 and E3, and by E_p the ‘gravitational potential energy’ that makes the difference between choices E1 and E2, then the relationship between the reference choices above can be represented as follows:

$$\begin{array}{c}
 \underbrace{U + E_k + E_p}_{\text{velocity \& clock of matter}} \\
 \underbrace{\quad \quad \quad}_{\text{zero velocity \& proper-time clock}} \\
 \underbrace{\quad \quad \quad}_{\text{zero velocity \& coordinate-time clock}}
 \end{array}$$

Each of the three differently-defined energies satisfies a balance law with different fluxes and supplies. Note that we must *not* use *all* of these differently defined energies at the same time. *One is enough*. Depending on the physical application or problem, one definition can be mathematically more convenient to use than another.

In these notes we shall adopt the *total energy* choice, because it has the following interesting and convenient features:

- It is defined anywhere, even where no matter is present. This is important when we need to study phenomena involving electromagnetic fields.
- Its supply is practically zero in coordinate systems typically used for physical phenomena on Earth or in the solar system, like [the ITRS](#) or [the ICRS](#) §2.5 p.56. In other words, this definition of energy practically *satisfies a conservation law* in these common coordinate systems.
- Its volume content typically contains terms that can be easily connected to the motion of matter and to the gravitational field.



An aspect still rarely discussed

Most physics textbooks today unfortunately do not mention the dependence of the energy definition on a reference velocity & clock; or worse they mix it up with the dependence on coordinate systems.

The dependence on a reference velocity & clock is important because it establishes an exact distinction between energies that does *not* depend on constitutive relations. For instance, the distinction between an ‘elastic energy’ and an ‘electromagnetic energy’ comes from the use of particular constitutive relations, and it’s only approximate: more exact formulae

show that such a clear-cut separation cannot be made. But the distinction of a ‘gravitational potential energy’ can always be made and does not depend on the choice of constitutive relations.

§ 11.3 Is energy conserved?

Energy is *balanced*, but not *conserved*, according to our [definitions of these terms](#) §5.3 p. 131. This is not a controversial statement: recall that many texts use the term ‘conservation’ in the sense of ‘balance’, that is, they are not excluding the presence of a volume supply. In fluid mechanics, for instance, the internal-energy definition is typically used, and as previously mentioned this energy definition satisfies a balance law, not a conservation law.

The statement “energy is conserved”, however, is often meant in another sense: if we take a control volume where nothing is flowing in or out – no energy, matter, electric charge, electromagnetic field, momentum – then isn’t its energy content constant in time?

The answer to this question is also *no*. As far as we know, there is constant creation of energy on cosmological scales, independently from how we choose the reference velocity & clock for the definition of energy. The total energy of the universe (provided such a notion can be mathematically defined) is therefore not constant. The reason is that the spacetime metric of the universe is not constant along the time direction; this is the so-called [expansion of the universe](#)³. An energy that is strictly conserved can only be defined if the metric is constant along some time direction. See [Carroll’s](#)⁴ and [Baez’s](#)⁵ interesting essays about this topic.

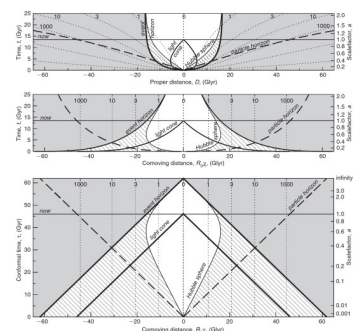
[C]osmologists have not done a very good job of spreading the word about something that’s been well-understood since at least the 1920’s: energy is not conserved in General Relativity [Carroll 2010](#)⁶

Something analogous is true for momentum and angular momentum: they are strictly conserved, that is, their supplies are zero, only if the spacetime metric remains the same in different spatial directions and under rotations. We have therefore this curious difference:

- On planetary scales the spacetime metric is not quite constant in space but almost constant in time; so momentum and angular momentum are not conserved, but energy approximately is.

$$\frac{DE}{Dt} = \frac{\sigma_{ij}}{\rho} \frac{\partial u_i}{\partial x_j} + \frac{1}{\rho} \frac{\partial}{\partial x_i} \left(k \frac{\partial T}{\partial x_i} \right)$$

Balance of internal energy in a classic text on fluid dynamics (Batchelor 2002). The first summand on the right (in the [red box](#)) is the energy *supply*.



Depiction of change of space-time metric on cosmological scales, in three different coordinate systems (from Davis & Lineweaver 2004).

- On cosmological scales the spacetime metric is approximately constant in space but not in time; so momentum and angular momentum are approximately conserved, but energy is not.

The fact that energy is not conserved, however, does not mean that we could find ways for our cars or laptops to magically operate by themselves. First, the amount of energy supply in our solar system and even in our galaxy is negligible. For devices operating in the solar system, any increase in energy content ultimately comes as an energy influx from the Sun. Second, the human problem of “using energy” is not about energy *creation* but about energy *conversion* from one form to another. This topic will be discussed in chapter 15.

§ 11.4 Forms of energy

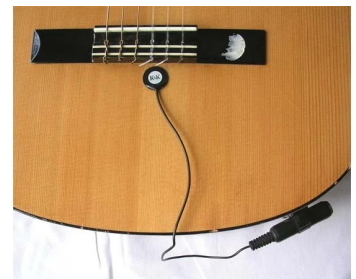
Once we have agreed on a definition of ‘total energy content’ by means of a reference velocity & clock, a distinction is usually further made between different “forms” of energy, for example ‘elastic’ or ‘electromagnetic’. This distinction as a different origin: it comes from particular *constitutive relations* that connect the total energy to other quantities like matter, temperature, electromagnetic field. There is a wild variety of such constitutive relations, which depend on the physical phenomenon to be described.

In general it is not possible to separate the dependence of total energy on other quantities into a sum of neatly distinct “pieces” like:

“mechanical energy” + “thermal energy” + “electromagnetic energy” + ...

and say “this is the energy contributed by matter”, “this is the energy contributed by the electromagnetic field”, and so on. *Energy is a property of the other quantities taken as a whole.* The absence of such a separation is the origin of many interesting and useful physical phenomena like [piezoelectricity](#)⁸ and [magnetostriction](#)⁹.

In some situations, however, it is possible to *approximately* express energy as a sum of terms that depend on particular quantities only. In these cases we can speak for instance of ‘elastic energy’, ‘radiation energy’, and similar expressions denoting the terms in the approximate sum.



The functioning of a guitar pickup is based on the fact that energy cannot be clearly separated into ‘mechanical’ and ‘electromagnetic’ ([image: Georg Feitscher](#)⁷).

§ 11.5 Internal energy, kinetic energy, gravitational potential energy

If a control volume contains matter, and any electric charge or electromagnetic field is negligible, then that control volume also contains an amount of total energy-mass that can be written in a very general form.



Energy-mass associated with matter

Take a coordinate system (t, x, y, z) close to the Earth's surface, roughly fixed with respect to the distant stars, and with z pointing upwards. Consider a control volume containing an amount of matter N but no electric charges or electromagnetic fields. This control volume must be such that the coordinate velocity $\mathbf{v}$, the [molar mass](#) $\blacktriangleright$ §7.4 p.193 ρ , and the z -coordinate of the matter within are approximately the same throughout the volume (this is true, for instance, if the volume is small enough).

Then the total energy-mass content in this control volume is given by

$$m_r c^2 + U + \frac{1}{2} m_r \mathbf{v}^2 + m_r g z \quad \text{with} \quad m_r = \rho N. \quad (11.2)$$

Remember that

$$\mathbf{v}^2 \equiv \mathbf{v} \cdot \mathbf{v} \equiv |\mathbf{v}|^2 \equiv v^2.$$

In this expression, c is the speed of light and g is the [acceleration of free fall](#) $\blacktriangleright$ §5.10 p.157. The other terms:

- $m_r c^2$ is the **rest energy-mass**, and in ordinary circumstances is vastly – say, 10^{15} times – larger than the other terms.
- U is called **internal thermodynamic energy**, or simply *internal energy*, and is usually connected to other quantities through further constitutive relations.
- $\frac{1}{2} m_r \mathbf{v}^2$ is called **kinetic energy**.
- $m_r g z$ is called **gravitational potential energy**.

Owing to the [equivalence of mass and energy](#) $\blacktriangleright$ §3.6 p.75, the expression above, divided by c^2 , represents also the total mass m within the control volume:

$$m = m_r + U/c^2 + \frac{1}{2} m_r \mathbf{v}^2/c^2 + m_r g z/c^2$$

but the first m_r term is so large that in practice we have $m \approx m_r$, with a relative error that can be smaller than 10^{-10} . For this reason, in applications where formula (11.2) is used, we just write m instead of m_r .

In many engineering applications, only energy-mass differences and changes are of importance, and it would be unwieldy to include the huge values coming from the rest-energy term. Therefore energy-mass is usually reported with respect to a ‘zero’ appropriately chosen, which can include part of gravitational and other energies; this zero is included in the $m_r c^2$ term. With this understanding we can write the energy-mass content as

$$E = U + \frac{1}{2} m \boldsymbol{v}^2 + m g z \quad (11.3)$$

omitting the constant, arbitrary amount $m_r c^2$ and using m instead of m_r . This E is what we call ‘energy’ rather than ‘energy-mass’.

The formulae above are valid in Newtonian approximation, for speeds smaller than the speed of light and weak gravitational fields, and negligible electromagnetic fields.

Here is an example of choice of a ‘zero’ reference. For water and steam, U is decreed to be zero at a particular physical condition called **triple point**¹⁰ (Wagner & Kretzschmar 2008). Therefore $m c^2$ is the total energy of a body of water in its triple-point state.

The general constitutive relation above applies to a large variety of physical phenomena. Its use often requires the further specification of a constitutive relation for the internal energy U . The latter can be wildly different depending on the kind and state of matter; this is again of interest in [materials science](#) ▶ §10.11 p. 259.

Zero of internal energy and conservation of matter

We must remember that $m = \rho N$, so the content of matter N and its balance should also enter our discussion. In phenomena where matter and antimatter are both present, the zero-point of internal energy cannot be redefined arbitrarily. Usually electromagnetic fields become important in such phenomena, so the general constitutive relation (11.2) may not be appropriate anyway.



Exercise 11.1

1. A small body of water of mass 2 kg, in its triple-point state, is moving at 3 m/s at a height of 250 m. How much is its energy, *with respect to its conventional zero*?
2. A large, static column of concrete is 10 m tall and has a rest mass of 24 000 kg. Its internal energy is $U = 10$ J. The acceleration of free fall is $g = 9.8$ N/kg. Can you apply formula (11.3), as-is, to calculate the total energy of the column?

§ 11.6 Internal and kinetic energy: observation scales

Given a control volume containing matter, in a particular coordinate system (t, x, y, z) , the amount of *total* energy contained therein is uniquely determined and agreed upon by all observers, even if one observer is making measurements on that volume with coarse instruments, and another observer is making measurements on a molecular scale.

But the separation of this total energy into internal, kinetic, gravitational *depends on the detail and resolution of observation*. For instance, a researcher who describes the matter within the volume as a continuous gas may measure a total energy $E = 4000$ J, and describe this as a sum of $U = 4000$ J internal energy, 0 J gravitational potential energy, and 0 J kinetic energy. Another researcher, who describes the matter within the same volume as a large number of atoms in motion, also measures a total energy $E = 4000$ J, but may describe this as a sum of $U = 0$ J internal energy, 0 J gravitational potential energy, and 4000 J kinetic energy, obtained by summing up $\frac{1}{2}m\mathbf{v}^2$ for all the atoms in the volume.

The reason of this difference is that the first observer does not measure any *visible* velocity in the gas: the measuring instruments average it out to zero. This observer therefore attributes the total energy completely to internal energy, and indeed still detects the atomic motion indirectly, as a *temperature* possessed by the gas. The second observer, on the other hand, can measure the velocities of the individual atoms, and therefore attributes all the total energy to kinetic energy. For this observer the gas has no temperature and no internal energy.

§ 11.7 The exchange between internal, kinetic, gravitational potential energies

Of importance in physics and engineering applications are not only changes in the total energy E , but also exchanges among its internal, kinetic, and gravitational terms, even when the total energy is constant. Here are some examples.

Bodies in motion. In the preceding chapters we have considered small bodies of matter such as a tennis ball. In many situations where the motion of a body is involved, its internal energy U is approximately constant, and therefore we only focus on its kinetic $\frac{1}{2}m\boldsymbol{v}^2$ and potential mgz energies. An exchange of energy can happen between them. For instance, in a tennis ball falling in a vacuum, the gravitational energy is decreasing while the kinetic energy is increasing at the same rate: the vertical coordinate of the ball is decreasing, and its downward velocity increasing.

In other situations the internal energy is not constant, and there are exchanges among all three energy components, as well as changes in the total energy. A ball bouncing on the floor is an example. We can clearly see that its vertical position z , and therefore its gravitational energy mgz , alternately decreases and increases. The same is true for its velocity $\boldsymbol{v}$ and kinetic energy $\frac{1}{2}m\boldsymbol{v}^2$, which are zero at the highest and lowest (bouncing) points. But we also observe that the highest vertical position of the ball gets lower and lower, until the ball rests on the floor, which means that its kinetic and gravitational energies are both zero. They have been converted partly in internal energy U of the ball, and partly transferred, via energy flux, to the internal energy of the floor.

Another example where this kind of energy conversion and energy flux are important is skydiving. The gravitational energy of a skydiver is obviously decreasing. If it were completely converted to kinetic energy, the skydiver would acquire a dangerously excessive falling speed. Instead, this energy is luckily mainly transferred, via friction, to the internal energy of the surrounding air, and partly converted into internal energy of the skydiver. For this reason the kinetic energy, and therefore the falling velocity, of the skydiver remain constant after some time.





Exercise 11.2

A skydiver jumps with zero initial velocity at an altitude of 3000 m. What would be the skydiver's velocity at 1000 m, if there were no changes in internal energy and no energy fluxes?

Compare the velocity you found and the typical velocity of 200 km/h that a skydiver may have at 1000 m, when the parachute is deployed.

Springs and rubber bands. In studying Hookean and non-Hookean springs and rubber bands in chapter 10, we said that they are usually modelled as objects without mass. They therefore have negligible kinetic and potential energies. All their energy is therefore internal energy.

Gases. Matter in a gaseous state is also often modelled as having negligible mass. All its energy is therefore modelled as internal energy. This is similar to the modelling of objects like springs and rubber bands. An important difference is that the internal energy of gases is typically highly dependent on temperature, whereas the the temperature dependence for springs can be neglected in many applications.

Solids and fluids. In modelling extended matter in a solid and fluid state, usually all three forms of energy must be taken into account. They can moreover be quite variable from one small control volume to another. This is why their modelling and simulation can be extremely complex.

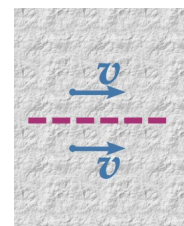
§ 11.8 Heat flux and mechanical power

We shall study some constitutive relations for the energy flux through a control surface, which are only valid under three conditions:

- there's matter on both sides of the control surface,
- this matter has velocity $\mathbf{v}$ (possibly zero) on both sides,
- there is no flux of matter through the control surface.

These three requirements might appear as contradictory: if there's no flux of matter, then how can matter have a non-zero velocity? So let's discuss them in more detail first.

One situation in which the conditions above can coexist is when the control surface is parallel to the direction of the velocity $\mathbf{v}$, as illustrated in



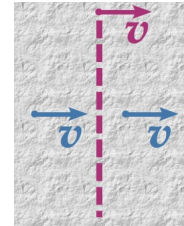
the side figure, where the control surface is represented by the **dashed red line**, and matter as a **greyish texture**. In this case the surface may be static; yet there's no flux of matter through it, and the matter around it is moving with velocity $\mathbf{v}$. Imagine for instance a fictitious horizontal surface in the middle of a water stream. Note that the matter on the two sides doesn't need to be of the same kind, though. For instance there could be water on one side of the control surface, and concrete on the other.

Another situation in which three conditions can coexist is when the control surface is also moving, with the same velocity $\mathbf{v}$ as the moving matter. See side figure. Obviously there's no flux of matter through the control surface, yet there's matter on both sides of it, moving with velocity $\mathbf{v}$. Note again that the matter on the two sides doesn't need to be of the same kind.

This situation occurs when a person pushes a door, the imaginary control surface being between hand and door; or when a hammer hits and bends a piece of hot iron, the imaginary surface being between hammer and iron; or when a bike-pump piston is pressed down vigorously, the imaginary surface being between the piston and the air within the pump.

Other situations in between the two above are also possible.

Now that these possible relationships between a control surface and matter around it are clearer, let's study an important constitutive formula for the flux of energy through a control surface



Heat flux and mechanical power

Fix a coordinate system (t, x, y, z) , and consider an imaginary control surface satisfying the following conditions:

- electric charges and electromagnetic fields are negligible;
- there's matter on both sides of the control surface;
- there is no flux of matter through the control surface;
- matter in contact with it has coordinate velocity $\mathbf{v}$, the same at every point of the surface;

the last condition is true, for instance, if the surface is small enough.

Now choose a crossing direction for the surface, in order to define fluxes. If $\mathbf{F}$ is the flux of momentum through this control surface, then the energy flux W through the surface is given by

$$W = \dot{Q} + \mathbf{F} \cdot \mathbf{v} . \quad (11.4)$$

The term $\dot{Q}$ is called the **heat flux** or *heating*; the term $\mathbf{F} \cdot \mathbf{v}$ is called the **mechanical power** or *working* transmitted by the force $\mathbf{F}$.

Heat flux $\dot{Q}$ and force $\mathbf{F}$ are typically specified by further constitutive relations or **boundary conditions** §4.14 p. 122 §6.5 p. 176. One particular boundary condition is the vanishing of the heat flux: $\dot{Q} = 0$ J/s. When this boundary condition applies, the energy flux is called **adiabatic**.

We see that the momentum *flux*, or *surface* force, $\mathbf{F}$ appears in the constitutive relation for the energy flux. Therefore, the presence of a momentum flux through a surface can contribute to the flux of energy through it: we say that “contact forces do work”.

In the integral expression for the balance of energy, the **time-integrated flux** §4.14 p. 122 $\int_{t_0}^{t_1} W(t) dt$ appears. Given the constitutive relation (11.4), this integral can be written as the sum of two integrals:

$$\int_{t_0}^{t_1} W(t) dt = \int_{t_0}^{t_1} \dot{Q}(t) dt + \int_{t_0}^{t_1} \mathbf{F}(t) \cdot \mathbf{v}(t) dt .$$

These integrals have special names:



Heat and mechanical work

The time integral of the heat flux between times t_0 and t_1 ,

$$\int_{t_0}^{t_1} \dot{Q}(t) dt ,$$

is called the **heat** transferred through the surface during that time interval.

The time integral of the mechanical power between times t_0 and t_1 ,

$$\int_{t_0}^{t_1} \mathbf{F}(t) \cdot \mathbf{v}(t) dt ,$$

is called the **mechanical work** done by the force $\mathbf{F}$ through that surface during that time interval.

It's important to keep in mind several points in applying the constitutive formula for energy flux, and in speaking about mechanical power and work. So get ready now for a string of warnings:

! Momentum supply does not contribute to energy flux

The momentum *supply*, or *volume* force, $\mathbf{G}$ does *not* appear in the constitutive relation for the energy flux. In other words, **volume forces do not contribute to energy flux**. In fact, since we're speaking of a control *surface*, such forces are not even clearly defined, since they refer to a volume.

In particular coordinate systems, some kinds of non-gravitational volume forces contribute to the energy *supply*. And if we choose different [definitions of total energy](#) ^{§11.2 p.320}, then gravitational volume forces may contribute to the energy supply as well.

! Tricky points in applying the energy-flux formula

In applying the formula above, keep always in mind the following:

- The control surface can be purely imaginary, with no real physical separation between the matter on its sides.
- The heat flux could be negative and the mechanical power positive, or vice versa.
- When we deal extended control surfaces, the conditions for applying the formula above often do **not** hold over the whole surface. For example, the velocity $\mathbf{v}$ or the heat flux $\dot{\mathbf{Q}}$ could be different on different parts the surface. To calculate the *total* energy flux in such cases, we must first divide the surface into parts for which the formula above can be applied, then add the results.

! Flux symmetry does not always apply to heat flux and mechanical power

The principle of [symmetry of flux](#) ^{§4.6 p.106} always applies to the flux of total energy W . But **symmetry of flux does not always apply to the heat flux $\dot{\mathbf{Q}}$ and the mechanical power $\mathbf{F} \cdot \mathbf{v}$ separately**.

Let's put this in symbols. Consider a control surface with sides a and b . For the heat flux $\dot{Q}_{a \rightsquigarrow b}$ in the crossing direction $a \rightsquigarrow b$, and the heat flux $\dot{Q}_{b \rightsquigarrow a}$ in the crossing direction $b \rightsquigarrow a$ it may happen that

$$\dot{Q}_{a \rightsquigarrow b} \neq -\dot{Q}_{b \rightsquigarrow a} .$$

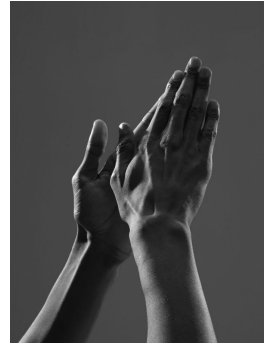
In fact in some situations the heat flux may be *positive in both crossing directions*! That is, positive heat flows to both sides. Just rub the palms of your hands against each other to verify this: both gets warmer, both received heat.

The same is true for the mechanical power $\mathbf{F} \cdot \mathbf{v}$. The reason why the symmetry of flux may break down is clearer in this case: *it happens when matter has two different velocities $\mathbf{v}_a, \mathbf{v}_b$ on the two sides of the control surface*, as it may happen when [kinetic friction](#) ^{§10.27 p.300} is present:

$$\mathbf{F}_{a \rightsquigarrow b} \cdot \mathbf{v}_b \neq -\mathbf{F}_{b \rightsquigarrow a} \cdot \mathbf{v}_a.$$

Note that the momentum flux does always satisfy the symmetry principle: $\mathbf{F}_{a \rightsquigarrow b} = -\mathbf{F}_{b \rightsquigarrow a}$.

Therefore be careful in drawing separate conclusions about heat flux and mechanical power on two sides of a control surface. We shall see examples of this curious possibility when discussing [surfaces of discontinuity](#) ^{§11.16 p.365}.

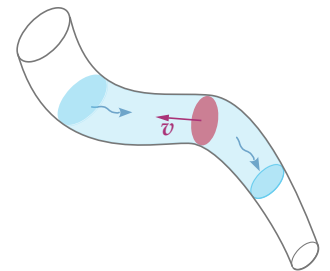


(Image: Leon Zheng ¹¹)



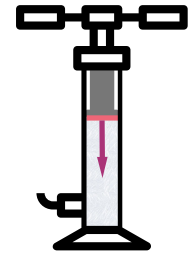
Exercise 11.3

1. Water is flowing downward in a pipe, as illustrated in the side figure by the [blue squiggly arrows](#). Take a control surface moving with velocity $\mathbf{v}$ as depicted in [red](#). Can we apply formula (11.4) for the energy flux through this control surface?
2. Suppose that there is no power $\mathbf{F} \cdot \mathbf{v}$ transmitted through a moving control surface. Does this mean that the momentum flux $\mathbf{F}$ is zero? Or does this mean that the matter's velocity $\mathbf{v}$ is zero?
3. A control surface, with a given crossing direction, is moving with velocity $[0, 2, -3]$ m/s. Through this surface we have a heat flux of -3 J/s and a momentum flux of $[1, -2, 1]$ N. How much is the energy flux through the surface?
4. Consider the control surface that separates air from the piston within a bike pump, schematized by the [red line](#) in the side illustration. The piston and the gas in contact with it, on the two sides of the control surface, are moving downward with a velocity $[0, 0, -0.5]$ m/s. The edownward flux of momentum is $[0, 0, -20]$ N. The energy flux



through this surface is adiabatic. How much is the downward flux of energy through this surface?

5. Consider again the control surface of question 3. above. We are now told that there is no matter flux through this surface, and the matter in contact with the surface on both sides is moving with the same velocity as the surface. How much is the energy flux through the surface?
6. Imagine a cylindrical *closed* control surface enveloping the air within the bike pump of question 4. above. The surface previously considered is part of this closed control surface. We are told that across the rest of the closed surface there is a total heat **efflux** of 2 J/s. How much is the total energy **influx** through the closed control surface?



Contrasting definitions of 'work'

The definition above is about the mechanical work **done by a surface force $\mathbf{F}$** – note the explicit mention of 'force'. This definition seems to be standard across all physics and engineering texts.

Many texts also give additional definitions of different kinds of power and work, which do not explicitly mention a specific force; for example 'net', or 'total', or 'internal', or 'stress' work and power. Unfortunately these definitions can be drastically different from text to text. For instance some texts define the 'total' or 'net' work on a control volume as follows:

net/total work $:=$ work done by surface forces on whole control surface .

But other texts instead call 'stress' work the definition above, and define 'net' or 'total' work as follows:

net/total work $:=$ (work done by surface forces on whole control surface) –
 (change in total kinetic energy) –
 (change in total gravitational potential energy) .

Still other texts call the above definition 'internal' work. There are other alternative definitions.

Therefore, whenever you peruse a new text for research or job purposes, and you read the simple word 'work', make sure to check the definition or formula of that term. You don't want to use incorrect values in your formulae!

$$\begin{aligned}\frac{d(U + K)}{dt} &= \dot{W} + \dot{Q}, \\ \frac{d(U + E_{\text{pot}} + K)}{dt} &= \dot{W}_{\text{stress}} + \dot{Q} \\ \frac{dU}{dt} &= \dot{W}_{\text{int}} + \dot{Q},\end{aligned}$$

Snippet from a text on thermomechanics (Müller & Müller 2009) showing three special versions of the balance of energy. The symbols $\dot{W}$, $\dot{W}_{\text{stress}}$, $\dot{W}_{\text{int}}$ stand for three different definitions of 'total mechanical power'. The heat flux is $\dot{Q}$, kinetic energy K , potential energy E_{pot} .

§ 11.9 Heat and power: observation scales

The total-energy content in a control volume depends on our coordinate system, but it does not depend on the spatial scale of our description; the separation into internal and kinetic energy, however, does depend on the spatial scale. A completely analogous situation, and for analogous reasons, occurs for the flux of total energy through a control surface, and for its separation into heat flux and mechanical power.

Consider two bodies of matter in contact with each other. For one researcher A, who observes these bodies macroscopically, there may be a flux of total energy $W = 1 \text{ J/s}$ through the contact surface, and no visible motion of matter. This researcher therefore considers this energy flux to consist completely in a heat flux $\dot{Q}_A = 1 \text{ J/s}$, and in no mechanical power $\mathbf{F}_A \cdot \mathbf{v}_A$ because the velocity $\mathbf{v}_A$ measured by A is zero. For another researcher B, who can keep track of molecular motions and velocities, the total-energy flux is still $W = 1 \text{ J/s}$; but this flux consists completely in 1 J/s of mechanical power, obtained by summing up $\mathbf{F}_B \cdot \mathbf{v}_B$ for all molecules at the contact surface; the heat flux $\dot{Q}_B$ is zero. In the notation just used, 'A' and 'B' refer, not to different control surfaces, but to different researchers, who measure the same control surface with instruments of different spatial resolutions.

Examples of heat and mechanical-power fluxes

Let us see some examples in which total-energy flux appears as heat flux, flux of mechanical-power, or both. *In all these examples we choose a coordinate system at rest with the ground.*

Holding a cup of hot tea. When we hold a cup of hot tea or coffee, we can feel a flux of energy from the cup to our hands. How much of it is heat flux, and how much is mechanical power?

Let's consider a control surface between our hands and the cup, and let's choose a hands $\rightsquigarrow$ cup crossing direction. There is obviously a momentum flux from our hands to the cup. The cup has a constant [supply of downward momentum from gravity](#)^{§10.3 p.231}, but it isn't falling. This means that there must be an influx of upward momentum: it comes from our hands. We can also feel the pressure that the cup exerts on our hands; by the symmetry of flux this means that there's a flux of momentum from our hands to the cup. So the total influx of momentum $\mathbf{F}$ is not zero. The

cup, however, is not moving: its velocity $\mathbf{v}$ is zero. The mechanical power $\mathbf{F} \cdot \mathbf{v}$ is therefore zero as well.

The energy flux through our control surface must therefore be completely in the form of heat flux $\dot{Q}$. The heat **influx** through the control surface is negative, because energy is flowing from the cup to our hands.

If we move the cup around, then its velocity is no longer zero, and some mechanical power $\mathbf{F} \cdot \mathbf{v}$ can be transmitted. It shows as an increase in the kinetic energy of the tea, which can even splash out of the cup.

Cooking. When we cook something we create a heat **influx** into the item being cooked. No mechanical power is transferred, since the velocity of matter is zero.

Spring and body. In a Hookean or non-Hookean spring or rubber band there is a momentum flux from one end of the spring to the body attached at that end. If the momentum influx for the body is $\mathbf{F}$, and the body is moving with velocity $\mathbf{v}$, then there is an energy flux into the body, possibly negative, in the form of mechanical power $\mathbf{F} \cdot \mathbf{v}$. This energy flux changes the kinetic energy and gravitational potential energy of the attached body.

By the symmetry of flux, this same amount of energy (possibly negative) flows out of the spring, whose internal energy changes. The change in internal energy can be verified by a [simple experiment](#)¹²: press an unstretched rubber band against your lips (which are especially sensitive to temperature changes), and quickly stretch it. Do you feel any change in the temperature of the rubber band? an increase or a decrease? Now allow some seconds for the stretched rubber band to reach ambient temperature again. Keeping it pressed against your lips, quickly un-stretch it. What temperature change do you feel?

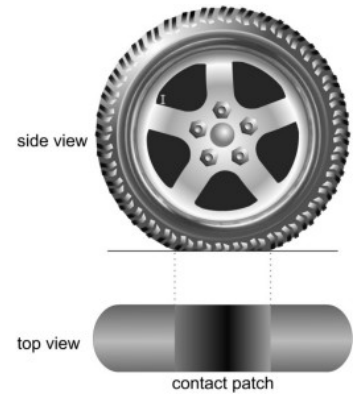
Gases. In describing and using matter in a gaseous state, usually heat flux and mechanical power must both be taken into account. Many physical theories and technologies are indeed focused on the problem of generating *effluxes of power* as large as possible from a body of matter, by providing *influxes of heat* to it.

Foot, wheel, track on ground. When we walk, there is a flux of momentum $\mathbf{F}$ from the ground to our bodies. This momentum points in the direction of our motion and upward. The vertical component nullifies the gravitational supply of momentum; the horizontal component is what keeps our forward

motion. (Obviously this is equivalent to a flux of momentum from our body to the ground, having opposite orientation.)

What about the flux of energy across the imaginary control surface that separates our foot from the ground? To calculate the mechanical power $\mathbf{F} \cdot \mathbf{v}$ we must consider the velocity $\mathbf{v}$ of our foot – more precisely, of the sole of our foot or shoe – as it's in contact with the ground. The foot actually isn't moving with respect to the ground while the flux of momentum occurs; its velocity $\mathbf{v}$ is the zero vector. This means that the mechanical power $\mathbf{F} \cdot \mathbf{v}$ is also zero. There may be a slight heat flux between foot and ground, but usually it is neglected, $\dot{Q} \approx 0 \text{ J/s}$, when considering the energies involved in the motion.

The flux of total energy between the ground and our body is therefore approximately zero (in the present coordinate system). The same conclusion is true also for a car. In particular, there is no flux of mechanical power or work done from the ground to the car, because the part of the tyre momentarily touching the ground, called 'contact area' or 'contact patch', has zero velocity in the present coordinate system. Similarly for a vehicle moving on tracks like a caterpillar or a bulldozer.



(Image: Rajamani 2012)

§ 11.10 Volumic heating and volumic mechanical power

✂ To be written in a later version

§ 11.11 Summary: constitutive relations for energy in the presence of matter

So far we have discussed two general constitutive relations: one for energy content E , formula (11.3); and one for energy flux W , formula (11.4). We also said that our definition of total energy [approximately satisfies a conservation law](#) ^{§11.2 p. 322}, so the energy supply $\mathcal{A}$ is zero. Let us rewrite here these constitutive relations, explicitly indicating their time dependence:

$$E(t) = U(t) + \frac{1}{2}m\mathbf{v}(t)^2 + mgz(t)$$

$$W(t) = \dot{Q}(t) + \mathbf{F}(t) \cdot \mathbf{v}(t)$$

$$\mathcal{A}(t) = 0 \text{ J/s}$$

Let us recall the conditions under which they are valid:

- The coordinate system is at rest with the ground, and with a vertical z -coordinate pointing upward.

- Matter is present within and directly outside the control volume.
- The control volume and surface are such that the vertical position z , the matter velocity $\mathbf{v}$ and molar mass ρ , and therefore also the rest mass m , are the same throughout.
- No matter flux occurs through the control surface.
- Electric charges or electromagnetic fields are negligible.
- The speeds involved are much lower than light's and the control volume is close to the Earth's surface.

In some situations we must divide a control volume or closed control surface into several parts where these conditions are valid.

Under these conditions, the formulae above can be used together with the balance of energy (11.1) in order to describe many common physical situations, with all sorts of materials, and to make predictions. Note how these constitutive formulae connect the fluxes and volume contents of energy, matter, momentum. They also involve two new auxiliary quantities: the internal energy U and the heat flux $\dot{Q}$.

Properly speaking the formulae above are *families* of constitutive relations. To have proper constitutive relations that give us actual numeric values, we need to specify how the internal energy U , the heat flux $\dot{Q}$, and the momentum flux $\mathbf{F}$ depend on other quantities. We are again in the domain of materials science. There are many constitutive relations for these quantities, many of which are mathematically extremely complex, involving integrals and partial derivatives. This is no surprise, because they reflect the complexity of the materials they model.

In forthcoming sections we'll focus on a set of simple constitutive relations for $\mathbf{F}$, U , $\dot{Q}$ which can be used in a more restricted but still quite broad range of physical situations, involving gases.



What if there's matter flux? Transport terms

We have repeated many times that the constitutive relation for the energy flux is through a control surface

$$W = \dot{Q} + \mathbf{F} \cdot \mathbf{v}$$

is only valid when there isn't any flux of matter through that surface. What if there *is* a flux of matter? how does that constitutive relation change?

There is indeed a more general formula that can be used even when there is a flux of matter through the control surface. To understand it, however, we must:

- give the formulae for the fluxes of matter, momentum, energy at the same time
- take into account the contents of matter, momentum, and energy in a thin control volume connected to the surface

- take into account the velocity of the control surface itself.

So the description is somewhat more complicated; you see why we choose to avoid this more general situation. But here are the fully general formulae, in case you're curious.

Choose a crossing direction for the surface, and consider the following quantities:

- $\mathbf{v}_s$ is the velocity of the surface,
- $\mathbf{e}_n$ is a *unit* vector orthogonal to the surface, having the same direction as the crossing direction,
- A is the area of the surface,
- V is the volume of a *thin control volume* in contact with the surface,
- $N, \mathbf{P}, E$ are the matter, momentum, energy contents in that thin control volume,
- $\mathbf{v}$ is the velocity of matter in contact with the surface.

Then the fluxes of matter, momentum, energy are given by the following general formulae, if no electromagnetic fields or electric charges are present and in Newtonian approximation:

$$\begin{aligned}\mathcal{J} &= A \mathbf{e}_n \cdot (\mathbf{v} - \mathbf{v}_s) \frac{N}{V} \\ \mathbf{F} &= A \mathbf{e}_n \cdot \boldsymbol{\sigma} + A \mathbf{e}_n \cdot (\mathbf{v} - \mathbf{v}_s) \frac{\mathbf{P}}{V} \\ W &= A \mathbf{e}_n \cdot \mathbf{q} + A \mathbf{e}_n \cdot \boldsymbol{\sigma} \cdot \mathbf{v} + A \mathbf{e}_n \cdot (\mathbf{v} - \mathbf{v}_s) \frac{E}{V}\end{aligned}\tag{11.5}$$

In these formulae, $\boldsymbol{\sigma}$ is a 3-by-3 matrix called the *stress tensor* or *pressure tensor*, and $\mathbf{q}$ a vector called *heat-flux vector*. The terms containing the expression $\mathbf{e}_n \cdot (\mathbf{v} - \mathbf{v}_s)$ are called *transport terms*, because they originate from the transport of matter through the surface. If matter is moving with the same velocity as the surface, $\mathbf{v} = \mathbf{v}_s$, then there is no flux of matter $\mathcal{J}$, the transport terms disappear, and we recover the simpler constitutive relations we are familiar with.



Exercise 11.4

From eqs (11.5), prove that when $\mathbf{e}_n \cdot (\mathbf{v} - \mathbf{v}_s) = 0$ (no matter flux), the energy flux W takes on the familiar expression $\dot{Q} + \mathbf{F} \cdot \mathbf{v}$, with $\dot{Q} := A \mathbf{e}_n \cdot \mathbf{q}$.



Symmetry between energy and momentum fluxes

Let's compare the formulae for energy flux and momentum flux in the presence of matter, which we've encountered so far:

$$\begin{aligned}\text{energy flux: } & \dot{Q} + \mathbf{F} \cdot \mathbf{v} \\ \text{momentum flux: } & \mathbf{F}\end{aligned}$$

where $\dot{Q}$ and $\mathbf{F}$ need to be specified by further constitutive relations.

Upon looking at the two formulae, you might wonder: Why does the full flux of momentum contribute to flux of energy, but full flux of energy doesn't contribute to the flux of momentum? Mathematically, why does ' $\mathbf{F}$ ' appear in both fluxes, but ' $\dot{Q}$ ' only in one?

You should feel proud if a question of this kind has appeared in your mind! This kind of questions has led to the discovery of new important scientific theories, like electromagnetics and relativity theory.

And the answer to your question is: *both fluxes, in fact, contribute to each other!*

We must remember that the formulae above are only Newtonian approximations, valid only in particular circumstances. A better approximation than the Newtonian is the following:

$$\text{energy flux: } \dot{Q} + \mathbf{F} \cdot \mathbf{v} + \dots$$

$$\text{momentum flux: } \mathbf{F} + \frac{1}{c^2} \dot{Q} \mathbf{v} + \dots$$

where $\mathbf{F}$ is the *stress vector or traction* §10.28 p.302 previously discussed. The tension $\mathbf{F}$ and heat flux $\dot{Q}$ represents fluxes of momentum and of energy measured in a coordinate system where matter is at rest.

The more precise formulae above show that parts of energy flux contribute to momentum flux, just like parts of momentum flux contribute to energy flux. The contribution of heat to momentum flux, however, is negligible in many situations, as clear from the very small factor $1/c^2$ containing the speed of light c . This is why this contribution is neglected in the Newtonian approximation. But there are situations in which the heat contribution to momentum flux becomes important; for instance with spacecrafts on very long journeys in the solar system and constantly receiving heat from the Sun.

The two fluxes can actually be written in an even more symmetrical way. The remaining asymmetry comes from the fact that energy flux and momentum flux have different physical dimensions. We can change the physical dimensions and units of momentum flux into those of an energy flux by multiplying it by c , so that the new momentum is $c\mathbf{F}$ and so on:

$$\text{energy flux: } \dot{Q} + c\mathbf{F} \cdot \frac{\mathbf{v}}{c} + \dots$$

$$\text{momentum flux (energy dimensions): } c\mathbf{F} + \dot{Q} \frac{\mathbf{v}}{c} + \dots$$

Now the symmetry is complete!

§ 11.12 Energy content, flux, supply depend on the coordinate system

We have repeated several times that the amount of energy in a given control volume and the flux of energy through a given control surface depend on the coordinate system adopted. Because of this dependence,

even the **explanations of energetic processes depend on the coordinate system**, and can be surprisingly different. This is a fact that surprises many, so let's illustrate it by means of a common and concrete example: a car accelerating on the road.

Consider a car that at coordinate time t_0 is at rest with respect to the ground, and at a later coordinate time t_1 is moving with respect to the ground. Thus the car and the ground have increased their motions with respect to each other between times t_0 and t_1 . We shall analyse the balances, contents, fluxes, and supplies of momentum and energy for this motion in two coordinate systems. These systems have the same coordinate time t , horizontal coordinate y , and vertical coordinate z in common, but two different horizontal coordinates x and x' :

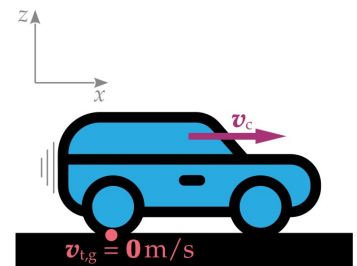
(t, x, y, z) : the ground is always at rest in these coordinates.

(t, x', y, z) : the car is always at rest in these coordinates.

In our analysis we shall only determine which quantities are positive, negative, or zero, without definite numerical values. What we want to understand are the change and exchange mechanisms for energy, and how they differ in different coordinates. Our main focus is the car, considered as a control volume. Denote by $\mathbf{P}$ its momentum, U its internal energy, E_k its kinetic energy. We use the common constitutive relations for momentum and for energy. For simplicity we imagine there's no air – even if the analysis of the energy exchanges with air is very fascinating. We also assume that heat exchanges between the car and ground are negligible.

Coordinates (t, x, y, z) : ground at rest, car moving

In these coordinates the ground is always at rest. The car is at rest at time t_0 and the velocity $\mathbf{v}_c(t)$ of its chassis is increasing in the positive- x direction until time t_1 . Note that the wheels have different velocities from the chassis, because they rotate.



Balance of momentum

The car has zero momentum at time t_0 , and momentum with positive- x direction at time t_1 . There is an influx of horizontal momentum $\mathbf{F}(t)$ with positive- x component, [friction](#) §10.27 p. 297, from the ground to the car. This influx from the ground makes the car acquire a net positive- x momentum. The momentum influx from the ground also has a positive- z component,

which cancels the negative- z gravitational momentum supply. Thus the vertical momentum of the car stays zero.

$$\underbrace{\mathbf{P}(t_1)}_{\rightarrow} = \underbrace{\mathbf{P}(t_0)}_{\text{zero}} + \int_{t_0}^{t_1} \underbrace{\mathbf{F}(t)}_{\nearrow} dt + \int_{t_0}^{t_1} \underbrace{\mathbf{G}(t)}_{\downarrow} dt$$

Balance of energy

The internal energy $U(t)$ of the car consists of chemical energy in the fuel or in the electric battery, plus other internal energy, for instance elastic, of the various other parts of the car; we shall neglect these. We know well that the chemical energy decreases between the initial and final times. The kinetic energy $E_k(t)$ of the car is zero at time t_0 and is positive at time t_1 . The gravitational potential energy of the car doesn't change because the car doesn't change its vertical position; let's set the zero of this gravitational energy content at the same vertical position as the car.

What about the energy flux between ground and car? The ground has zero velocity, and also the parts of the tyres in contact with it have instantaneously zero velocity $\mathbf{v}_{t,g} = 0$ m/s. Therefore there is *no flux of mechanical power between ground and car* in this coordinate system.

Therefore

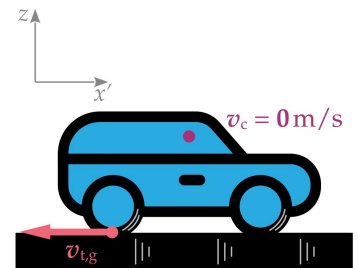
$$\underbrace{U(t_1)}_{\text{lower}} + \underbrace{E_k(t_1)}_{\text{positive}} = \underbrace{U(t_1)}_{\text{higher}} + \underbrace{E_k(t_1)}_{\text{zero}} + \int_{t_0}^{t_1} \underbrace{\mathbf{F}(t) \cdot \mathbf{v}_{t,g}(t)}_{\text{zero}} dt$$

In this coordinate system we therefore describe the energy process as follows:

(t, x, y, z): The chemical internal energy of the car's engine is being converted into kinetic energy; no mechanical energy is being released from the car to the ground.

Coordinates (t, x', y, z): ground moving, car at rest

In these coordinates the chassis of the car is always at rest, although its wheels are in motion. The ground is at rest at time t_0 and its velocity increases in the negative- x' direction until time t_1 .



Balance of momentum

The car has constant zero momentum. The influx of momentum is essentially the same as in the previous coordinate system; the differences become relevant only when the speeds involved are comparable to light's. Thus there is an increasing influx of horizontal momentum $\mathbf{F}(t)$ from the ground, having positive x' and z components. The vertical component cancels out the vertical component of the gravitational supply of momentum. But where does the *horizontal* component of the ground force go, given that the car momentum stays zero? It goes to compensate a *horizontal component* of gravitational supply of momentum, having a negative- x' direction. The supply of momentum is different in this coordinate system. The horizontal component is called **inertial force** ^{§10.3 p. 230}, but there is no real distinction between it and the gravitational force.

$$\underbrace{\mathbf{P}(t_1)}_{\text{zero}} = \underbrace{\mathbf{P}(t_0)}_{\text{zero}} + \int_{t_0}^{t_1} \underbrace{\mathbf{F}_{\text{ground}}(t)}_{\nearrow} dt + \int_{t_0}^{t_1} \underbrace{\mathbf{G}(t)}_{\nwarrow} dt$$

Balance of energy

The decrease in the chemical internal energy $U(t)$ of the fuel or battery is essentially the same as in the previous coordinate system, again neglecting differences relevant when close to light's speed. The kinetic energy of the car is zero at all times: $E_k(t) = 0$ J.

What about the energy flux between ground and car? In the present coordinate system the ground has a velocity in the negative- x direction, and also the parts of the tyres in contact with it have the same velocity $\mathbf{v}_{t,g}$ negative in the x' -direction. The momentum flux $\mathbf{F}(t)$ has a positive- x' component instead. Therefore there is a *loss of mechanical power from the car to the ground* in this coordinate system. Note that the kinetic energy of the ground is indeed increasing, because the ground moves faster and faster in the negative- x' direction.

Therefore

$$\underbrace{U(t_1)}_{\text{lower}} + \underbrace{E_k(t_1)}_{\text{zero}} = \underbrace{U(t_1)}_{\text{higher}} + \underbrace{E_k(t_1)}_{\text{zero}} + \int_{t_0}^{t_1} \underbrace{\mathbf{F}(t) \cdot \mathbf{v}_{t,g}(t)}_{\text{negative}} dt$$

In this coordinate system we therefore describe the energy process as follows:

(t, x', y, z) : The chemical internal energy of the car's engine is being transferred to the ground as mechanical power; the kinetic energy is always zero.

Comparison

Regarding momentum, we have seen that the contents and supplies of momentum are quite different in the two coordinate systems. This fact is usually not perceived as surprising.

Regarding energy, we have seen that the kinetic energy of the car is different in the two coordinate systems. This fact is usually not perceived as surprising either.

But we have also seen that what happens to the internal chemical energy is completely different in the two coordinate systems. In (t, x, y, z) it's converted into kinetic energy, so "it stays within the car" in a manner of speaking. In (t, x', y, z) it's transferred to the ground (which increases its kinetic energy), so "it doesn't stay within the car". This difference is often perceived as surprising. Yet it makes sense, because mechanical power $\mathbf{F} \cdot \mathbf{v}$ explicitly depends on a coordinate velocity, which in turn is completely dependent on the choice of coordinates.

Analyses similar to the one above can be made for all mechanical systems, with the same conclusion about coordinate-dependence. This shows that we must take "explanations" about energy flowing here or there always with a grain of salt: they may be true in one coordinate system, but false in another.



Exercise 11.5

The movement of a bicycle with respect to the ground is connected with the burning of internal chemical energy in the biker's muscles. Intuitively we would say that energy flows from the biker's feet to the back wheel; but along which path does it flow there?

Analyse, in the same way as for the car example, the flow of mechanical energy, at a given time instant t , through the three control surfaces shown by the blue lines in the side figure:

- (1) across the upper, taut part of the chain; note that there is a *tensile force* ^{§4.12 p. 119} here;
- (2) across the lower, slack part of the chain; note that, approximately, there is no surface force here;



(Image: Bikerumor¹³)

- (3) between the ground and the back wheel;
- (4) consider control surfaces across other rigid parts on which the back wheel is mounted.

Do the analysis in three different coordinate systems:

- (a) coordinate system where the ground is at rest;
- (b) coordinate system where main body of the bike is at rest;
- (c) coordinate system where, momentarily, the *upper, taut* part of the chain is at rest.

§ 11.13 Rigid motion and rigid bodies

Choose a rectangular coordinate system, and consider a body of matter; for instance a tennis ball, or some arbitrarily defined part within a block of concrete or within a parcel of water. In ordinary circumstances we can associate a **coordinate velocity** ^{§4.15 p.124} to the matter at each point in the body, or more precisely to the matter in any small control volume within the body.

Now imagine to compute the **physical distances** ^{§2.4 p.54} between all pairs of points in the body at a given instant of time t . Then check if these distances change as you follow, for a very short time lapse Δt , each point as it moves with its velocity.



Rigid motion

If the network of distances at time t between all pairs of points in a body of matter does not change during a very small time lapse Δt , then we say that the body is undergoing an **instantaneously rigid motion** at time t .

If the network of distances does not change between two times t_0 and t_1 , then we say that the body undergoes a **rigid motion** between those two times.

Clearly in a rigid motion the body has a constant volume and shape; that is, it isn't deforming.

We don't actually need to consider the distances between each point and every other point in the body: it's enough if we consider those between a point and, loosely speaking, its immediate neighbours. In fact for some pairs of points the **physical distance may not even be defined** ^{§2.4 p.54}.

When we study a body of matter in circumstances in which it only undergoes rigid motions, then we call it a *rigid body*. For instance, a block of concrete, a block of ice, a bar of hard iron, a pen undergo only rigid motions in ordinary situations, so we may call them rigid bodies in these situations. Not so with a parcel of water or air in a cylinder. No body is truly rigid. Whether we call it ‘rigid’ or not depends on the circumstances and the precision required. Moreover, under extreme conditions a body may no longer be rigid: we may melt the block of concrete and break the pen. A tennis ball may be considered as a rigid body as long as it’s freely flying; but not as it bounces on the floor or is hit by a tennis racket. In general it’s better to speak of a ‘rigid motion of a body’ rather than of a ‘rigid body’.

Rigid motions have special properties as regards the balance of energy.



Balance of energy for matter in rigid motion

For matter undergoing a rigid motion, the balance of energy take on a special form:

- The heat flux contributes only to changes in internal energy.
- The internal energy can only change through heat fluxes.
- The mechanical power contributes only to changes in kinetic & gravitational potential energy.
- Kinetic & gravitational potential energy can only change through mechanical power.

Let’s express these conditions mathematically. Consider a control volume containing all or part of a body undergoing rigid motion, and for which the velocity $\mathbf{v}$ is the same throughout, even if it can change with time. This will be approximately true at least if the control volume is small enough. The mass and internal energy contained in the control volume are m and U , the total momentum influx and heat flux across the closed control surface are $\mathbf{F}$ and $\dot{Q}$. Use a vertical coordinate z . Then, in integral expression:

$$U(t_1) = U(t_0) + \int_{t_0}^{t_1} \dot{Q}(t) dt \quad (11.6a)$$

$$\frac{1}{2}m\mathbf{v}(t_1)^2 + mgz(t_1) = \frac{1}{2}m\mathbf{v}(t_0)^2 + mgz(t_0) + \int_{t_0}^{t_1} \mathbf{F}(t) \cdot \mathbf{v}(t) dt \quad (11.6b)$$

or equivalently in differential expression:

$$\frac{dU(t)}{dt} = \dot{Q}(t) \quad (11.7a)$$

$$\frac{d}{dt} \left[\frac{1}{2} m \mathbf{v}(t)^2 + m g z(t) \right] = \mathbf{F}(t) \cdot \mathbf{v}(t) \quad (11.7b)$$

In other words, mechanical and thermal phenomena gets completely separated under a rigid motion. This separation does *not* occur when the motion is not rigid: in general, mechanical power and changes in internal energy are related to each other.

Moreover, it turns out that under a rigid motion **the equations relating kinetic & gravitational potential energy and mechanical power, (11.6b) and (11.7b), can be derived from the balance of momentum.**

The special separation between “thermal” and “mechanical” energy changes under rigid motions has an important consequence. *If we are not interested in the changes of the internal energy of a body in rigid motion, then we do not need to consider the balance of energy.* The balance of momentum is all we need.

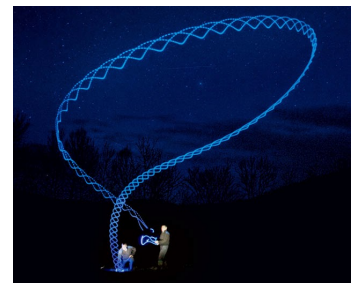
This fact explain why we were able to solve many problems of motion in the previous chapters: many motions we considered, such as the flight of a tennis balls or the fall of a block of material, were practically rigid, and we were not investigating the internal energies of those bodies. The balance of momentum was therefore sufficient to describe those motions. Springs and rubber bands do *not* undergo rigid motions, but for them we implicitly assumed that their internal energy was constant, and no heat fluxes occurred.



Exercise 11.6

1. Suppose someone tells you the positions that all matter points of a body have at a given time t (but you aren't told about other times). From these positions you can calculate all distances at time t . From this information can you tell whether that body is undergoing an instantaneously rigid motion?
2. Consider a body undergoing a rigid motion. Does this mean that
 - all of its points have the *same velocity*?
 - all of its points have the *same mass*?

Think of a boomerang for instance.



Long exposure of the trajectory of a boomerang outfitted with LEDs¹⁴

§ 11.14 Constitutive relations for ideal gases

There is a set of simple constitutive relations for energy, momentum flux, and mechanical power, which can be used as approximations for many gases, especially when they are rarefied, that is, when their amount of matter per volume N/V is low. We say that these constitutive relations apply to **ideal gases**. When we say that some material can be modelled as an ideal gas – or simply say “it’s an ideal gas”, we mean that we can model it with good enough approximation using these relations.

The constitutive relations for ideal gases involve constants and a couple of functions that can be different depending on the kind of gas. So there isn’t just one ideal gas, but a family of them. Some texts speak of “the” ideal gas because they discuss properties that are common to all ideal gases.

In discussing the constitutive relations below, we consider a closed control surface that encloses an amount of some ideal gas. We assume that there is no flux of the gas through the enclosing surface, and more generally that the conditions listed in the [previous summary](#) ^{§11.11 p.337} are satisfied. These conditions are approximately true for many practical cases, like air under compression or expansion in a bike pump.

If we want to model a body of gas accurately, however, we need to use a large number of such control volumes, each one enough small that the conditions above are approximately satisfied within it. This way we can try to simulate many interesting physical behaviours that ideal gases can have, such as turbulence or shock waves.

Pressure (momentum flux) of an ideal gas

In chapter 10 we discussed the notion of [internal pressure](#) ^{§10.17 p.283} and the [ideal-gas law with viscosity](#) ^{§10.18 p.287}: the constitutive relation that connects pressure, amount, temperature, volume, and rate of change of volume of an ideal gas, rewritten here:

$$p = \frac{RNT}{V} - \mu \frac{1}{V} \frac{dV}{dt}$$

where R is the molar gas constant.

If the surface through which this momentum flux occurs has area A , the magnitude of the force is therefore $|\mathbf{F}| = A p$.

Internal energy of an ideal gas

Next we discuss a constitutive relation that connects the amount of internal energy in the control volume with the amount of ideal gas and its temperature:



Internal energy of an ideal gas

If a small control volume contains an amount N of ideal gas at uniform temperature T , then it also contains an amount of internal energy U given by

$$U = C N T \quad (11.8)$$

where the constant C is called **molar heat capacity** and depends on the kind of ideal gas.

This formula is very important: for an ideal gas, the absolute temperature is a direct measure of the internal energy, so it can often be used as a proxy for the latter. For air, when it can be modelled as an ideal gas, the value of the molar heat capacity is approximately $C_{\text{air}} \approx 20 \text{ J}/(\text{mol K})$.



Warnings about the internal-energy formula

- The particular constitutive relation (11.8) is valid *only for an ideal gas*. The internal energy of a generic material is related to other quantities besides amount of matter and temperature.
- The constant C is also called ‘molar heat capacity at constant volume’, but despite this name *the constitutive relation above is also valid when the volume of the gas is changing!*
- Some books express the formula above saying “the internal energy of an ideal gas depends only on temperature”. This statement is somewhat vague and easy to misunderstand. First, the internal energy clearly “depends” also on the amount of matter N . Second, if in some application we express the amount of matter or the temperature as functions of other quantities, such as volume, then the internal energy also becomes a function of those quantities. The formula above can be simply expressed as follow: if we know the amount (and kind) of matter in a volume, and the temperature in that volume, then we also know the amount of internal energy therein.

Heat flux between sides at different temperatures

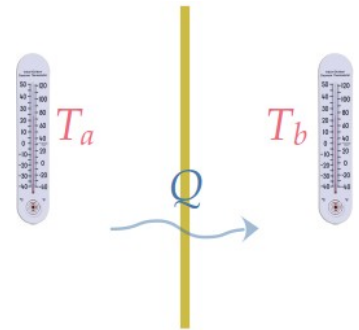
Lastly we discuss a constitutive relation that connects heat flux and temperature. This relation applies to many physical phenomena and materials: not only to ideal gases, but to many other fluids and solids as well.

Newton's law of cooling

Consider a control surface of area A and with two sides a and b . If the matter on side a has temperature T_a everywhere close to the surface, and the matter on side b has temperature T_b everywhere close to the surface, then the heat flux $\dot{Q}_{a \rightsquigarrow b}$ in the $a \rightsquigarrow b$ crossing direction is given by **Newton's 'law of cooling'**:

$$\dot{Q}_{a \rightsquigarrow b} = Ah (T_a - T_b) \quad (11.9)$$

where h is called the *surface coefficient of heat transfer* and depends on the particular physical conditions of the matter on the two sides of the surface. This coefficient is usually positive.



The constitutive formula above with $h > 0$ says that if the temperature on side a is higher than the one on side b , then a positive heat flux occurs from a to b . In other words, positive heat flows from the hotter to the colder side of the surface.

Newton's law of cooling implies that we can approximately consider temperature as having a jump or discontinuity between the two sides of the surface. In situations where this approximation is too gross, another constitutive equation is often used: **Fourier's law of heat conduction**, which we can write as

$$\dot{Q} = -Ak \frac{\partial T}{\partial x}$$

In this expression we imagine the surface to be orthogonal to the yz directions, and the crossing direction to be the positive- x direction. The derivative $\frac{\partial T}{\partial x}$ is the gradient of the temperature, expressing how much the temperature changes from one point to another very close one.

Let's make clear that Newton's law of cooling and Fourier's law *are not universal*. There are physical situations and materials for which the heat flux is connected to temperature in more complex ways; and not only to temperature, but also to momentum flux, matter flux, electromagnetic

quantities. **Thermoelectric coolers**¹⁵ are an example application of these more complex constitutive relations for the heat flux.



Heat can flow from cold to hot

Some texts say that “heat cannot flow from cold to hot”, and present this vague statement as a consequence of, or equivalent to, the second law of thermodynamics. This statement is actually *not* true, from several points of view.

The fact that heat can flow from cold to hot is very clear in devices such as refrigerators and freezers, which would not be possible otherwise:

It is obvious that, on a macroscopic scale, heat flows from a cold source to a hotter sink in a refrigerator: thus the idea that the second law requires heat to invariably flow from hot to cold is belied by the fact that refrigerators do work. (Astarita 1990)

Even more interesting examples, in which Fourier’s law does not always apply, occur for materials such as polymers ✂ add text and reference

Other common assumptions about ideal gases

Besides the particular constitutive relations for momentum flux, internal energy, and heat flux just discussed, it is typically assumed that a not-too-large volume of ideal gas has a negligible mass. This is usually a reasonable assumption (see exercise below).

If the mass m of a volume of ideal gas is assumed to be zero, then four other quantities are zero as well in that volume, at all times:

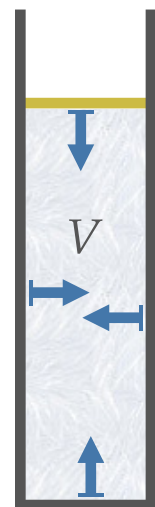
- the total momentum content $\mathbf{P} = m\mathbf{v}$
- the momentum supply from gravity $\mathbf{G} = m\mathbf{g}$
- the kinetic energy $\frac{1}{2}m\mathbf{v}^2$
- the gravitational potential energy $m\mathbf{g}z$

This assumption has an important consequence for the balance of momentum of a control volume containing an ideal gas: we have

$$\frac{d\mathbf{P}(t)}{dt} = \mathbf{F}(t) + \mathbf{G}(t) \quad \Rightarrow \quad \mathbf{F}(t) = 0$$

$=0 \qquad \qquad =0$

that is, the *total* momentum influx is always zero. This means that the momentum influxes of opposite parts of the surface cancel out perfectly, as schematized in the margin figure.



Another important consequence of assuming that a volume of ideal gas has zero mass is that **its total energy is purely internal energy**:

$$E = \underbrace{U}_{=CNT} + \underbrace{\frac{1}{2}m\boldsymbol{v}^2}_{=0} + \underbrace{mgz}_{=0} = CNT$$

which also means that any energy influx or efflux only changes the internal energy of the gas.

The masslessness assumption is of course inadequate in some conditions like extremely fast compressions or expansions. In such conditions also the ideal-gas law (10.13) must be modified.



Exercise 11.7

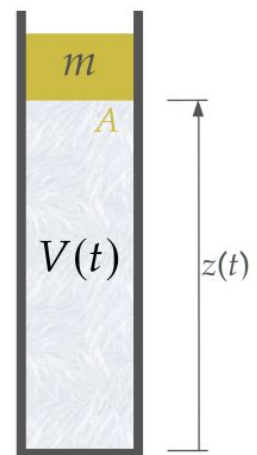
Calculate the mass of a litre, that is 10^{-3} m^3 , of air. Use the following information:

- the **molar mass** $\blacktriangleright$ §7.4 p. 193 ρ of air is around 0.03 kg/mol ; this means that an amount N of air has mass $m = \rho N$;
- the pressure p of air is around 10^5 N/m^2 ;
- take a thermodynamic temperature of air $T = 300 \text{ K}$ (around 27°C);
- pressure, volume, temperature, and amount of air are related by the **ideal-gas law** $\blacktriangleright$ §10.18 p. 287 (10.13).

§ 11.15 Simulation: ideal gas & piston with variable temperature

We previously discussed and simulated the motion of a **movable piston in contact with an ideal gas** $\blacktriangleright$ §10.19 p. 288. The setup is shown again in the margin figure. In our previous discussion we considered the thermodynamic temperature of the gas to be constant in time. Since also the amount of gas is constant in time, the pressure and therefore the force on the piston could change only with the length z of the chamber.

The constitutive equation (11.8) for the **internal energy of an ideal gas** $\blacktriangleright$ §11.14 p. 349 says that if its temperature changes then also its internal energy changes. If we want to consider the possibility of a variable temperature, we must therefore take care of the balance of energy. Let's modify our initial setup and simulation so that it allows for changes in time of energy and temperature. We do this step by step, reconsidering all control volumes, balance laws, and constitutive relations involved.



The chamber walls should be imagined indefinitely high: the piston never gets out.

Control volumes & surfaces

As usual we must choose a set of control volumes and surfaces to describe the physical system, so that we can define the contents, fluxes, and supplies of any relevant balance laws.

A natural choice is a control volume coinciding with the piston, and another control volume coinciding with the gas. The two control volumes have one horizontal surface in common, of area A , where the piston is in contact with the gas. Therefore there will be fluxes of some quantities between the two control volumes. By the [principle of symmetry of flux](#) ^{§4.6 p. 106}, a flux X of any quantity from one side to the other of this surface is equivalent to a flux $-X$ from the other side to the first.

The control volume of the piston doesn't change in size but is movable. Its position is determined by the coordinate $z(t)$. The size $V(t)$ of the gas's control volume can instead change with time, and is related to $z(t)$ by

$$V(t) = A z(t). \quad (11.10)$$

The piston's vertical velocity $v(t)$ is related to the coordinate $z(t)$ by

$$\frac{dz(t)}{dt} = v(t)$$

and is positive if the piston is moving upward. We also define the rate of change of the volume $V(t)$, which is related to the velocity $v(t)$:

$$\frac{dV(t)}{dt} = \frac{dA z(t)}{dt} = A v(t) \quad (11.11)$$

Let's now write down the main balances for the chosen control volumes of this physical system.

Conservation of matter

The geometric relations (11.10) and (11.11) have a geometric character. But they are also deeply connected to the balance of matter, because the shapes and positions of the control volumes and surfaces are chosen so that there are no matter fluxes, and the balance of matter is automatically satisfied. We therefore don't need to worry about its balance, as long as we use those geometric relations.

The constant amount of matter in the piston has a rest mass-energy m , and the constant amount of ideal gas in the chamber is N .

Balances of momentum

Piston. Let's first write the balance of momentum for the control volume of the piston. Adopt this notation:

P_{pis} : total momentum content of the piston.

F_{pis} : total momentum **influx**, or surface force, to the piston.

G_{pis} : total momentum **supply**, or volume force, to the piston.

The balance law for this control volume is then

$$\frac{dP_{\text{pis}}(t)}{dt} = F_{\text{pis}}(t) + G_{\text{pis}}(t).$$

Now let's determine the relevant constitutive relations that connect the content, flux, supply above to other quantities; and the relevant boundary conditions.

The momentum content is connected to the velocity $\boldsymbol{v}(t)$ by Newton's formula for momentum: $P_{\text{pis}}(t) = mv(t)$. The signs are correct: positive velocity means upward movement, and this corresponds to positive momentum.

For the *total* momentum influx $F_{\text{pis}}(t)$ we cannot use a single constitutive relation, because different conditions hold on different parts of the control surface. We must therefore apply the [principle of extensivity](#) §3.2 p. 69 and calculate the total influx as the sum of the influxes through partial surfaces, for which we have different constitutive relations or boundary conditions.

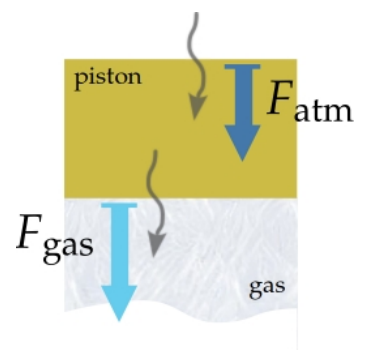
We must consider three partial surfaces:

Side: There usually are [friction forces](#) §10.27 p. 297 between the side of the piston and the walls of the chamber, but they are sometimes made negligible, for instance by the use of lubricants. Here we assume that such forces are zero. This is a boundary condition.

Top: If there's atmosphere above the piston, it exerts a downward force F_{atm} . This is also a boundary condition. See side figure.

Bottom: This surface is in common between the control volumes of the piston and of the gas. Therefore the principle of flux symmetry applies here. Denote by F_{gas} the **efflux** of momentum from the side with the piston to the side with the gas. The **influx** of momentum into the piston's control volume is then $-F_{\text{gas}}$.

For the moment we don't have any constitutive relation for the influx $-F_{\text{gas}}(t)$, but we shall find one when we examine the control volume of the gas.



The *total influx* of momentum to the piston is therefore

$$F_{\text{pis}}(t) = F_{\text{atm}} - F_{\text{gas}}(t) \quad (11.12)$$

For the total supply of momentum we use the constitutive relation

$$G_{\text{pis}} = -m g, \quad (11.13)$$

with a negative sign because this supply is directed downward.

The momentum equations for the piston's control volume are therefore

$$\begin{aligned} \frac{dP_{\text{pis}}(t)}{dt} &= F_{\text{pis}}(t) + G_{\text{pis}}(t) \\ P_{\text{pis}}(t) &= mv(t) \end{aligned} \quad (11.14)$$

$$F_{\text{pis}}(t) = F_{\text{atm}} - F_{\text{gas}}(t) \quad G_{\text{pis}} = -m g.$$

Ideal gas. We are considering the ideal gas to be practically massless. Therefore its momentum content and gravitational supply are zero. We saw that when the gas is considered massless, the balance of momentum just becomes a requirement that [the total influx of momentum be zero](#) §11.14 p.351. We must therefore simply ensure that this requirement is satisfied.

Recall that in the case of a gas the momentum influx through a small surface, being a pressure, always has an inward orientation and a direction orthogonal to the surface.

Different conditions apply to different parts of the gas's control surface; of particular importance is the part in common with the piston. We must therefore apply again the principle of extensivity. Consider three partial surfaces:

Side: The influxes on opposite parts of the side surface are equal in magnitude but opposite in orientation, and cancel each other out. The total surface force through the side surface is therefore zero.

Top: The gas is in contact with the piston at the top surface. We already denoted the **influx** of momentum to the gas by F_{gas} .

We have a constitutive relation for this influx: the [ideal-gas law](#) ▶ §10.18 p. 287:

$$F_{\text{gas}}(t) = -A p(t)$$

with $p(t) = \frac{RNT(t)}{V(t)} - \mu \frac{1}{V(t)} \frac{dV(t)}{dt}, \quad V(t) = A z(t)$

$$\Rightarrow F_{\text{gas}}(t) = -\frac{RNT(t)}{z(t)} + \mu \frac{A}{z(t)} v(t).$$

The first equation has a minus sign because the surface force points towards the gas, therefore it has a negative vertical component in this case. Note how this constitutive relation brings into play the volume $V(t)$ and the thermodynamic temperature $T(t)$ of the ideal gas, both of which can change with time.

Bottom: It is usually assumed that the chamber containing the ideal gas rests on some support. Through the corresponding surface there is therefore an influx of upward-pointing momentum, as was the case in our analysis of the [books on a table](#) ▶ §10.4 p. 233. Since the total influx of momentum to the gas must be zero, the momentum influx through the bottom must be equal and opposite to the one through the top surface. We assume that this is automatically satisfied, and therefore don't need to keep track of the momentum flux at the bottom.

The only equation for the total influx of momentum to the gas's control volume is therefore

$$F_{\text{gas}}(t) = -\frac{RNT(t)}{z(t)} + \mu \frac{A}{z(t)} v(t). \quad (11.15)$$

Balances of energy

Piston. Let's first consider the control volume of the piston. We are treating the piston as a *rigid* body. Its motion is therefore [completely determined by the balance of momentum](#) ▶ §11.13 p. 345.

Do we need to keep track of the balance of energy for the piston? Let's reason about this, considering some possible goals:

- If we needed to keep track of the internal energy or temperature of the piston, then we should use the balance of energy for it. In that case we would also need to find some constitutive relations or boundary conditions for the heat flux between piston and gas, and possibly for the internal energy of the piston.

- We shall need to keep track of the balance of energy for the gas. This means that we shall need to find constitutive relations or boundary conditions for the heat flux to the gas, including heat flux from the piston. In turn this may mean that we have to keep track of the internal energy of the piston.

So if there is a non-zero heat flux between piston and gas, then we need to keep track of the balance of energy for the piston.

In the present case let's simplify our problem by assuming that the surface between gas and piston is thermally insulated, so the energy flux through it [adiabatic](#) ^{§11.8 p.330}. With this assumption we don't need to keep track of the balance of energy for the piston.

Ideal gas. Adopt this notation:

E : total energy content of the gas.

W : total energy **influx** to the gas.

U : internal energy of the gas.

The balance law for this control volume is then

$$\frac{dE(t)}{dt} = W(t).$$

Now let's determine the relevant constitutive relations that connect the content and flux of energy to other quantities.

For the total-energy content $E(t)$ we have the usual constitutive relation for matter in terms of internal, kinetic, and gravitational potential energy. But owing to the negligible-mass assumption for the ideal gas, the total energy [amounts only to its internal energy](#) ^{§11.14 p.351} $U(t)$; no kinetic or potential energies: $E(t) = U(t)$.

For the internal energy $U(t)$ we have a the [ideal-gas constitutive relation](#) ^{§11.14 p.349}, which connects it with the amount of gas N of gas and to its thermodynamic temperature $T(t)$:

$$E(t) = U(t) = CNT(t). \quad (11.16)$$



Uniform conditions in the gas

Keep in mind that this constitutive relation can only be used when the temperature and the amount of matter per volume are approximately the same everywhere in the control volume. If we have very compression

or decompression, this assumption won't be true. What counts as 'fast' in this physical situation? The relevant speed turns out to be the speed of sound in the gas. For air it is approximately 340 m/s. Therefore, as long as the piston is moving with a velocity well below that, our approximation of uniform temperature and uniform gas distribution shouldn't be too bad.

For the *total* energy influx W we can use the general constitutive relation that expresses it as a [sum of heat flux and mechanical power](#) ^{§11.8 p. 330}. But this constitutive relation cannot be used for the gas's surface as a whole, because different heat fluxes, velocities, and surface forces occur on different parts of it. We therefore need to apply the principle of extensivity.

We must consider three partial surfaces:

Side: There could be a heat flux through the side walls, for which we could use [Newton's law of cooling](#) ^{§11.14 p. 350}. Note that as the piston moves up or down, the area of the side walls changes, and this should be taken into account in Newton's law of cooling. For simplicity let's assume that the heat flux through the side walls is zero, that is, the energy flux is [adiabatic](#) ^{§11.8 p. 330}.

Let's focus on the mechanical power now. There is a momentum influx $\mathbf{F}_{\text{side}}$, or surface force, on side walls. But according to the ideal-gas law this surface force is *orthogonal* to the control surface. As the piston moves up or down, we assume that the gas is approximately moving only vertically. This means that, on the side walls, the mechanical power $\mathbf{F}_{\text{side}} \cdot \mathbf{v}_{\text{side}}$ is approximately zero, because $\mathbf{F}_{\text{side}}$ and $\mathbf{v}_{\text{side}}$ are orthogonal there.

The energy influx through the bottom surface is therefore

$$W_{\text{side}}(t) \approx 0 \text{ J/s}.$$

Bottom: Let's consider the possibility of a heat flux $\dot{Q}_{\text{bot}}(t)$ through the bottom surface, for which we can use Newton's law of cooling holds as a constitutive relation. Since the area A of this surface is constant, this flux is

$$\dot{Q}_{\text{bot}}(t) = A h [T_{\text{env}} - T(t)],$$

where T_{env} is the temperature of the environment. In principle this temperature could be under our control, and we could choose its time dependence. For simplicity let's assume that it's kept constant.

Regarding the mechanical power, the momentum influx $\mathbf{F}_{\text{bot}}$ through the bottom wall is directed upward. The velocity of the gas $\mathbf{v}_{\text{bot}}$ at the bottom surface must be zero, otherwise the gas would detach from it and a vacuum would form, and this is not what's typically observed. Since the gas velocity $\mathbf{v}_{\text{bot}}$ at this surface is zero, then the mechanical power is also zero: $\mathbf{F}_{\text{bot}} \cdot \mathbf{v}_{\text{bot}} = 0 \text{ J/s}$.

The energy influx through the bottom surface is therefore

$$W_{\text{bot}}(t) = \dot{Q}_{\text{bot}}(t) + \mathbf{F}_{\text{bot}} \cdot \mathbf{v}_{\text{bot}} = A h [T_{\text{env}} - T(t)].$$

Top: We could consider a heat flux through the top surface, between gas and piston. The analysis would be similar to the one for the bottom surface, but we would need to consider the temperature of the piston, and in turn possibly its energy balance, as previously discussed. For simplicity let's assume that this heat flux is zero: $\dot{Q}_{\text{top}} = 0 \text{ J/s}$.

Let's analyse the mechanical power. The momentum influx $F_{\text{gas}}(t)$ through the top surface is has a negative vertical component, and has magnitude given by the ideal-gas law as in formula (11.15). The velocity of the ideal gas at this surface must be equal to that of the surface itself, $v(t)$, otherwise the gas would detach from the piston and a vacuum would form – and then we would need very different constitutive relations. Force and velocity are therefore parallel, and at the top surface there is a non-zero influx of mechanical power $F_{\text{gas}}(t) \cdot v(t)$. The energy influx across this surface is therefore

$$W_{\text{top}}(t) = F_{\text{gas}}(t) \cdot v(t)$$

where $F_{\text{gas}}(t)$ is given by formula (11.15).

Summing up the influxes from all partial surfaces, the total energy influx into the gas's control volume is therefore

$$W(t) = W_{\text{bot}} + W_{\text{top}} \quad \text{with}$$

$$W_{\text{bot}} = A h [T_{\text{env}} - T(t)] \tag{11.17}$$

$$W_{\text{top}} = F_{\text{gas}}(t) \cdot v(t), \quad F_{\text{gas}}(t) = -\frac{RNT(t)}{z(t)} + \mu \frac{A}{z(t)} v(t).$$

The energy equations for the gas's control volume are therefore

$$\frac{dE(t)}{dt} = W(t)$$

$$E(t) = CNT(t)$$

$$W(t) = W_{\text{bot}} + W_{\text{top}} \quad \text{with} \quad (11.18)$$

$$W_{\text{bot}} = A h [T_{\text{env}} - T(t)]$$

$$W_{\text{top}} = F_{\text{gas}}(t) \cdot v(t), \quad F_{\text{gas}}(t) = -\frac{RNT(t)}{z(t)} + \mu \frac{A}{z(t)} v(t).$$

We finally have all the equations and boundary conditions to describe and numerically time-integrate our physical system. We only need to specify the initial conditions and some additional constants. Let's write them all together here:

$$\text{balances:} \quad \left\{ \begin{array}{l} \frac{dz(t)}{dt} = v(t) \\ \frac{dP_{\text{pis}}(t)}{dt} = F_{\text{pis}}(t) + G_{\text{pis}}(t) \\ \frac{dE(t)}{dt} = W(t) \end{array} \right.$$

$$\text{constit. relations:} \quad \left\{ \begin{array}{l} P_{\text{pis}}(t) = mv(t) \\ F_{\text{pis}}(t) = F_{\text{atm}} - F_{\text{gas}}(t) \\ G_{\text{pis}} = -m g \\ E(t) = CNT(t) \\ W(t) = W_{\text{bot}} + W_{\text{top}} \quad \text{with} \\ W_{\text{bot}} = A h [T_{\text{env}} - T(t)] \quad W_{\text{top}} = F_{\text{gas}}(t) \cdot v(t) \\ F_{\text{gas}}(t) = -\frac{RNT(t)}{z(t)} + \mu \frac{A}{z(t)} v(t). \end{array} \right.$$

$$\text{boundary conditions:} \quad G_{\text{pis}} = -m g, \quad F_{\text{atm}}, \quad T_{\text{env}}$$

$$\text{physical constants:} \quad A, \quad m, \quad g, \quad N, \quad C, \quad h, \quad R, \quad \mu.$$

Note that there isn't a clear distinction between some constitutive relations and boundary conditions. For instance $G_{\text{pis}} = -m g$ is a constitutive

relation, but it can also be considered a boundary condition because it's constant and known beforehand in this setup.



Exercise 11.8

1. Use the [basic script-writing strategy](#) ^{§10.15 p. 269} to write a script that simulates the system of ideal gas & piston.
2. How can we choose the [state variables](#) ^{§10.15 p. 274} for the system?
3. Simulate the system with the following numerical constants, boundary conditions, and initial conditions:

$$\begin{aligned}
 A &= 0.03 \text{ m}^2, & m &= 1 \text{ kg}, & g &= 9.8 \text{ N/kg}, \\
 N &= 0.04 \text{ mol}, & C &= 20 \text{ J/(mol K)}, \\
 h &= 8 \times 10^3 \text{ J/(K s m}^2\text{)}, & \mu &= 4 \times 10^{-5} \text{ N s/m}^2 \\
 R &= 8.314\,462\,618\,153\,24 \text{ J/(mol K)}, \\
 T_{\text{env}} &= 296.15 \text{ K}, & F_{\text{atm}} &= -A \cdot 10^5 \text{ N/m}^2 \\
 t_0 &= 0 \text{ s}, & t_1 &= 2 \text{ s}, & \Delta t &= 0.0001 \text{ s} \\
 z(t_0) &= 1 \text{ m}, & v(t_0) &= 0 \text{ m/s}, & T(t_0) &= 296.15 \text{ K}
 \end{aligned}$$

Plot the position $z(t)$ of the piston and the temperature $T(t)$ of the ideal gas as functions of time. What do you observe?

4. Simulate again with the same values as before but a surface coefficient of heat transfer $h = 0 \text{ J/(K s m}^2\text{)}$, that is, assuming that all energy fluxes are adiabatic. Plot again position and temperature against time. What do you observe?
5. Keeping all energy fluxes adiabatic, simulate now by setting the viscosity coefficient $\mu = 400 \text{ N s/m}^2$. What do you observe?
6. Play and simulate with other values!

Table 11.1 on page 363 presents an example script for simulating the system of ideal gas & piston, as proposed in the exercise above, and plotting $z(t)$ and $T(t)$. The resulting plot is shown at the end of the script on page 364.

The plot shows that the piston's position and the temperature of the gas oscillate in time, with slightly less than two oscillations per second. The oscillations are initially of a couple of centimetres for the piston and a

couple of degrees for the temperature; but they are quickly damped and tend to settle on some constant values – their *equilibrium values*.

Table 11.1 Script for ideal gas & piston

```

1  %% Simulation of ideal gas & mass piston in 1D
2  %% Coordinates (t, z)
3
4  %% Constants
5  R = 8.31446261815; % N m/(K*mol): molar gas constant
6  g = 9.8;          % N/kg: gravitational acceleration
7  C = 20;           % J/(K mol): molar heat capacity
8  mu = 4e-5;        % N s/m^2: gas viscosity
9  h = 8e3;          % J/(K m^2): heat-transfer coefficient
10 N = 0.04;          % mol: amount of ideal gas
11 A = 0.03^2;        % m^2: area of piston
12 m = 1;             % kg: mass of piston
13
14 %% Initial conditions. State: (z, v, T)
15 t = 0;             % s: time
16 z = 1;             % m: position of piston
17 v = 0;             % m/s: velocity of piston
18 T = 296.15;        % K: temperature of gas
19
20 %% Boundary conditions
21 Gpis = -m * g;      % N: gravity supply of momentum to piston
22 Fatm = -1e5 * A;    % N: force on piston by atmosphere
23 Tenv = 273.15 + 23; % K: temperature of environment
24 % other fluxes are zero
25
26 %% Time-iteration parameters
27 t1 = 2;             % s: final time
28 dt = 0.0001;        % s: time step
29
30 %% Plotting
31 dtplot = t1/360;    % time interval between plots
32 tplot = dtplot;      % time for next plot
33 figure
34 subplot(2, 1, 1); plot(t, z, 's', 'color', '#4477AA')
35 xlabel('\it t/s'); ylabel('\it z/m')
36 axis('tight'); xlim([0, t1]); grid on; hold on
37 subplot(2, 1, 2); plot(t, T, 'v', 'color', '#EE6677')
38 xlabel('\it t/s'); ylabel('\it T/K')
39 axis('tight'); xlim([0, t1]); grid on; hold on
40

```

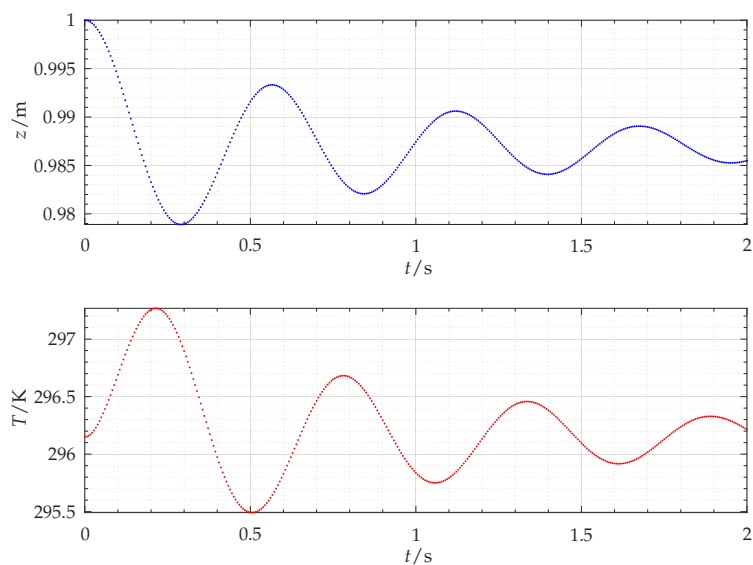
Download Octave version
[idealgas_piston.m](#)¹⁶

Download Python version
[idealgas_piston.py](#)¹⁷

```

41 %% Numerical time integration
42 while t < t1
43     %% constitutive relations
44     Fgas = -(R * N * T / z - mu * A * v / z);
45     Fpis = -Fgas + Fatm;
46     Ppis = m * v;
47     Phibot = A * h * (Tenv - T);
48     Phitop = Fgas * v;
49     Phi = Phibot + Phitop;
50     E = C * N * T;
51
52     %% step forward in time with balance laws
53     t = t + dt;
54     Ppis = Ppis + (Fpis + Gpis) * dt;
55     z = z + v * dt;
56     E = E + Phi * dt;
57
58     %% constitutive relations: find new state variables
59     T = E / (C * N);
60     v = Ppis / m;
61
62     %% plot
63     if t > tplot
64         subplot(2, 1, 1); plot(t, z, 's', 'color', '#4477AA')
65         subplot(2, 1, 2); plot(t, T, 'v', 'color', '#EE6677')
66         pause(0)
67         tplot = tplot + dtplot;
68     end
69 end

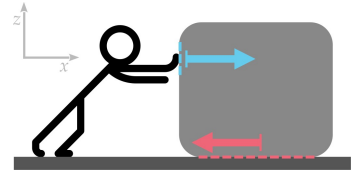
```



§ 11.16 Surfaces of discontinuity

From our discussion about [friction](#) §10.27 p. 297, we know what happens from the point of view of momentum when a person is pushing a heavy object, say a crate, on the floor. The person is providing a constant influx of horizontal momentum $\mathbf{F}_t$ to the crate, but the floor is also providing horizontal-momentum influx $\mathbf{F}_f$, kinetic friction, having opposite orientation. Suppose that the influx by the person is equal and opposite to the friction: $\mathbf{F}_t = -\mathbf{F}_f$. Then, according to Newton's constitutive equation for momentum $\mathbf{P} = m\mathbf{v}$, the crate moves with constant horizontal velocity $\mathbf{v}_{\text{crate}}$, because its time-rate change of horizontal momentum is zero:

$$\frac{d\mathbf{P}}{dt} = \mathbf{F}_t + \mathbf{F}_f = -\mathbf{F}_f + \mathbf{F}_t = 0 \text{ N}.$$



Let us consider what happens from the point of view of energy. Through the part of control surface where the person pushes the crate there's an influx of mechanical power $\mathbf{F}_t \cdot \mathbf{v}_{\text{crate}}$. Let's say that there's no heat flux there. The total-energy influx through that part of surface is then

$$W_p = \mathbf{F}_t \cdot \mathbf{v}_{\text{crate}} > 0.$$

This energy influx is positive because the force $\mathbf{F}_t$ exerted by the person and the velocity $\mathbf{v}_{\text{crate}}$ have the same orientation.

What about the contact surface of crate and floor? There should be an energy flux through there too. In fact, the motion of the crate is [rigid](#) §11.13 p. 345; therefore the positive mechanical power coming from the person should increase the kinetic or the gravitational potential energy of the crate. But these two energies are constant; this means that there must be a negative influx of mechanical power somewhere else. Obviously it must occur through the surface between crate and floor. What kind of energy flux occurs there? is it only mechanical power? is it also heating?

The constitutive relation $W = \dot{Q} + \mathbf{F} \cdot \mathbf{v}$ does *not* apply to the control surface between crate and floor, because the velocity of matter $\mathbf{v}$ is not the same on both sides of the surface. On the upper side, the matter constituting the crate is moving with horizontal velocity $\mathbf{v}_{\text{crate}}$; on the lower side, the matter constituting the floor is at rest, with zero velocity.

This contact surface is an example of *surface of discontinuity*:

Surface of discontinuity

If a physical quantity X which is a function of position $\mathbf{r}$ has two different limit values as we consider positions closer and closer to the two sides of a small control surface, then the latter is called a **surface of discontinuity for the quantity X** .

Note that it doesn't make sense to speak of surface of discontinuity for a quantity which is not a function of position, such as the volume contents of matter, momentum, energy, and so on. Volume contents are related to particular control volumes, not to positions. Quantities for which it makes sense to speak of surfaces of discontinuities are for example velocity, temperature, pressure, matter density, energy density, and similar.

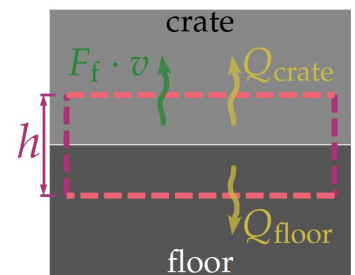
A control surface may be a surface of discontinuity for some quantity but not for another. In our present example, for instance, the matter velocity is different on the two sides, but the temperature of the floor and of the bottom of the crate might be the same. This contact surface is then a surface of discontinuity for the velocity but not for the temperature.

We shall now learn a common and important technique to study surfaces of discontinuity. This technique allows us to extend the application of some constitutive relations also to cases where they cannot be applied at first. Our discussion of the technique is largely intuitive, but it can be made mathematically rigorous.

Let's return to the example with the crate and the floor. Zoom in on the imaginary control surface that separates them. Replace this surface with a very thin imaginary control volume; see the side figure. Two horizontal sides (dashed light-red lines) of this control volume have the same extension as the original surface and are parallel to it. One of them lies completely within the crate, and the other completely within the floor. The lateral sides (dashed dark-red lines) of the control volume have a very small height h .

Consider the fluxes of momentum for this control volume. Thanks to extensivity, they can be separated into three contributions:

Surface within crate: Horizontal momentum (having leftward orientation in the figure), as friction, is coming from the floor and the lower parts of the crate, and is passed on to the upper parts of the crate. Through this surface there is therefore an *efflux* of momentum equal to $\mathbf{F}_f$.



All matter close to this surface has velocity $\mathbf{v}_{\text{crate}}$, on both sides, because this surface is completely within the crate. For the energy efflux W_{crate} through this surface we can therefore use the constitutive relation

$$W_{\text{crate}} = \dot{Q}_{\text{crate}} + \mathbf{F}_f \cdot \mathbf{v}_{\text{crate}}$$

where $\dot{Q}_{\text{crate}}$ is the heat efflux, possibly zero.

Surface within floor: Horizontal momentum (having rightward orientation in the figure), as friction, is coming from the crate and the upper parts of the floor, and is passed on to the lower parts of the floor. Through this surface there is therefore an efflux of momentum equal to $-\mathbf{F}_f$.

All matter close to this surface has zero velocity, on both sides, because this surface is completely within the floor. For the energy efflux W_{floor} through this surface we can therefore use the constitutive relation for matter, but with zero mechanical power, owing to the zero velocity:

$$W_{\text{floor}} = \dot{Q}_{\text{floor}}$$

where $\dot{Q}_{\text{floor}}$ is the heat efflux, possibly zero.

Side surfaces: We imagine to take the height h to be smaller and smaller: so small that the area of the side control surfaces is negligible. All fluxes through these surfaces are, intuitively, negligible in the limit.

Now let's examine the balance of energy for this thin control volume. Owing to the very small height h , the volume is extremely small. Therefore the energy content E and its time-rate of change dE/dt are, intuitively, negligible. The balance of energy then yields

$$\begin{aligned} \frac{dE}{dt} &= -W_{\text{crate}} - W_{\text{floor}} \\ 0 \text{ J/s} &= -(\dot{Q}_{\text{crate}} + \mathbf{F}_f \cdot \mathbf{v}_{\text{crate}}) - \dot{Q}_{\text{floor}} \end{aligned}$$

from which we find

$$\begin{aligned} W_{\text{crate}} &= -W_{\text{floor}} \\ \dot{Q}_{\text{crate}} + \mathbf{F}_f \cdot \mathbf{v}_{\text{crate}} &= -\dot{Q}_{\text{floor}} \end{aligned}$$

This is a very interesting result. Zoom out from the thin control volume just analysed, so that it looks like the initial control surface between crate and floor. What happens at this control surface is the following:

- The [symmetry of flux](#) ^{§4.6 p.106} is still valid for the flux of *total-energy*: the flux from floor to crate, W_{crate} , and the flux from crate to floor, W_{floor} , are equal in magnitude but opposite: $W_{\text{crate}} = -W_{\text{floor}}$.
- However, the energy flux appears as *heat flux plus mechanical power* $\dot{Q}_{\text{crate}} + \mathbf{F}_f \cdot \mathbf{v}_{\text{crate}}$ on one side (crate) of the surface, but only as *heat flux* $\dot{Q}_{\text{floor}}$ on the other side (floor).

In other words, the contact surface acts as a sort of converter between mechanical power and heating. When we discuss the [balance of entropy](#) ^{chapter 15 p.399} we shall see that this surface can be considered as a peculiar sort of *thermodynamic engine*. We shall also prove that positive influx of mechanical power on one side can be converted into positive outflux of heat on the other side – but the opposite conversion *cannot* happen.

This conversion, at a contact surface, between mechanical power and heating makes sense from a molecular point of view. Roughly speaking, the *visible* kinetic energy of the macroscopic, coordinated part of the motion of the molecules that make up the crate – what we call ‘the velocity of the crate’ – is transformed into *invisible* kinetic energy of a microscopic, uncoordinated motion of the molecules of crate and floor. This is why objects typically become warmer under friction.

But note an amazing fact: we found out about this conversion between mechanical power and heating by applying the balance laws, *without invoking any molecular picture!*

The result above also shows, in accordance with a [previous warning](#) ^{§11.8 p.332}, that the principle of extensivity only holds for the flux of total-energy and other total quantities, but not necessarily for the individual parts into which we may decompose those quantities; like heat flux and mechanical power in the case of energy flux.

General case

The example with the crate & floor generalizes to situations where the matter velocity is different on the two sides a and b of a control surface, possibly zero or non-zero on either side. Call $\mathbf{v}_a$ the velocity of matter on side a , and $\mathbf{F}_a$, $\dot{Q}_a$, the momentum flux and heat flux appearing on side a ;

analogously for side b . Then we have:

$$\mathbf{F}_a = -\mathbf{F}_b, \quad \dot{Q}_a + \mathbf{F}_a \cdot \mathbf{v}_a = -(\dot{Q}_b + \mathbf{F}_b \cdot \mathbf{v}_b)$$

but possibly with $\mathbf{v}_a \neq \mathbf{v}_b, \quad \dot{Q}_a \neq -\dot{Q}_b, \quad \mathbf{F}_a \cdot \mathbf{v}_a \neq -\mathbf{F}_b \cdot \mathbf{v}_b.$
(11.19)



Exercise 11.9

Consider a person pushing a crate on a conveyor belt or on moving walkway, in a direction opposite to the belt. Using the technique of the “thin control volume” and proceeding in a way analogous to the discussion above, analyse this situation and prove the formulae and inequalities (11.19).

§ 11.17 Constitutive relations for rates of formation: temperature dependence

✂ To be written

URLs for chapter 11

1. <https://education.nationalgeographic.org/resource/industrial-revolution-and-technology/>
2. <http://hyperphysics.phy-astr.gsu.edu/hbase/gpot.html>
3. <https://www.jpl.nasa.gov/edu/news/2023/7/24/exploring-the-mystery-of-our-expanding-universe>
4. <https://www.preposterousuniverse.com/blog/2010/02/22/energy-is-not-conserved>
5. https://math.ucr.edu/home/baez/physics/Relativity/GR/energy_gr.html
6. <https://www.preposterousuniverse.com/blog/2010/02/22/energy-is-not-conserved>
7. https://commons.wikimedia.org/wiki/images/File:Piezoelectric_pickup1.jpg
8. <http://hyperphysics.phy-astr.gsu.edu/hbase/Solids/piezo.html>
9. <http://hyperphysics.phy-astr.gsu.edu/hbase/Solids/magstrict.html>
10. <http://hyperphysics.phy-astr.gsu.edu/hbase/thermo/pvtsur.html>
11. <https://ca.pinterest.com/pin/713890978409863838>
12. <https://edu.rsc.org/experiments/rubber-band-experiment/460.article>
13. <https://bikerumor.com/review-specialized-s-works-ares-road-shoes-are-great-just-not-for-the-obvious-reasons>
14. <https://imgur.com/BzcZ0v1>
15. <https://www.energy.gov/energysaver/thermoelectric-coolers>
16. https://pglpm.github.io/7wonders/scripts/idealgas_piston.m
17. https://pglpm.github.io/7wonders/scripts/idealgas_piston.py

Balance of angular momentum

Around the world, around the world

Daft Punk [2005a](#)

§ 12.1 Formulation and generalities



Balance of angular momentum

Volume content: $\mathbf{L}$ Flux: $\mathbf{M}$ Supply: $\mathcal{T}$

$$\mathbf{L}(t_1) = \mathbf{L}(t_0) + \underbrace{\int_{t_0}^{t_1} \mathbf{M}(t) dt + \int_{t_0}^{t_1} \mathcal{T}(t) dt}_{\text{integral expression}} \quad \underbrace{\frac{d\mathbf{L}(t)}{dt} = \mathbf{M}(t) + \mathcal{T}(t)}_{\text{differential expression}} \quad (12.1)$$

The balance of angular momentum – or rotational momentum or moment of momentum – is no less important and no less present in everyday phenomena than the balances of momentum or energy. Like the balance of momentum, it is at the core of applications where motion and stability are important. It is crucial in studying, predicting, and planning the motion of artificial satellites around Earth and of celestial bodies, from planets to galaxies.

Yet the balance of angular momentum is mentioned less often, and may *seem* less often, used than the balance of momentum. This happens because of several related reasons. The most important constitutive relations involving angular momentum require slightly more complicated mathematics, in particular they need explicit integration of functions and the use of vector cross-products or bivectors. Because of this, the consequences of the balance of angular momentum are often pre-emptively baked-in into constitutive relations; we shall soon see an example with the Hookean spring.

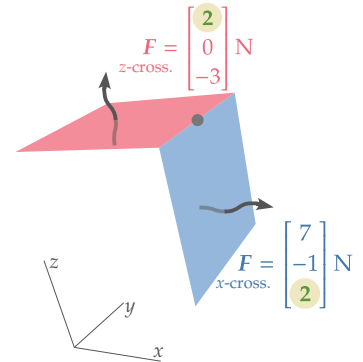
It moreover turns out that *the balance of angular momentum can be expressed without the time derivative of a volume content, or without the time-integrals of a flux and of a supply*. It can instead be expressed as a sort of symmetry condition; we can call this the **tensorial expression** of this balance:

Balance of angular momentum, tensorial expression

Take a small control surface passing through some point P , of area A , parallel to the zx directions, and with a positive- y crossing direction. Take another small control surface also passing through P and of area A , parallel to the xy directions, and with a positive- z crossing direction. Then the flux of z -momentum F_z through the zx control surface must be equal to the flux of y -momentum F_y through the xy control surface. Two more analogous laws hold by exchanging y & x or z & x in the statement above:

$$\begin{aligned} F_z \big|_{y^+\text{-crossing}} &= F_y \big|_{z^+\text{-crossing}} \\ F_z \big|_{x^+\text{-crossing}} &= F_x \big|_{z^+\text{-crossing}} \\ F_x \big|_{y^+\text{-crossing}} &= F_y \big|_{x^+\text{-crossing}} \end{aligned} \quad (12.2)$$

These equations only hold if the momentum fluxes are uniform throughout the surfaces.



The x -component of the momentum flux crossing in the z -direction must equal the z -component of the momentum flux crossing in the x -direction. In this picture each equals 2 N .

In this alternative formulation we don't even need to speak of a volume content, a flux, or a supply of angular momentum; and we don't use derivatives or integrals.

The two expressions of the balance of angular momentum can be proven to be completely equivalent, provided that the balance of momentum is satisfied. The integral and differential expressions (12.1) involve time-integration or time-differentiation, but they apply to control volumes of arbitrary shape, size, motion. The tensorial expression (12.2) does not involve integration or differentiation in time, but it can usually only be applied to very small control surfaces.

Each expression of the balance of angular momentum is used in particular applications and disciplines. The integral and differential expressions are typically used in applications with small bodies or approximately **rigid bodies** ▶ §11.13 p.345, or in applications where forces are given as boundary

conditions rather than via constitutive relations. The tensorial expression is typically used in materials science and fluid dynamics, where constitutive relations for the momentum flux $\mathbf{F}$ are given in such a way that the tensorial expression of the balance of angular momentum is automatically satisfied. This is again one case in which the balance of angular momentum plays its role in an invisible way.

§ 12.2 Definitions of angular momentum

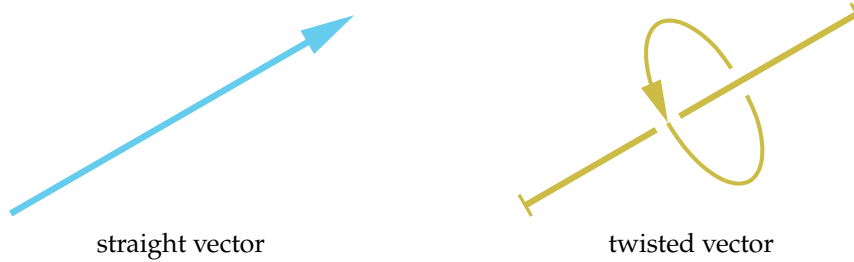
Just like [total energy](#) ^{§11.2 p.320}, also angular momentum can be defined in different ways. In fact, the freedom we have in its definition is more visible than the one we have in defining total energy. Most important is that **the definitions of angular momentum content, flux, supply depend on the choice of a reference position**. We could make this clear in the notation for momentum: if the reference position is $\mathbf{r}$, we could then use the symbol ' $\mathbf{L}_{\mathbf{r}}$ ' instead of the generic ' $\mathbf{L}$ '. There is no standard notation, for instance some texts use ' $\mathbf{L}^{[\mathbf{r}]}$ '.

The reference point is typically chosen to be **fixed with respect to the coordinate system**, that is, its coordinates do not change with time. Although it is possible to choose a reference point which depends on time, $\mathbf{r}(t)$, such choice leads to much more complicated equations. In the present notes *we shall use the origin of the coordinate system as reference point*, that is, the position $[0, 0, 0]$; and we shall simply use the symbols $\mathbf{L}$, $\mathbf{M}$, $\mathcal{T}$ for this particular choice.

§ 12.3 Angular momentum as a twisted vector

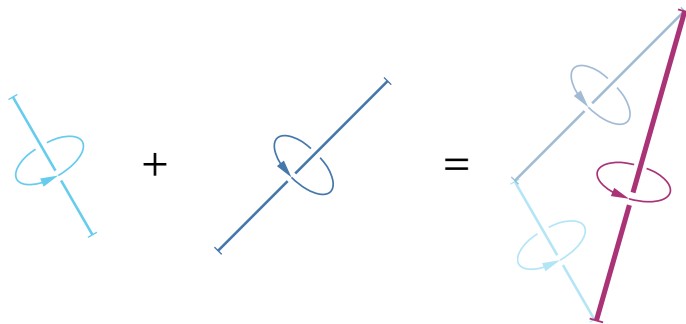
In order to represent content, flux, and supply of angular momentum we can use a kind of vectors different from the arrow-like ones with which you are probably familiar. These vectors of a different kind are called **twisted vectors**, or *pseudo-vectors* or *axial vectors* or *outer-oriented vectors*, whereas the traditional ones are called *polar*, or *straight*, or *inner-oriented* vectors.

A twisted vector represent an object related to rotation. Like a traditional straight vector it has a magnitude and a direction, that is, the line on which it lies. But its orientation is different: it is not along the direction, but *around* around it. Here is the graphical difference between a traditional straight vector and a twisted one:

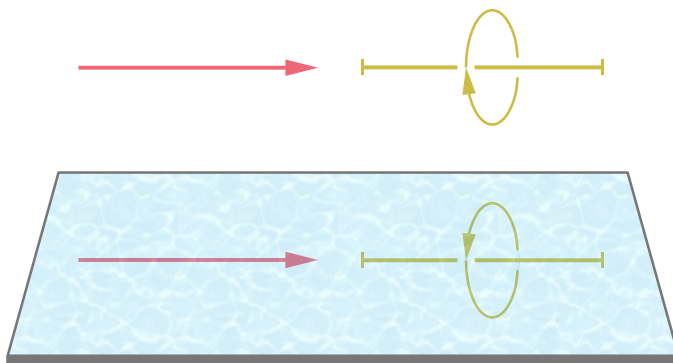


In both cases a segment indicate the direction of the vector, and the length of the segment represents the magnitude. The orientation of the twisted vector makes it immediately clear that some rotation is involved, and immediately indicates the direction of the rotation.

We can do with twisted vectors the same mathematical operation that we do with straight ones. The sum of two twisted vectors, for instance, is done with the parallelogram rule:



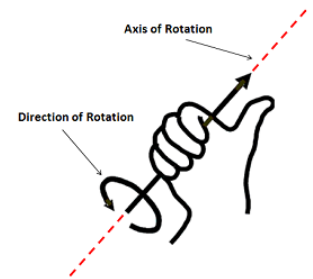
Ordinary vectors and twisted vectors behave very differently if we look at their images through a mirror parallel to their axis: the orientation of ordinary vectors appears unchanged, whereas the orientation of twisted vectors appears *reversed*:



this phenomenon reflects the behaviour of rotations under reflections.

Owing to intellectual inertia, many books are afraid of using twisted vectors, and rely on ordinary vectors instead, introducing the ‘right-hand rule’ to determine the sense of rotation from the orientation of the ordinary vector. If you’ve ever asked yourself “why the right hand, and not the left hand?”, the answer is that it’s purely a convention indeed; one could have introduced a left-hand rule instead. Using twisted vectors we don’t need these arbitrary conventions and mnemonics: the sense of rotation is unequivocally indicated.

Use whichever vector representation you prefer!



§ 12.4 Constitutive relations for angular momentum

The peculiar capacity of the balance of angular momentum to be expressed in two different ways has tight connections with the constitutive relations for angular momentum. In fact its constitutive relations could almost be taken as *definitions* of the content, flux, and supply of angular momentum. They are used in essentially all situations and theories: with matter and with electromagnetic fields, in General Relativity and in Newtonian approximation. The only exception are physical phenomena with particular kinds of materials variously called *polar*, *micropolar*, or *with intrinsic spin*.

These general constitutive relations for content, flux, and supply have common form, which involves the content, flux, supply of momentum, as well as the position vector in a given coordinate system. We therefore present all of them here.

Choose a coordinate system (t, x, y, z) ; it does not need to be [Cartesian](#) ^{§2.6 p. 60}. In the following we shall speak of momentum; keep it can be the momentum associated to matter, or to electromagnetic field, or both – there are no restriction whatsoever.



Angular-momentum content

Take a *small* control volume having position vector $\mathbf{r} = [x, y, z]$ – it’s necessary for the control volume to be small, otherwise we could not associate just one position to it. Denote by $\mathbf{P} = [P_x, P_y, P_z]$ the content of momentum in this small control volume.

The content of angular momentum $\mathbf{L} = [L_x, L_y, L_z]$, defined *with respect to the origin of coordinates* is given by the vector product

$$\mathbf{L} = \mathbf{r} \times \mathbf{P} \quad \text{or equivalently} \quad \begin{cases} L_x = y P_z - z P_y \\ L_y = z P_x - x P_z \\ L_z = x P_y - y P_x \end{cases} \quad (12.3a)$$

The components $[L_x, L_y, L_z]$ can alternatively be denoted $[L_{yz}, L_{zx}, L_{xy}]$, as some texts do. The latter notation makes the formulae above easier to remember:

$$\begin{cases} L_{yz} = y P_z - z P_y \\ L_{zx} = z P_x - x P_z \\ L_{xy} = x P_y - y P_x \end{cases} \quad (12.3b)$$

Choose whichever you prefer.



Angular-momentum flux

Take a *small* control surface having position vector $\mathbf{r} = [x, y, z]$ – it's necessary for the control surface to be small, otherwise we could not associate just one position to it. Denote by $\mathbf{F} = [F_x, F_y, F_z]$ the flux of momentum, or surface force, through this small control surface.

The flux of angular momentum $\mathbf{M} = [M_x, M_y, M_z]$, defined *with respect to the origin of coordinates*, is also called **surface torque** and is given by the vector product

$$\mathbf{M} = \mathbf{r} \times \mathbf{F} \quad \text{or equivalently} \quad \begin{cases} M_x = y F_z - z F_y \\ M_y = z F_x - x F_z \\ M_z = x F_y - y F_x \end{cases} \quad (12.4)$$

Analogously to angular-momentum content, you can denote the components by $[M_{yz}, M_{zx}, M_{xy}]$ instead.

Angular-momentum supply

Take a *small* control volume having position vector $\mathbf{r} = [x, y, z]$. Denote by $\mathbf{G} = [G_x, G_y, G_z]$ the supply of momentum, or volume force, in this small control volume.

The supply of angular momentum $\mathcal{T} = [\mathcal{T}_x, \mathcal{T}_y, \mathcal{T}_z]$, defined *with respect to the origin of coordinates*, is also called **volume torque** and is given by the vector product

$$\mathcal{T} = \mathbf{r} \times \mathbf{G} \quad \text{or equivalently} \quad \begin{cases} \mathcal{T}_x = y G_z - z G_y \\ \mathcal{T}_y = z G_x - x G_z \\ \mathcal{T}_z = x G_y - y G_x \end{cases} \quad (12.5)$$

As usual you can denote the components by $[\mathcal{T}_{yz}, \mathcal{T}_{zx}, \mathcal{T}_{xy}]$ instead.

So the content, flux, and supply of angular momentum for *small* control volumes or surfaces are obtained by the cross-product of position $\mathbf{r}$ and the corresponding momentum quantities. What's "small" is determined by the distances and velocities involved in the phenomenon under study. For example, billiard balls cannot be considered "small" when we study their collisions on a billiard table: their rotations lead to special motions; on the other hand, a planet can in some circumstances be considered as "small" when we study its motion in the solar system.

The constitutive relations above therefore *cannot be used directly for extended control volumes or surfaces*. An extended control volume or surface doesn't have a definite, single position vector $\mathbf{r}$. The only way to apply the constitutive relations above to an extended control volume or surface is therefore by [extensivity](#)^{§3.2 p.69}: divide it into parts *so small* that each of them can be characterized by a single position vector. This means an infinite number of small parts: we are performing a *three-dimensional integration*, possibly over a complicated shape.

But the difficulties do not end here. If the control volume or surface is moving, rotating, deforming, then the position vectors $\mathbf{r}(t)$ of its small parts will be changing with time. This means that we may need to perform the three-dimensional integration anew at every instant t .

It is because of these complications with three-dimensional integration that the alternative tensorial expression (12.2) of the balance of momentum is used when studying materials. But there is a special case where the constitutive relations above may be slightly simplified: to study the behaviour

of materials that always undergo *rigid motion* ^{§11.13 p.345}, that is, *rigid bodies*. We shall study these simplifications in the next sections.



Exercise 12.1

1. Consider a GPS satellite as a point that has, at a given instant, the following position and momentum content in a geocentric coordinate system (t, x, y, z) :

$$\mathbf{r} = [1.4, 1.6, 0] 10^7 \text{ m}, \quad \mathbf{P} = [-4.0, 3.5, 0] 10^6 \text{ N s}.$$

Calculate the satellite's angular momentum $\mathbf{L}$ with respect to the origin of the coordinate system (remember that you should obtain a vector with three components).

2. Assume that the GPS satellite has the following time-dependent position vector and momentum:

$$\mathbf{r}(t) = [\cos(\omega t), \sin(\omega t), 0] \cdot 2.1 \times 10^7 \text{ m}$$

$$\mathbf{P}(t) = [-\sin(\omega t), \cos(\omega t), 0] \cdot 5.4 \times 10^6 \text{ N s}$$

$$\text{with } \omega = 1.5 \times 10^{-4} \text{ s}^{-1}.$$

Calculate the time dependence of the satellite's angular momentum $\mathbf{L}(t)$ with respect to the origin of the coordinate system. Try to explain the peculiar result you obtain.

3. A flying tennis ball, considered as a point, has the following time-dependent position vector in a coordinate system (t, x, y, z) :

$$\mathbf{r}(t) = [vt, 0, -\frac{1}{2}gt^2 + z_0]$$

$$\text{with } v = 1 \text{ m/s}, \quad g = 9.8 \text{ m/s}^2, \quad z_0 = 3 \text{ m}.$$

- (a) Find its time-dependent momentum $\mathbf{P}(t)$ by using Newton's constitutive relation for momentum, assuming a mass-energy $m = 0.058 \text{ kg}$.
- (b) Then find its time-dependent angular momentum $\mathbf{L}(t)$ with respect to the origin of the coordinates.
- (c) Represent the angular momentum you found as a straight or twisted vector.

§ 12.5 Angular-momentum flux for a simplified piece of material

In chapter 10 about the balance of momentum, we studied the fluxes of momentum from a control volume consisting of a *piece of material having several simplifying properties* › §10.12 p. 260. Let us list those properties again:

- Momentum flux only occurs through two separate parts of the control surfaces, usually diametrically opposite.
- The shape and dimensions of the closed control surface can be summarized by the distance between the two parts.
- There is no flux of matter through the control surface.
- The matter within the closed control surface has negligible mass-energy, and therefore the momentum content is practically zero at all times.

We also claimed that the simplifying assumptions above have two important consequences:

- C1 The fluxes $\mathbf{F}_a$ and $\mathbf{F}_b$ must have equal magnitudes and directions, and opposite orientations, that is,

$$\mathbf{F}_a = -\mathbf{F}_b .$$

- C2 If the partial control surfaces a and b are enough small, then the fluxes $\mathbf{F}_a$ and $\mathbf{F}_b$ must each have the same direction as the line connecting the two partial surfaces:

$$\mathbf{F}_a , \mathbf{F}_b , \mathbf{r}_a - \mathbf{r}_b \text{ are parallel}$$

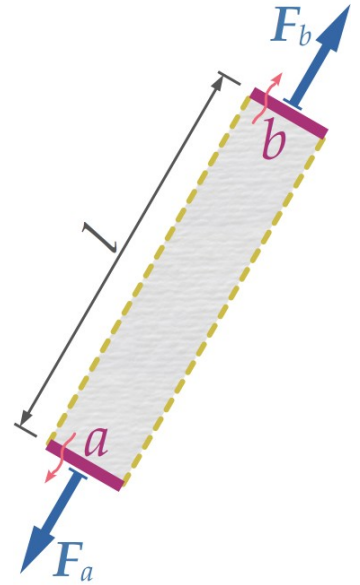
this condition can equivalently be written in two other ways:

$$(\mathbf{r}_a - \mathbf{r}_b) \times \mathbf{F}_a = (\mathbf{r}_a - \mathbf{r}_b) \times \mathbf{F}_b = \mathbf{0} \text{ N m}$$

or

$$\pm \frac{\mathbf{F}_a}{|\mathbf{F}_a|} = \pm \frac{\mathbf{F}_b}{|\mathbf{F}_b|} = \pm \frac{\mathbf{r}_a - \mathbf{r}_b}{|\mathbf{r}_a - \mathbf{r}_b|} .$$

Let us see how these two consequences come about, very simply, from the balance of momentum, the balance of angular momentum, *Newton's constitutive relation* › §10.2 p. 228 for the momentum flux of matter, and the almost universal constitutive relations for *angular-momentum flux* › §12.4 p. 376, eq. (12.4), and for *angular-momentum supply* › §12.4 p. 377, eq. (12.5).



The balances of momentum and of angular momentum for the piece of material are

$$\frac{d\mathbf{P}}{dt} = \mathbf{F} + \mathbf{G}, \quad \frac{d\mathbf{L}}{dt} = \mathbf{M} + \mathcal{T}.$$

By Newton's constitutive relation for the momentum flux of matter, if the mass content m of the control volume is always zero, then also the momentum content and its rate of change are always zero:

$$\mathbf{P} = m\mathbf{v} \text{ and } m \equiv 0 \text{ kg} \implies \mathbf{P} \equiv 0 \text{ N s} \implies \frac{d\mathbf{P}}{dt} \equiv 0 \text{ N}.$$

But by the constitutive relation for angular-momentum content, in every small sub-volume at position $\mathbf{r}$ the angular momentum is also zero:

$$\mathbf{L} = \mathbf{r} \times \mathbf{P} \text{ and } \mathbf{P} \equiv 0 \text{ N s} \implies \mathbf{L} \equiv 0 \text{ N m s} \implies \frac{d\mathbf{L}}{dt} \equiv 0 \text{ N m}$$

everywhere. Similar reasoning with the supplies for momentum and for angular momentum:

$$m = 0 \text{ kg} \implies \mathbf{G} = \mathbf{0} \text{ N}, \quad \mathcal{T} = \mathbf{0} \text{ N m}.$$

This means that the balances of momentum and of angular momentum for our piece of material are

$$\mathbf{F} = \mathbf{0} \text{ N}, \quad \mathbf{M} = \mathbf{0} \text{ N m},$$

where $\mathbf{F}$ and $\mathbf{M}$ are the total fluxes of momentum and angular momentum into the material. But these occur only at the surfaces at positions $\mathbf{r}_a$ and $\mathbf{r}_b$:

$$\mathbf{0} \text{ N} = \mathbf{F} = -\mathbf{F}_a - \mathbf{F}_b, \quad \mathbf{0} \text{ N m} = \mathbf{M} = -\mathbf{r}_a \times \mathbf{F}_a - \mathbf{r}_b \times \mathbf{F}_b.$$

From the first equation we obtain

$$\mathbf{F}_a = -\mathbf{F}_b,$$

which is consequence C1 above. Replacing $\mathbf{F}_b$ with $-\mathbf{F}_a$ in the second equation we find

$$\mathbf{r}_a \times \mathbf{F}_a - \mathbf{r}_b \times \mathbf{F}_a = \mathbf{0} \text{ N m} \implies (\mathbf{r}_a - \mathbf{r}_b) \times \mathbf{F}_a = \mathbf{0} \text{ N m}.$$

Now recall: when is the cross-product between two vectors equal to zero? When the two vectors are *aligned*, that is, parallel or anti-parallel. Therefore the equation above says that $\mathbf{r}_a - \mathbf{r}_b$ and $\mathbf{F}_a$ are parallel. Then we have found consequence C2:

$$\mathbf{F}_a, \mathbf{F}_b, \mathbf{r}_a - \mathbf{r}_b \text{ are parallel}.$$

§ 12.6 Centre of mass-energy

The centre of mass-energy can be defined for any control volume having a non-zero energy-mass content: therefore for control volumes that contain matter, or that contain only electromagnetic fields, or both.



Centre of mass-energy

Consider a coordinate system (t, x, y, z) and a control volume having non-zero mass-energy content m at some time t .

Divide the control volume into smaller and smaller sub-volumes, which we can number $1, 2, 3, \dots$. When they are small enough, we can associate with them the positions $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \dots$. Call $m_1, m_2, m_3, \dots$ their mass-energy contents; some of these may be zero, if some sub-volumes are empty. By extensivity we have that $m = m_1 + m_2 + m_3 + \dots$.

The **centre of mass-energy** $\mathbf{r}_c(t)$ of the control volume at coordinate time t is defined as

$$\mathbf{r}_c := \frac{1}{m} \lim_{\substack{\text{smaller} \\ \text{sub-volumes}}} [m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2 + m_3 \mathbf{r}_3 + \dots]. \quad (12.6)$$

In the definition of the centre of mass-energy, *all* mass-energy is to be counted: not only the rest mass-energy, but also internal energy, kinetic energy, and so on.

For a control volume containing matter in most everyday situations, counting only the rest mass-energy is an extremely good approximation. In calculating the centre of mass-energy of bodies in the Solar System, however, the kinetic and gravitational potential energies of the bodies are also included (Park et al. 2021; Folkner et al. 2014). For a control volume containing electromagnetic fields, all the electromagnetic energy must be considered.

We have an intuitive understanding of the centre of mass-energy of objects in everyday life. When we try to balance an object on our hands or fingers, we're trying to keep the object's centre of mass-energy directly on the vertical above the balancing point. For control volumes of particular shapes, the centre of mass-energy *may lie outside the control volume itself*; think for example about the centre of mass-energy of a horseshoe.

The centre of mass-energy in general does not stay in a fixed position with respect to the control volume, but moves around as the volume

$$\mu_i^* = GM_i \left(1 + \frac{1}{2c^2} v_i^2 - \frac{1}{2c^2} \sum_{j \neq i} \frac{GM_j}{r_{ij}} \right)$$

Expression for the mass-energy of a Solar System object, used to compute the centre of mass-energy of the Solar System (Park et al. 2021). In the second term of the sum you may recognize the kinetic energy divided by c^2 ; the third term is the gravitational potential energy.



changes its shape and the mass-energy within flows in different directions.

The centre of mass-energy of a body in [rigid motion](#) ^{§11.13 p.345}, however, does stay in a fixed position with respect to that body.



Exercise 12.2

In the [ICRS coordinate system](#) ^{§2.5 p.56} centred on the Sun, the Earth and Moon at coordinate time t have positions

$$\mathbf{r}_E(t) = \begin{bmatrix} -2.759 \\ 13.20 \\ 5.727 \end{bmatrix} \cdot 10^{10} \text{ m}, \quad \mathbf{r}_M(t) = \begin{bmatrix} -2.744 \\ 13.17 \\ 5.710 \end{bmatrix} \cdot 10^{10} \text{ m}.$$

Earth's mass-energy is $5.972 \times 10^{24} \text{ kg}$ and Moon's is $7.346 \times 10^{22} \text{ kg}$.

1. Calculate the centre of mass-energy of the Earth-Moon system at the same coordinate time, in this system of coordinates.
2. Considering that Earth's mean radius is 6371 km, determine how many kilometres above or below Earth's surface is that centre of mass-energy located.

§ 12.7 Angular momentum and rigid motion

The constitutive relations (12.3a), (12.4), (12.5) cover almost all physical phenomena, even those involving electromagnetic fields. But the use of these constitutive relations typically requires the division of any extended control volume into very small parts – a three-dimensional spatial integration – and then the monitoring of each of these parts through time. This procedure can be computationally expensive in simulations.

In the case of matter undergoing a [rigid motion](#) ^{§11.13 p.345}, the constitutive relations for angular momentum can sometimes be applied without the need of a three-dimensional spatial integration. We shall now discuss these simplified forms of constitutive relations. In order to do so we need to use the *centre of mass-energy*, introduced earlier, and two new concepts: the *angular velocity* and the *moment of inertia*.

§ 12.8 Angular velocity

Given a coordinate system (t, x, y, z) , we can define the angular velocity of a position vector $\mathbf{r}(t)$ with respect to another position vector $\mathbf{r}_0(t)$, relative to the coordinate system. It represents how fast, in terms of $\frac{\text{angle}}{\text{time}}$, the first position vector is instantaneously rotating with respect to the second point.



Angular velocity

In a Cartesian coordinate system (t, x, y, z) , take a point having position vector $\mathbf{r}(t)$ at coordinate time t , and another position vector $\mathbf{r}_0(t)$ distinct from the first. Consider their vector difference $\mathbf{r}(t) - \mathbf{r}_0(t)$, which can be seen as a vector going from the second point to the first.

After a short time lapse Δt the vector $\mathbf{r}(t + \Delta t) - \mathbf{r}_0(t + \Delta t)$ may get slightly shorter or longer, and it may get slightly *rotated* with respect to the coordinates (x, y, z) . The axis of rotation is perpendicular to the plane formed by $\mathbf{r}(t) - \mathbf{r}_0(t)$ and $\mathbf{r}(t + \Delta t) - \mathbf{r}_0(t + \Delta t)$.

The **angular velocity** $\boldsymbol{\omega}(t)$ of the position $\mathbf{r}(t)$ with respect to the position $\mathbf{r}_0(t)$ at time t is a vector defined as follows:

magnitude: the angle between $\mathbf{r}(t) - \mathbf{r}_0(t)$ and $\mathbf{r}(t + \Delta t) - \mathbf{r}_0(t + \Delta t)$, divided by Δt ;

direction: the instantaneous axis of rotation;

orientation: the sense of rotation;

all of them observed at time t .

Angular velocity has physical dimension time^{-1} , and unit rad/s.

The definition above is only valid in situations where we can use the Newtonian approximation. For the most general definition, which is mathematically more complicated and holds in any coordinate system, we must consider how the [metric](#) ^{§3.13 p.87} gets deformed around the position $\mathbf{r}(t)$ in the time lapse Δt .

The angular velocity is the velocity that the vector pointing from $\mathbf{r}_0$ to $\mathbf{r}$ has in a direction perpendicular to itself, also called *tangential velocity*, scaled by the length of that vector.

The angular-velocity vector

$$\boldsymbol{\omega} \equiv [\omega_x, \omega_y, \omega_z]$$

can also be represented as an *angular-velocity matrix* $\mathbf{\Omega}$:

$$\mathbf{\Omega} = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

having a zero diagonal and elements of opposite signs on opposite sides of the diagonal.

If we consider the position vectors $\mathbf{r}_1(t), \mathbf{r}_2(t), \mathbf{r}_3(t), \dots$ of different points of a body of matter, and one given reference position $\mathbf{r}_0(t)$, then in general the angular velocities of the points with respect to $\mathbf{r}_0(t)$ are all different and change with time in different ways. But for a set of points undergoing a rigid motion something special happens: *all points have the same angular velocity*, and in particular the same angular velocity with respect to the centre of mass-energy.



Exercise 12.3

A point with time-dependent position $\mathbf{r}(t)$ is rotating around the fixed point $\mathbf{r}_0 = [2, -3, 0]$ m. The trajectory of $\mathbf{r}(t)$ is parallel to the zx plane. The rotation is clockwise if we look at $\mathbf{r}(t)$ from some point far away on the positive- y axis. The point $\mathbf{r}(t)$ does two complete rotations every second.

1. Write the angular-velocity vector corresponding to this rotation.
2. Write the corresponding angular-velocity matrix.

§ 12.9 Angular velocity and rigid motion

In order to describe the motion of a generic body of matter, like a parcel of water, we need to specify the position or velocity of each of its points, because any two points can move far apart or close together. This means that an infinity of positions needs to be specified.

Recall that if a body of matter undergoes a **rigid motion** ^{§11.13 p.345}, then all distances between its points remain constant in time. In other words, the body can move around and rotate, but it doesn't change shape or expand or contract. We call a body that only undergoes a rigid motion a *rigid body*.

To describe the motion of a rigid body we do not need an infinity of positions, but only *six* parameters, thanks to its rigidity. Three of these

six parameters correspond to the motion of one fixed point in the rigid body. The remaining three correspond to the rotation of the rigid body with respect to the coordinate system.

These six parameters can be defined in different but equivalent ways. We choose the following: the centre of mass-energy:

$$\mathbf{r}_c(t) \equiv [x_c(t), y_c(t), z_c(t)] ,$$

from which we can calculate the velocity

$$\mathbf{v}_c(t) := \frac{d\mathbf{r}_c(t)}{dt} \equiv \left[\frac{dx_c(t)}{dt}, \frac{dy_c(t)}{dt}, \frac{dz_c(t)}{dt} \right] ;$$

and the *angular velocity* of any point of the rigid body with respect to its centre of mass-energy:

$$\boldsymbol{\omega}(t) \equiv [\omega_x(t), \omega_y(t), \omega_z(t)] .$$

Together, they allow us to find the velocity of any point of the rigid body:



Velocity of a point in a rigid body

If $\mathbf{r}(t)$ is the position vector of a point of a rigid body at time t in a coordinate system (t, x, y, z) , then its velocity $\mathbf{v}(t)$ is given by

$$\mathbf{v}(t) = \mathbf{v}_c(t) + \boldsymbol{\omega}(t) \times [\mathbf{r}(t) - \mathbf{r}_c(t)] \quad (12.7)$$

where $\mathbf{r}_c(t)$ and $\mathbf{v}_c(t)$ are the position and velocity of the centre of mass-energy, and $\boldsymbol{\omega}(t)$ is the angular velocity.

The equation above tells us that if we want to track how the position $\mathbf{r}(t)$ of any point of a rigid body changes in time, then we only need to track how the centre of mass-energy $\mathbf{r}_c(t)$ and the angular velocity $\boldsymbol{\omega}(t)$ change in time. If we have them, we can use the formula above to calculate the new position at any time. This is easily understood if imagine to [iteratively update](#) ^{§6.7 p.181} the position $\mathbf{r}(t)$ at small time steps by using the [finite-difference approximation](#) ^{§6.3 p.173}:

$$\begin{aligned} \mathbf{r}(t + \Delta t) &\approx \mathbf{r}(t) + \mathbf{v}(t) \Delta t \\ &\approx \mathbf{r}(t) + \{ \mathbf{v}_c(t) + \boldsymbol{\omega}(t) \times [\mathbf{r}(t) - \mathbf{r}_c(t)] \} \Delta t . \end{aligned}$$

§ 12.10 Moment of inertia

The *moment of inertia* or *tensor of inertia* is a mathematical object that encodes some information about the distribution of mass-energy within a control volume. This mathematical object is represented by a *matrix*. It is defined with respect to a coordinate system and a reference point, which for our purpose we'll choose as the centre of mass-energy.



Moment of inertia

In a coordinate system (t, x, y, z) , the **moment of inertia** or **tensor of inertia** $\mathbf{J}_c(t)$ at time t of a set of points *with respect to their centre of mass-energy* is given by a matrix:

$$\mathbf{J}_c(t) \equiv \begin{bmatrix} I_{xx} & I_{xy} & I_{zx} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{zx} & I_{yz} & I_{zz} \end{bmatrix} \quad (12.8)$$

where all components depend on time. This matrix has equal elements on opposite sides of the diagonal, so it can have at most six different components (three on the diagonal and three on one side of the diagonal).

If all coordinates have physical dimension length, then the components of the moment of inertia have dimension mass · length² and unit kg m².

The moment of inertia is **calculated**¹ from the distribution of the mass-energy in the rigid body in the chosen coordinate system. It therefore also depends on the shape of the body. For instance, the moment of inertia for a solid ball of radius R and total mass-energy m , uniformly distributed, is

$$\mathbf{J}_{c,\text{ball}} = \frac{2}{5}mR^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

the centre of the ball being also its centre of mass-energy. In this particular case the moment of inertia does not depend on time.

Given its relation with mass-energy and possibly with matter, we suspect that the moment of inertia must be somehow related to their balances as well. Indeed its time derivative is given by a special equation:

Time derivative of moment of inertia

The time derivative of the moment of inertia, defined with respect to the coordinate system (t, x, y, z) and the centre of mass-energy, is given by:

$$\frac{d}{dt} \mathbf{J}_c(t) = \boldsymbol{\Omega}(t) \mathbf{J}_c(t) - \mathbf{J}_c(t) \boldsymbol{\Omega}(t) \quad (12.9)$$

where $\boldsymbol{\Omega}$ is the angular-velocity matrix. The right side of this formula has two multiplications of two matrices. This equation is essentially a consequence of the balances of matter and mass-energy.

§ 12.11 Constitutive relations for a rigid body

Equipped with the concepts of centre of mass-energy, angular velocity, and moment of inertia, we can state the constitutive relation for the momentum content, energy content, and angular-momentum content of a rigid body.

Momentum of a rigid body

Take a coordinate system (t, x, y, z) , and a control volume with total mass-energy m . If the masses in the control volume are undergoing a rigid motion with centre of mass-energy $\mathbf{r}_c(t)$, then the momentum $\mathbf{P}(t)$ in the control volume is given by

$$\mathbf{P}(t) = m \mathbf{v}_c(t) \quad (12.10)$$

where $\mathbf{v}_c$ is the velocity of the centre of mass-energy.

The momentum for a rigid body is therefore the same as if the control volume were just a small point of mass-energy m moving with velocity $\mathbf{v}_c$.

The constitutive relation for the angular momentum is more complicated:

Angular momentum of a rigid body

Take a coordinate system (t, x, y, z) , and a control volume with total mass-energy m . If the masses in the control volume are undergoing a rigid motion with centre of mass-energy $\mathbf{r}_c(t)$ and angular velocity $\boldsymbol{\omega}(t)$, and if their moment of inertia is $\mathbf{J}_c(t)$, then the angular momentum $\mathbf{L}(t)$

in the control volume is given by

$$\mathbf{L}(t) = \mathbf{r}_c(t) \times m\mathbf{v}_c(t) + \mathbf{J}_c(t) \boldsymbol{\omega}(t), \quad (12.11)$$

where $\mathbf{v}_c$ is the velocity of the centre of mass-energy, and $\mathbf{J}_c \boldsymbol{\omega}$ is the matrix multiplication of the matrix $\mathbf{J}_c$ and the column-vector $\boldsymbol{\omega}$.

The angular momentum for a rigid body is therefore the sum of two parts. One part is analogous to the standard constitutive relation ' $\mathbf{r} \times \mathbf{P}$ ', if the control volume were just a small point with position $\mathbf{r}_c$ and momentum content $m\mathbf{v}_c$. The other part, given by a matrix multiplication, takes care of the fact that the total momentum is actually "spread out" over an extended control volume.



Total energy of a rigid body

Take a coordinate system (t, x, y, z) close to Earth's surface and at rest with respect to it, with the z -coordinate pointing upwards. Consider a control volume with total mass-energy m . If the masses in the control volume are undergoing a rigid motion with centre of mass-energy $\mathbf{r}_c(t)$, then the total-energy $E(t)$ in the control volume is given by

$$E(t) = U(t) + \frac{1}{2}m\mathbf{v}_c(t)^2 + \frac{1}{2}\boldsymbol{\omega}(t) \cdot \mathbf{J}_c(t) \boldsymbol{\omega}(t) + mgz_c(t) \quad (12.12)$$

where U is the internal energy in the control volume, $\mathbf{v}_c$ is the velocity of the centre of mass-energy, and z_c its vertical coordinate.

The total energy for a rigid body is therefore, as usual, the sum of an internal energy, a kinetic energy, and a gravitational potential energy. The kinetic energy consists in two parts: one part is the kinetic energy of a small control volume having the same mass m as the rigid body, and the velocity $\mathbf{v}_c$ of its centre of mass-energy; the other part takes care of the kinetic energy of all small masses that constitute the rigid body. The gravitational potential energy is simply that of a small control volume having the same mass as the rigid body, and vertical coordinate of its centre of mass-energy.

§ 12.12 Numerical time integration for rigid body

Now that we have the constitutive relations for momentum and angular-momentum content of a rigid body, we can get a glimpse of how they could be used to keep track of its motion. In concrete applications, rigid-body motion is predicted and simulated with a wide range of different techniques, some of which use fascinating mathematics like the *quaternions*², generalizations of complex numbers.

The discussion to follow is only meant to give you a glimpse that the concepts introduced so far can indeed be used to study and simulate the motion of a rigid body; and that we cannot avoid using them or other concepts equivalent to them.

The constitutive relations and balances that we have discussed are summarized below. We write the balances and other time derivatives in an approximate finite-difference from appropriate for time iteration:

$$\mathbf{P}(t) = m\mathbf{v}_c(t) \quad \text{constit. relation for momentum content}$$

$$\mathbf{L}(t) = \mathbf{r}_c(t) \times m\mathbf{v}_c(t) + \mathbf{J}_c(t) \boldsymbol{\omega}(t) \quad \text{constit. relation for angular-momentum content}$$

$$\mathbf{P}(t + \Delta t) \approx \mathbf{P}(t) + [\mathbf{F}(t) + \mathbf{G}(t)] \Delta t \quad \text{balance of momentum}$$

$$\mathbf{L}(t + \Delta t) \approx \mathbf{L}(t) + [\mathbf{M}(t) + \mathcal{T}(t)] \Delta t \quad \text{balance of angular-momentum}$$

$$\mathbf{r}_c(t + \Delta t) \approx \mathbf{r}_c(t) + \mathbf{v}_c(t) \Delta t \quad \text{update of mass-energy centre}$$

$$\mathbf{J}_c(t + \Delta t) \approx \mathbf{J}_c(t) + [\boldsymbol{\Omega}(t) \mathbf{J}_c(t) - \mathbf{J}_c(t) \boldsymbol{\Omega}(t)] \Delta t \quad \text{update of moment of inertia}$$

keep in mind that the angular-velocity vector $\boldsymbol{\omega}$ and matrix $\boldsymbol{\Omega}$ have the [same numerical components](#) ^{§12.8 p.383}, so if we know the one we also know the other.

It isn't difficult to see that if we know the values of the [state variables](#) ^{§10.15 p.274}

$$\mathbf{r}_c, \quad \mathbf{v}_c, \quad \boldsymbol{\omega}, \quad \mathbf{J}_c$$

at a time instant $\mathcal{T}$, and we also have appropriate constitutive relations for the surface and volume forces $\mathbf{F}$, $\mathbf{G}$ and torques $\mathbf{M}$, $\mathcal{T}$, then it's possible to calculate the values of the state variables a short time lapse Δt later. See the exercise below.



Exercise 12.4

For this exercise, keep in mind the following:

- The angular velocity $\boldsymbol{\omega}$ can be calculated from $\mathbf{L}, \mathbf{r}_c, \mathbf{v}_c, \mathbf{J}_c, m$ by the following formula:

$$\boldsymbol{\omega} = \mathbf{J}_c^{-1} (\mathbf{L} - \mathbf{r}_c \times m \mathbf{v}_c)$$

where $\mathbf{J}_c^{-1}$ is the inverse of the matrix $\mathbf{J}_c$. You don't need to explicitly calculate anything with this formula, but only know that $\boldsymbol{\omega}$ can be calculated this way.

- Assume that the forces $\mathbf{F}, \mathbf{G}$ and torques $\mathbf{M}, \mathbf{T}$ are known at all times, so you don't need to worry about them.

Given this information, and given the starting values

$$\mathbf{r}_c(t), \quad \mathbf{v}_c(t), \quad \boldsymbol{\omega}(t), \quad \mathbf{J}_c(t):$$

1. Find out how the formulae for rigid-body motion summarized above could be used to calculate the values a time lapse later

$$\mathbf{r}_c(t + \Delta t), \quad \mathbf{v}_c(t + \Delta t), \quad \boldsymbol{\omega}(t + \Delta t), \quad \mathbf{J}_c(t + \Delta t).$$

In other words, build a time-stepping scheme like those shown in formula (6.8) on page 180, or (10.7) on page 255.

2. You know the starting position $\mathbf{r}(t)$ of one point in the rigid body. Using the position-update relation

$$\mathbf{r}(t + \Delta t) \approx \mathbf{r}(t) + \mathbf{v}(t) \Delta t,$$

and the relationship (12.7) on page 385 for the velocity of a point in a rigid body, update your timestepping scheme to include the updated position $\mathbf{r}(t + \Delta t)$ of the point.

3. Suppose you are given the initial values

$$\mathbf{r}_c(t), \quad \mathbf{v}_c(t), \quad \boldsymbol{\omega}(t), \quad \mathbf{J}_c(t), \quad \mathbf{r}_1(t), \quad \mathbf{r}_2(t), \quad \mathbf{r}_3(t)$$

where $\mathbf{r}_1(t), \mathbf{r}_2(t), \mathbf{r}_3(t)$ are the initial positions of three points in the rigid body. Based on your previous results, can you track the positions of all the three points? (Remember that forces and torques are assumed to be known.)

4. Based on your previous results, would you be able to track the points of the rigid body *if the balance of angular momentum $\mathbf{L}(t + \Delta t) \approx \mathbf{L}(t) + [\mathbf{M}(t) + \mathbf{T}(t)] \Delta t$ were not available?* That is, is it possible to follow the motions of the points in a rigid body just by using the balance of momentum?

§ 12.13 Constitutive relations for forces and torques on a rigid body

We managed to find constitutive relations for the contents of momentum, angular momentum, and total energy in a rigid body.

The tricky point are the constitutive relations for surface and volume forces, and surface and volume torques. They also need spatial integration over extended surfaces and volumes. Unfortunately there aren't many simplifications for such integrations. Somewhat simple formulae can be found for forces of very special kind, for instance forces that act on one point only of the surface of the rigid body; or forces that are constant everywhere, like the gravitational force close to Earth's surface; or forces that have constant magnitude and a direction always perpendicular to the surface of the rigid body.

Gravitational torque on rigid body near a planet's surface

Take a coordinate system (t, x, y, z) fixed with the ground, with z as vertical upward coordinate. A control volume of a rigid body containing an amount of mass-energy m has a supply of angular momentum, the **gravitational torque**, given by

$$\mathcal{T} = \mathbf{r}_c \times m \mathbf{g}, \quad \text{with} \quad \mathbf{g} = -g \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad (12.13)$$

where $\mathbf{r}_c$ is the centre of mass-energy and $\mathbf{g}$ is the [acceleration of free fall](#) §10.3 p.231. As usual we are taking the origin of the coordinates as reference point.

Therefore this volume torque has the same expression as if the control volume were a small one at the centre of mass-energy. This results comes from the fact that the gravitational force $\mathbf{G}$ near a planet's surface is approximately constant in space. An analogous result holds in fact for any other surface or volume force which is constant over the whole rigid body. But if we are enough far away from a planet's surface, so that the gravitational force cannot be considered approximately constant, then the result above is no longer true.

Total surface torques can be simplified in some important cases:



Surface torque from air pressure on rigid body

Take a coordinate system (t, x, y, z) . If on the closed control surface there is a pressure or a tension having

- constant magnitude,
- direction always orthogonal to the surface,

then the **total torque of this surface force is zero**.

This is the case for *atmospheric pressure*, whose torque we can therefore neglect if the rigid body has enough small vertical extension. This is the reason why atmospheric pressure is usually neglected when simulating rigid bodies.

Friction from air, however, does not satisfy the conditions above, and therefore must be accounted for in some situations.

One more case in which we can manage to handle surface or volume torques in a simple way is when they are concentrated in a very small area or very small volume. In that case the standard constitutive relations (12.4), (12.5) can be applied directly, without integration. Note that we may need to keep track of their application points.

URLs for chapter 12

1. <https://scienceworld.wolfram.com/physics/MomentofInertia.html>
2. <https://mathworld.wolfram.com/Quaternion.html>

Balance of boost momentum

> *At least I searched in many books and they didn't even mention that. [...] Is Noether's theorem not valid in such cases or what?*

It's valid, all right.

> *and if Noether's theorem still works in these cases, why the heck do books not talk about the conserved quantities corresponding to such symmetries?*

Either 1) the textbook writers are too stupid to have thought about this issue, or 2) they have decided it's better to let everybody figure this out for themselves, or 3) they feel the answer is not sufficiently important to waste a precious paragraph on it. I don't know. When I become king of the universe, I will make all books on mechanics mention this issue.

J. Baez 2006

§ 13.1 Formulation and generalities

Balance of boost momentum or energy-mass moment

Volume content: $\mathbf{r} E - t \mathbf{P}$ Flux: $\mathbf{r} W - t \mathbf{F}$ Supply: $\mathbf{r} \mathcal{A} - t \mathbf{G}$

$$(t_1) = (t_0) + \int_{t_0}^{t_1} (t) dt + \int_{t_0}^{t_1} (t) dt \qquad \frac{d(t)}{dt} = (t) + (t) \qquad (13.1)$$

integral expression differential expression

✂ To be written in a later version

Momentum and energy: remarks

I hold in fact

(1) That small portions of space *are* in fact of a nature analogous to little hills on a surface which is on the average flat; namely, that the ordinary laws of geometry are not valid in them.

(2) That this property of being curved or distorted is continually being passed on from one portion of space to another after the manner of a wave.

(3) That this variation of the curvature of space is what really happens in that phenomenon which we call the *motion of matter*, whether ponderable or etherial.

(4) That in the physical world nothing else takes place but this variation, subject (possibly) to the law of continuity.

W. K. Clifford [1876](#)

§ 14.1 Common misunderstandings on momentum, energy, angular momentum

Momentum of what? Energy of what?

✂ To be written in a later version

Balance of entropy

Their various “second laws” sound more like warnings or threats than principles of a rational science.

C. A. Truesdell, III 1984

§ 15.1 Formulation and generalities



Balance of entropy

Volume content: S Flux: Π

$$S(t_1) \geq S(t_0) + \int_{t_0}^{t_1} \Pi(t) \, dt \qquad \frac{dS(t)}{dt} \geq \Pi(t) \qquad (15.1)$$

integral expression differential expression

The balance of entropy expresses what’s commonly called “second law of thermodynamics”. Entropy and its balance are successfully used in many applications, but our understanding of them and of their physical foundation is still incomplete. This state of affairs is reflected in the many and wildly different presentations of entropy and its balance: different in wording, mathematical formulation, scope, and sometimes even in physical consequences.

Many textbooks only present limited and special cases of properties and uses of entropy and its balance, and unfortunately they often make these limited, special cases appear as more general, or of broader application, than they actually are. Such textbooks also typically restrict themselves to situations where there the contents of quantities in a control volume do not change with time, and the fluxes are zero. We call this a situation of *equilibrium*. The discipline that studies equilibrium situations is called *thermostatistics*.

In these lecture notes, entropy and its balance are presented from a point of view, actively developed and used since the 1960s, having the following features:

- It has been used for many years in concrete technological applications (some examples at NASA: Chang & Haddad 1971; Hughes et al. 1986; Turon et al. 2004; Diosady et al. 2018; Kato & Rose 2020), and in the study complex materials such as polymers and mixtures.
- It has led to new physical constitutive relations, or to the physical and mathematical foundation of existing ones, from first principles.
- It is formulated with the same mathematics, and at the same mathematical level, as the physics of matter, momentum, angular momentum, energy, and electromagnetism.
- It includes time-dependent phenomena and is fully connected with phenomena involving the other six basic quantities.

In the technical literature the entropy balance above, used in its full generality, goes under the name of **Clausius-Duhem inequality**¹.

Thermodynamic entropy and statistical entropy

One added difficulty is that entropy and its balance can also be approached from a completely different direction, especially when we study physical systems on small scales. It's the approach of *statistical mechanics*², which considers physical situations in where we lack information about initial conditions, or boundary conditions, or constitutive relations. In statistical mechanics, a conceptually different entropy appears, not as a physical quantity, but as a measure of our lack of information about the physical system, in the strict technical sense of **Information Theory**³. Also this entropy satisfies a balance law very similar to (15.1) above.

One of the reasons for the bewilderment which is sometimes felt at an unheralded appearance of the term entropy is the superabundance of objects which bear this name. On the one hand, there is a large choice of macroscopic quantities (functions of state variables) called entropy, on the other hand, a variety of microscopic quantities, similarly named, associated with the logarithm of a probability or the mean value of the logarithm of a density. Each one of these concepts is suited for a specific purpose. More confusing, however, than the lack of imagination in terminology is the fact that several of these distinct concepts, different in meaning and in numerical value, may be significant in a single problem. (Grad 1961 §1 p. 323)

We thus have a **thermodynamic entropy** and a **statistical entropy**. Their fascinating relation is only partly understood, and still the object of some debate. In the present notes we shall focus on *thermodynamic* entropy.

Entropy depends on the observation scale

The entropy content in a control volume and the entropy flux across a control surface are not uniquely defined; that is, several choices are possible, each one correct in a specific situation. This non-uniqueness is actually two-fold. In the present section we discuss a first sense in which entropy is non-unique; in a later section we'll discuss a second sense.

First let's make clear that the entropy content in a given control volume, and the entropy flux through a given control surface, at a given coordinate time, *do not depend on the coordinate system* chosen. In this regard they are like the content & flux of matter, magnetic flux, and electric charge; and unlike momentum, angular momentum, and energy.

On the other hand, entropy content & flux *do depend on the detail and scale of observation* of a physical phenomenon. In this regard they are unlike all other six fundamental quantities, whose *total* content and flux do not depend on the observation scale.

As an example, take a control volume containing air. This control volume could be studied, described, and measured in three different ways: (a) as containing a continuous, fluid amount of matter of one kind: 'air'; (b) as containing a continuous, fluid mixture of amounts of matter of different kinds: nitrogen, oxygen, and several others; (c) as containing a bunch of molecules in motion.

The *energy* content measured within this control volume, at a particular time instant, will be exactly the same in all three cases. What changes among them is the [division of this total energy into internal and kinetic](#) ^{§11.6 p. 327}, but the total is the same.

The *entropy* content assigned to this control volume, on the other hand, will be different in each case.

A given object of study cannot always be assigned a unique value, its "entropy". It may have many different entropies, each one worthwhile. The proper choice will depend on the interests of the individual, the particular phenomena under study, the degree of precision available or arbitrarily decided upon, or the method of description which is employed; and each of these criteria is largely subject to the discretion of the individual. [. . .]

For another example we turn to aerodynamics. The existence of diffusion between oxygen and nitrogen somewhere in a wind tunnel will usually be of no interest. Therefore the aerodynamicist uses an entropy which does not recognize the separate existence of the two elements but only that of "air". In other circumstances, the possibility of diffusion between elements with a much smaller mass ratio (e.g., 238/235) may be considered quite relevant.

(Grad 1961 §1 pp. 323, 325)

We shall now see that this peculiar dependence of entropy on the details and scale of observation actually makes a lot of sense when we understand how the entropy balance is used.

§ 15.2 The physical role of the balance of entropy

The entropy balance (15.1) has an apparent peculiarity, compared to the general form of a [balance law](#) ^{§5.5 p. 137}: it is not an equality

$$\dots = \dots$$

but an *inequality*

$$\dots \geq \dots$$

Why? and what are the consequences of this peculiarity?

Many texts and media try to summarize the meaning of the entropy balance and its inequality sign in simple terms. But the reality is that this inequality leads to an amazing variety of very different phenomena, and cannot be summarized in words.

This should not be surprising. Take the balance of momentum for example. It leads to all sorts of motions and deformations of objects, but also to the stability of objects: from the extremely complicated motion of the atmosphere to the stillness of a pen resting on a table. This balance law could not be summarized in some simple sentence, like “objects spontaneously fall downward”. First, such a sentence would be false in many situations: just look at the motion of rocks expelled by a volcano, or at cosmological expansion. Second, it would be useless for precise predictions and numerical simulations.

The same remark holds true for the balance of entropy. This balance also leads to a wild variety of physical consequences. The mixing of two liquids can be said to be a consequence of the balance of entropy (together with the other balances). But also the appearance of life on Earth is a consequence of the balance of entropy; and all complex physical mechanisms underlying life are consequences of the balance of entropy, too (together with the other balances). Any simplistic summaries, like common ones mentioning “tendency to disorder” or “spontaneously doing this or that” or similar, are (a) simply *false* in many physical situations; (b) vague: what does ‘to tend’ mean? how is ‘disorder’ defined and quantified? to what initial values of position, velocity, and so on, does ‘spontaneous’ refer to? what’s ‘spontaneous’ and what’s not? (c) useless for quantitative predictions.



The mixing of two liquids and the growing of life are both consequences of the second law of thermodynamics.

§ 15.3 Irreversibility

From a qualitative point of view, the inequality sign in the balance of entropy expresses *irreversibility*; but we must understand this word in the right way.

Consider first a generic balance or conservation law; let's take conservation of matter for instance:

$$N(t_1) = N(t_0) + \int_{t_0}^{t_1} \mathcal{J}(t) dt$$

Suppose a control volume contains an amount 20 mol of matter. During an interval of time we provide a net flow of matter $\int_{t_0}^{t_1} \mathcal{J}(t) dt = 3 \text{ mol}$ into the control volume. By the conservation law, at the end we have an amount 23 mol of matter in the control volume. This also means that in principle we could revert to an amount of 20 mol by providing a *negative* amount of the same flow as before, -3 mol . We can, in principle, change the matter content in the volume between 20 mol and 23 mol by providing a net flow of $+3 \text{ mol}$ or its opposite -3 mol . If we consider a balance with a supply, the conclusion remains the same, if we invert also the sign of the supply.

Now consider the balance of entropy with a *strict* inequality:

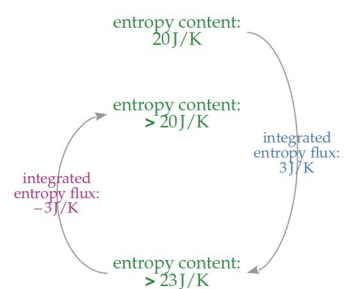
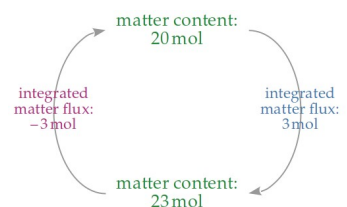
$$S(t_1) > S(t_0) + \int_{t_0}^{t_1} \Pi(t) dt$$

Suppose a control volume contains an amount 20 J/K of entropy. During an interval of time we provide a net flow of entropy $\int_{t_0}^{t_1} \Pi(t) dt = 3 \text{ J/K}$ into the control volume. By the inequality above, at the end we *cannot* have an amount 23 J/K of entropy in the control volume: we must have a larger amount, for instance 23.1 J/K. If we want to revert to an entropy content of 20 J/K, then simply reversing the total flow to -3 J/K won't be enough. In fact, even -3.1 J/K won't be enough, because by the inequality above we must have

$$\text{new entropy amount} > 23.1 \text{ J/K} - 3.1 \text{ J/K}$$

and the new entropy content would be more than 20 J/K.

This is what we call **irreversibility**: we cannot revert to a given entropy content in the volume *by inverting the entropy flux*.





'Irreversibility' does not mean that we cannot revert to a given entropy content

Note that the example above does *not* imply that we can never have again an entropy content of 20 J/K in the control volume. It is only showing that we cannot achieve this simply by providing an entropy flux opposite in sign to the flux we provided before. The original entropy content can still in principle be re-established; but in order to achieve this we need to provide a larger negative entropy flux than before.

And the example does *not* imply that the entropy content of a control volume can only increase. The entropy content can very well decrease, provided an enough large negative entropy flow is provided.

In cases where the balance of entropy holds with an equal sign:

$$S(t_1) = S(t_0) + \int_{t_0}^{t_1} \Pi(t) dt$$

then we can of course revert to a given entropy content by inverting the entropy flux.



Reversible and irreversible processes

We call a physical change, process, transformation, or phenomenon between two times t_0 and t_1 **reversible** if it satisfies the balance of entropy with an equal '=' sign, at least approximately.

We call it instead **irreversible** if it satisfies the balance of entropy with a strict inequality '>' sign.

There is no clear-cut division between 'reversible' and 'irreversible' processes. A process may lead to an entropy content that is *slightly* larger than the one we would have obtained with a perfectly reversible process; but the difference may be so small, with respect to our approximations, that it can be neglected. In this case the process is called reversible for all practical purposes. Arguably there are in fact no *exactly* reversible processes in nature.

§ 15.4 Entropy balance as a meta-law

One important consequence of the inequality sign in the balance of entropy is that this balance *cannot* be used in a [numerical time-integration](#) ^{§6.3 p. 171} scheme. If we rewrite it in an approximate form for a short timestep Δt :

$$S(t + \Delta t) \gtrsim S(t) + \Pi(t) \Delta t$$

this relation doesn't tell us the value of the entropy content at time $t + \Delta t$, but only an approximate minimum value that this content could have. As far as we know, it could be much larger than this minimum. For example, if a closed control surface at some time t contains an amount of entropy $S(t) = 10 \text{ J/K}$, and the net entropy flux at that time is $\Pi(t) = 2 \text{ J/(K s)}$, then we can say that a time $\Delta t = 0.1 \text{ s}$ after the content should be

$$\begin{aligned} S(t + \Delta t) &\gtrsim S(t) + \Pi(t) \Delta t \\ &\gtrsim 10 \text{ J/K} + 2 \text{ J/(K s)} \cdot 0.1 \text{ s} \\ &\gtrsim 10.2 \text{ J/K} \end{aligned}$$

This means that maybe $S(t + \Delta t) \approx 10.2 \text{ J/K}$, or maybe $S(t + \Delta t) = 1000 \text{ J/K}$, or maybe even larger values – the entropy balance doesn't really tell us. This law, therefore, apparently does not allow us to “drive forward in time” a physical system.

In fact, when we simulate or predict the behaviour of any physical system, it turns out that we can in principle always do without the balance of entropy. We only need to use the other six balances together with any relevant constitutive equations. The entropy content S itself may appear in these constitutive relations, and is often a useful quantity. But the balance of entropy is not used.

What is the physical role of this balance law, then?

The answer is that *the balance of entropy is a 'meta-law'*. It is not a law directly about physical phenomena, but a *law about laws* of physical phenomena. Roughly speaking, this meta-law determines which constitutive relations are physically admissible and which are not admissible.

The subtlety arises, of course, because the second law is an inequality, and therefore only other inequalities can be deduced from it by a purely algebraic procedure. However, [...] one can in fact deduce from the second law, when coupled with appropriate constitutive assumptions, consequences which are equations, not inequalities; the procedure is more a logical than an algebraic one.

(Astarita 1990 §2.4 p. 46)

In very simple cases this meta-law leads to restrictions on the values of physical coefficients. For example, it's a consequence of the balance of entropy that the viscosity coefficient μ in the [ideal-gas law](#) › §10.18 p. 287 and the surface coefficient of heat transfer h in [Newton's law of cooling](#) › §11.14 p. 350 must be positive. We shall see a simple example of how this kind of restrictions come about.

But in more complex cases this meta-law leads to much more powerful results. For example, it can dictate that some constitutive relations cannot contain particular physical quantities as variables. The discussion of these complex and more fascinating cases unfortunately requires more advanced mathematics, so we can only get a dim glimpse of them in these notes.

Considering the role of the entropy balance as a meta-law that decides which constitutive relations are admissible and which aren't, and considering that [constitutive relations heavily depend](#) › §5.9 p. 154 on the scales of time & space and on the measurement precision with which we describe a physical phenomenon, it then makes sense that the entropy content and the entropy flux should also [depend on details and scale of observation](#) › §15.1 p. 401, as discussed in the previous sections.

🧐 If you feel adventurous, check the simple examples discussed in Astarita's book, in the section cited in the quote, or in chapter 2 of Samohýl & Pekař 2014.

§ 15.5 Entropy depends on a reference state

Besides [depending on the observation scale](#) › §15.1 p. 401, entropy additionally depends on the choice of a *reference state* › §10.15 p. 274 of a control volume; that is, on a particular set of values for the state variables. The reference state is a state to which we assign zero entropy by convention.

For many physical phenomena and materials, a change of the reference state simply shifts all entropy values by a constant amount, so our entropy is defined except for a 'zero' of its measurement scale. But there are also important and quite common phenomena and materials for which a change of reference state leads to a *different function* for the entropy – not just the addition or subtraction of a constant. An ordinary paper clip is a simple example. The non-uniqueness of entropy is also a consequence of the peculiar inequality sign in the balance of entropy.



The entropies that can be defined for an ordinary paper clip do not differ simply by a constant.

First, there are bodies which have two entropies [...] whose difference is nonconstant [...]. Second, it can happen that the body does not have a smooth entropy. These two peculiarities are related to the fact that for the entropy we have only an inequality. (Šilhavý 1997 §7.6)

But this non-uniqueness is not a problem. Any one of the entropies, defined with respect to different reference states, can be used for instance as one of the state variables. The prediction of the physical system's behaviour will be the same.

§ 15.6 Examples of constitutive relations

We have emphasized that the balance of entropy (together with the other balances) imposes restrictions on the possible constitutive relations between the physical quantities used to model a physical phenomenon. Among such quantities are also the entropy content S and entropy flux Π which enter this very balance (if you think about it, it's extremely intriguing that a physical law manages to determine the expressions of the very terms it contains).

Entropy flux, heat, temperature

The most important constitutive relation for entropy is the one that relates its flux Π with the [heat flux](#) ^{§11.8 p. 329} $\dot{Q}$ and the thermodynamic temperature T .



Entropy flux

Consider a control surface, possibly moving, with an assigned crossing direction, and satisfying the following conditions:

- the surface is in contact with matter on both sides
- no flux of matter through the surface
- no chemical reactions (transformations between matter types) occur across the surface
- no electromagnetic phenomena involved
- the temperature T is the same on every part of the surface
- the heat flux through the surface is $\dot{Q}$

Then across the control surface there is an entropy flux Π given by

$$\Pi = \frac{\dot{Q}}{T} \quad (15.2)$$

This relation has wide applicability, but pay close attention to the conditions for its validity. In particular, it is not correct if there is a net flux of matter through the surface. Nor is it correct if there are opposite fluxes of *different*

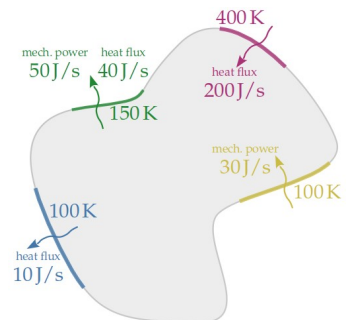
kinds of matter that cancel each one out – so the *net* flux is zero – but chemical reactions are occurring between these different matter kinds. It is also important that the temperature be well-defined and be uniform, that is, have the same value, on the surface and at least in a small spatial region on each side of the surface. If the temperature is not uniform, the surface is usually divided into smaller parts, so that the temperature can be considered uniform in each part separately, and then the total flux is obtained by summation, thanks to [extensivity](#) §3.2 p. 69.



Exercise 15.1

1. At a particular time instant, across a closed control surface there is a flux of energy as illustrated in the side picture:

- **through one part** of the surface there is a heat influx of 200 J/s and no flux of mechanical power; the temperature around that part is 400 K
- **through one part** there is a 40 J/s heat efflux and a 50 J/s efflux of mechanical power; the temperature around that part is 150 K
- **through one part** there is a 10 J/s heat efflux and no flux mechanical power; the temperature around that part is 100 K
- **through one part** there is a 30 J/s influx of mechanical power and no heat flux; the temperature around that part is 100 K
- **through the rest** of the surface there are no energy fluxes of any kind.



Each part of the surface satisfies the conditions of the constitutive relation (15.2).

How much is the *net influx* of entropy through the whole closed control surface?

2. Suppose that through a particular control surface, at a given time, there is zero *net* energy flux. The surface satisfies the conditions of the constitutive relation (15.2). Can we say that the entropy flux through that surface is also zero?

Entropy of an ideal gas

In chapter 11 we discussed constitutive relations for [the pressure \$p\$ and the internal energy \$U\$](#) ^{§11.14 p.348} of an control volume containing an ideal gas:

$$p = \frac{RNT}{V} - \mu \frac{1}{V} \frac{dV}{dt}, \quad U = C N T$$

where N is the amount of gas, T the temperature of the gas, assumed uniform, V is the volume, and R, C, μ are the molar gas constant, molar heat capacity, and viscosity coefficient.

Under the same conditions of validity for the constitutive relations above, we also have a constitutive relation for the entropy content S of the volume of ideal gas:



Entropy of an ideal gas

If a small control volume contains an amount N of ideal gas at uniform temperature T , then it also contains an amount of entropy S given by

$$S = CN \ln \frac{T}{T_0} - RN \ln \frac{N/V}{N_0/V_0} \quad (15.3)$$

where R and C are the molar gas constant and the gas's molar heat capacity, and T_0, N_0, V_0 are arbitrary reference temperature, amount of matter, and volume.

§ 15.7 Thermal engines

Recall that we are completely free in our choice of a [control volume](#) ^{§4.3 p.98}: it can have any size and shape, and can move and deform in any way. This freedom is extremely powerful: we can for example imagine a control volume that wraps a very complex engine with moving parts. Through the surface of this control volume we can keep track of any exchanges of matter, momentum, energy between the engine and its exterior; in particular, exchanges of heat and of mechanical power. And whatever happens within the engine, that is, within our imaginary control volume, must obey the seven universal balance laws.

This powerful freedom in choosing a control volume, when combined with the balances of entropy and energy, can lead to amazingly general physical results. Thanks to these results we can for example prevent

wasting our efforts in technological ideas that would eventually turn out to be unfeasible. We shall now see a couple of examples.

First of all let's define what we mean by 'engine'.



Cyclic thermal engine

A **cyclic thermal engine** is a device that can absorb or emit heat and mechanical work, and that can operate repeatedly, in principle forever. The ability to operate forever means that at recurring points in time the device must find itself in the same **physical state** ^{§10.15 p. 274}, so as to start over.

A thermal engine can also receive or release matter, momentum, angular momentum, and electromagnetic quantities.

Note the requirement of operating *cyclically*. A device consisting of an electric battery actuating a mechanical arm, for instance, is *not* a cyclic thermal engine, because it will cease operating once the battery is exhausted. Some of the theorems that we discuss below are not valid without the condition of cyclic operation.



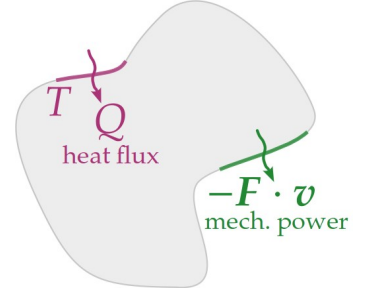
Exercise 15.2

1. Take a control volume wrapping a car. Can this control volume be considered as a cyclic thermal engine? Why or why not?
2. Take a control volume wrapping the engine of a traditional internal-combustion car: the system of pistons, valves, shafts, but excluding the fuel. Can this control volume be considered as a cyclic thermal engine? Are there any fluxes of matter during one cycle of operation?

§ 15.8 One-temperature thermal engines: heaters

Would it be possible to build a cyclic thermal engine that takes some energy in the form of heat from an inlet *at constant temperature*, and releases energy in the form of work, say by lifting an object?

The operation of such an engine is schematized in the side figure. Imagine to wrap the engine, no matter how complex it is, in a closed control surface, defining a control volume. A part of the control surface, in **red** in the figure, delimits the inlet through which the engine receives a heat flux $\dot{Q}(t)$, possibly variable in time. The temperature T at the inlet is *constant* in time. Another part of the control surface, in **blue** in the figure, delimits the movable components through which the engine is releasing mechanical power $-\mathbf{F}(t) \cdot \mathbf{v}(t)$, where $\mathbf{F}(t)$ is the influx of momentum through that part, and $\mathbf{v}(t)$ is the velocity of the matter set into motion; both can vary with time. The expression for the mechanical power has a minus sign because it's the power *we* receive from the engine, so it's an *efflux* for the engine. Through the rest of the control surface there, in **grey**, there are *no exchanges of heat or mechanical power*, but there may be fluxes of matter, momentum, angular momentum, and electromagnetic quantities. At the end of a cycle the volume contents of all these quantities are, by definition of 'cycle', the same as at the beginning. From the balance laws, the time-integrated fluxes of all these quantities must be therefore be zero over a cycle.



But note that the time-integrated flux of heat $\dot{Q}$ and the mechanical power $-\mathbf{F} \cdot \mathbf{v}$ can both be non-zero over a cycle: the balance of energy requires only that their *sum* be zero.

The engine starts a cycle at time t_0 and operates until time t_1 , at which time its physical state is exactly the same as at t_0 . The cycle is complete, and the engine is ready to start a new operation cycle. Let's introduce symbols for the time-integrated amount of heat we provide to the engine and the total amount of work we *receive* from it:

$$\Delta Q := \int_{t_0}^{t_1} \dot{Q}(t) dt \quad \Delta W := - \int_{t_0}^{t_1} \mathbf{F}(t) \cdot \mathbf{v}(t) dt$$

The net amount of total energy flowing into the control volume between t_0 and t_1 is therefore

$$\int_{t_0}^{t_1} W(t) dt = \int_{t_0}^{t_1} \dot{Q}(t) dt + \int_{t_0}^{t_1} \mathbf{F}(t) \cdot \mathbf{v}(t) dt \equiv \Delta Q - \Delta W$$

We already remarked that this time-integrated flux must be zero: from the balance of energy and the condition of cyclic operation:

$$E(t_1) = E(t_0) + \int_{t_0}^{t_1} W(t) dt \equiv E(t_0) + \Delta Q - \Delta W$$

we find

$$\Delta W = \Delta Q \quad (15.4)$$

that is, the mechanical work produced in a cycle must be equal to the total amount of heat provided in a cycle, as expected.

The balance of energy doesn't require more than this. According to the balance of energy this engine is therefore admissible, as long as "heat in = work out".

Let's see what the balance of entropy says. We have

$$\begin{aligned} S(t_1) &\geq S(t_0) + \int_{t_0}^{t_1} \Pi(t) dt = S(t_0) + \int_{t_0}^{t_1} \frac{\dot{Q}(t)}{T} dt && \text{constit. relation for entropy flux} \\ &= S(t_0) + \frac{1}{T} \int_{t_0}^{t_1} \dot{Q}(t) dt && \text{temperature was assumed constant} \\ &= S(t_0) + \frac{\Delta Q}{T} && \text{notation for total heat inflow} \\ &\Rightarrow S(t_1) \geq S(t_0) + \frac{\Delta Q}{T} \end{aligned}$$

Also in this case the initial and final entropy content in a cycle must be the same: $S(t_1) = S(t_0)$. We find

$$\frac{\Delta Q}{T} \leq 0 \quad \Rightarrow \quad \Delta Q \leq 0 \quad (15.5)$$

because $T > 0$

This is a remarkable result: *it's impossible to give a positive net amount of heat, in a cycle, to such an engine.* Otherwise the entropy-balance law would be broken.

Together with the equality (15.4) obtained from the balance of energy, we also find that

$$\Delta W \leq 0 \quad (15.6)$$

that is, *it's impossible to receive a positive net amount of work, in a cycle, from such an engine.*

We conclude that *an engine designed in this way cannot give us any positive net work.* No matter which kind of ingenious technology or materials we tried to use, we would never be able to cyclically gain positive mechanical work from it.

Note that the opposite use, though, is physically possible: we can, cyclically, provide positive work to the engine and receive positive heat

out at a constant temperature. Indeed this is how many heating systems operate.

It's important to understand correctly what's possible and what's not. We *can* of course provide positive heat to the device, at a constant temperature, as much as we please. The result above says that *we won't be able to return the device to its initial state* as long as we do so. The incompatibility is between:

- positive net amount of heat,
- constant temperature,
- cyclic operation.

But note that the converse is possible: we can *extract* a positive net amount of heat from the device, at constant temperature, as much as we please, and return the device to its initial state.



Exercise 15.3

An important result of this section is the inequality (15.5), which says that this engine cannot absorb a positive amount of heat in a cycle. Now think of when, in a cold day, you are standing still in front of a fire, or under the sun. There's no doubt that you are *absorbing* a positive amount of heat, and the temperature could be constant and uniform. Why isn't this situation in contradiction with inequality (15.5)?

§ 15.9 Two-temperature thermal engines: bounds on mechanical work

Let's try a different design, where one of the conditions in the previous section is dropped. The ability to operate cyclically is very convenient, so we keep it. What happens if we let heat be exchanged at different temperatures?

Let's modify the design so that the exchange of heat happens at two different temperatures. This could be done by changing the temperature of one part of the surface over time (within a cycle), or by allowing heat to be exchanged at two different inlets, having constant but different temperatures. We choose the second option because it's easier to analyse and leads to the same results as the first.

The new engine design is schematized in the side figure. An influx of heat $\dot{Q}^+(t)$ occurs through a part of the closed control surface, in **dark red**, at constant temperature T^+ ; another influx of heat $\dot{Q}^-(t)$ occurs through another part of the surface, in **light red**, at constant temperature T^- . Either flux can be positive or negative. For definiteness let's say that

$$T^+ > T^-$$

but we are not making assumptions about relative magnitudes of $\dot{Q}^+(t)$ and $\dot{Q}^-(t)$. Through another, movable part of the control surface, in **blue**, the engine is releasing mechanical power $-\mathbf{F}(t) \cdot \mathbf{v}(t)$, which can also be positive or negative. Through the rest of the surface, in **grey**, there may be fluxes of matter and other quantities, except heat and mechanical power; the time-integrated fluxes of such quantities are zero over an operation cycle.

Consider a cycle of the engine between times t_0, t_1 . Employ similar symbols as before:

$$\begin{aligned} \Delta Q^+ &:= \int_{t_0}^{t_1} \dot{Q}^+(t) dt & \Delta Q^- &:= \int_{t_0}^{t_1} \dot{Q}^-(t) dt \\ \Delta W &:= - \int_{t_0}^{t_1} \mathbf{F}(t) \cdot \mathbf{v}(t) dt \end{aligned}$$

so that the net amount of energy flowing into the control volume in this cycle is

$$\int_{t_0}^{t_1} W(t) dt = \Delta Q^+ + \Delta Q^- - \Delta W$$

The balance of energy applied to the engine's control volume leads to:

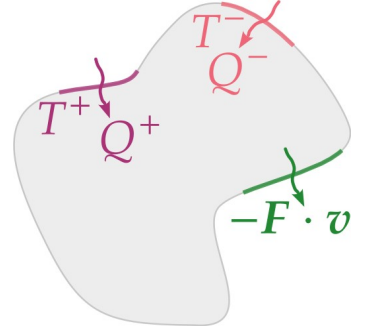
$$\begin{aligned} E(t_1) &= E(t_0) + \Delta Q^+ + \Delta Q^- - \Delta W, & E(t_1) &= E(t_0) \\ \implies & \Delta W = \Delta Q^+ + \Delta Q^-. \end{aligned} \quad (15.7)$$

That is, the net work released in a cycle must equal to the net heat provided, as expected. Obviously we would like ΔW to be positive; let's see if this is possible with our new engine design.

The time-integrated flux of entropy for the engine's control volume is

$$\int_{t_0}^{t_1} \Pi(t) dt = \int_{t_0}^{t_1} \left[\frac{\dot{Q}^+(t)}{T^+} + \frac{\dot{Q}^-(t)}{T^-} \right] dt \equiv \frac{\Delta Q^+}{T^+} + \frac{\Delta Q^-}{T^-}$$

where we have again used our short symbols for the time-integrated heat, and taken into account that the temperatures are constant.



The balance of entropy, during a cycle, therefore says

$$\begin{aligned} S(t_1) &\geq S(t_0) + \frac{\Delta Q^+}{T^+} + \frac{\Delta Q^-}{T^-}, & S(t_1) &= S(t_0) \\ \implies &\frac{\Delta Q^+}{T^+} + \frac{\Delta Q^-}{T^-} \leq 0. \end{aligned}$$

For this new engine, the entropy balance is not saying that the net amount of heat provided in a cycle must be negative or zero. It looks like the new engine design might work.

With a little algebra, and recalling that a thermodynamic temperature is always positive, so that $T^+ > T^- > 0$, we can rewrite the inequality above as follows:

$$\Delta Q^- \leq -\frac{T^-}{T^+} \Delta Q^+. \quad (15.8)$$

This inequality has several interesting consequences. Suppose for instance that the net amount of heat ΔQ^+ provided at the higher temperature is strictly positive. The fraction T^-/T^+ is also positive. Then ΔQ^- must be strictly negative, because it must be smaller than a strictly negative quantity. The entropy balance therefore says that *in a cycle, if the net amount of heat provided at the higher temperature is positive, then the net amount provided at the lower temperature must be negative*.

But the more interesting consequence of the inequality above concerns our main question: does this new two-temperature design allow us to receive positive net work ΔW from the engine, in a cycle? To answer this question, add ΔQ^+ to both sides of the inequality:

$$\Delta Q^+ + \Delta Q^- \leq \Delta Q^+ - \frac{T^-}{T^+} \Delta Q^+.$$

The left side is equal to the net work gained in a cycle, from formula (15.7), which comes from the balance of energy. The right side can be rewritten with a little algebra. We obtain the following important inequality:

$$\Delta W \leq \left(1 - \frac{T^-}{T^+}\right) \Delta Q^+ \quad (15.9)$$

This is the combined consequence of the balances of energy and entropy for this cyclic thermal engine. The factor $1 - \frac{T^-}{T^+}$ on the right side is called the **efficiency** of the thermal engine. Since thermodynamic temperature

cannot be negative, the efficiency cannot be greater than 1; and since $T^- < T^+$ by design, the efficiency cannot be negative:

$$0 < 1 - \frac{T^-}{T^+} < 1.$$

Does the inequality (15.9) allow us to get positive net work in a cycle? Yes! The thermal-efficiency factor is positive. If the heat ΔQ^+ provided to the engine at the higher temperature is also positive, then the right side of that inequality is positive. The net work can therefore be positive too, as long as it's less than or equal to the positive quantity on the right side. For instance,

$$\text{if } T^- = 300 \text{ K}, T^+ = 400 \text{ K}, \Delta Q^+ = 1000 \text{ J} \quad \text{then} \quad \Delta W \leq 250 \text{ J}.$$

and we could therefore obtain, say, 249 J of mechanical work at every cycle. Our new engine design is successful!

Note the upper bound ' $\Delta W \leq \dots$ ' in formula (15.9), on how much work we can gain in a cycle. We can only get strictly less than the heat we provide at the highest temperature. At first, one might wonder why this must be the case, because we could also provide heat at the lower temperature, heat which could be converted into work as well. But we must remember the inequality (15.8) for ΔQ^- : if ΔQ^+ is positive – and we need it to be positive if we want to gain positive work – then ΔQ^- must be negative. So we're losing some energy as heat there. If we try to eliminate that heat loss by making ΔQ^- zero, then the inequality (15.8) says that ΔQ^+ must be zero too – and then the work gained is zero or negative. In fact if ΔQ^- were zero then we would effectively be back to our previous ineffective one-temperature engine design.

Let's see how to maximize the amount of work ΔW that we can gain, for a given amount of heat ΔQ^+ provided to the engine. Inequality (15.9) shows that there are two ways, both of which increase the efficiency; the two are not mutually exclusive:

- decrease as much as possible the temperature T^- at which heat is released from the engine;
- increase as much as possible the temperature T^+ at which heat is provided to the engine.

It is amazing that we can predict how much work can be obtained from such an engine, without knowing or needing to specify what kind of technology, materials, and way of operation the engine could be based

upon. You see the strength of the consequences of the “ $\geq$ ” sign in the entropy balance.

The thermal-engine example above also hints at the role of the balance of entropy as a meta-law about constitutive relations. In a real application and construction of an engine, the heat flux $\dot{Q}$ and momentum flux $\mathbf{F}$ will be specified by constitutive relations; think for instance about [Newton’s law of cooling](#) ^{§11.14 p.350} for $\dot{Q}$, or about the [ideal-gas law](#) ^{§10.18 p.287} for $\mathbf{F}$. But if a limitation such as the maximal work efficiency (15.9), which we can rewrite in full as

$$-\int_{t_0}^{t_1} \mathbf{F}(t) \cdot \mathbf{v}(t) \, dt \leq \left(1 - \frac{T^-}{T^+}\right) \int_{t_0}^{t_1} \dot{Q}^+(t) \, dt ,$$

is to be universally valid, then the specific mathematical formulae for $\dot{Q}$ and $\mathbf{F}$ cannot be arbitrary. Their mathematical expressions must have severe restrictions.



Exercise 15.4

1. Find an upper bound for the amount of work gained similar to formula (15.9), but in terms of T^+ , T^- , and the heat ΔQ^- provided at the lower temperature. (*Hint: use formula (15.7) to express ΔQ^+ in terms of ΔW and ΔQ^- ; replace in formula (15.9); solve for ΔW .*)
2. A two-temperature engine receives, in a cycle, an amount of heat of 500 J at a temperature of 400 K. You want to gain 300 J of work from this engine over a cycle. What’s the maximum temperature T^- at which heat can be discharged from the engine, in order to achieve your goal?

§ 15.10 Example application: friction coefficient

In a previous discussion about [surfaces of discontinuity](#) ^{§11.16 p.365} we analysed the flux of total-energy at the surface of contact between floor and a crate pushed or dragged on it.

URLs for chapter 15

1. https://encyclopediaofmath.org/wiki/Clausius-Duhem_inequality
2. <https://plato.stanford.edu/entries/statphys-statmech/>
3. <https://plus.maths.org/content/information-surprise>

Postface to the teacher

“What you do in this world is a matter of no consequence,”
returned my companion, bitterly. “The question is, what
can you make people believe that you have done? . . .”

Sherlock Holmes (A. C. Doyle) [1887](#)

The majority of everyday and forefront technologies is based on physical phenomena at the intersection of traditional categories such as “mechanics”, “electromagnetics”, “thermodynamics”. In many cases it is not clear whether these phenomena should be labelled as belonging to one category rather than another. Sometimes such labelling is artificial, more misleading than helpful. How to prepare students in physics or engineering for the myriad of possible physics specializations and applications lying ahead? The problem nowadays is not only that the students may not know yet which physics field they’ll want to pursue, but also that they’ll likely need some knowledge of all other fields anyway.

I believe that the best approach to this problem is by teaching them physical notions and physical laws that are common to as many physical phenomena as possible, and can be used in as many physical applications as possible. Notions and laws that the students will always be able to use afterwards, and upon which they can gradually build more specialized knowledge.

As pointed out in the [Preface](#), *we do actually have such notions and laws:*

- what we can call ‘matter’ or ‘substance’ or ‘particle number’
- energy
- momentum
- angular momentum (including boost momentum)
- magnetic flux
- electric charge
- entropy

These quantities and their balances are usually recognized and emphasized in books on continuum physics, non-relativistic as well as relativistic. The classic treatise *The Classical Field Theories* ([1960](#)), for instance, states this

programme very clearly (as explained in chapter 15, the balance of entropy is a *metalaw* rather than a law):

Motion, stress, energy, entropy, and electromagnetism are the concepts upon which field theories are constructed. Certain laws of *conservation* or *balance* are laid down as relating these quantities in all cases. These basic principles, which are in integral form, in regions where the variables change sufficiently smoothly are equivalent to differential *field equations* [. . .]. The field equations [. . .] form an underdetermined system, insufficient to yield specific answers unless further equations are supplied. Within the embracing concept of the balanced fields, it is possible to define *ideal materials* by certain further conditions. These defining conditions are called *constitutive equations*.

Besides being common to *all* physical phenomena, and to *all* our main physical theories, the notions and laws above have outstanding pedagogical features:

- They are few.
- They are easy to understand intuitively, because they express the idea of a “budget” or balance.
- They can be mathematically expressed by a common formula that is simple and yet can be directly applied, *as-is*, to all physical phenomena, theories, and disciplines.
- And this formula is directly and intuitively connected with the ideas of *prediction* and *simulation*.

Thanks to these amazing features, the students can be taught to recognize these notions and their mathematical “budgets” in the great variety of physical phenomena around them.

What will be left to learn – probably a lifelong learning – is the great variety of ways in which these seven basic notions can be mathematically connected to one another. It is this latter variety of *constitutive relations*, which depend on the physical phenomenon, that the students will thereafter specialize into. But note the advantage of having established a foundation of a few simple universal notions: whenever our future engineers and physicists will need to deal with physical phenomena outside their specialization, they will only have to learn new ways to connect the physical quantities that *they already know*.

From a pedagogical point of view, one could hardly ask for a better internal simplicity out of a discipline. The present, very imperfect notes try to exploit this pedagogical potential.

Many current physics textbooks take a different approach. They introduce the students to a mixture of some notions and laws that are common to all physical phenomena, but together with others notions and laws that are specific to a particular physical field: the Newtonian mechanics of *point-particles*, with its intuition of momentum as “mass times velocity”. In my opinion this approach has many grave drawbacks:

- The mixture of general and discipline-specific notions is often not clearly separated, and this lack of separation leads to confusion. Momentum as “mass times velocity” is not universal. “ $\mathbf{F} = m\mathbf{a}$ ” breaks down in many physical phenomena. Then which law, if any, is still valid in its stead? The “law of action and reaction” is not universal either. Then can any similar law be universally extended? What happens to angular momentum and its balance then? If an electromagnetic field exerts a force on a mass, is the mass also exerting a force on the electromagnetic field? how can the latter sustain a force?
- This approach unnecessarily suppresses some useful intuitions that many students have from everyday life, such as the distinction between contact forces and body forces. The students will later have to recover these intuitions as they study more general phenomena.
- This particular mixture of notions and laws is today used almost only in textbooks; very little in physical and engineering applications. Students who will research and work in actual mechanics – buildings, bridges, aeroplanes, fluid flow, and generally extended bodies – will need to amend their *point-particle* intuitions to a *continuum* one. Students who will research and work in electromagnetics will need to amend their intuition of momentum as “mass times velocity”. Students who will research in particle physics will also need to amend their intuition of momentum in several different ways. Students who will work in General Relativity will need all these amendments at once. The increasing importance of amending our Newtonian intuition of space and time was discussed in chapter 2.

Newtonian point-mass mechanics is less and less used in astronomy as well. Ephemerides use post-Newtonian approximations of General

“The plot for Cesium [...] characterizes the best orbiting clocks in the GPS system. What this means is that after initializing a Cesium clock, and leaving it alone for a day, it should be correct to within [...] 4 nanoseconds. Relativistic effects are huge compared to this.”

Ashby 2003

Relativity (Park et al. 2021). NASA and the [Jet Propulsion Laboratory](#)⁴ by default include relativistic effects (Moyer 2000) when they plan or calculate trajectories for Earth, Moon, and beyond. The same general-relativistic formulae are used to calculate and plan spacecraft dynamics, for navigating in [cis-lunar](#)⁵ or geocentric space as well as for interplanetary missions: the same software is used for navigating in both regimes (Park & Chodas 2024).

- The students typically end up with the understanding that “every phenomenon is just a consequence of Newton’s laws”, of the second law in particular. This is clearly false: this law is just one out of six or seven that determine the evolution of physical phenomena.
- A fault, not of this approach per se, but of the way it is taught, is that students often remain confused about what this all-powerful second law *exactly* say, and about its precise mathematical expression. Is it “ $\mathbf{F} = m\mathbf{a}$ ”? or is it “ $\mathbf{F} = d\mathbf{P}/dt$ ”? Why do many extra added terms suddenly appear when this “second law” of Newton’s is used in continuum mechanics? is it still the same law?
- Science and education have had the noble tradition of founding their teachings on the notions of the theories that proved to be most correct. That is how we got Newtonian mechanics and electromagnetics in our schools. Today we know that Newtonian mechanics and some of its intuitions are only approximate; whereas the predictions and explanations offered by General Relativity (and quantum theory) keep on being beautifully confirmed. It’s time we continue our noble tradition and replace those Newtonian notions that are only approximate with more exact ones.

The student is, in other words, introduced to notions and a physical understanding that are fuzzy, are partially incorrect according to our present understanding of physics, and moreover cannot be used as-is, but will instead require revisions – some of which are quite drastic (I say this out of my own experience as a physics student, researcher, and teacher).

One might hear the argument that the teaching approach via Newtonian point-mass mechanics is closer to our “everyday intuition”. But that’s a topsy-turvy argument. Our everyday intuition *comes from* that teaching approach. As an analogy, someone in the 16th century could have said that it’s better to teach the [geocentric model](#)⁶ of the solar system, than the heliocentric one, because the former is closer to everyday intuition. Yet our

children today quickly develop a heliocentric intuition, simply because it's the one that enters our education from the very start. Similar arguments could be made for other concepts, such as energy or the electromagnetic field, that once were not part of everyday intuition, but today are.

Validity of the mathematical form of the balance laws in General Relativity

These notes state several times that the mathematical form of the equation for balance, for instance

$$\frac{dE(t)}{dt} = W(t) + \mathcal{A}(t) \quad (15.10)$$

or its integral expression, is also valid in General Relativity. I would like to give a brief explanation, if not a proof, of this fact for those who are not familiar with General Relativity.

✂ To be continued

The balance equations of the first six quantities are commonly used in numerical relativity for magneto-hydrodynamical problems. Here's an example reproduced from the textbook by Baumgarte & Shapiro [2010](#):

Box 5.1 The relativistic MHD equations

The coupled set of relativistic MHD equations can be written in conservative form as follows:

$$\partial_t \rho_* + \partial_j (\rho_* v^j) = 0, \quad (5.168)$$

$$\partial_t \tilde{S}_i + \partial_j (\alpha \sqrt{\gamma} T^j_i) = \frac{1}{2} \alpha \sqrt{\gamma} T^{ab} g_{ab,i}, \quad (5.169)$$

$$\partial_t \tilde{\tau} + \partial_i (\alpha^2 \sqrt{\gamma} T^{0i} - \rho_* v^i) = s_{\tilde{\tau}}, \quad (5.170)$$

$$\partial_t \tilde{B}^i + \partial_j (v^j \tilde{B}^i - v^i \tilde{B}^j) = 0, \quad (5.171)$$

where we have the balances of matter, momentum, energy, and magnetic flux; the balance of angular momentum is satisfied implicitly by the symmetry of the four-stress tensor T^{ab} , and the balance of charge by the use of a specific constitutive relation. In another example, Komissarov [2005](#) merges together the balances of momentum and energy:

The evolution equations of ideal MHD include the continuity equation,

$$\partial_t(\alpha\sqrt{\gamma}\rho u^t) + \partial_i(\alpha\sqrt{\gamma}\rho u^i) = 0, \quad (2)$$

the energy–momentum equations,

$$\partial_t(\alpha\sqrt{\gamma}T^t_v) + \partial_i(\alpha\sqrt{\gamma}T^i_v) = \frac{1}{2}\partial_v(g_{\alpha\beta})T^{\alpha\beta}\alpha\sqrt{\gamma}, \quad (3)$$

and the induction equation,

$$\partial_t(B^i) + e^{ijk}\partial_j(E_k) = 0. \quad (4)$$

✂ To be continued

URLs for chapter *Postface to the teacher*

4. <https://www.jpl.nasa.gov>
5. <https://csp.aerospace.org/papers/cislunar-development-what-build-and-why>
6. <https://cmb.physics.wisc.edu/pub/tutorial/briefhist.html>

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Believe nothing, O monks, merely because you have been told it, or because it is traditional, or because you yourselves have imagined it. Do not believe what your teacher tells you merely out of respect for the teacher.

(Attributed to Gautama Buddha)

("de X" is listed under D, "van X" under V, and so on, regardless of national conventions.)

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